12. Installation Instruction

Attached accessories

No.	Accessory part	Qty.
1	Adjustable Feet 	4
2	Drain Elbow 	1
3	Packing for drain 	1
4	Network Adaptor (CZ-TAW1B) 	1

Optional Accessories

No.	Accessories part	Qty.
5	Remote Controller Case	1
6	Extension Cable (CZ-TAW1-CBL)	1
7	Optional PCB (CZ-NS5P)	1

Field Supply Accessories (Optional)

No.	Part		Model	Specifications	Maker
i	2-way valve kit *Cooling model	Electromotoric Actuator	SFA21/18	AC230V, 12 VA	Siemens
		2-port Valve	VXI46/25		Siemens
ii	Room thermostat	Wired	PAW-A2W-RTWIRED	AC230V	-
		Wireless	PAW-A2W-RTWIRELESS		
iii	Mixing valve	-	167032	AC230V, 6VA	Caleffi
iv	Pump	-	Yonos 25/6	AC 230V, 0.6 A max	Wilo
v	Buffer tank sensor	-	PAW-A2W-TSBU	-	-
vi	Outdoor sensor	-	PAW-A2W-TSOD	-	-
vii	Zone water sensor	-	PAW-A2W-TSHC	-	-
viii	Zone room sensor	-	PAW-A2W-TSRT	-	-
ix	Solar sensor	-	PAW-A2W-TSSO	-	-
x	Wall mounting bracket	-	-	-	-

■ It is recommended to purchase the field supply accessories listed in above table.

12.1 Indoor Unit (For WH-ADC0509L3E51 only)

12.1.1 Select the Best Location

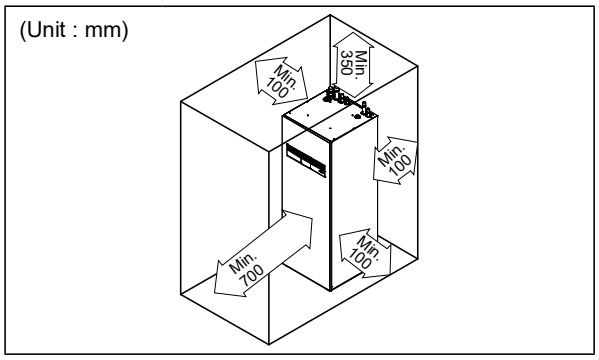
Before choosing the installation site, obtain user approval.

- Install the Tank Unit in indoors with frost free weather proof location only.
- Must install on a flat horizontal and solid hard surface.
- There should not be any heat source or steam near the Tank Unit.
- A place where air circulation in the room is good.
- A place where drainage can be easily done (e.g. Utility room).
- A place where Tank Unit's operation noise will not cause discomfort to the user.
- A place where Tank Unit is far from door way.
- A place where accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated below from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- Secure the Tank Unit to prevent it being knocked over accidentally or during earthquakes.

Please avoid installations which expose the Tank Unit to any of the following conditions:

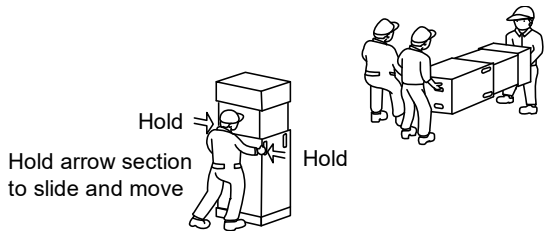
- Extraordinary environment conditions; installation in frost or exposure to unfavorable weather conditions.
- Voltage input exceeding the specified voltage.

12.1.1.1 Required Space for Installation



12.1.1.2 Transport and Handling

- Be careful during transportating the unit so that it is not damaged by impact.
- Only remove the packaging material once it has reached desired installation location.
- It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.
- The Tank Unit can be transported either in vertical or horizontal.
  - If it transported in horizontal, make sure Front of packaging material (printed with "FRONT") must facing upwards.
  - If it transported in vertical, use the hand holes on sides, slide and move to the desired location.
- Fix the Adjustable Feet 1 , if the Tank unit installed on a uneven surface.

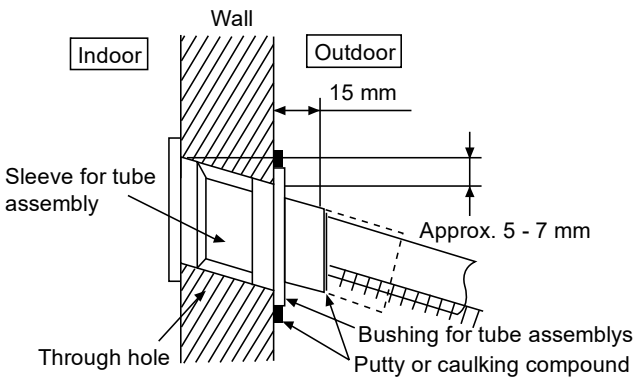


12.1.2 To Drill a Hole in the Wall and Install a Sleeve of Piping

1. Make a through hole. (Check pipe diameter and insulation thickness)
2. Insert the piping sleeve to the hole.
3. Fix the bushing to the sleeve.
4. Cut the sleeve until it extrudes about 15 mm from the wall.

⚠ CAUTION
When the wall is hollow, please be sure to use the sleeve for tube assembly to prevent dangers caused by mice biting the connection cable.

5. Finish by sealing the sleeve with putty or caulking compound at the final stage.

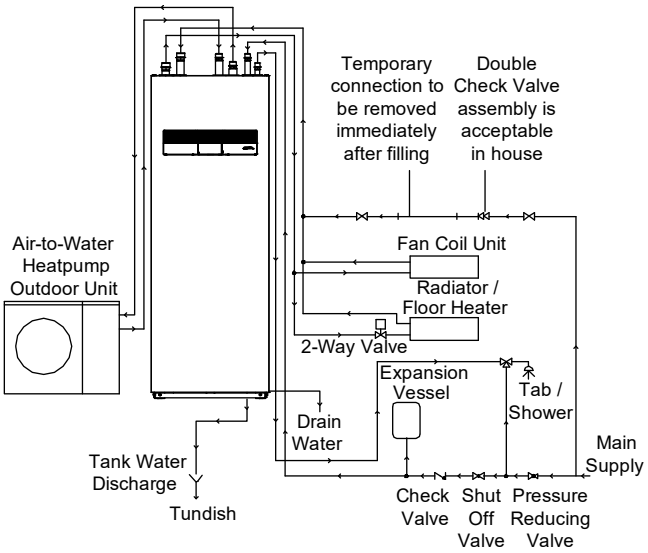


12.1.3 Piping Installation

12.1.3.1 Water Quality Requirement

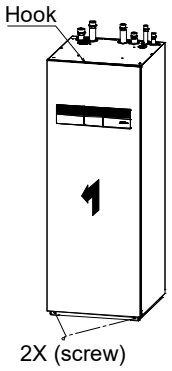
Must use water that complies with European Drinking Water Directive (EU)2020/2184. The lifespan of the Tank Unit will be shorter if groundwater (include spring water and well water) is used.
The Tank Unit shall not be used with the tap water containing contaminants such as salt, acid, and other impurities which may corrode the tank and its component.
Use sanitized water that is free of Legionella and other bacteria and microorganisms. If the water contains Legionella bacteria, it may harm the health of the user.

12.1.3.2 Typical Piping Installation



12.1.3.3 Access to Internal Components

⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

*The remote control cable is connected to the front panel, so be careful when removing the panel.

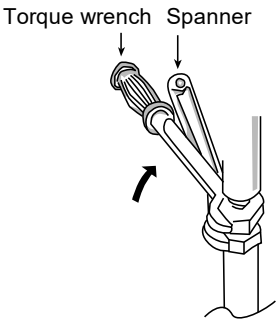
Open and Close Front Plate ⑱

1. Remove the 2 mounting screws of Front Plate ⑱.
2. Slide it upwards to unhook the Front Plate ⑱ hook.
3. Reverse above steps 1~2 for close it.

12.1.3.4 Water Piping Installation

- Please engage a licensed water circuit installer to install this water circuit.
- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not use worn out tube or detachable hose-set.
- Do not apply excessive force to pipes that may damage the pipes.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Further tighten the nuts with torque wrench in specified torque as stated in the table.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall.
- If non-brass metallic piping is used for installation, make sure to insulate the pipes to prevent galvanic corrosion.
- Do not connect galvanised pipes, this will cause galvanic corrosion.
- Use correct nut for all Tank Unit tube connections and clean all tubes with tap water before installation. See Tube Position Diagram for detail.

Tube Connector	Nut Size	Torque
ⓐ & ⓑ	RP 1¼"	117.6 N•m
ⓒ & ⓓ	RP ¾"	58.8 N•m
ⓔ & ⓕ	RP 1"	88.2 N•m



⚠ CAUTION
Do not overtighten, overtightening may cause water leakage.

- Make sure to insulate the water circuit pipes to prevent reduction of heating capacity.
- After installation, check the water leakage condition in connection area during test run.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Protection From Frost:
  If the Tank Unit is being exposed to frost while power supply failure or pump operating failure, drain the system. When water is idle inside the system, freezing up is very likely to happen which could damage the system. Make sure the power supply is turned off before draining. Heater Assembly ⑩ may be damaged under dry heating.
- Corrosion Resistance:
  Duplex stainless steel is naturally corrosion resistant to mains water supply. No specific maintenance is required to maintain this resistance. However, please note that Tank Unit is not guaranteed for use with a private water supply.
- It is recommended to use a tray (field supply) to collect water from the Tank Unit if water leakage occur.

Recommended piping installation sequence:
(a) → (c) → (e) → (f) → (b) → (d)

(A) Space Heating/Cooling Pipework

- Connect Tank Unit Tube Connector ⓐ to outlet connector of Zone 1 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓑ to inlet connector of Zone 1 Panel/Floor heater.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Refer below table for the rated flow rate of each particular Outdoor Unit.

Model		Rated Flow Rate (l/min)	
		Cool	Heat
WH-ADC0509L3E51, WH-ADC0509L3E5AN1	WH-WDG05LE5*	14.3	14.3
	WH-WDG07LE5*	20.1	20.1
	WH-WDG09LE5*	23.5	25.8

*Do not install automatic air purge valves on indoor water pipes. In the unlikely event that the R290 refrigerant leaks into the water circuit, there is a risk that the refrigerant will leak indoors.

(B) Circulating Pipework

- Connect Tank unit Tube Connector ⓕ to outdoor unit inlet water socket.
- Connect Tank Unit Tube Connector ⓔ to Outdoor unit outlet water socket.
- Failure to connect results in an error stop the system.

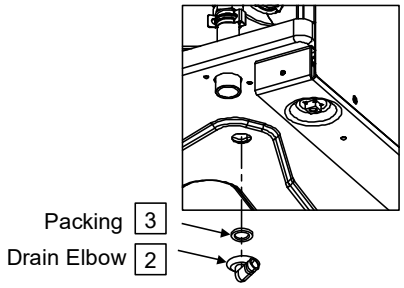
Model	Water piping between outdoor unit and indoor unit			
	Inner diameter	Maximum length	Insulator thickness	Maximum Elevation
WH-WDG05LE5*	ø20	30 m	30 mm or more	10 m
WH-WDG07LE5*	ø25			
WH-WDG09LE5*				

(C) Domestic Hot Water Tank Pipework

- It's strongly recommended to install an expansion vessel (field supply) in the Domestic Hot Water Tank circuit. Refer Typical Piping Installation section to locate the expansion vessel.
  - Recommended pre-charge pressure of the expansion vessel (field supply) = 3.5bar (0.35MPa)
- In high water pressure or water supply is above 5bar, please install the Pressure Reducing Valve for water supply. If the pressure higher than that, it might damage the Tank Unit.
- A Pressure Reducing Valve (field supply) with below specification is strongly advised to be installed along the line of the tube connector ⓒ of Tank Unit. Refer Typical Piping Installation section to locate both of these valves.
  Recommended Pressure Reducing Valve specifications:
  - Set pressure: 3.5bar (0.35MPa)
- Must connect a faucet to Tank Unit Tube Connector ⓓ and main water supply, in order to supply water with appropriate temperature for shower or tap usage. Failure to do so might cause scalding.
- Failure to connect the tube appropriately might causing the Tank Unit malfunction.

(D) Drain Elbow and Hose Installation

- Fix the Drain Elbow [2] and Packing [3] to the bottom of Drain Water Hole ⓗ.



- Use inner diameter 17 mm drain hose in the market, fix to Drain Elbow [2] .
- This hose must to be installed in a continuously downward direction and in a frost-free environment. Improper drain piping may cause water leakage hence damage the furnitures.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Guide the drain hose to outdoor as illustrated.

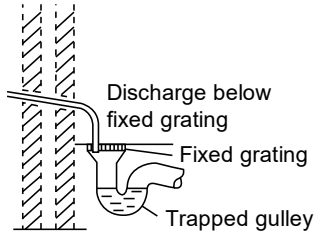


Illustration of guide drain hose to outdoor

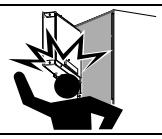
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulphuric gas etc.
- If necessary, use hose clamp to further tighten the hose at drain hose connector to prevent leakage.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.
- If drain hose is in the room (where dew may form), please increase the insulation by using POLY-E FOAM with thickness 6 mm or above.

(E) Domestic Hot Water Tank Discharge (Drain Tap) and Safety Relief Valve Pipework

- Safety Relief Valve 8bar (0.8MPa) incorporated in Domestic Hot Water Tank.
- Drain Tap and Safety Relief Valve discharge fittings share the same drainage outlet.
- Use R½" male connector for this drainage outlet connection (Tube connector ⓖ).
- Piping must always be installed in a continuously downward direction. It must not be longer than 2m, with no more than 2 elbows, and must not allow condensation to build up or freezing to occur.
- The pipe from this drainage outlet fitting must not be shut off. The discharge must be freed.
- The end of this pipework must be in such a way so that the outlet is visible and can not cause any damage. Keep away from electrical components.
- It is recommended to fit a tundish into this ⓖ pipework. Tundish should be visible and positioned away from frost environment and electrical components.

12.1.4 Connect the Cable to the Tank Unit

⚠ WARNING
This section is for authorized and licensed electrician only. Work behind the Control Board Cover ⑤ secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

⚠ CAUTION	
Please take extra precaution when open the control board cover ⑤ and control board ⑥ for unit installation and servicing. Failure to do so may cause injury.	

12.1.4.1 Fixing of Power Supply Cable and Connecting Cable

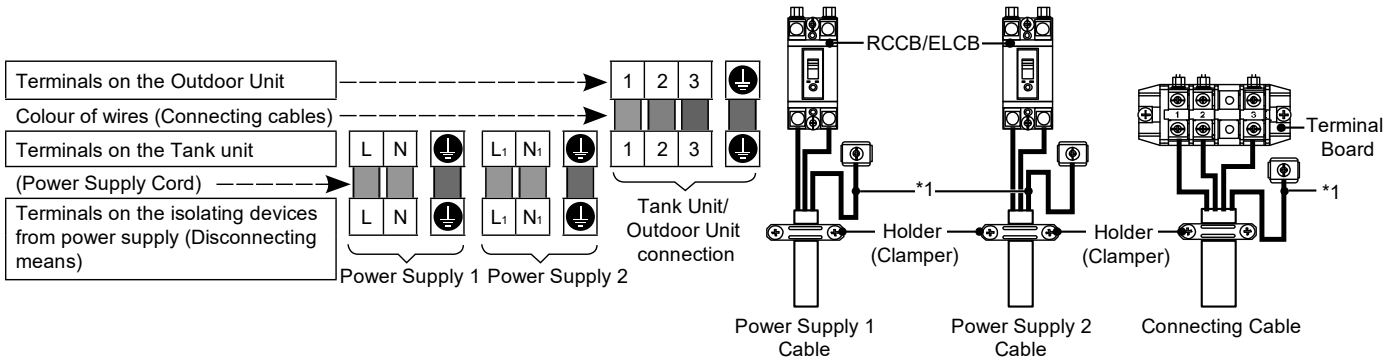
1. Connecting cable between Tank Unit and Outdoor Unit shall be approved polychloroprene sheathed flexible cord, type designation 60245 IEC 57 or heavier cord. See below table for cable size requirement.

Model		Connecting Cable Size
Tank Unit	Outdoor Unit	
WH-ADC0509L3E51, WH-ADC0509L3E5AN1	WH-WDG05LE5* WH-WDG07LE5* WH-WDG09LE5*	4 × min 2.5 mm^2

   - Ensure the colour of wires of Outdoor Unit and the terminal no. are the same to the Tank Unit respectively.
   - Earth wire shall be longer than the other wires as shown in the figure for the electrical safety in case of the slipping out of the cord from the Holder (Clamper).
2. An isolating device must be connected to the power supply cable.
   - Isolating device (disconnecting means) should have minimum 3.0 mm contact gap.
   - Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means). See below table for cable size requirement.

Model		Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
Tank Unit	Outdoor Unit				
WH-ADC0509L3E51, WH-ADC0509L3E5AN1	WH-WDG05LE5* WH-WDG07LE5* WH-WDG09LE5*	1	3 × min 2.5 mm^2	25A	30mA, 2P, type A
		2	3 × min 1.5 mm^2	15/16A	30mA, 2P, type AC

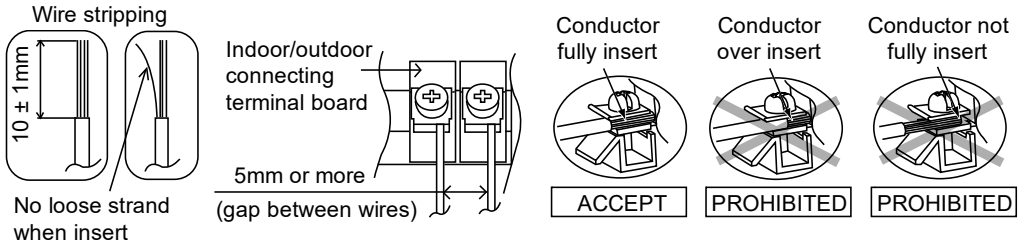
3. To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through a bushing (located at the bottom of Control Board ⑥) before terminal board. The bushing must be used and must not be removed.



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25}

*1 - Earth wire must be longer than other cables for safety reasons

12.1.4.2 Wire Stripping and Connecting Requirement



12.1.4.3 Connecting Requirement

For Tank Unit WH-ADC0509L3E51, WH-ADC0509L3E5AN1 with WH-WDG05LE5*, WH-WDG07LE5*, WH-WDG09LE5*

- The equipment's Power Supply 1 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-3 and can be connected to current supply network.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-3 and can be connected to current supply network.

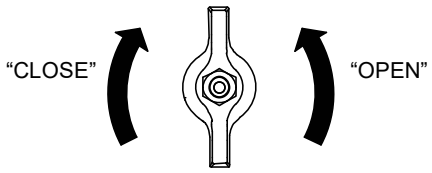
12.1.5 Charging and Discharging the Water

- Make sure all the piping installations are properly done before carry out below steps.

12.1.5.1 Charge the Water

For Domestic Hot Water Tank

1. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⑨ to "CLOSE".

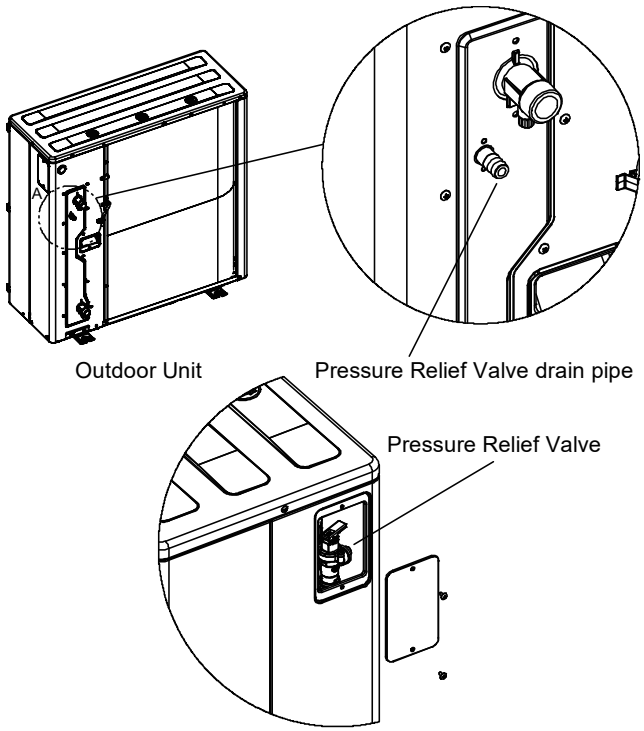


Domestic Hot Water Tank Discharge (Drain Tap) ⑨

2. Set all Tap / Shower "OPEN".
3. Start filling water to the Domestic Hot Water Tank via Tube Connector ⓒ.
   After 20~40min, water should flow out from Tap / Shower.
   Else, please contact your local authorized dealer.
4. Check and make sure no water leaking at the tube connecting points.
5. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⑨ to "OPEN" for 10 seconds to release air from this pipeline. Then set it "CLOSE".
6. Turn the Safety Relief Valve ㉕ knob counterclockwise slightly and hold for 10 seconds to release air from this pipeline. Then recover the knob to original position.
7. Ensure Step 5 & 6 is carried out each time after charging water to Domestic Hot Water Tank.
8. To prevent back pressure from happening to the Safety Relief Valve ㉕, do turn the Safety Relief Valve ㉕ knob counterclockwise.

For Space Heating / Cooling

1. Start filling water (with pressure more than 1 bar (0.1MPa)) to the Space Heating/Cooling circuit via Tube Connector ⓐ.
2. Stop filling water if the free water flow through Pressure Relief Valve drain pipe. (Check the Outdoor Unit)
3. Turn ON the Tank Unit.
4. Remote control menu → Installer setup → Service setup → pump maximum speed → Turn on the pump.
5. Make sure Water Pump ④ is running.
6. Check and make sure no water leaking at the tube connecting points.



12.1.5.2 Discharge the Water

For Domestic Hot Water Tank

1. Turn OFF power supply.
2. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⑨ to “OPEN”.
3. Open Tap / Shower to allow air inlet.
4. Turn the Safety Relief Valve ㉕ knob counterclockwise slightly and hold it until all air is released from this pipeline. Then recover the knob to original position after ensured the pipeline is emptied.
5. After discharge, set Domestic Hot Water Tank Discharge (Drain Tap) ⑨ to “CLOSE”.

12.1.6 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings.

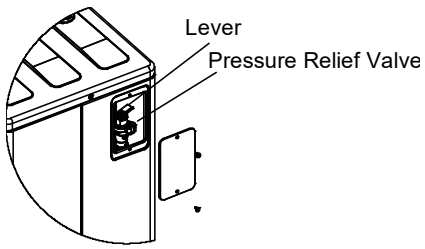
12.1.6.1 Check Water Pressure

*(1 bar = 0.1MPa)
Water pressure should not lower than 0.5 bar (with inspects the Water Pressure from Remote Controller).
If necessary add water into Tank Unit (via Tube Connector ⓐ).

12.1.6.2 Check Pressure Relief Valve

*Pressure Relief Valve is mounted in the Outdoor Unit.

1. Confirm that the pressure relief valve is working properly, Pull the lever horizontal direction.
2. Release the lever when water comes out of the drain pipe of the pressure relief valve.
   (While the air continues to come out of the drain pipe, keep raising the lever to completely discharge the air.)
3. Confirm that the water from the drain pipe stops.
4. If water is leaking, pull the lever several times and return it to make sure the water stops.
5. If water keeps coming out of the drain, drain water.
   Turn off the system and contact your local authorized dealer.



12.1.6.3 Check Air Accumulation

- Open the air vent plugs on the heating panel, fan convector, etc., and remove the air accumulated in the equipment and piping.
- If the outdoor unit and the indoor unit are installed on different floors, open the air vent plug on the water plug of the outdoor unit and the air vent plug on the heater bottle inside the indoor unit to remove the air. (be careful, water will come out)

12.1.6.4 Expansion Vessel ⑬ Pre Pressure Checking

For Space Heating / Cooling

- Expansion Vessel ⑬ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit’s piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m. (Extra pump may be required)

12.1.6.5 Check RCCB/ELCB

Ensure the RCCB/ELCB set to “ON” condition before check RCCB/ELCB.
Turn on the power supply to the Tank Unit.
This testing could only be done when power is supplied to the Tank Unit.

⚠ WARNING
Be careful not to touch parts other than RCCB/ELCB test button when the power is supplied to Tank Unit. Else, electrical shock may happen. Before obtaining access to terminals, all supply circuits must be disconnected.

- Push the “TEST” button on the RCCB/ELCB. The lever would turn down and indicate “0”, if it functions normal.
- Contact authorized dealer if the RCCB/ELCB malfunction.
- Turn off the power supply to the Tank Unit.
- If RCCB/ELCB functions normal, set the lever to “ON” again after testing finish.

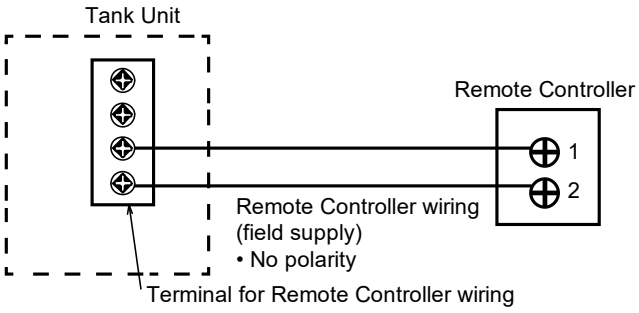
12.1.7 Installation of Remote Controller as Room Thermostat

- Remote Controller ① mounted to the Tank Unit can be moved to the room and serve as Room Thermostat.

12.1.7.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1. By the window, etc. exposed to direct sunlight or direct air.
  2. In the shadow or backside of objects deviated from the room airflow.
  3. Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4. Location near heat source.
  5. Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

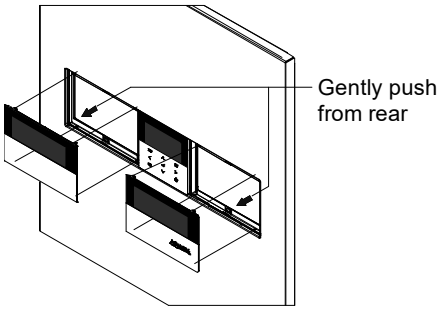
12.1.7.2 Remote Controller Wiring



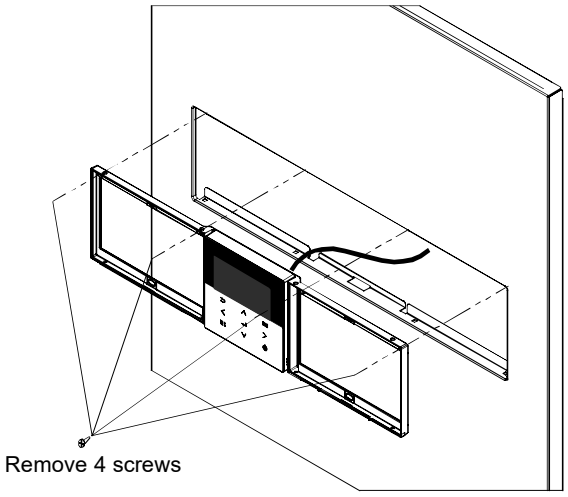
- Remote Controller cable shall be (2 × min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals of Tank Unit (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.
- When using the 2nd. Remote Controller (option), connect it to the terminal of the tank unit by tightening it together.

12.1.7.3 Remove the Remote Controller from Tank Unit

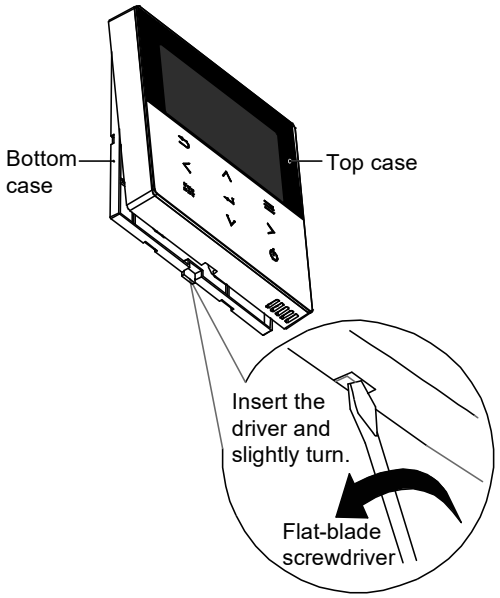
1. Remove both Left Decoration Panel ② and Right Decoration Panel ③ from Front Plate ⑱ with gently push the panels from back.



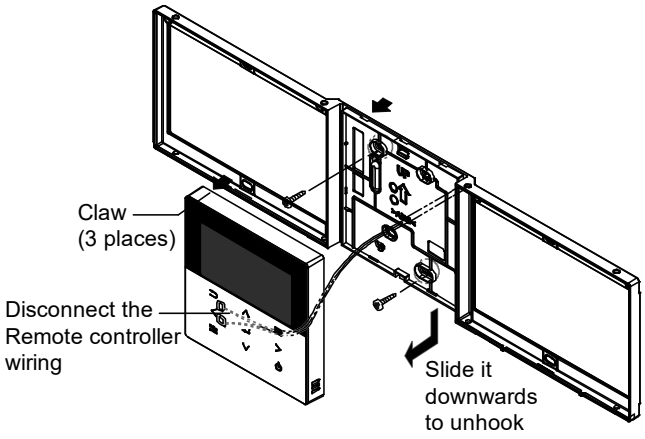
2. Remove the 4 screws and take out the holder with Remote Controller ①.



3. Remove the top case from the bottom case.

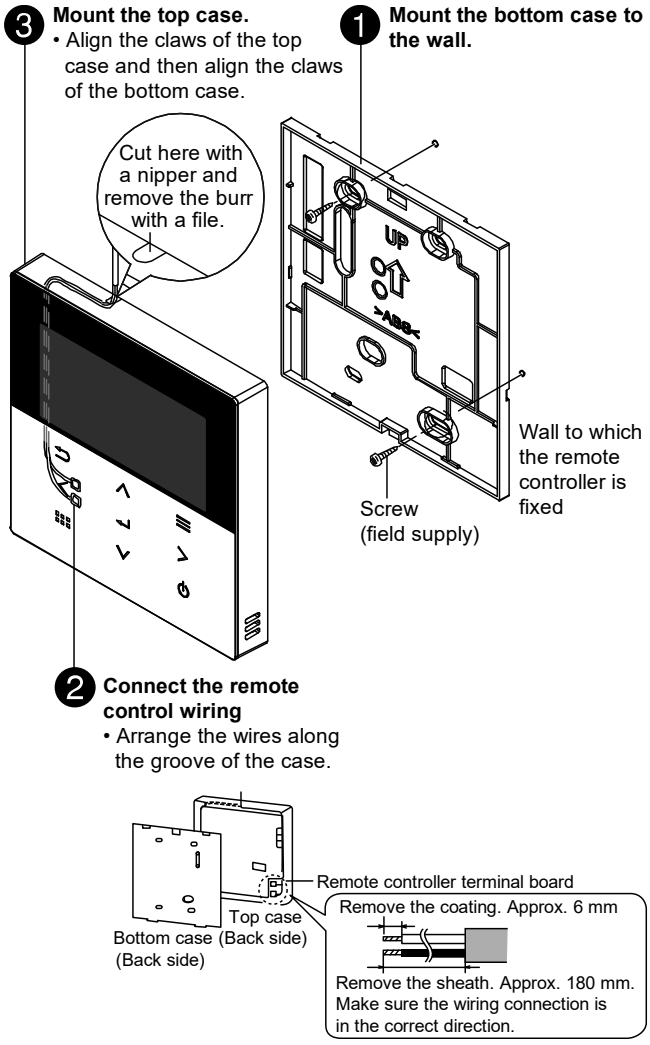


4. Remove the wiring between Remote controller ① and Tank Unit terminal.

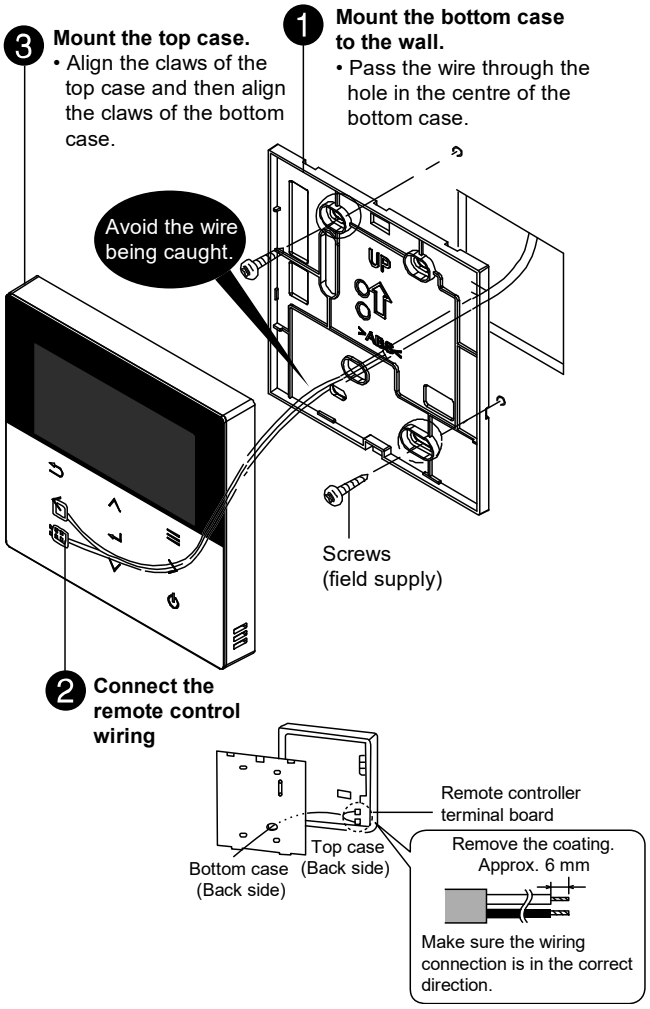


12.1.7.4 Mounting the Remote Controller

For exposed type
Preparation: Make 2 holes for screws using a driver.



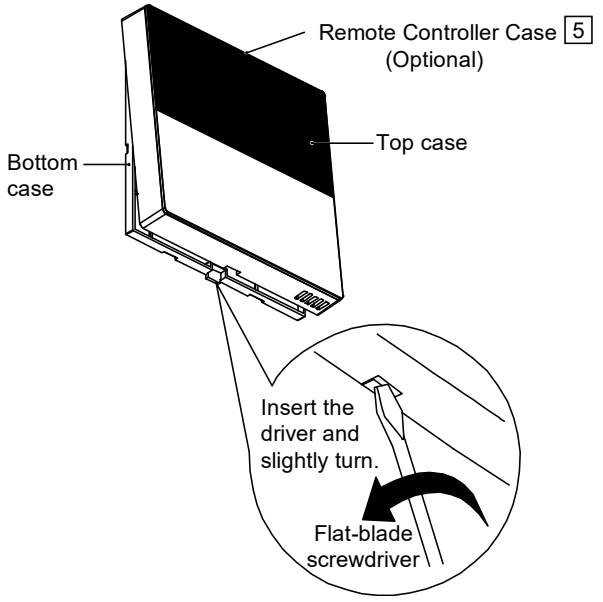
For embedded type
Preparation: Make 2 holes for screws using a driver.



12.1.7.5 Replace the Remote Controller Cover

- Replace the existing Remote Controller with Remote Controller Case [5] to close the hole left after remove the Remote Controller.

1. Refer Section “Remove The Remote Controller From Tank Unit” for remove Remote Controller.
2. Remove the top case from the bottom case of Remote Controller Case [5] .



3. Reverse the steps 1 to 4 of section “Remove The Remote Controller From Tank Unit” to fix Remote Controller Case [5] on Tank Unit.

12.1.8 Test Run

1. Before test run, make sure below items have been checked:-
   a) Pipework are properly done.
   b) Electric cable connecting work are properly done.
   c) Tank Unit is filled up with water and trapped air is released.
   d) Please turn on the power supply after filling the tank until full.
2. Switch ON the power supply of the Tank Unit. Set the Tank Unit RCCB/ELCB to “ON” condition. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run. Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

3. For normal operation, Water Pressure reading should be in between 0.5 bar and 3 bar (0.05 MPa and 0.3 MPa). If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4. Remove the electric anode cover to check the electric anode PCB ㉘. (AN model only) Confirm the LED is green. If the LED is red, confirm the tank is full of water. If the LED is OFF, please set the electric anode “YES” in the system set up of the R/C.
5. After test run, please clean the Magnetic Water Filter Set ⑨ and Water Filter Set ㉚. Reinstall it after finish cleaning.

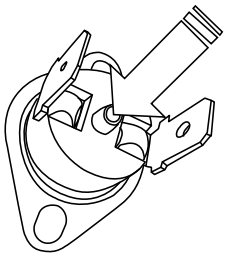
12.1.8.1 Check Water Flow of Water Circuit

Select Installer setup → Service setup → Pump maximum speed → Air purge
Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]
*If there is no flow or H62 is displayed, stop operating the pump and release the air (see Checking for Air Accumulation 12.1.6.3).

12.1.8.2 Reset Overload Protector ⑫

Overload Protector ⑫ a serves the safety purpose to prevent the water over heating. When the Overload Protector ⑫ a trip at high water temperature, take below steps to reset it.

1. Take out the cover.
2. Use a test pen to push the centre button gently in order to reset the Overload Protector ⑫.
3. Fix the cover to the original fixing condition.

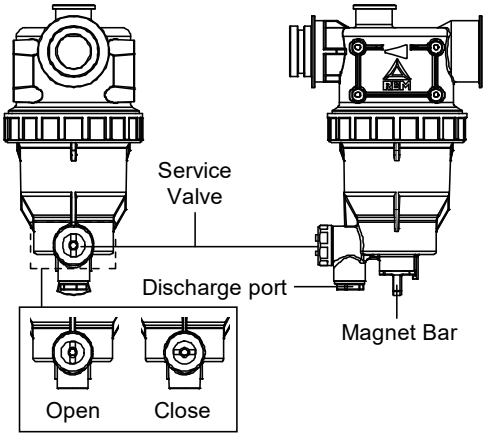


12.1.9 Maintenance

- In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

12.1.9.1 Maintenance for Magnetic Water Filter Set ⑨

1. Turn OFF power supply.
2. Place a container below Magnetic Water Filter Set ⑨.
3. Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑨.
4. By using Allen key (8mm), remove the Cap of Discharge Port.
5. By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
6. Reinstall the Cap of Discharge Port and Magnet Bar.
7. Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 12.1.5 for details.)
8. Turn ON power supply.



12.1.9.2 Maintenance for Safety Relief Valve ㉕

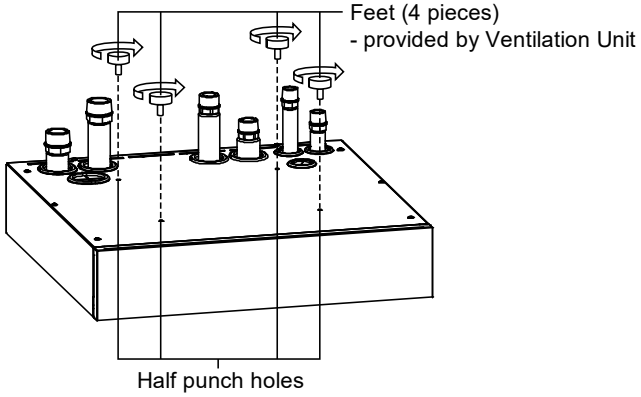
- It is strongly recommended to operate the valve by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

12.1.9.3 Installation of Ventilation Unit on Top of Tank Unit (Optional)

- For installation works of Ventilation Unit on top of Tank Unit, refer to the Ventilation Unit Installation Manual.

⚠ CAUTION
Before install Ventilation Unit, fix the Feet that provided by Ventilation Unit to the half punch holes on Top Panel of Tank Unit, Otherwise, heavy Ventilation Unit may fall and cause injury.



12.2 Appendix

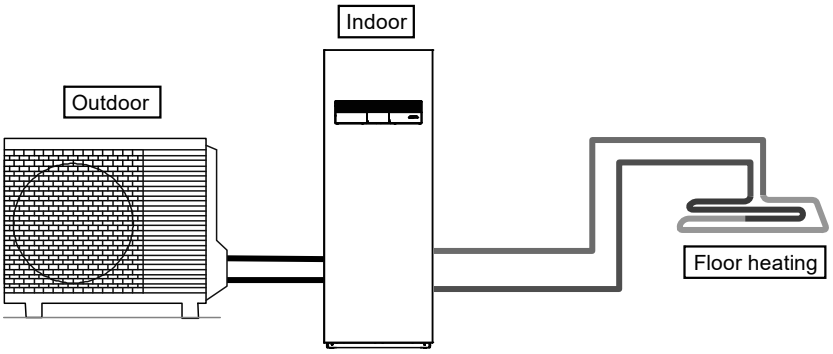
12.2.1 Variation of System

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.
(NOTE) : For this model, both external room thermistor of Zone 1 and external room thermostat of Zone 1 must always be connected to main indoor PCB only regardless of Optional PCB (CZ-NS5P) connection.

12.2.1.1 Introduce Application Related to Temperature Setting

12.2.1.1.1 Temperature Setting Variation for Heating

1. Remote Controller



Setting of remote controller

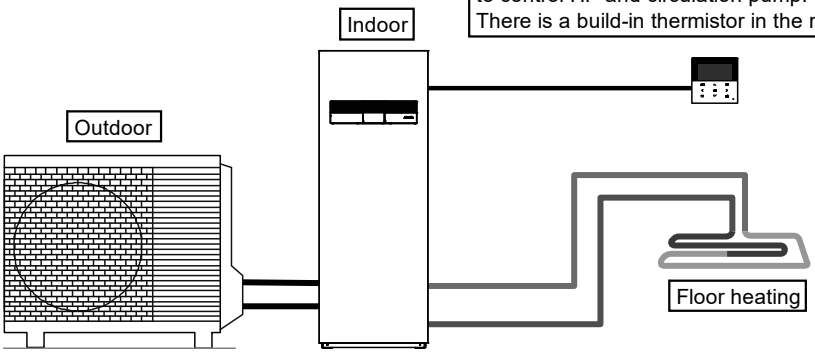
Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Water temperature

Connect floor heating or radiator directly to the Tank Unit.
Remote controller is installed on Tank Unit.
This is the basic form of the most simple system.

2. Room Thermostat

Tank Unit receive Room Thermo signal (ON/OFF) from Remote controller to control HP and circulation pump.
There is a build-in thermistor in the remote controller.



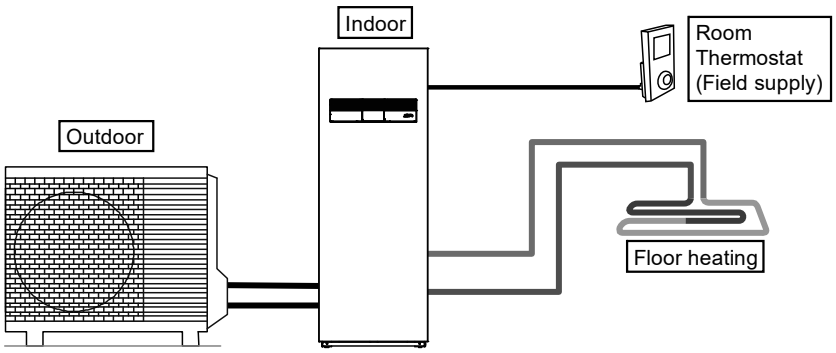
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
Internal

Connect floor heating or radiator directly to the Tank Unit.
Remove remote controller from Tank Unit and install it in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat

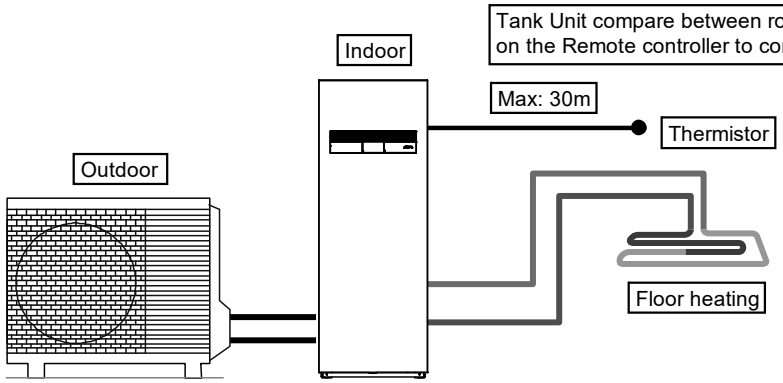


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No
Zone & Sensor:
Room thermostat
(External)

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



Setting of remote controller

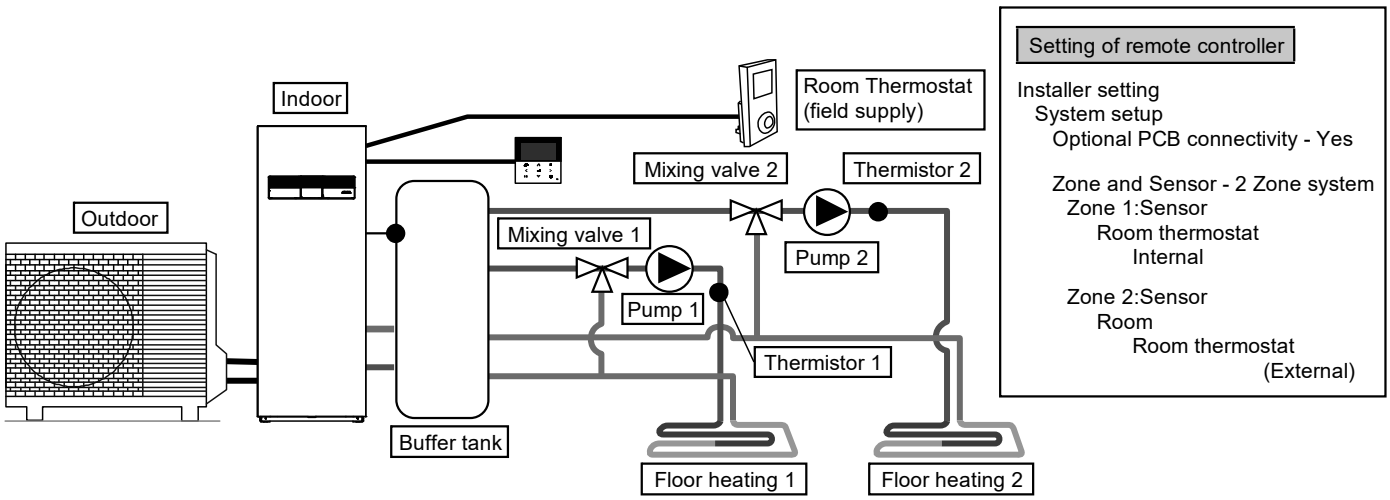
Installer setting
System setup
Optional PCB connectivity - No
Zone & Sensor:
Room thermistor

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, compensation curve is shifted according to the thermo ON/OFF situation.
- (Example) If room temperature increasing speed is;
very slow → shift up the compensation curve
very fast → shift down the compensation curve

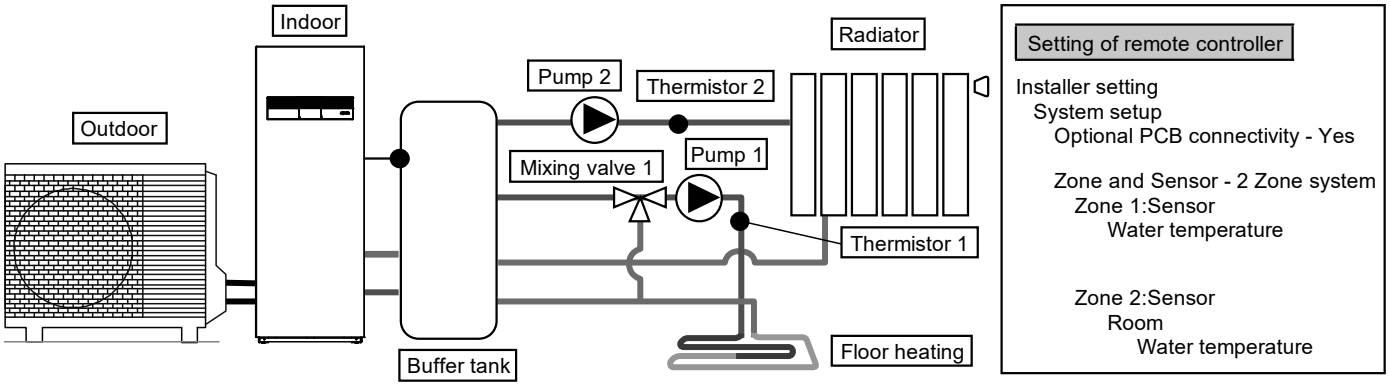
12.2.1.1.2 Examples of Installations

Floor heating 1 + Floor heating 2



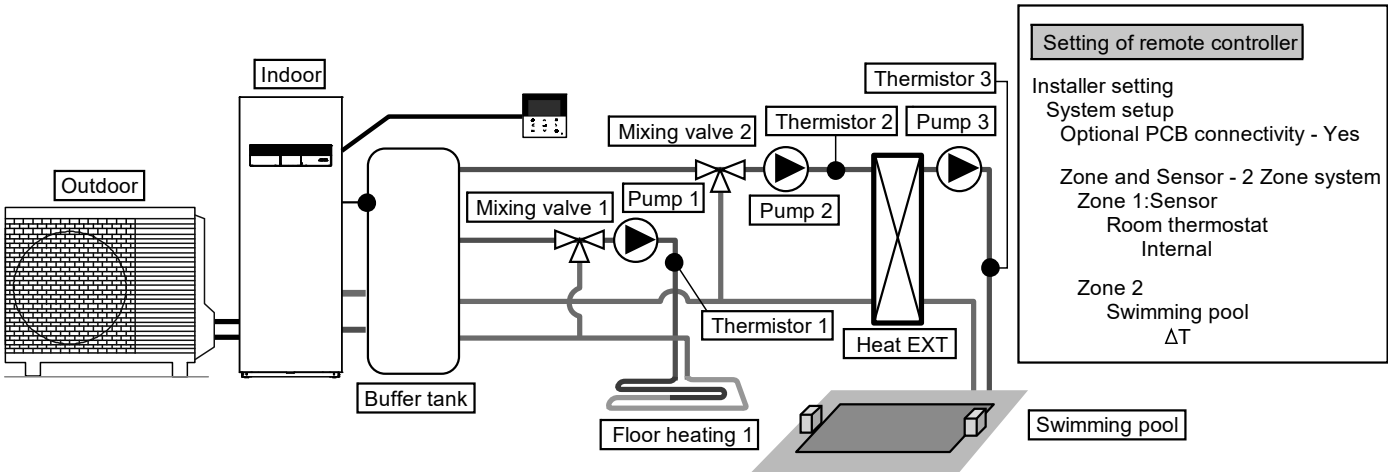
Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Remove remote controller from Tank Unit, install it in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Floor heating + Radiator



Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
Remote controller is installed on Tank Unit.
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the Optional PCB (CZ-NS5P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

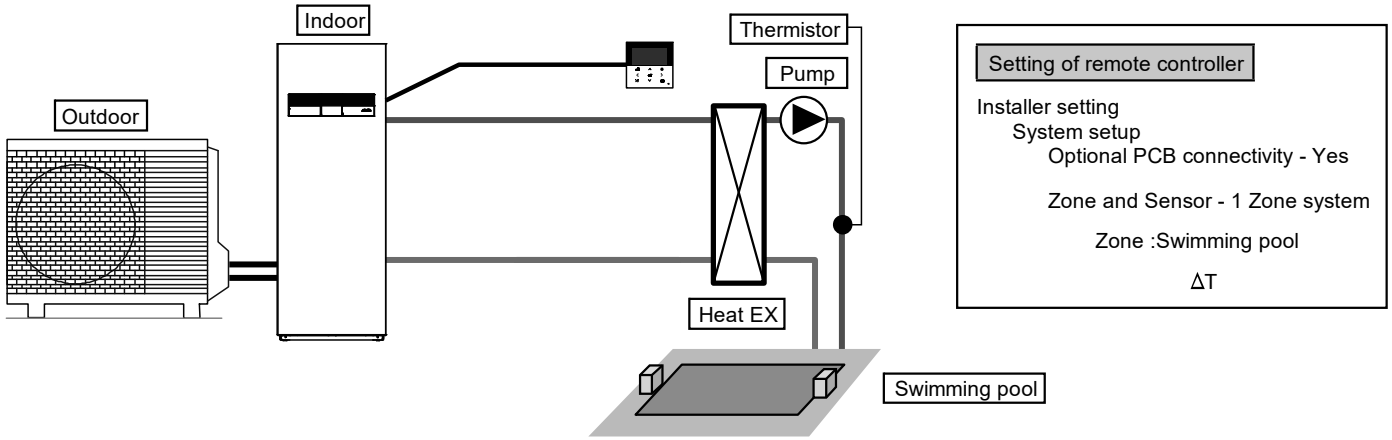
Floor heating + Swimming pool



Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Remove remote controller from Tank Unit and install in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the Optional PCB (CZ-NS5P).

❊ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

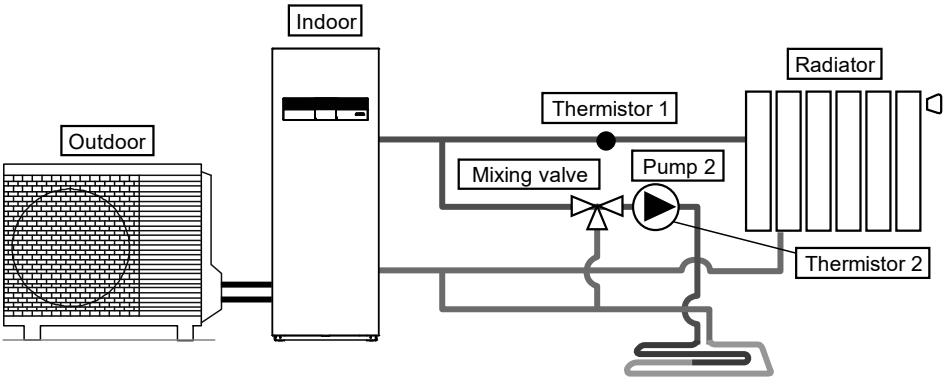
Swimming pool only



This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Tank Unit without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Remove remote controller from Tank Unit and install in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the Optional PCB (CZ-NS5P).

In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



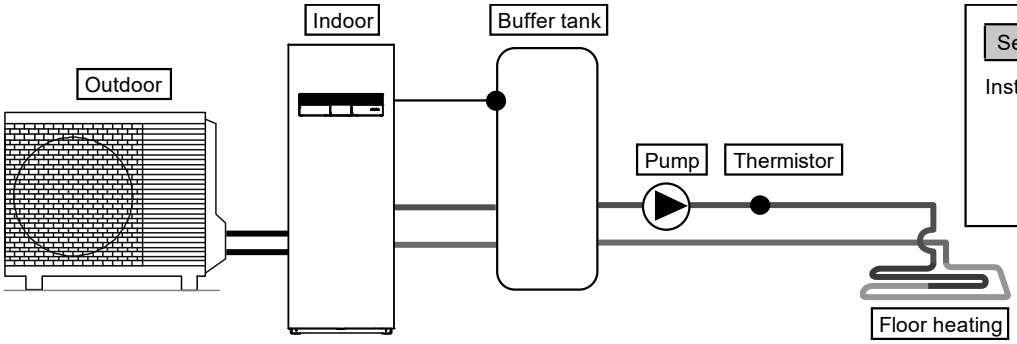
Setting of remote controller

Installer setting
- System setup
  - Optional PCB connectivity - Yes
  - Zone and Sensor - 2 Zone system
    - Zone 1:Sensor
      - Water temperature
    - Zone 2:Sensor
      - Room
        - Water temperature
- Operation setup
  - Heat
    - ΔT for heating ON – 1°C
  - Cool
    - ΔT for cooling ON – 1°C

This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Tank Unit served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the Optional PCB (CZ-NS5P).

(NOTE)
- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
- Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

Buffer tank connection

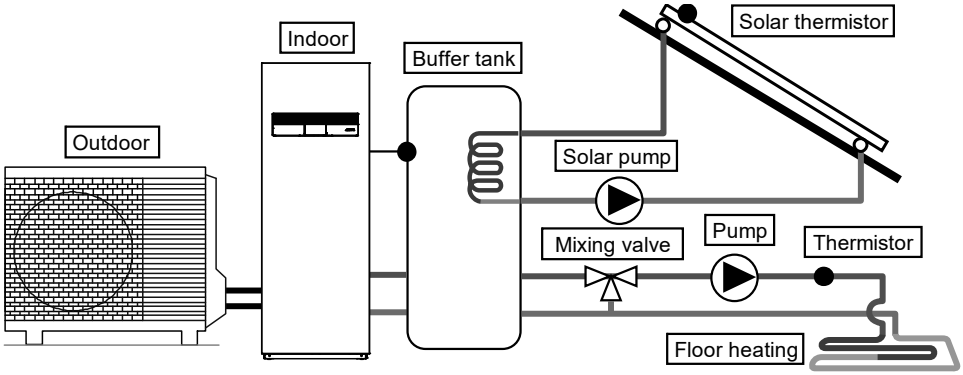


Setting of remote controller

Installer setting
- System setup
  - Optional PCB connectivity - No
  - Buffer Tank connection - Yes
    - ΔT for buffer tank

This is an application that connects the buffer tank to the Tank Unit.
Buffer tank’s temperature is detected by buffer tank thermistor (specified by Panasonic).
Without connection of Optional PCB, external pump can be used for circulation in the floor heating circuit.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

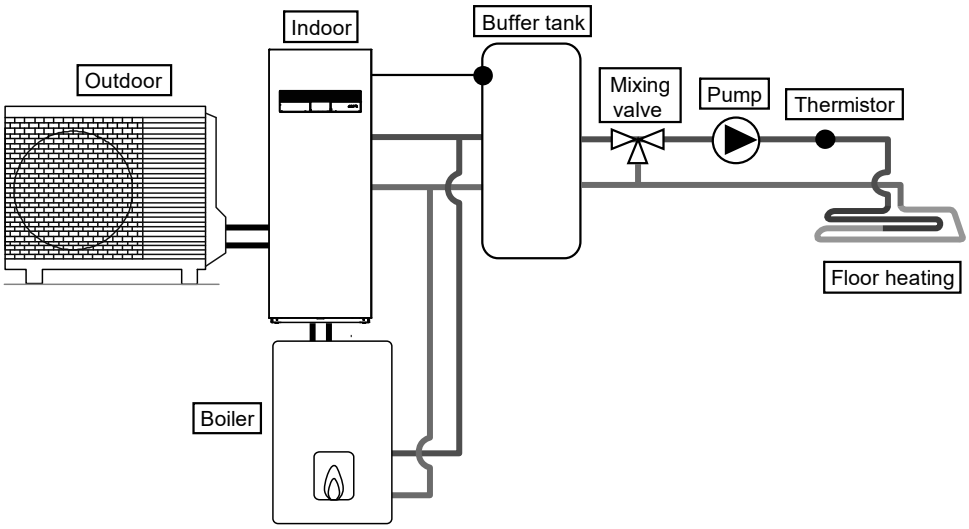
Buffer tank + Solar



Setting of remote controller
Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank
Solar connection - Yes
Buffer tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the buffer tank to the Tank Unit before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Boiler connection



Setting of remote controller
Installer setting
System setup
Optional PCB connectivity - Yes
Bivalent - Yes
Turn ON: outdoor temp.
Control pattern

This is an application that connects the boiler to the Tank Unit, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
Besides that, an application that connects to the DHW tank's circuit to heat up tank's hot water is also possible.
Boiler output can be control by either SG ready input from optional PCB or Auto control by 3 modes selection pattern. (Operation setting of boiler shall be responsible by installer.)
This system requires Optional PCB (CZ-NS5P) for SG ready input control.
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when select Advanced Parallel setting.)
Note : Buffer tank thermistor must be connected to main indoor PCB only.

⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

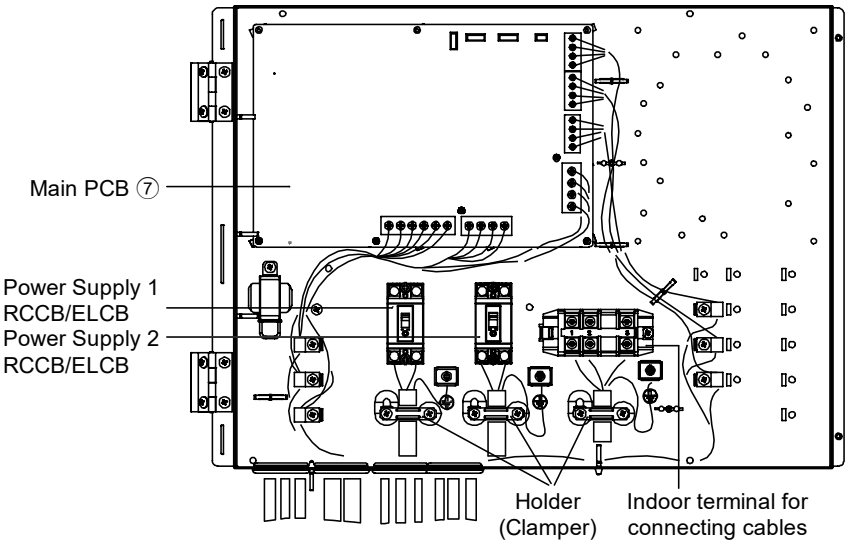
⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the Tank Unit does NOT exceed 70°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

12.2.2 How to Fix Cable

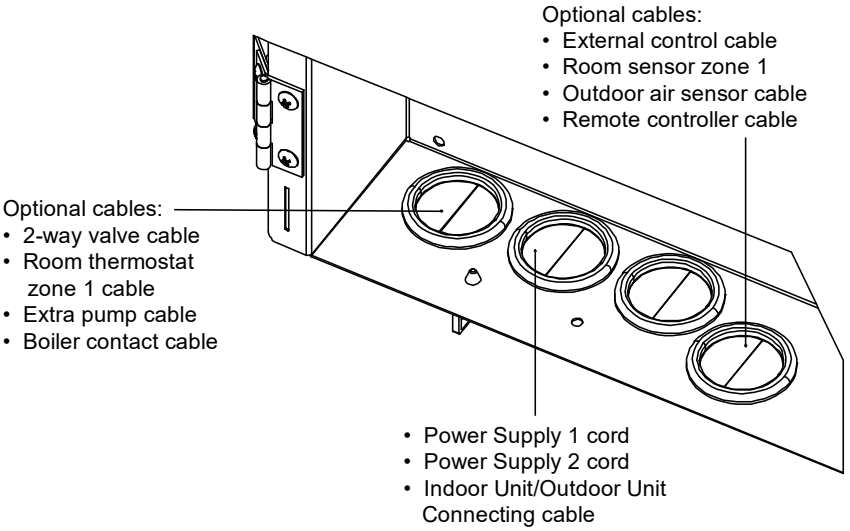
12.2.2.1 Connecting with External Device (Optional)

- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB ⑦

1. Two-way valve shall be spring and electronic type, refer to “Field Supply Accessories” table for details. Valve cable shall be (3 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
   *note: - Two-way Valve shall be CE marking compliance component.
   - Maximum load for the valve is 12VA.
2. Room thermostat cable must be (4 or 3 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
3. Extra pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
4. Boiler contact cable shall be (2 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
5. External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 × min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
   * note: - Switch used shall be CE compliance component.
   - Maximum operating current shall be less than $3A_{rms}$.
6. Room sensor zone 1 cable shall be (2 × min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.
7. Outdoor air sensor cable shall be (2 × min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.

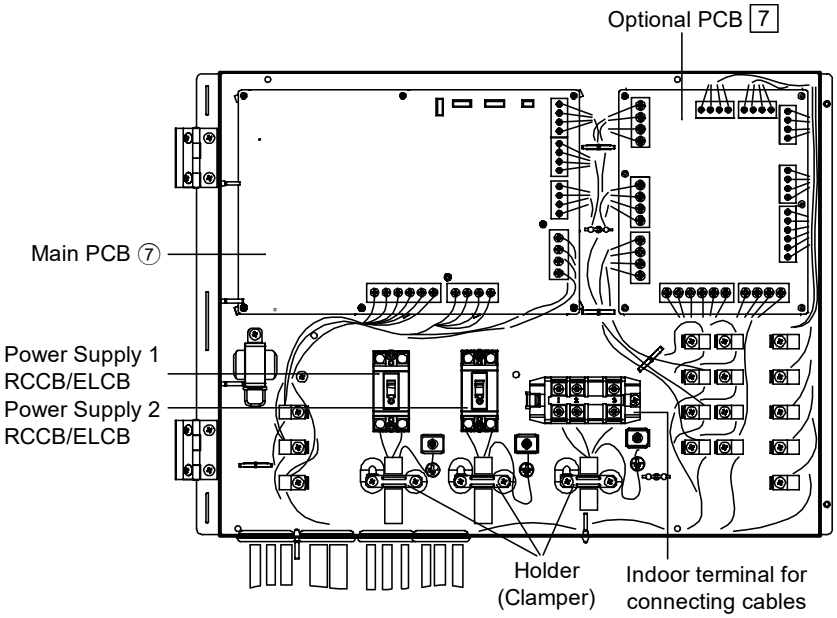


How to guide the optional cables and power supply cord
(view without internal wiring)

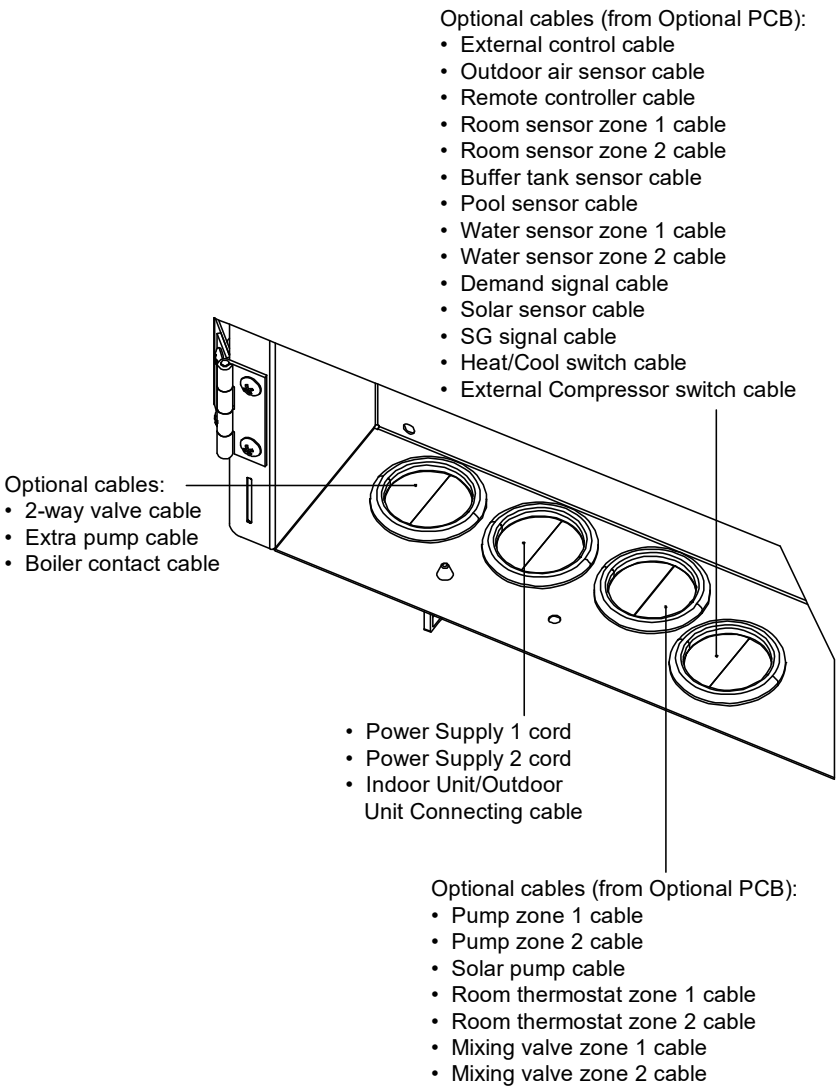


- For connection to Optional PCB [7]

1. By connecting Optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in Optional PCB. Temperature of each zone can be controlled independently by remote controller.
2. Pump zone 1 and zone 2 cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
3. Solar pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
4. Pool pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
5. Room thermostat zone 1 and zone 2 cable shall be (4 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
6. Mixing valve zone 1 and zone 2 cable shall be (3 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
7. Room sensor zone 1 and zone 2 cable shall be (2 × min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
8. Buffer tank sensor, pool water sensor and solar sensor cable shall be (2 × min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
9. Water sensor zone 1 and zone 2 cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
10. Demand signal cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
11. SG signal cable shall be (3 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
12. Heat/Cool switch cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
13. External compressor switch cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide the optional cables and power supply cord
(view without internal wiring)

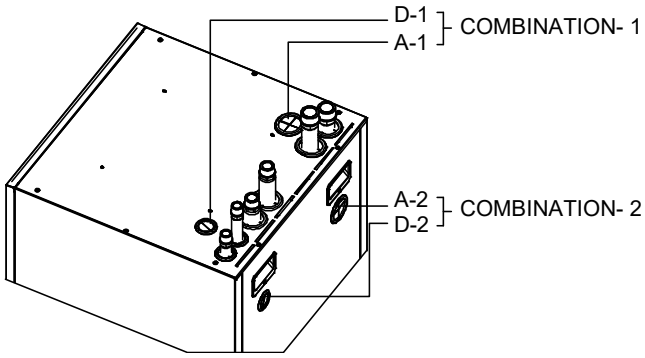


Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

12.2.2.2 Guide Optional Cables and Power Supply Cords to Bushings

⚠ CAUTION
Wire guiding shall be free from hot surfaces. Else, cable insulator damage and electrical shock may happen.
Wire ways shall be smooth and free from sharp edges. Else, cable insulator damage and electrical shock may happen.

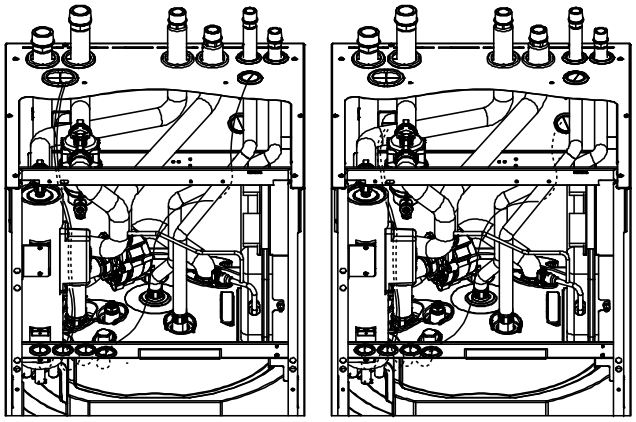
- Use either “COMBINATION-1” or “COMBINATION-2” for guiding Optional Cables and Power Supply Cords to Bushings.



- A-1 and A-2 Bushings are for:
  - Power Supply 1 cord
  - Power Supply 2 cord
  - Indoor Unit/Outdoor Unit Connecting cable
  - Pump zone 1 cable
  - Pump zone 2 cable
  - Solar pump cable
  - Room thermostat zone 1 cable
  - Room thermostat zone 2 cable
  - Mixing valve zone 1 cable
  - Mixing valve zone 2 cable
  - 2-way valve cable
  - Extra pump cable
  - Boiler contact cable

- D-1 and D-2 Bushings are for:
  - External control cable
  - Outdoor air sensor cable
  - Remote controller cable
  - Room sensor zone 1 cable
  - Room sensor zone 2 cable
  - Buffer tank sensor cable
  - Pool sensor cable
  - Water sensor zone 1 cable
  - Water sensor zone 2 cable
  - Demand signal cable
  - Solar sensor cable
  - SG signal cable
  - Heat/Cool switch cable
  - External Compressor switch cable

- Ensure all sensor cables are not touching with Front Panel
- Guide the wiring inside the unit like below figure. Once all wiring work done, tie the cable / cord with the banding strap (field supply), to prevent them touching with hot surfaces such as Heater Assembly, bare copper pipes and etc.



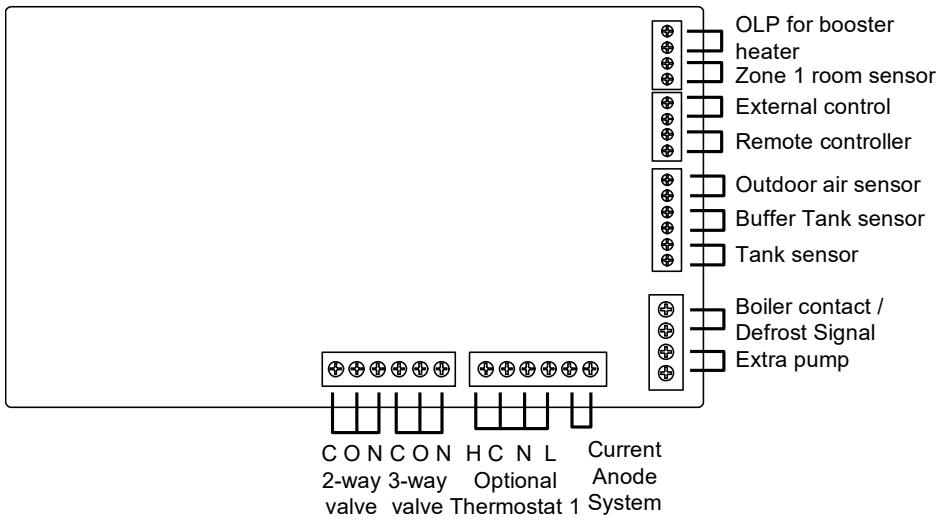
Wiring for “COMBINATION-1” Wiring for “COMBINATION-2”

12.2.2.3 Connecting Cables Length

When connecting cables between Tank Unit and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Mixing valve	50
Room thermostat	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Room sensor	30
Outdoor air sensor	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

12.2.2.4 Connection of the Main PCB



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

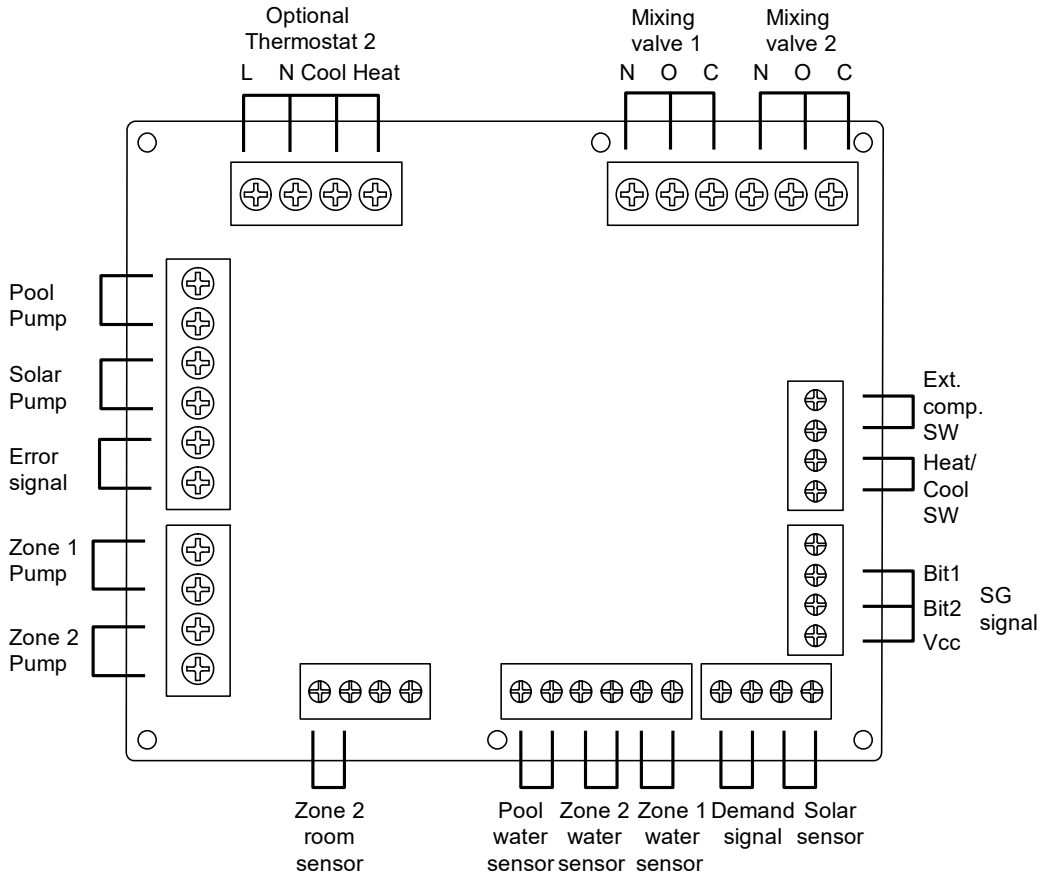
■ Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)	AC230V, 12 VA
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)	AC230V, 12 VA
Extra pump	AC230V (Used when Tank Unit pump capacity is insufficient)	AC 230V, 0.6 A max
Boiler contact / Defrost signal	Dry contact (System setup necessary)	

■ Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT
Outdoor air sensor	PAW-A2W-TSOD (Total cable length shall be 30m or less)

12.2.2.5 Connection of Optional PCB (CZ-NS5P)



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.OFF, Short=Comp.ON (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

■ Outputs

Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s	AC230V, 6VA
Pool pump	AC230V	AC 230V, 0.6 A max
Solar pump	AC230V	AC 230V, 0.6 A max
Zone pump	AC230V	AC 230V, 0.6 A max

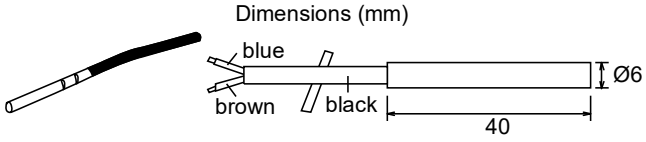
■ Thermistor inputs

Zone room sensor	PAW-A2W-TSRT
Buffer tank sensor	PAW-A2W-TSBU
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

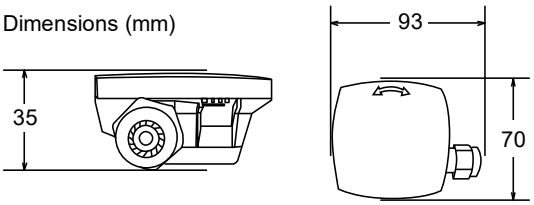
12.2.2.6 Recommended External Device Specification

- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.

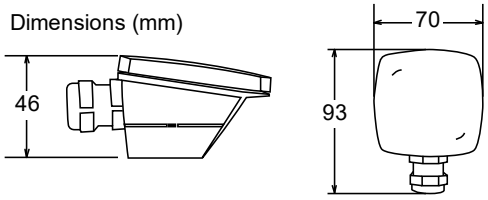
1. Buffer tank sensor: PAW-A2W-TSBU
   Use for measurement of the buffer tank temperature.
   Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



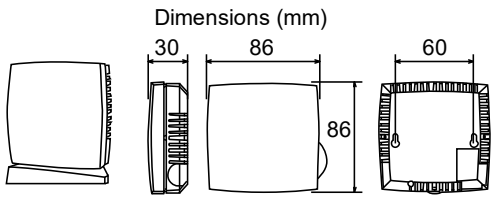
2. Zone water sensor: PAW-A2W-TSHC
   Use to detect the water temperature of the control zone.
   Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



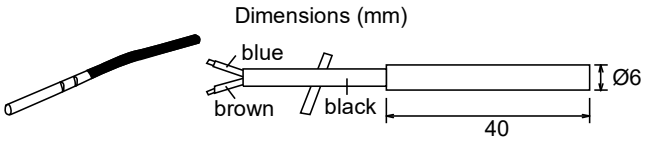
3. Outdoor sensor: PAW-A2W-TSOD
   If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
   In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



4. Room sensor: PAW-A2W-TSRT
   Install the room temperature sensor to the room which requires room temperature control.



5. Solar sensor: PAW-A2W-TSSO
   Use for measurement of the solar panel temperature.
   Insert the sensor into the sensor pocket and paste it on the solar panel surface.

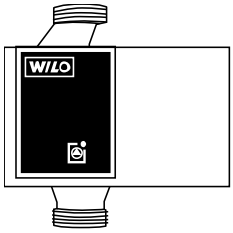


6. Please refer to the table below for sensor characteristic of the sensors mentioned above.

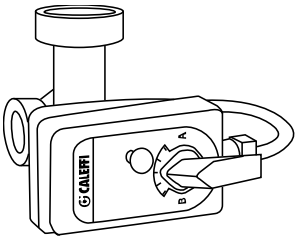
Temperature (°C)	Resistance (kΩ)
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo



- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

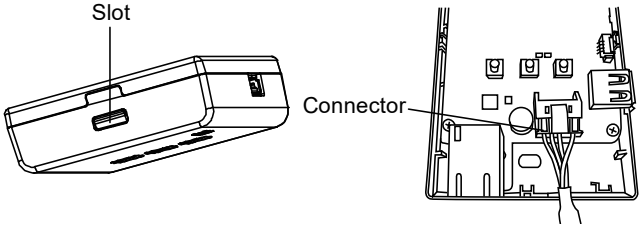


⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

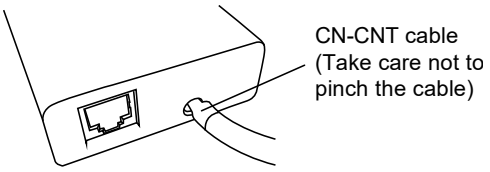
12.2.2.7 Network Adaptor [4] Installation

1. **Remove the Control Board Cover ⑤, then connect the cable included with this adaptor to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Tank Unit so that there is no pinching.
   - If an Optional PCB has been installed in the Tank Unit, connect to the CN-CNT connector of the Optional PCB.

2. **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**

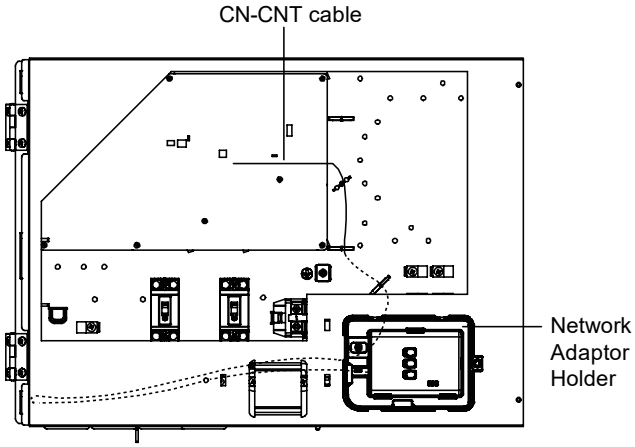


3. **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**

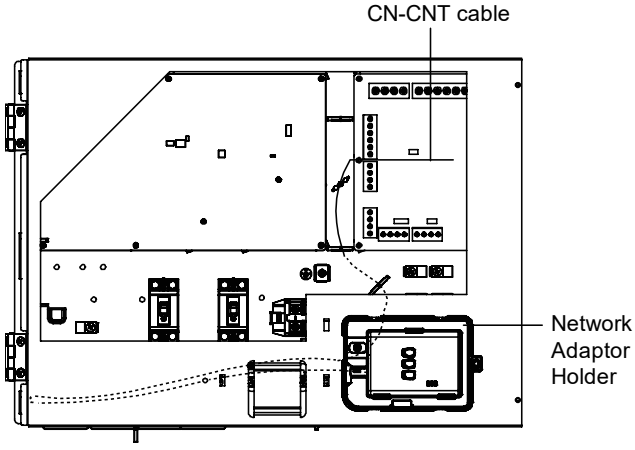


4. **Fix the Network Adaptor [4] to Network Adaptor Holder.**
   Guide the cable as shown in the diagram so that external forces cannot act on the connector in the adaptor.

Connection examples:



Without Optional PCB

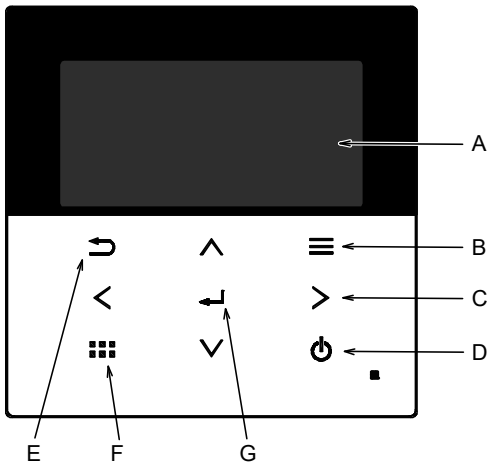


With Optional PCB

12.2.3 System Installation

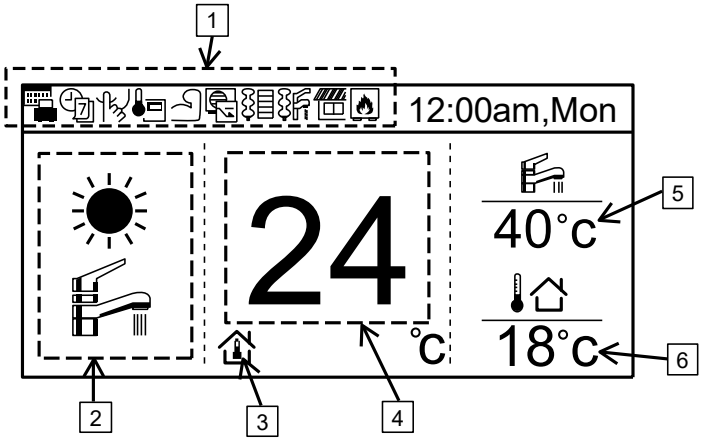
12.2.3.1 Remote Controller Outline

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.



Name	Function
A: Main screen	Display information
B: Menu	Open/Close main menu
C: Triangle (Move)	Select or change item
D: Operate	Start/Stop operation
E: Back	Back to previous item
F: Quick Menu	Open/Close Quick menu
G: OK	Confirm

LCD Display
(Actual - Dark background with white icons)



	Name	Function
1:	Function icon	Display set function/status


Holiday mode


Weekly timer


Quiet mode


Remote controller room thermostat


Powerful mode


Demand control


Room heater


Tank heater


Solar


Boiler

2:	Mode	Display set mode/current status of mode


Heating


Cooling


Auto


Hot water supply


Auto heating


Auto cooling


Heat pump operating

3:	Temp setting	 Set room temp

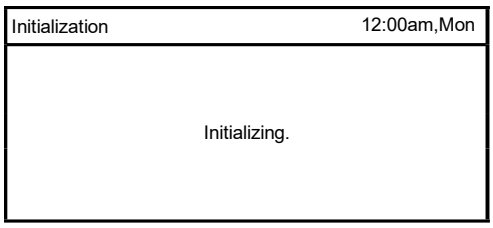

Compensation curve


Set direct water temp

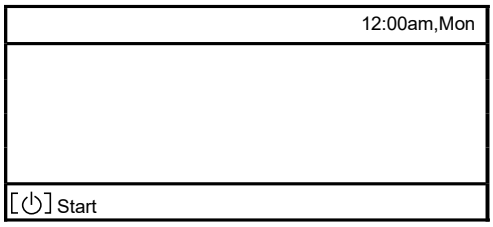

Set pool temp

4:	Display Heat temp	Display current heating temperature (it is set temperature when enclosed by line)
5:	Display tank temp	Display current tank temperature (it is set temperature when enclosed by line)
6:	Outdoor temp	Display outdoor temp

First time of power ON (Start of installation)



When power is ON, firstly initialization screen appears (10 sec)



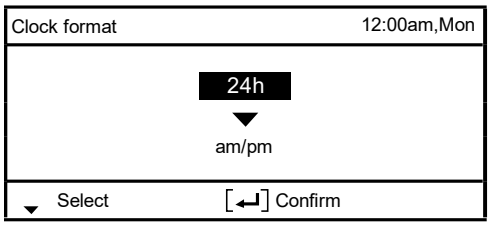
When initialization screen ends, it turns to normal screen.



When any button is pressed, language setting screen appears.
(NOTE) If initial setting is not performed, it does not go into menu.

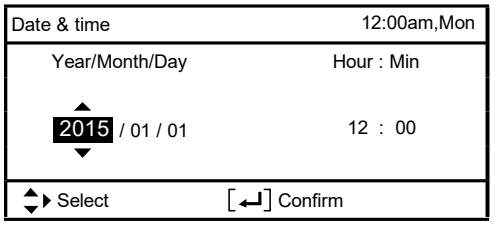
When there are two remote controllers installed from the beginning, the first remote controller to set and confirm language will be recognised as main remote controller.

Set language & confirm



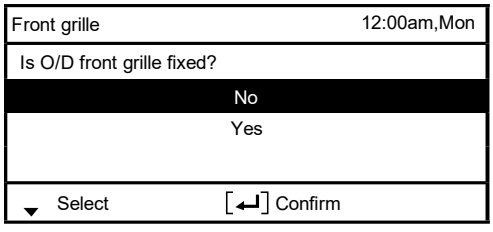
When language is set, setting screen of time display appears (24h/am/pm)

Set time display & confirm

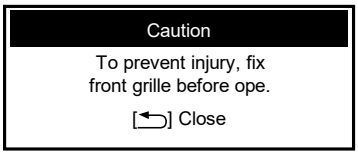


YY/MM/DD/Time setup screen appears

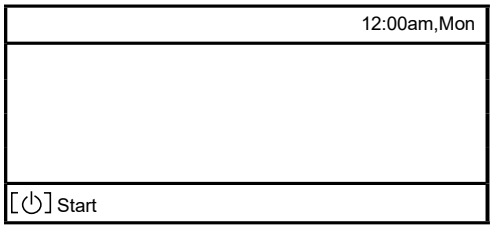
Set YY/MM/DD/Time & confirm



If set No & confirm, a caution message will be displayed to ensure outdoor front grille is installed before proceed to operate the unit.

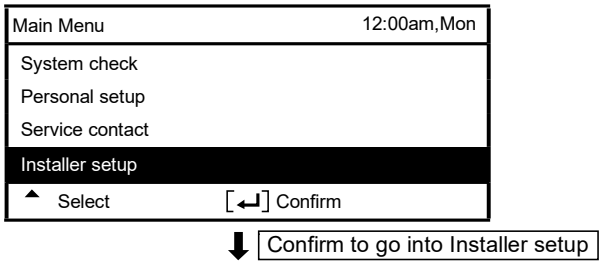


Set Yes & confirm if outdoor front grille has been installed

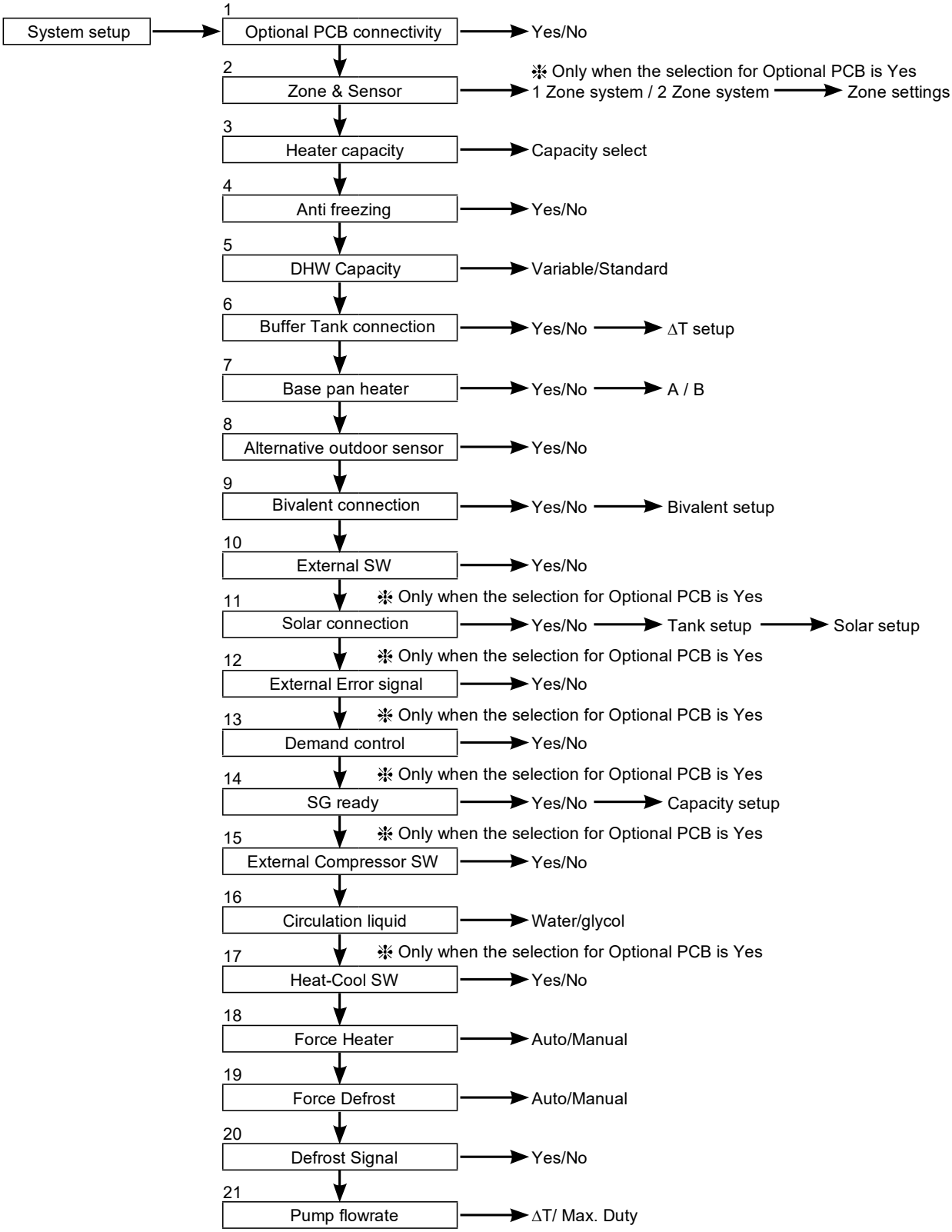


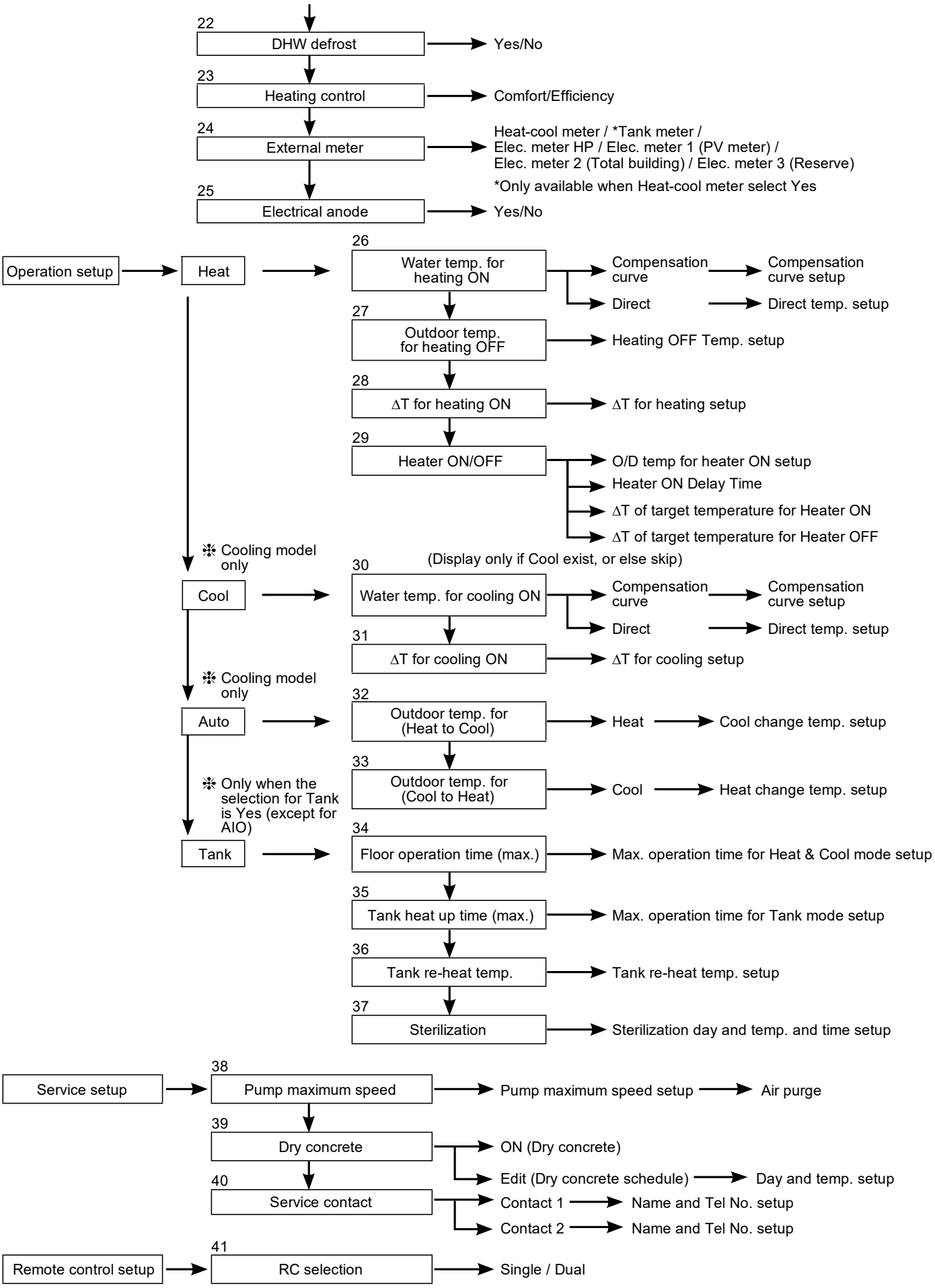
Back to initial screen

Press menu, select Installer setup



12.2.3.2 Installer Setup



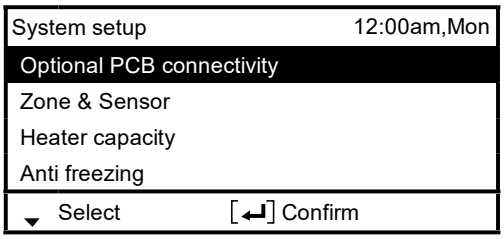

22
DHW defrost
Yes/No
23
Heating control
Comfort/Efficiency
24
External meter
Heat-cool meter / *Tank meter /
Elec. meter HP / Elec. meter 1 (PV meter) /
Elec. meter 2 (Total building) / Elec. meter 3 (Reserve)
*Only available when Heat-cool meter select Yes
25
Electrical anode
Yes/No
Operation setup
Heat
26
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
27
Outdoor temp. for heating OFF
Heating OFF Temp. setup
28
ΔT for heating ON
ΔT for heating setup
29
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
❇ Cooling model only
Cool
(Display only if Cool exist, or else skip)
30
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
31
ΔT for cooling ON
ΔT for cooling setup
❇ Cooling model only
Auto
32
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
33
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
❇ Only when the selection for Tank is Yes (except for AIO)
Tank
34
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
35
Tank heat up time (max.)
Max. operation time for Tank mode setup
36
Tank re-heat temp.
Tank re-heat temp. setup
37
Sterilization
Sterilization day and temp. and time setup
Service setup
38
Pump maximum speed
Pump maximum speed setup
Air purge
39
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
40
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup
Remote control setup
41
RC selection
Single / Dual

12.2.3.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install Optional PCB.
Please select Yes after installing Optional PCB.

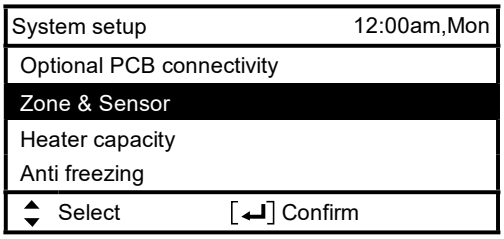
- 2-zone control
- Pool
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

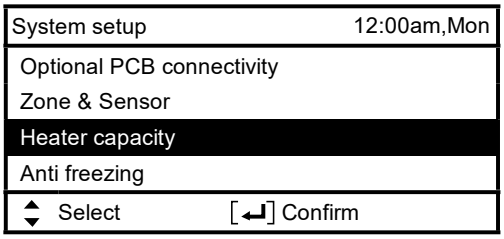
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

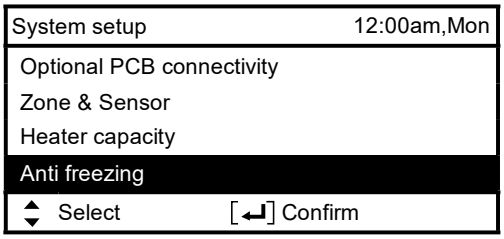
(NOTE) There are models which cannot select Heater capacity.



4. Anti freezing | Initial setting: Yes

Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

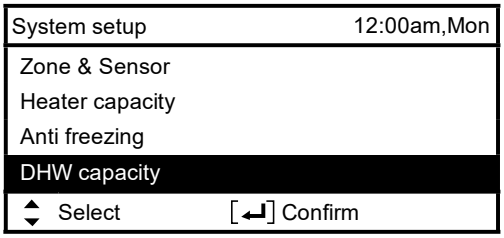
(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



5. DHW Capacity | Initial setting: Variable

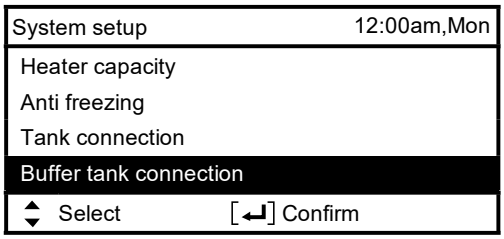
Select “Variable”
- The heat pump runs with efficient boiling, which is an energy-saving mode. However, if hot water usage is high and the tank water temperature is low, the heat pump will switch to fast heat-up mode, which uses higher heating capacity to heat up the tank. When “Efficiency” is selected in the “23. Heating control” setting, the tank heat-up process will be delayed until a much lower tank water temperature is reached, which improves the efficiency of the heating process. To ensure sufficient hot water, select either “Standard” in this setting or “Comfort” in the “23. Heating control” setting.

Select “Standard”
- The heat pump operates at its rated heating capacity during tank heating.



6. Buffer Tank connection | Initial setting: No

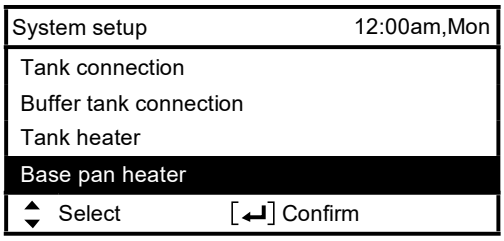
Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
If the buffer tank capacity is not so large, please set larger value for ΔT.



7. Base pan heater | Initial setting: No

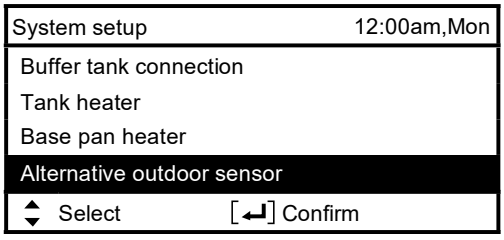
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



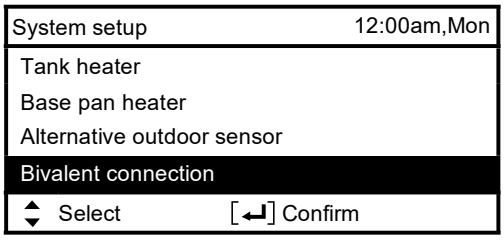
8. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



9. Bivalent connection | Initial setting: No

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.



After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)

1) SG ready (Only available to set when optional PCB set to YES)
   - SG Ready input from optional PCB terminal control ON/OFF of boiler and heat pump as below condition

SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

* This bivalent SG ready input is sharing same terminal as [14. SG ready] connection. Only one of these two setting can be set at the same time.
When one is set, another setting will reset to not set.

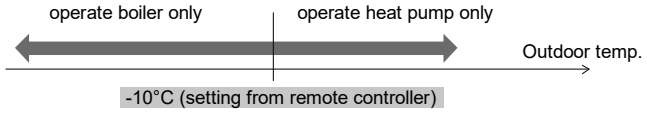
2) Auto

There are 3 different modes in the boiler auto pattern operation. Movement of each modes are shown below.

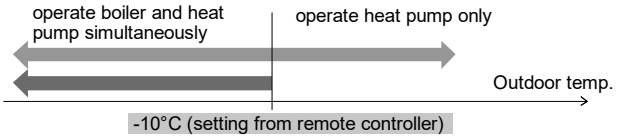
① Alternative (switch to boiler operation when drops below setting temperature)
② Parallel (allow boiler operation when drops below setting temperature)
③ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is “ON”, “boiler contact” is “ON”, “_”(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

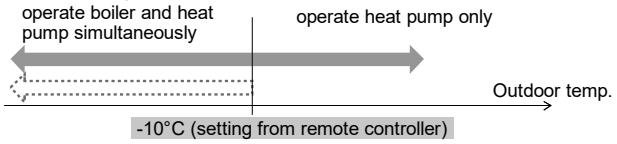


Parallel mode

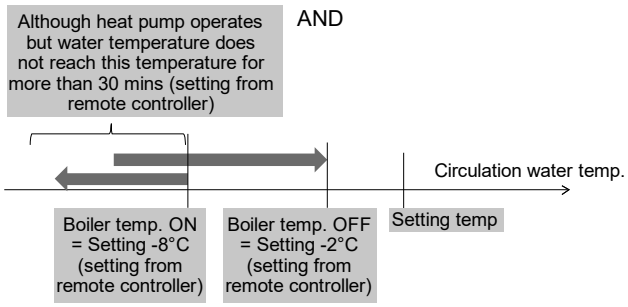


Advanced Parallel mode

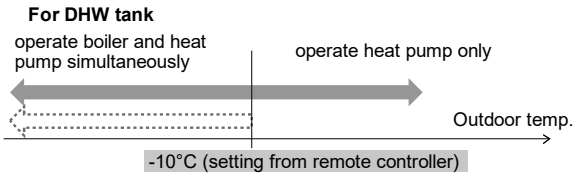
For heating



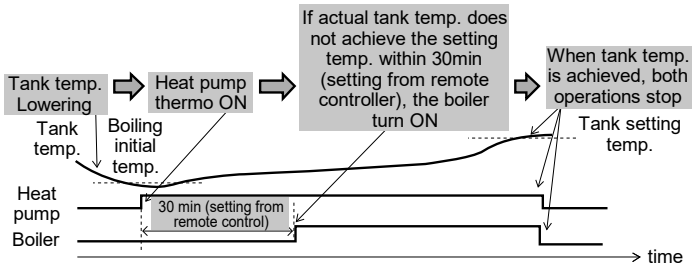
AND



For DHW tank



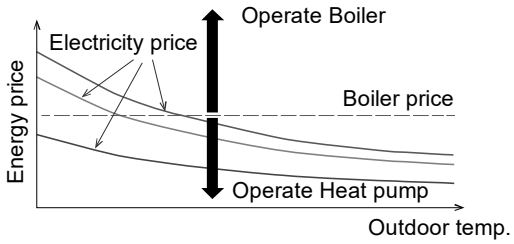
AND



In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of “Heating/Tank” mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

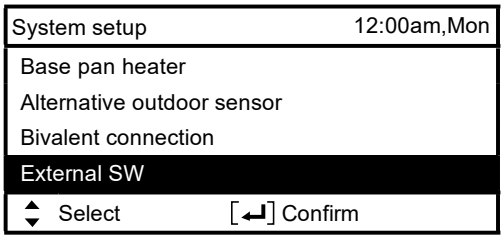
3) Smart

There are Energy price (both electricity and boiler) and Schedule to be set on remote controller.
Operation setting of Energy price and Schedule shall be responsible by installer.
Based on these settings, system will calculate the final price for both electricity and boiler.
When final price of Electricity is lower than Boiler’s, heat pump will operate.
When final price of Electricity is higher than Boiler’s, boiler will operate.



10. External SW | Initial setting: No

Able to turn ON/OFF the operation by external switch.

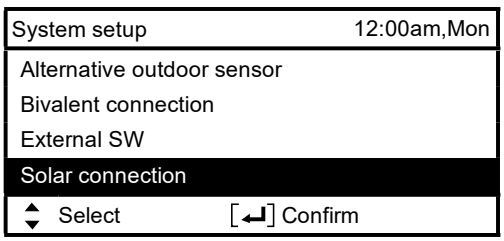


11. Solar connection | Initial setting: No

Set when solar water heater is installed.

Setting include items below.

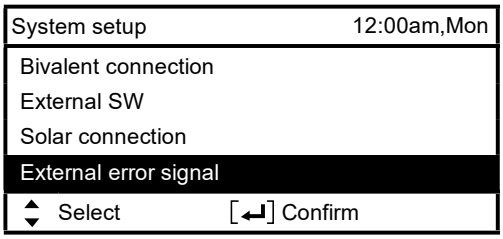
① Set either buffer tank or DHW tank for connection with solar water heater.
② Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to operate the solar pump.
③ Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to stop the solar pump.
④ Anti-freezing operation start temperature (please change setting based on usage of glycol.)
⑤ Solar pump stop operation when it exceeds high limit temperature (when tank temperature exceed designated temperature (70~90°C))



12. External Error Signal | Initial setting: No

Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

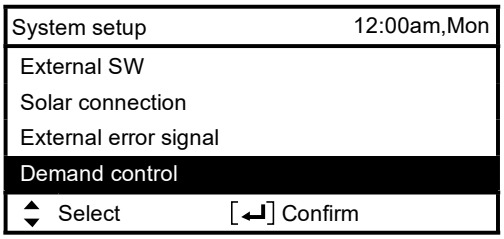
(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off “close” from the display, error signal will still remain ON.



13. Demand control | Initial setting: No

Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.



Analog input [v]	Rate [%]
0.0	not activate
0.1 ~ 0.6	
0.7	10
0.8	not activate
0.9 ~ 1.1	10
1.2	15
1.3	10
1.4 ~ 1.6	15
1.7	20
1.8	15
1.9 ~ 2.1	20
2.2	25
2.3	20
2.4 ~ 2.6	25
2.7	30
2.8	25
2.9 ~ 3.1	30
3.2	35
3.3	30
3.4 ~ 3.6	35
3.7	40
3.8	35

Analog input [v]	Rate [%]
3.9 ~ 4.1	40
4.2	45
4.3	40
4.4 ~ 4.6	45
4.7	50
4.8	45
4.9 ~ 5.1	50
5.2	55
5.3	50
5.4 ~ 5.6	55
5.7	60
5.8	55
5.9 ~ 6.1	60
6.2	65
6.3	60
6.4 ~ 6.6	65
6.7	70
6.8	65
6.9 ~ 7.1	70
7.2	75
7.3	70

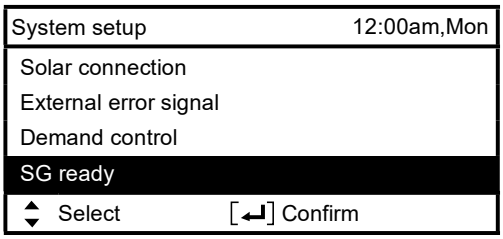
Analog input [v]	Rate [%]
7.4 ~ 7.6	75
7.7	80
7.8	75
7.9 ~ 8.1	80
8.2	85
8.3	80
8.4 ~ 8.6	85
8.7	90
8.8	85
8.9 ~ 9.1	90
9.2	95
9.3	90
9.4 ~ 9.6	95
9.7	100
9.8	95
9.9 ~	100

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

14. SG ready | Initial setting: No

Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2



Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Set by SG ready setting of remote controller

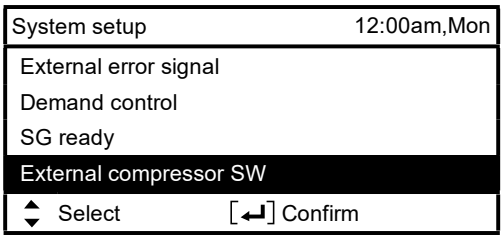
(When SG ready set to YES, Bivalent control pattern will set to Auto.)
(NOTE) Does not display if there is no Optional PCB.

15. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

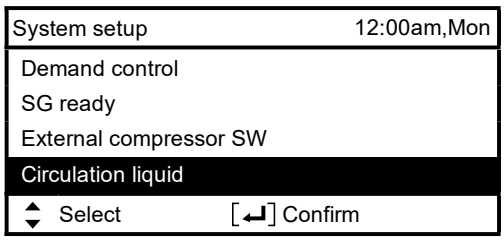


16. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

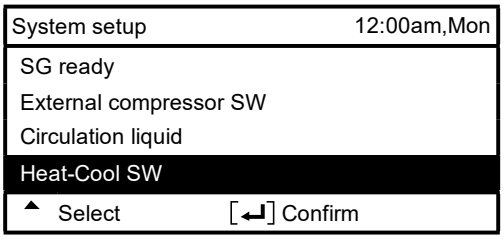


17. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

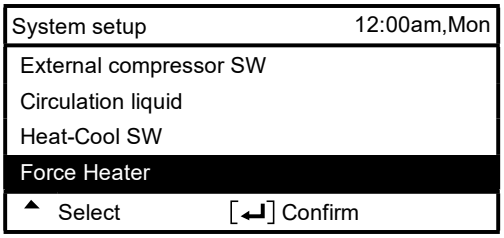


18. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

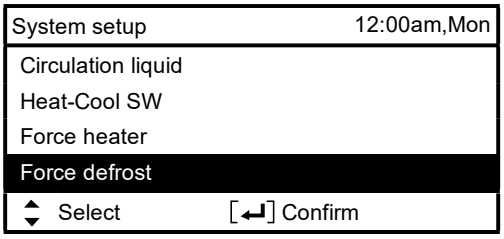
Heater source will ON during force heater mode.



19. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

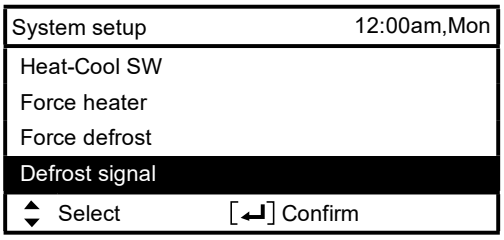
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition. (Even auto is selected, user still can turn on force defrost through quick menu)



20. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).

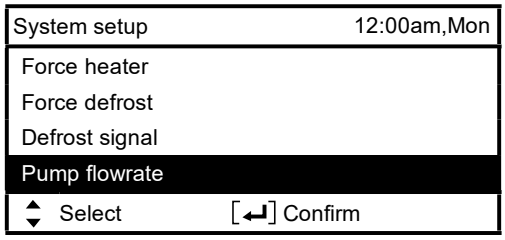


21. Pump flowrate

Initial setting: ΔT

If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

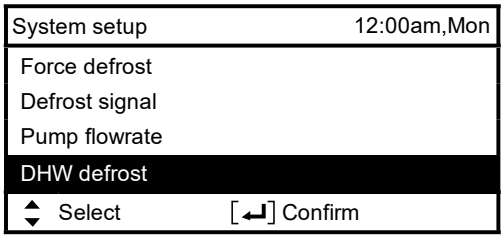
If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



22. DHW defrost

Initial setting: Yes

When DHW defrost set to YES, hot water of domestic hot water tank will be used during defrost cycle.
When DHW defrost set to NO, hot water of floor heating circuit will be used during defrost cycle.



23. Heating control

Initial setting: Comfort

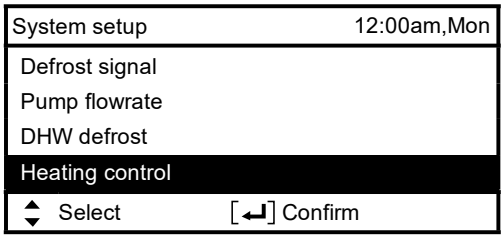
Compressor frequency control can be selected from two modes: Comfort and Efficiency.

Select “Comfort”
- The compressor operates at maximum frequency at the upper zone limit and reaches the set temperature faster.

Select “Efficiency”
- The compressor frequency is gradually increased to reduce power consumption.

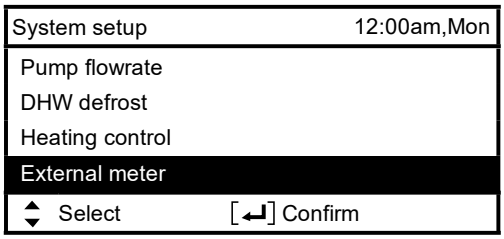
When “Variable” is selected in the “5. DHW capacity” setting, the tank heat-up process will be delayed until a much lower tank water temperature is reached, which improves the efficiency of the heating process. To ensure sufficient hot water, select either “Comfort” in this setting or “Standard” in the “5. DHW capacity” setting.

It is a function for Heating + Tank operation.



24. External meter

Initial setting : [Heat-cool meter : No]
[Tank meter : No] *only available when Heat-cool meter select Yes
[Elec. meter HP : No]
[Elec. meter 1 (PV meter) : No]
[Elec. meter 2 (Total building) : No]
[Elec. meter 3 (Reserve) : No]



There are two systems for generation meter connection : one generation meter system (Heat-cool meter) or two generation meter system (Heat-cool meter and Tank meter)
Both systems can provide all generation data of heating, cooling and DHW directly from external meter.
If Heat-cool meter is set to Yes, it will read from external meter for heat pump’s energy generation data during heating, cooling and DHW operation[*1].
If Heat-cool meter is set to No, it will base on unit’s calculation for heat pump’s energy generation data during heating, cooling and DHW operation.
If Tank meter is set to Yes, it will read from external meter for heat pump’s energy generation data during DHW operation[*1].
If Elec. meter HP is set to Yes, it will read from external meter for heat pump’s energy consumption data.
If Elec. meter HP is set to No, it will base on unit’s calculation for heat pump’s energy consumption data.
If Elec. meter 1 (PV meter) is set to Yes, it will read from external meter for energy generation data of solar system and display it on Cloud system.
If Elec. meter 2 (Building) is set to Yes, it will read from external meter for energy consumption data of the building and display it on Cloud system.
If Elec. meter 3 (Reserve) is set to Yes, it will read from external meter for energy consumption data obtained from reserved electricity meter and display it on Cloud system.

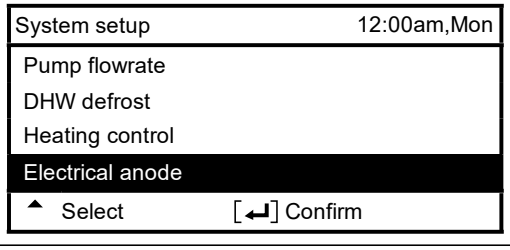
[*1] Set Heat-cool meter to Yes and set Tank meter to No when 1 generation meter system is installed.
Set Heat-cool meter to Yes and set Tank meter to Yes when 2 generation meter system is installed.

Remark : Elec. meter HP refers to Electricity meter that measures Heat Pump unit’s consumption.
Elec. meter 1 / 2 / 3 refers to Electricity meter no. 1 / no. 2 / no. 3

25. Electrical anode

For WH-ADC0509L3E5AN1 model, initial setting : Yes
For other models, initial setting : No

When Electrical anode set to YES, anode will be turned on.
When Electrical anode set to NO, anode will not be turned on.



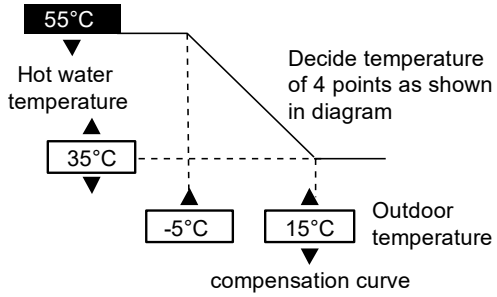
12.2.3.4 Operation Setup

Heat

26. Water temp. for heating ON | Initial setting: compensation curve

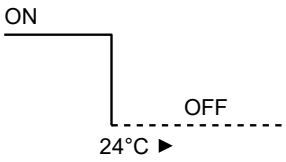
Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



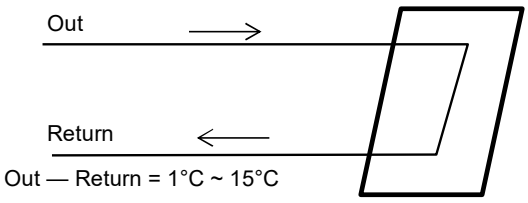
27. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C



28. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

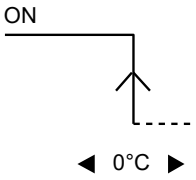


29. Heater ON/OFF

a. Outdoor temp. for heater ON | Initial setting: 0°C

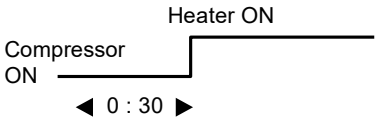
Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.



b. Heater ON delay time | Initial setting: 30 minutes

Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

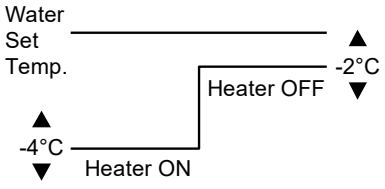


c. Heater ON: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp | Initial setting: -2°C

Set water temperature for heater to turn off at heat mode.
Setting range is -3°C ~ 0°C

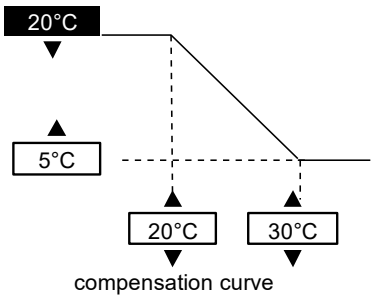


Cool

30. Water temp. for cooling ON | Initial setting: compensation curve

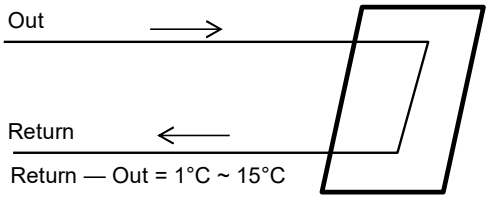
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



31. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

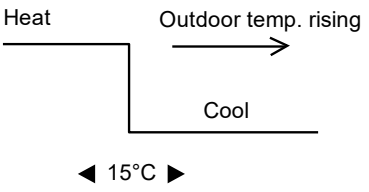


Auto

32. Outdoor temp. for (Heat to Cool) | Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 11°C ~ 25°C

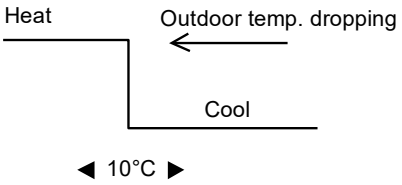
Timing of judgement is every 1 hour



33. Outdoor temp. for (Cool to Heat) | Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 14°C

Timing of judgement is every 1 hour

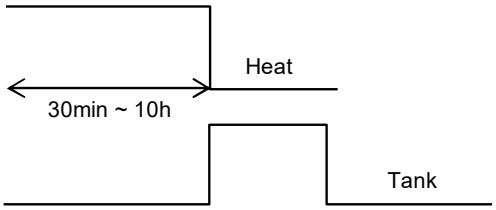


Tank

34. Floor operation time (max.) Initial setting: 8h

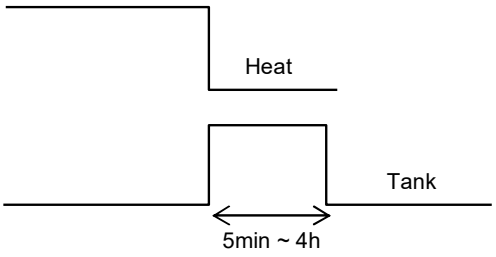
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



35. Tank heat up time (max.) Initial setting: 60min

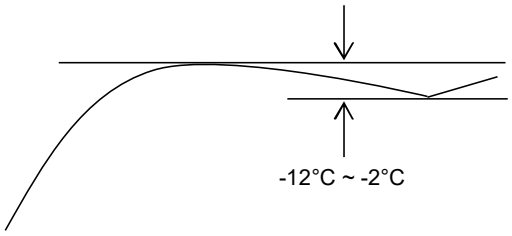
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



36. Tank re-heat temp. Initial setting: -8°C

Set temp to perform reboil of tank water.

Setting range is -12°C ~ -2°C

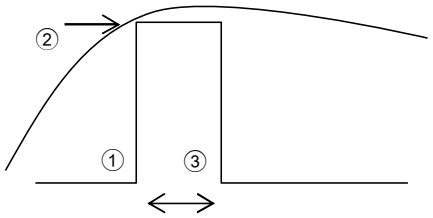


37. Sterilization Initial setting: 65°C 10min

Set timer to perform sterilization.

① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.



12.2.3.5 Service Setup

38. Pump maximum speed | Initial setting: Depend on model

Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.

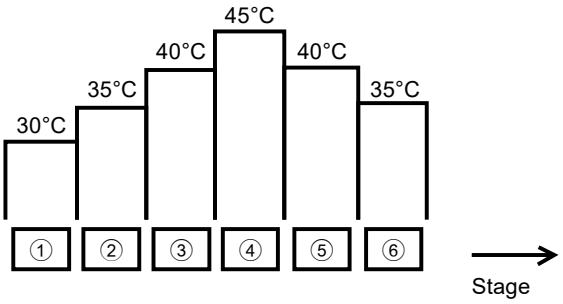
Service setup		12:00am,Mon
Flow rate	Max. Duty	Operation
88:8 L/min	0xCE	▲ Air Purge
◀▲ Select		

39. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



40. Service contact

Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)

Service setup	12:00am,Mon
Service contact:	
Contact 1	
Contact 2	
▲ Select	[↵] Confirm

Contact-1: Bryan Adams		
	ABC/ abc	0-9/ Other
A B C D E F G H I J K L M N O P Q R		
S T U V W X Y Z a b c d e f g h i		
j k l m n o p q r s t u v w x y z		
▼▶ Select	[↵] Enter	

12.2.3.6 Remote Control Setup

41. RC selection | Initial setting: Single

Set to “Single” when only one remote controller is installed.
Set to “Dual” when two remote controllers are installed.

RC selection	12:00am,Mon
Single	
▼	
Dual	
▼ Select	[↵] Confirm

12.3 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

12.3.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

12.3.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 5 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it.
In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer.
When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

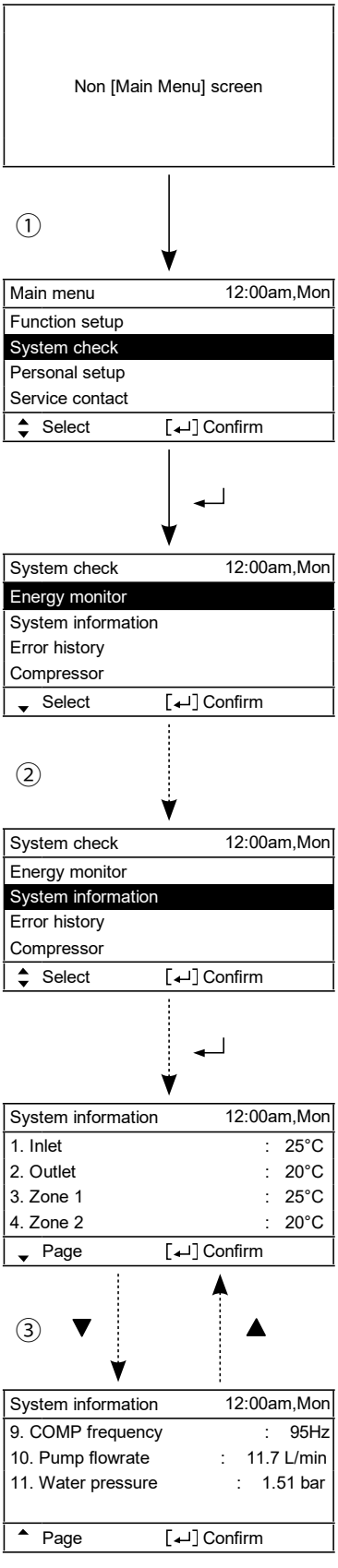
③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

12.3.3 Check Water Pressure from Remote Controller

1. Press ☰ SW and scroll to “System check”.
2. Press ↵ and scroll to “System information”.
3. Press ↵ and search for “Water pressure”.



Screens shown are for illustration purposes only.

12.4 Indoor Unit (For WH-ADC0509L3E5UK1 only)

12.4.1 Select the Best Location

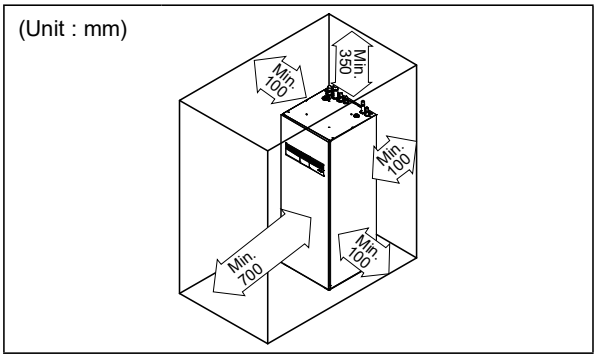
Before choosing the installation site, obtain user approval.

- Install the Tank Unit in indoors with frost free weather proof location only.
- Must install on a flat horizontal and solid hard surface.
- There should not be any heat source or steam near the Tank Unit.
- A place where air circulation in the room is good.
- A place where drainage can be easily done (e.g. Utility room).
- A place where Tank Unit's operation noise will not cause discomfort to the user.
- A place where Tank Unit is far from door way.
- A place where accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated below from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- Secure the Tank Unit to prevent it being knocked over accidentally or during earthquakes.

Please avoid installations which expose the Tank Unit to any of the following conditions:

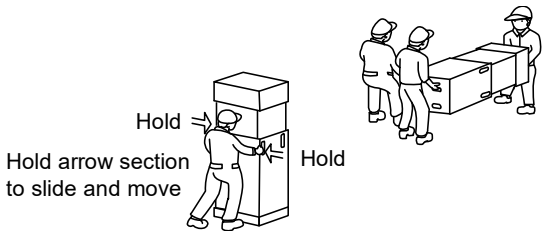
- Extraordinary environment conditions; installation in frost or exposure to unfavorable weather conditions.
- Voltage input exceeding the specified voltage.

12.4.1.1 Required Space for Installation



12.4.1.2 Transport and Handling

- Be careful during transportating the unit so that it is not damaged by impact.
- Only remove the packaging material once it has reached desired installation location.
- It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.
- The Tank Unit can be transported either in vertical or horizontal.
  - If it transported in horizontal, make sure Front of packaging material (printed with "FRONT") must facing upwards.
  - If it transported in vertical, use the hand holes on sides, slide and move to the desired location.
- Fix the Adjustable Feet [1] , if the Tank unit installed on a uneven surface.



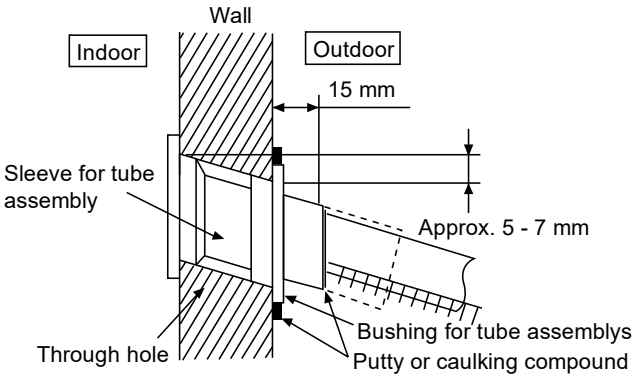
12.4.2 To Drill a Hole in the Wall and Install a Sleeve of Piping

1. Make a through hole. (Check pipe diameter and insulation thickness)
2. Insert the piping sleeve to the hole.
3. Fix the bushing to the sleeve.
4. Cut the sleeve until it extrudes about 15 mm from the wall.

⚠ CAUTION

When the wall is hollow, please be sure to use the sleeve for tube assembly to prevent dangers caused by mice biting the connection cable.

5. Finish by sealing the sleeve with putty or caulking compound at the final stage.

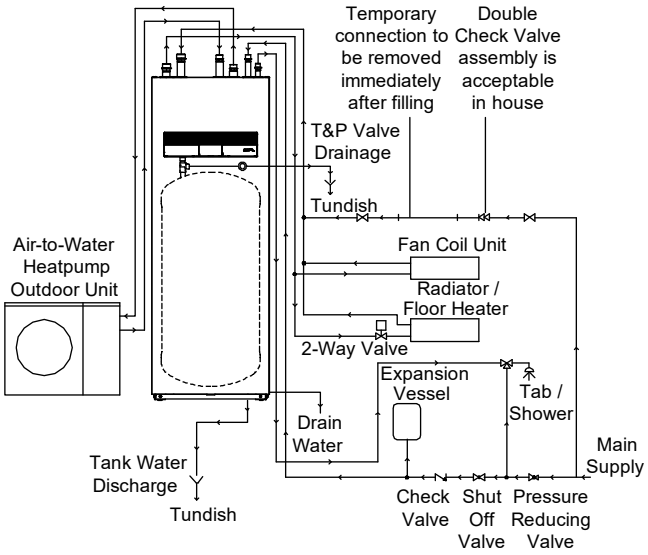


12.4.3 Piping Installation

12.4.3.1 Water Quality Requirement

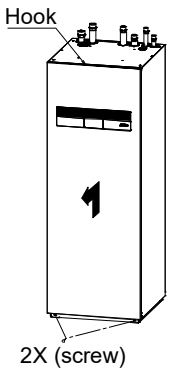
Must use water that complies with European Drinking Water Directive (EU)2020/2184. The lifespan of the Tank Unit will be shorter if groundwater (include spring water and well water) is used.
The Tank Unit shall not be used with the tap water containing contaminants such as salt, acid, and other impurities which may corrode the tank and its component.
Use sanitized water that is free of Legionella and other bacteria and microorganisms. If the water contains Legionella bacteria, it may harm the health of the user.

12.4.3.2 Typical Piping Installation



12.4.3.3 Access to Internal Components

⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

*The remote control cable is connected to the front panel, so be careful when removing the panel.

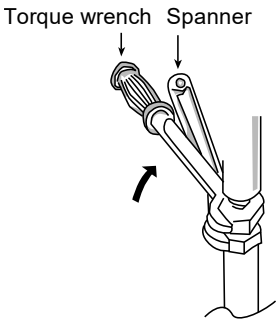
Open and Close Front Plate ⑱

1. Remove the 2 mounting screws of Front Plate ⑱.
2. Slide it upwards to unhook the Front Plate ⑱ hook.
3. Reverse above steps 1~2 for close it.

12.4.3.4 Water Piping Installation

- Please engage a licensed water circuit installer to install this water circuit.
- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not use worn out tube or detachable hose-set.
- Do not apply excessive force to pipes that may damage the pipes.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Further tighten the nuts with torque wrench in specified torque as stated in the table.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall.
- If non-brass metallic piping is used for installation, make sure to insulate the pipes to prevent galvanic corrosion.
- Do not connect galvanised pipes, this will cause galvanic corrosion.
- Use correct nut for all Tank Unit tube connections and clean all tubes with tap water before installation. See Tube Position Diagram for detail.

Tube Connector	Nut Size	Torque
ⓐ & ⓑ	RP 1¼"	117.6 N•m
ⓒ & ⓓ	RP ¾"	58.8 N•m
ⓔ & ⓕ	RP 1"	88.2 N•m



⚠ CAUTION
Do not overtighten, overtightening may cause water leakage.

- Make sure to insulate the water circuit pipes to prevent reduction of heating capacity.
- After installation, check the water leakage condition in connection area during test run.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Protection From Frost:
  If the Tank Unit is being exposed to frost while power supply failure or pump operating failure, drain the system. When water is idle inside the system, freezing up is very likely to happen which could damage the system. Make sure the power supply is turned off before draining. Heater Assembly ⑩ may be damaged under dry heating.
- Corrosion Resistance:
  Duplex stainless steel is naturally corrosion resistant to mains water supply. No specific maintenance is required to maintain this resistance. However, please note that Tank Unit is not guaranteed for use with a private water supply.
- It is recommended to use a tray (field supply) to collect water from the Tank Unit if water leakage occur.

Recommended piping installation sequence:
(a) → (c) → (e) → (f) → (b) → (d)

(A) Space Heating/Cooling Pipework

- Connect Tank Unit Tube Connector ⓐ to outlet connector of Zone 1 Panel/Floor heater.
- Connect Tank Unit Tube Connector ⓑ to inlet connector of Zone 1 Panel/Floor heater.
- Failure to connect the tube appropriately might cause the Tank Unit malfunction.
- Refer below table for the rated flow rate of each particular Outdoor Unit.

Model		Rated Flow Rate (l/min)	
Tank Unit	Outdoor Unit	Cool	Heat
WH-ADC0509L3E5UK1	WH-WDG05LE5*	14.3	14.3
	WH-WDG07LE5*	20.1	20.1
	WH-WDG09LE5*	23.5	25.8

*Do not install automatic air purge valves on indoor water pipes. In the unlikely event that the R290 refrigerant leaks into the water circuit, there is a risk that the refrigerant will leak indoors.

(B) Circulating Pipework

- Connect Tank unit Tube Connector ⓕ to outdoor unit inlet water socket.
- Connect Tank Unit Tube Connector ⓔ to Outdoor unit outlet water socket.
- Failure to connect results in an error stop the system.

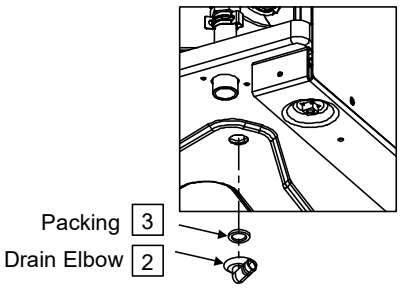
Model	Water piping between outdoor unit and indoor unit			
	Inner diameter	Maximum length	Insulator thickness	Maximum Elevation
WH-WDG05LE5*	ø20	30 m	30 mm or more	10 m
WH-WDG07LE5*	ø25			
WH-WDG09LE5*				

(C) Domestic Hot Water Tank Pipework

- It's strongly recommended to install an expansion vessel (field supply) in the Domestic Hot Water Tank circuit. Refer Typical Piping Installation section to locate the expansion vessel.
  - ■ Expansion Vessel type and specifications:
    - Size : Not more than ¾"
    - Pre-charge pressure : 3.5 bar (0.35MPa)
- If secondary return circuits are used then an additional expansion vessel may be required.
- In high water pressure or water supply is above 5 bar, please install the Pressure Reducing Valve for water supply. If the pressure higher than that, it might damage the Tank Unit.
- A Pressure Reducing Valve (field supply) and Pressure Relief Valve (field supply) with below specification must be installed along the line of the tube connector ⓒ of Tank Unit. Refer Typical Piping Installation section to locate both of these valves.
  - ■ Pressure Reducing Valve type and specifications:
    - Size : Not more than ¾"
    - Set Pressure : 3.5 bar (0.35MPa)
  - ■ Pressure Relief Valve type and specifications:
    - Size : Not more than ¾"
    - Set Pressure : 8 bar (0.8MPa)
- The pressure after pressure reducing valve is less than 3.5 bar (0.35MPa).
- Must connect a faucet to Tank Unit Tube Connector ⓓ and urban water supply, in order to supply water with appropriate temperature for shower or tap usage. Failure to do so might cause scalding.
- Failure to connect the tube appropriately might causing the Tank Unit malfunction.

(D) Drain Elbow and Hose Installation

- Fix the Drain Elbow [2] and Packing [3] to the bottom of Drain Water Hole ⓗ.



- Use inner diameter 17 mm drain hose in the market, fix to Drain Elbow [2].

- This hose must to be installed in a continuously downward direction and in a frost-free environment. Improper drain piping may cause water leakage hence damage the furnitures.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Guide the drain hose to outdoor as illustrated.

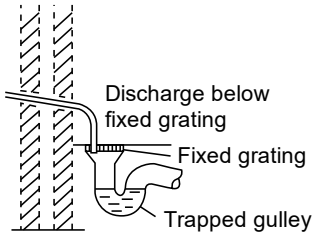


Illustration of guide drain hose to outdoor

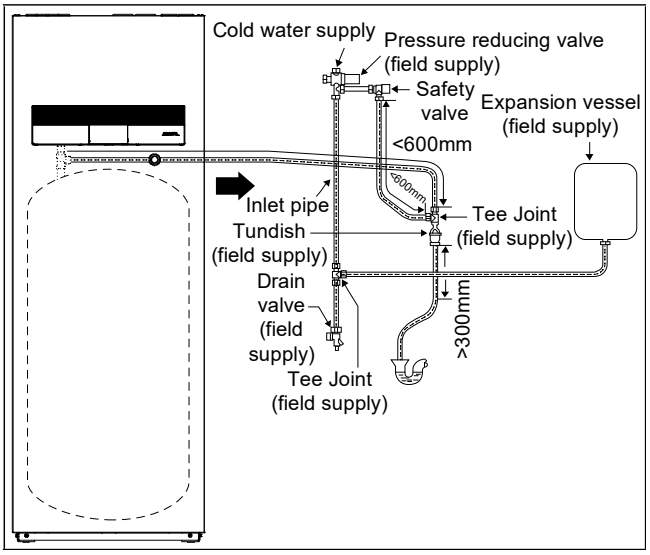
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulphuric gas etc.
- If necessary, use hose clamp to further tighten the hose at drain hose connector to prevent leakage.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.
- If drain hose is in the room (where dew may form), please increase the insulation by using POLY-E FOAM with thickness 6 mm or above.

(E) Domestic Hot Water Tank Discharge (Drain Tap) and Safety Relief Valve Pipework

- Use R½" male connector for this drainage outlet connection (Tube connector ⑨).
- Piping must always be installed in a continuously downward direction. It must not be longer than 2m, with no more than 2 elbows, and must not allow condensation to build up or freezing to occur.
- The pipe from this drainage outlet fitting must not be shut off. The discharge must be freed.
- The end of this pipework must be in such a way so that the outlet is visible and can not cause any damage. Keep away from electrical components.
- It is recommended to fit a tundish into this ⑨ pipework. Tundish should be visible and positioned away from frost environment and electrical components.

(F) Temperature and Pressure Relief Valve

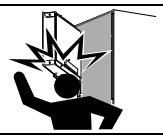
The Temperature and Pressure Relief Valve need appropriate discharge pipework. In accordance with Building Regulations a tundish must be fitted into the pipework within 600mm of the safety device. Due to the distance between the two safety devices it may be necessary to fit each safety device with its own tundish before run the pipework together to a safety discharge. The Rear Plate ㉒ has a Bushing so that the connection can be made to the factory fitted Temperature and Pressure Relief Valve. Then connect the Temperature and Pressure Relief Valve to the discharge pipework (Ø15mm). The following instructions are a requirement of UK Building Regulations and must be adhere to. For the other countries please refer to local legislation. If there is any doubt the insulation procedure, always contact local building office.



1. Connect the tundish and route the discharge pipe as shown above. Tundish should be visible and positioned away from frost environment and electrical components.
2. The tundish should be fitted vertically and as close to the safety device as possible and within 600mm of the device.
3. The tundish should be visible to users and positioned away from electrical devices.
4. The discharge pipe from the tundish should be terminate in a safe place where there is no risk to person nearby to the discharge, be of metal construction and:
   A) Be at least one pipe size larger than the nominal outlet size of the safety device unless its total equivalent hydraulic resistance exceeds that of a straight pipe 9m long. Bends must be taken into account in calculating the flow resistance.
   B) Have a vertical section of pipe at least 300mm long, below the tundish before any elbows or bends in the pipework.
   C) Be installed with a continuous fall.
   D) Have discharges visible at both the tundish and final point of discharge.

12.4.4 Connect the Cable to the Tank Unit

⚠ WARNING
This section is for authorized and licensed electrician only. Work behind the Control Board Cover ⑤ secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

⚠ CAUTION	
Please take extra precaution when open the control board cover ⑤ and control board ⑥ for unit installation and servicing. Failure to do so may cause injury.	

12.4.4.1 Fixing of Power Supply Cable and Connecting Cable

1. Connecting cable between Tank Unit and Outdoor Unit shall be approved polychloroprene sheathed flexible cord, type designation 60245 IEC 57 or heavier cord. See below table for cable size requirement.

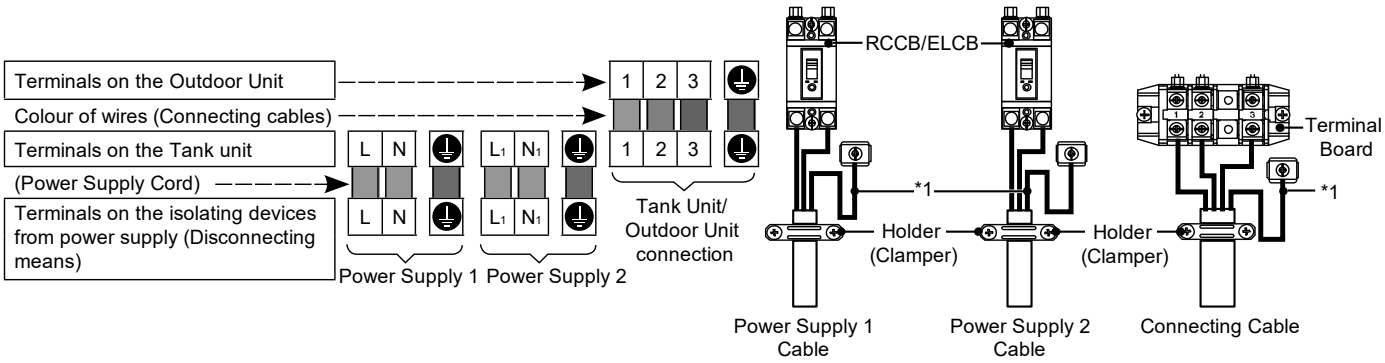
Model		Connecting Cable Size
Tank Unit	Outdoor Unit	
WH-ADC0509L3E5UK1	WH-WDG05LE5* WH-WDG07LE5* WH-WDG09LE5*	4 × min 2.5 mm^2

- Ensure the colour of wires of Outdoor Unit and the terminal no. are the same to the Tank Unit respectively.
- Earth wire shall be longer than the other wires as shown in the figure for the electrical safety in case of the slipping out of the cord from the Holder (Clamper).

2. An isolating device must be connected to the power supply cable.
   - Isolating device (disconnecting means) should have minimum 3.0 mm contact gap.
   - Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means). See below table for cable size requirement.

Model		Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
Tank Unit	Outdoor Unit				
WH-ADC0509L3E5UK1	WH-WDG05LE5* WH-WDG07LE5* WH-WDG09LE5*	1	3 × min 2.5 mm^2	25A	30mA, 2P, type A
		2	3 × min 1.5 mm^2	15/16A	30mA, 2P, type AC

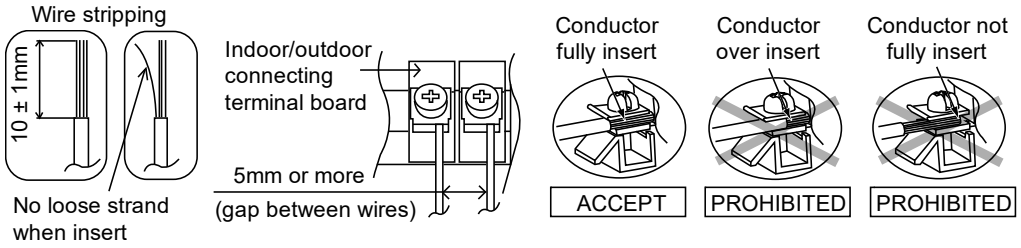
3. To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through a bushing (located at the bottom of Control Board ⑥) before terminal board. The bushing must be used and must not be removed.



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25}

*1 - Earth wire must be longer than other cables for safety reasons

12.4.4.2 Wire Stripping and Connecting Requirement



12.4.4.3 Connecting Requirement

For Tank Unit WH-ADC0509L3E5UK1 with WH-WDG05LE5*, WH-WDG07LE5*, WH-WDG09LE5*

- The equipment's Power Supply 1 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-3 and can be connected to current supply network.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-3 and can be connected to current supply network.

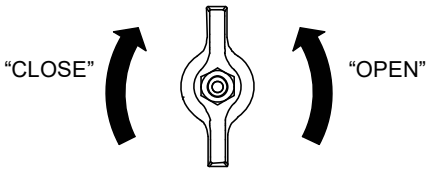
12.4.5 Charging and Discharging the Water

- Make sure all the piping installations are properly done before carry out below steps.

12.4.5.1 Charge the Water

For Domestic Hot Water Tank

1. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to "CLOSE".

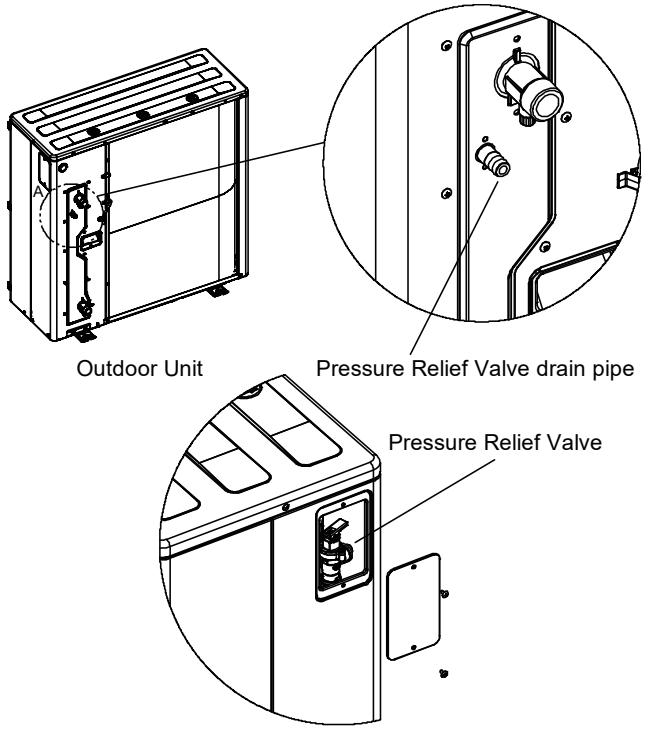


Domestic Hot Water Tank Discharge (Drain Tap) ⓖ

2. Set all Tap / Shower "OPEN".
3. Start filling water to the Domestic Hot Water Tank via Tube Connector ⓒ.
   After 20~40min, water should flow out from Tap / Shower.
   Else, please contact your local authorized dealer.
4. Check and make sure no water leaking at the tube connecting points.

For Space Heating / Cooling

1. Start filling water (with pressure more than 1 bar (0.1MPa)) to the Space Heating/Cooling circuit via Tube Connector ⓐ.
2. Stop filling water if the free water flow through Pressure Relief Valve drain pipe. (Check the Outdoor Unit)
3. Turn ON the Tank Unit.
4. Remote control menu → Installer setup → Service setup → pump maximum speed → Turn on the pump.
5. Make sure Water Pump ④ is running.
6. Check and make sure no water leaking at the tube connecting points.



12.4.5.2 Discharge the Water

For Domestic Hot Water Tank

1. Turn OFF power supply.
2. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to "OPEN".
3. Open Tap / Shower to allow air inlet.
4. After discharge, set Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to "CLOSE".

12.4.6 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings.

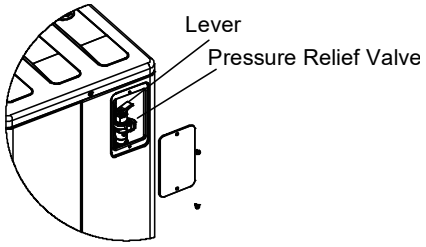
12.4.6.1 Check Water Pressure

*(1 bar=0.1MPa)
Water pressure should not lower than 0.5 bar (with inspects the Water Pressure from Remote Controller).
If necessary add water into Tank Unit (via Tube Connector ⓐ).

12.4.6.2 Check Pressure Relief Valve

*Pressure Relief Valve is mounted in the Outdoor Unit.

1. Confirm that the pressure relief valve is working properly, Pull the lever horizontal direction.
2. Release the lever when water comes out of the drain pipe of the pressure relief valve.
   (While the air continues to come out of the drain pipe, keep raising the lever to completely discharge the air.)
3. Confirm that the water from the drain pipe stops.
4. If water is leaking, pull the lever several times and return it to make sure the water stops.
5. If water keeps coming out of the drain, drain water.
   Turn off the system and contact your local authorized dealer.



12.4.6.3 Check Air Accumulation

- Open the air vent plugs on the heating panel, fan convector, etc., and remove the air accumulated in the equipment and piping.
- If the outdoor unit and the indoor unit are installed on different floors, open the air vent plug on the water plug of the outdoor unit and the air vent plug on the heater bottle inside the indoor unit to remove the air. (be careful, water will come out)

12.4.6.4 Expansion Vessel ⑬ Pre Pressure Checking

For Space Heating / Cooling

- Expansion Vessel ⑬ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit's piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m. (Extra pump may be required)

12.4.6.5 Check RCCB/ELCB

Ensure the RCCB/ELCB set to "ON" condition before check RCCB/ELCB.
Turn on the power supply to the Tank Unit.
This testing could only be done when power is supplied to the Tank Unit.

⚠ WARNING
Be careful not to touch parts other than RCCB/ELCB test button when the power is supplied to Tank Unit. Else, electrical shock may happen. Before obtaining access to terminals, all supply circuits must be disconnected.

- Push the "TEST" button on the RCCB/ELCB. The lever would turn down and indicate "0", if it functions normal.
- Contact authorized dealer if the RCCB/ELCB malfunction.
- Turn off the power supply to the Tank Unit.
- If RCCB/ELCB functions normal, set the lever to "ON" again after testing finish.

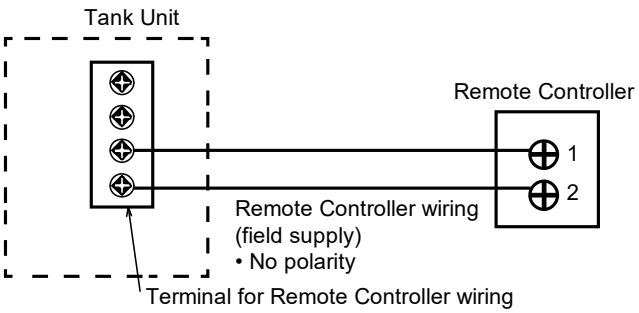
12.4.7 Installation of Remote Controller as Room Thermostat

- Remote Controller ① mounted to the Tank Unit can be moved to the room and serve as Room Thermostat.

12.4.7.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1. By the window, etc. exposed to direct sunlight or direct air.
  2. In the shadow or backside of objects deviated from the room airflow.
  3. Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4. Location near heat source.
  5. Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

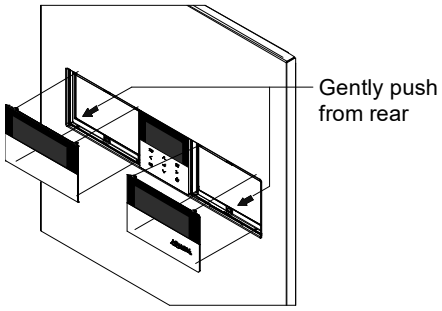
12.4.7.2 Remote Controller Wiring



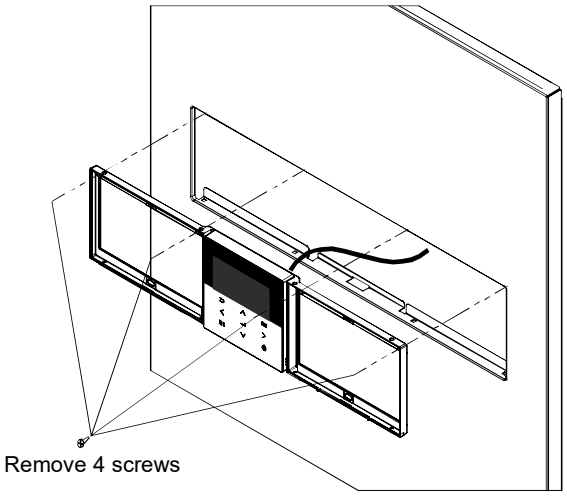
- Remote Controller cable shall be (2 × min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals of Tank Unit (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.
- When using the 2nd. Remote Controller (option), connect it to the terminal of the tank unit by tightning it together.

12.4.7.3 Remove the Remote Controller from Tank Unit

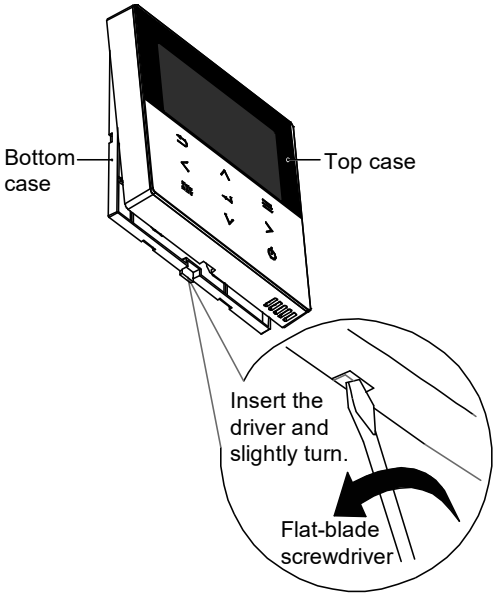
1. Remove both Left Decoration Panel ② and Right Decoration Panel ③ from Front Plate ⑱ with gently push the panels from back.



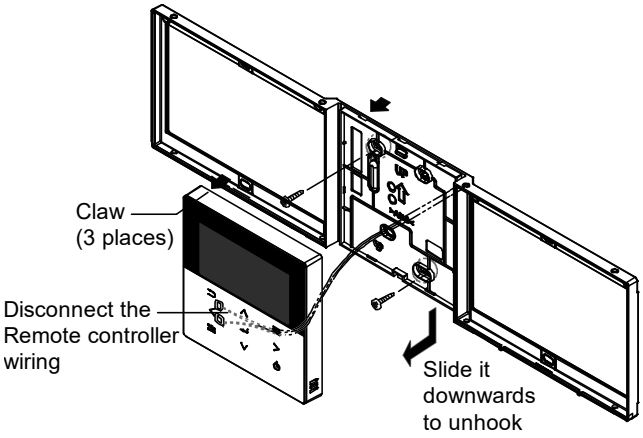
2. Remove the 4 screws and take out the holder with Remote Controller ①.



3. Remove the top case from the bottom case.



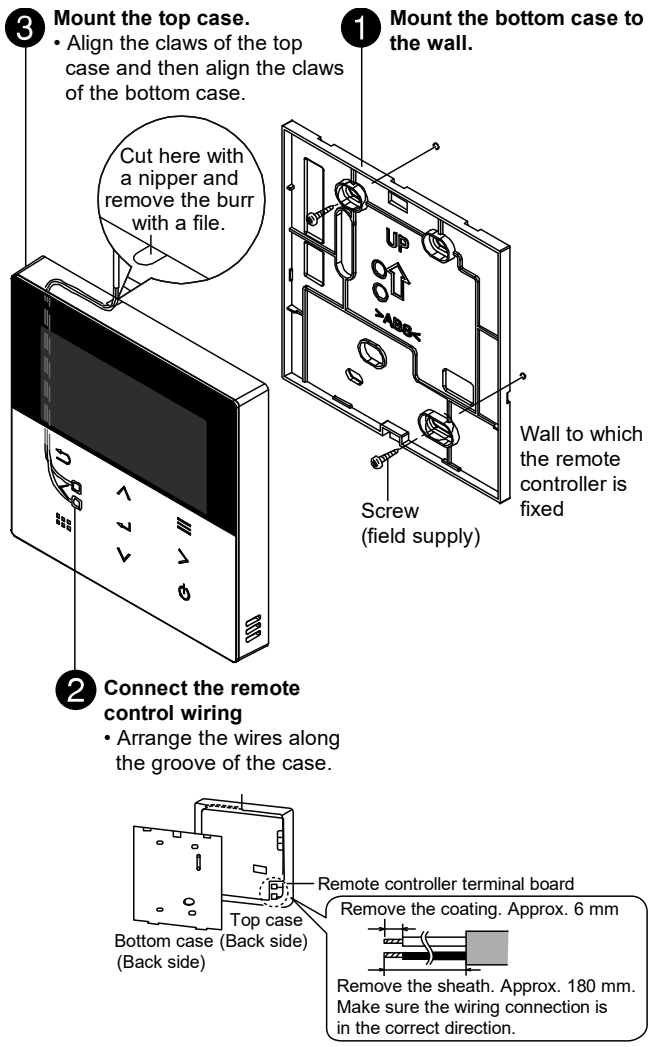
4. Remove the wiring between Remote controller ① and Tank Unit terminal.



12.4.7.4 Mounting the Remote Controller

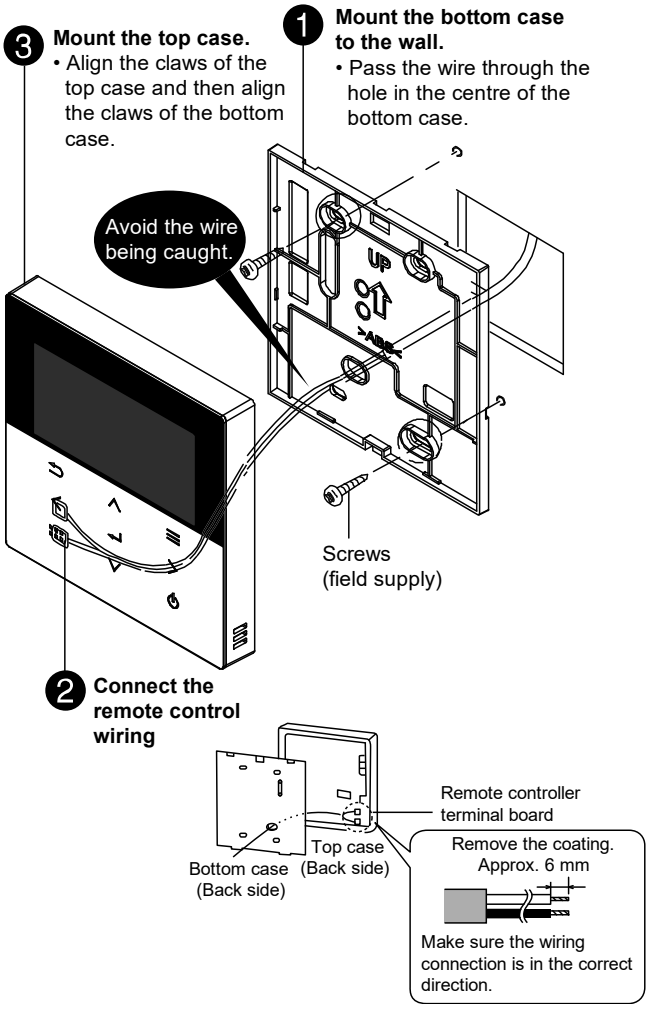
For exposed type

Preparation: Make 2 holes for screws using a driver.



For embedded type

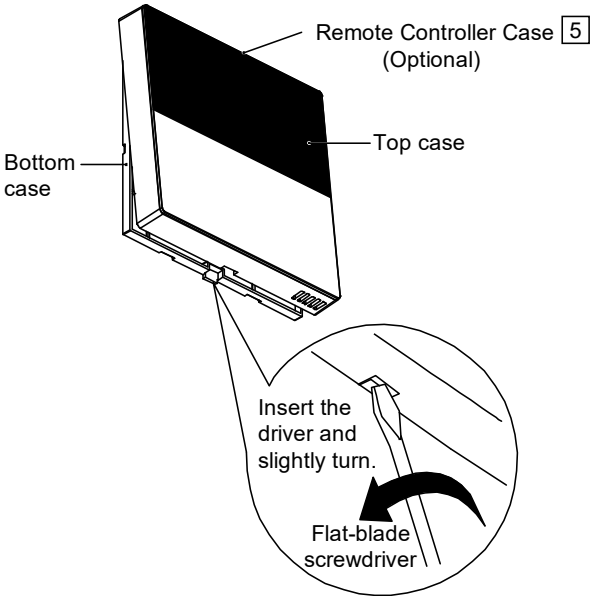
Preparation: Make 2 holes for screws using a driver.



12.4.7.5 Replace the Remote Controller Cover

- Replace the existing Remote Controller with Remote Controller Case [5] to close the hole left after remove the Remote Controller.

1. Refer Section “Remove The Remote Controller From Tank Unit” for remove Remote Controller.
2. Remove the top case from the bottom case of Remote Controller Case [5] .



3. Reverse the steps 1 to 4 of section “Remove The Remote Controller From Tank Unit” to fix Remote Controller Case [5] on Tank Unit.

12.4.8 Test Run

1. Before test run, make sure below items have been checked:-
   a) Pipework are properly done.
   b) Electric cable connecting work are properly done.
   c) Tank Unit is filled up with water and trapped air is released.
   d) Please turn on the power supply after filling the tank until full.
2. Switch ON the power supply of the Tank Unit. Set the Tank Unit RCCB/ELCB to “ON” condition. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run.
  Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

3. For normal operation, Water Pressure reading should be in between 0.5 bar and 3 bar (0.05 MPa and 0.3 MPa) If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4. After test run, please clean the Magnetic Water Filter Set ⑨ and Water Filter Set ㉚. Reinstall it after finish cleaning.

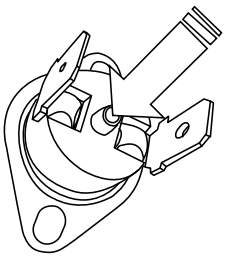
12.4.8.1 Check Water Flow of Water Circuit

Select Installer setup → Service setup → Pump maximum speed → Air purge.
Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]
*If there is no flow or H62 is displayed, stop operating the pump and release the air (see Checking for Air Accumulation 12.4.6.3).

12.4.8.2 Reset Overload Protector ⑫

Overload Protector ⑫ a serves the safety purpose to prevent the water over heating. When the Overload Protector ⑫ a trip at high water temperature, take below steps to reset it.

1. Take out the cover.
2. Use a test pen to push the centre button gently in order to reset the Overload Protector ⑫.
3. Fix the cover to the original fixing condition.

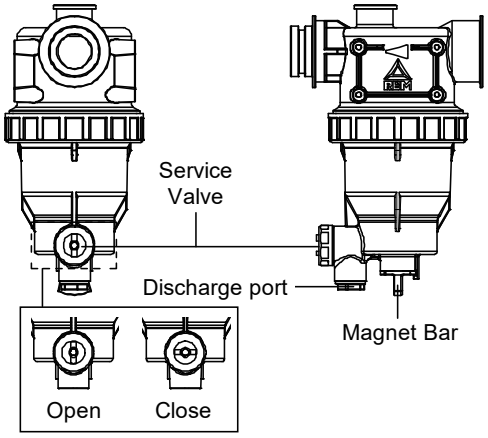


12.4.9 Maintenance

- In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

12.4.9.1 Maintenance for Magnetic Water Filter Set ⑨

1. Turn OFF power supply.
2. Place a container below Magnetic Water Filter Set ⑨.
3. Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑨.
4. By using Allen key (8mm), remove the Cap of Discharge Port.
5. By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.
6. Reinstall the Cap of Discharge Port and Magnet Bar.
7. Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 12.4.5 for details.)
8. Turn ON power supply.



12.4.9.2 Maintenance for Temperature and Pressure Relief Valve ㉕

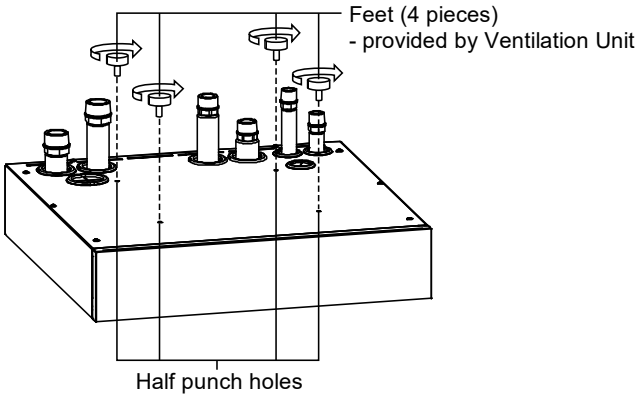
- Manually operate the Temperature and Pressure Relief Valve ㉕ by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

12.4.9.3 Installation of Ventilation Unit on Top of Tank Unit (Optional)

- For installation works of Ventilation Unit on top of Tank Unit, refer to the Ventilation Unit Installation Manual.

⚠ CAUTION
Before install Ventilation Unit, fix the Feet that provided by Ventilation Unit to the half punch holes on Top Panel of Tank Unit, Otherwise, heavy Ventilation Unit may fall and cause injury.



12.4.9.4 Technical Data

• Model	WH-ADC0509L3E5UK1
• Tank Capacity - Nominal - Actual	 120 L 120 L
• Rate Heat Exchanger Volume	6 L
• Weight - Empty - Full	 79 kg 206 kg
• Maximum operating Pressure - Primary - Secondary	 3.0 bar 3.5 bar
• Maximum Operating Temperature	75°C
• Operating Pressure - Tank Unit - Expansion Relief Valve	 3.5 bar 8.0 bar
• Maximum Working / Design Pressure - Space Heating / Cooling - Tank Circuit	 3.0 bar 10.0 bar
• Standing Heat Loss	1.15 kWh/24h
• Reheat Time	53 min
• Temperature and Pressure Relief Valve - Size - Pressure Relief - Temperature Relief	 ½" × 15mm 10 bar 90°C - 95°C
• Primary Heater Pressure Drop	0.2 bar
• Primary Flow Rate (Nominal)	9.2 - 25.8 L/min
• Primary Heating Power Input / Flow Rate	23.1 kW/ 15 L/min
• Hot Water Capacity as per EN 12897:2016	110 L

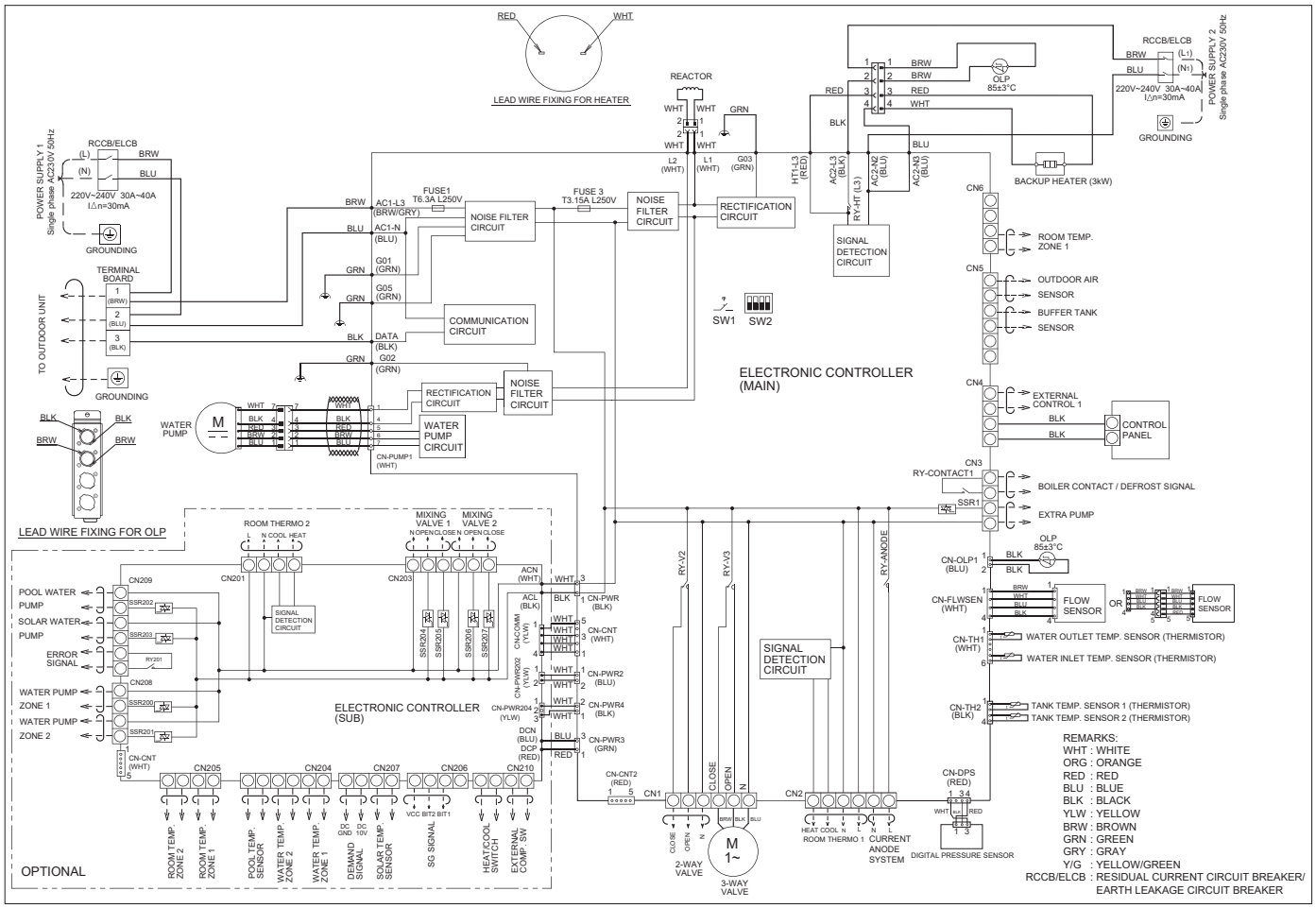
This Tank Unit comply with EN 12897:2016.

12.4.9.5 PAW-G3KIT-1 - Panasonic G3 22MM Unvented Kit

Part Code	Description	WRAS Approval Number
533117H CST	Inlet Control Set – 3.5 Bar PRV and 6 Bar SRV	1907367
AI-002402	½" Drain Cock	1811800
TI-10396151504	15mm × ½" × 15mm Tee	1804353
TI-1130031	22mm × ¾" MI DZR Elbow	1804353
PA-2822362151001	¾" F × ¾" M 1000mm Hose	2201803
DE 18 (7303030)	18 Litre Potable Vessel with Integral Bracket	2209753
2209753	15 × 22mm Straight Tundish	1901349

12.4.9.6 Control of Water Temperature : PCB, Limit Thermostat

WIRING DIAGRAM



12.5 Appendix

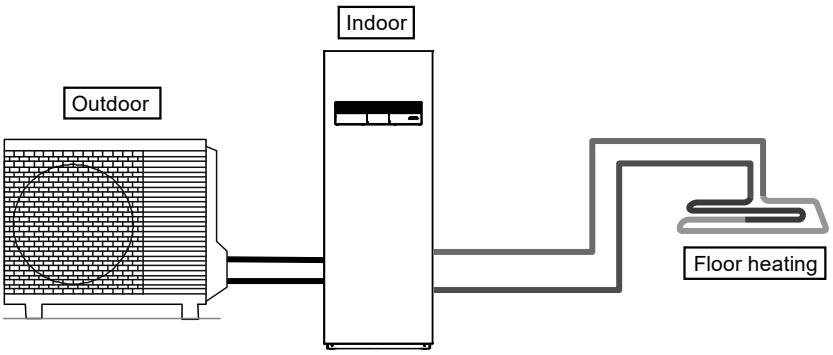
12.5.1 Variation of System

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.
(NOTE) : For this model, both external room thermistor of Zone 1 and external room thermostat of Zone 1 must always be connected to main indoor PCB only regardless of Optional PCB (CZ-NS5P) connection.

12.5.1.1 Introduce Application Related to Temperature Setting

12.5.1.1.1 Temperature Setting Variation for Heating

1. Remote Controller



Setting of remote controller

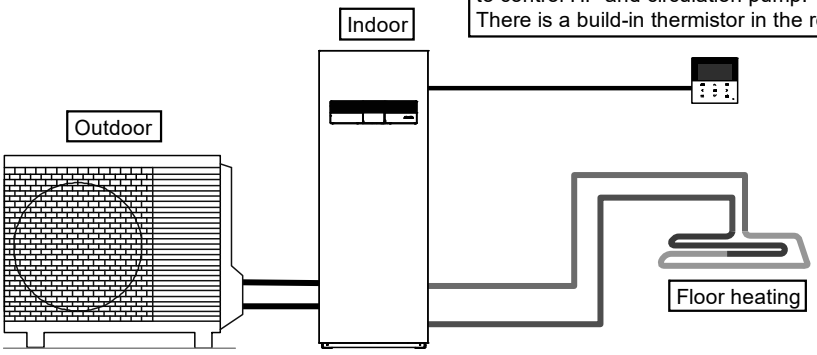
Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Water temperature

Connect floor heating or radiator directly to the Tank Unit.
Remote controller is installed on Tank Unit.
This is the basic form of the most simple system.

2. Room Thermostat

Tank Unit receive Room Thermo signal (ON/OFF) from Remote controller to control HP and circulation pump.
There is a build-in thermistor in the remote controller.



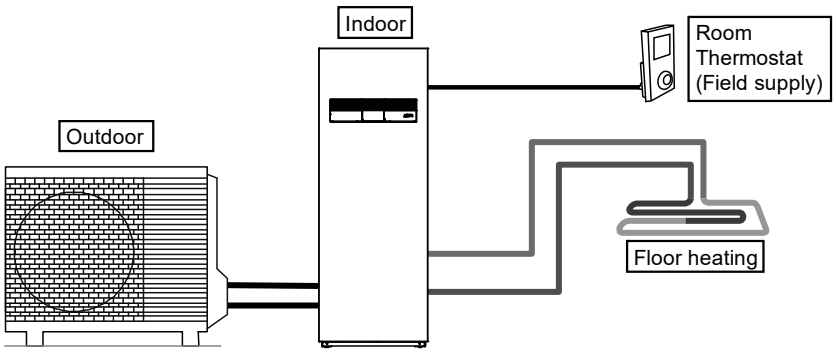
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
Internal

Connect floor heating or radiator directly to the Tank Unit.
Remove remote controller from Tank Unit and install it in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat

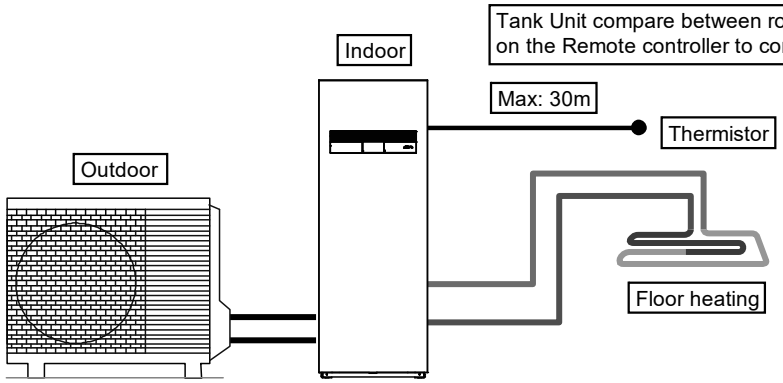


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No
Zone & Sensor:
Room thermostat
(External)

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



Setting of remote controller

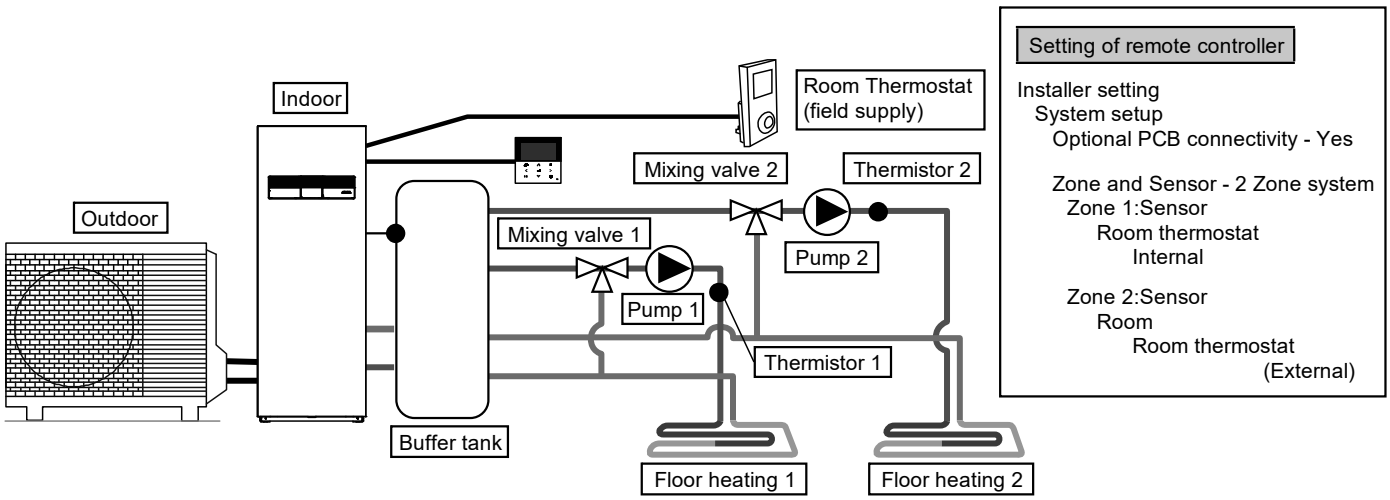
Installer setting
System setup
Optional PCB connectivity - No
Zone & Sensor:
Room thermistor

Connect floor heating or radiator directly to Tank Unit.
Remote controller is installed on Tank Unit.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, compensation curve is shifted according to the thermo ON/OFF situation.
- (Example) If room temperature increasing speed is;
very slow → shift up the compensation curve
very fast → shift down the compensation curve

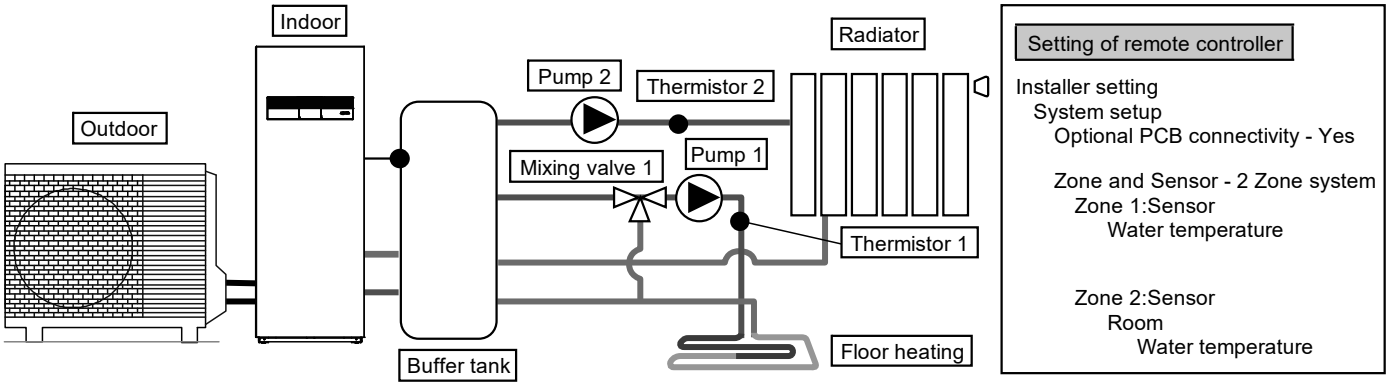
12.5.1.1.2 Examples of Installations

Floor heating 1 + Floor heating 2



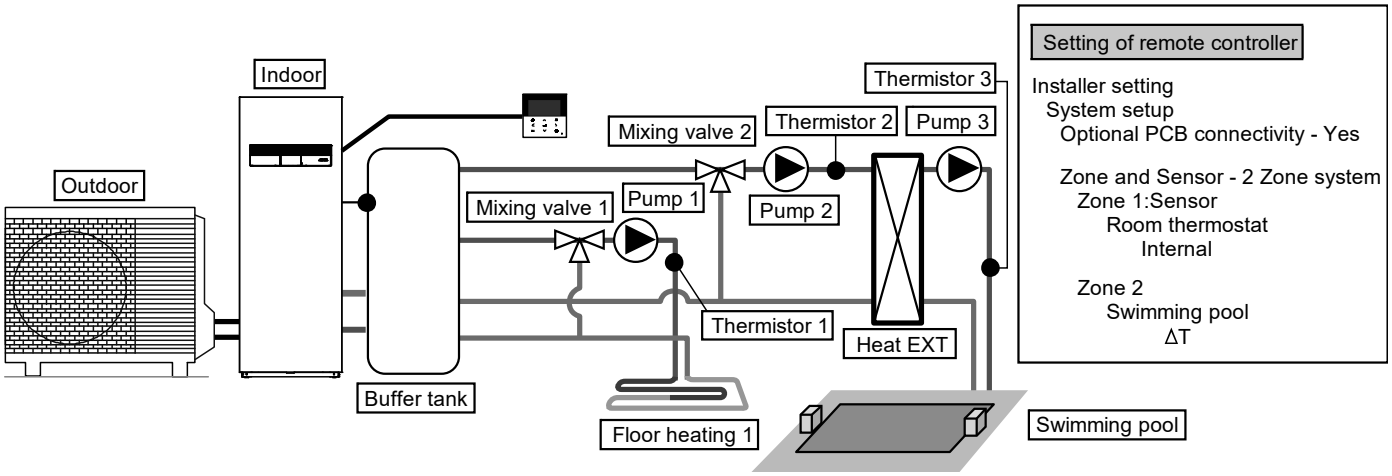
Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Remove remote controller from Tank Unit, install it in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Floor heating + Radiator



Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
Remote controller is installed on Tank Unit.
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the Optional PCB (CZ-NS5P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

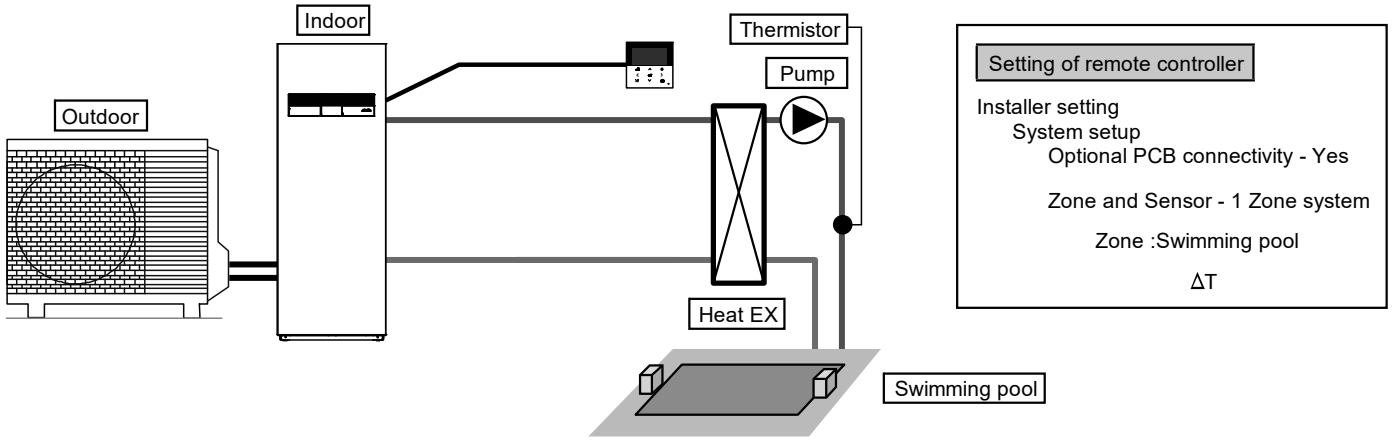
Floor heating + Swimming pool



Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Remove remote controller from Tank Unit and install in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the Optional PCB (CZ-NS5P).

❊ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

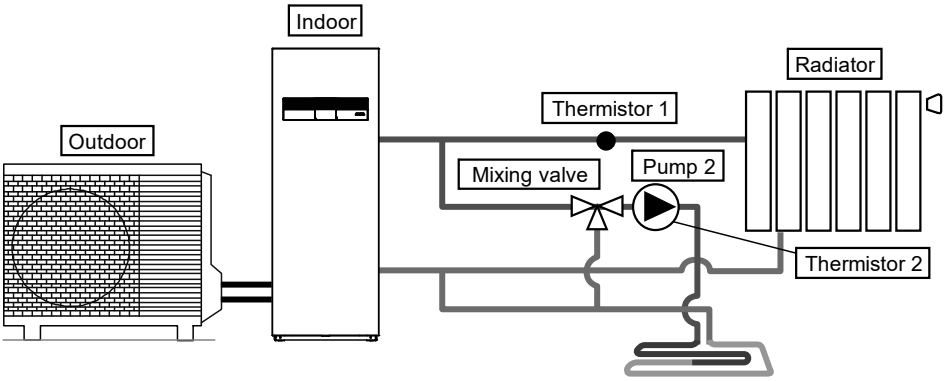
Swimming pool only



This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Tank Unit without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Remove remote controller from Tank Unit and install in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the Optional PCB (CZ-NS5P).

In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



Setting of remote controller

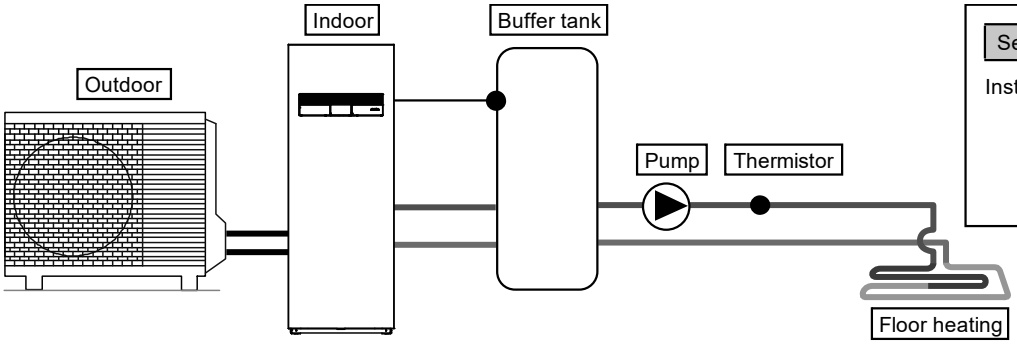
Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1:Sensor
Water temperature
Zone 2:Sensor
Room
Water temperature
Operation setup
Heat
ΔT for heating ON – 1°C
Cool
ΔT for cooling ON – 1°C

This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Tank Unit served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the Optional PCB (CZ-NS5P).

(NOTE)

- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
- Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

Buffer tank connection

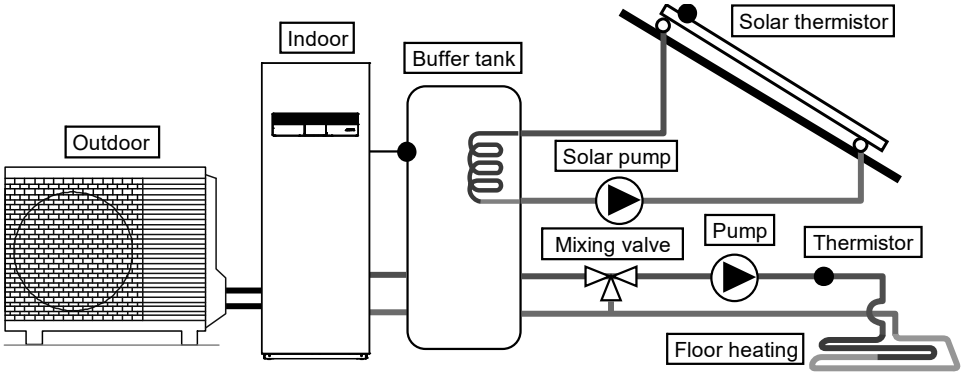


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No
Buffer Tank connection - Yes
ΔT for buffer tank

This is an application that connects the buffer tank to the Tank Unit.
Buffer tank’s temperature is detected by buffer tank thermistor (specified by Panasonic).
Without connection of Optional PCB, external pump can be used for circulation in the floor heating circuit.
Note : Buffer tank thermistor must be connected to main indoor PCB only.

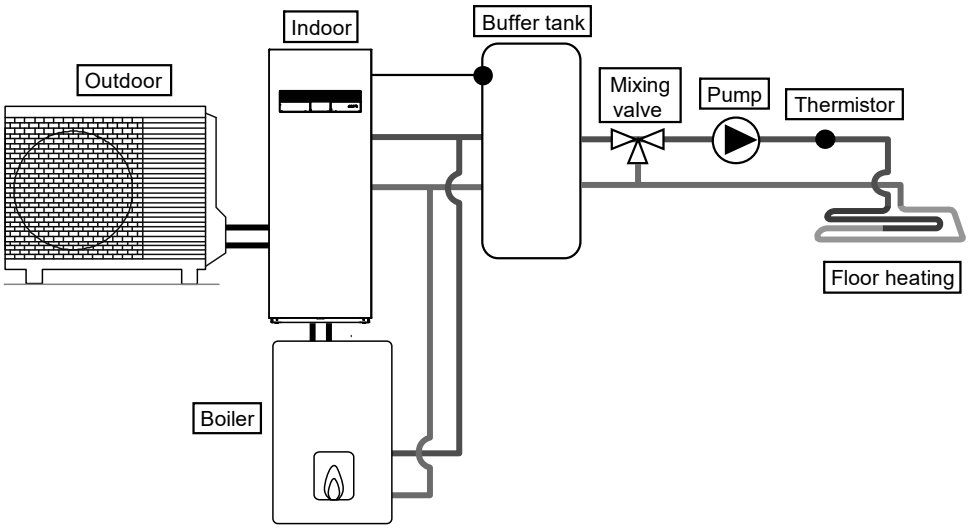
Buffer tank + Solar



Setting of remote controller
Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank
Solar connection - Yes
Buffer tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the buffer tank to the Tank Unit before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires Optional PCB (CZ-NS5P).
Note : Buffer tank thermistor must be connected to main indoor PCB only.

Boiler connection



Setting of remote controller
Installer setting
System setup
Optional PCB connectivity - Yes
Bivalent - Yes
Turn ON: outdoor temp
Control pattern

This is an application that connects the boiler to the Tank Unit, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
Besides that, an application that connects to the DHW tank's circuit to heat up tank's hot water is also possible.
Boiler output can be control by either SG ready input from optional PCB or Auto control by 3 modes selection pattern. (Operation setting of boiler shall be responsible by installer.)
This system requires Optional PCB (CZ-NS5P) for SG ready input control.
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when select Advanced Parallel setting.)
Note : Buffer tank thermistor must be connected to main indoor PCB only.

⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

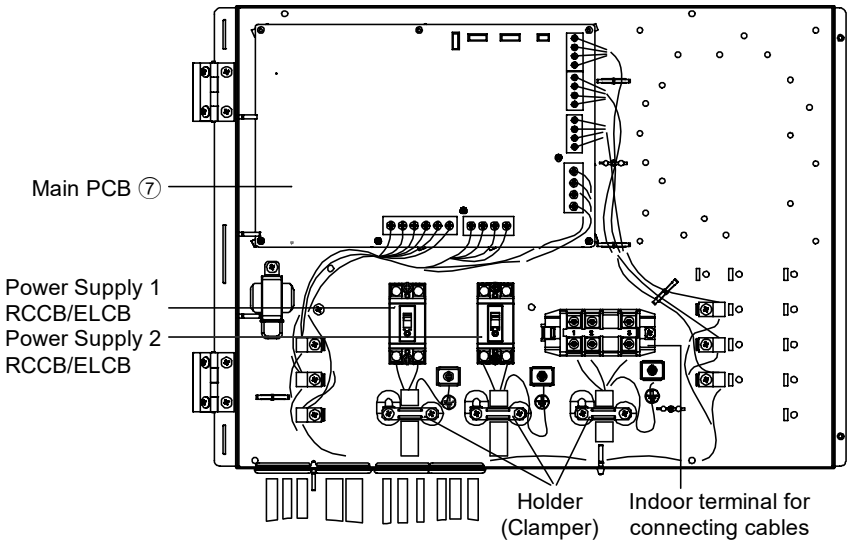
⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the Tank Unit does NOT exceed 70°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

12.5.2 How to Fix Cable

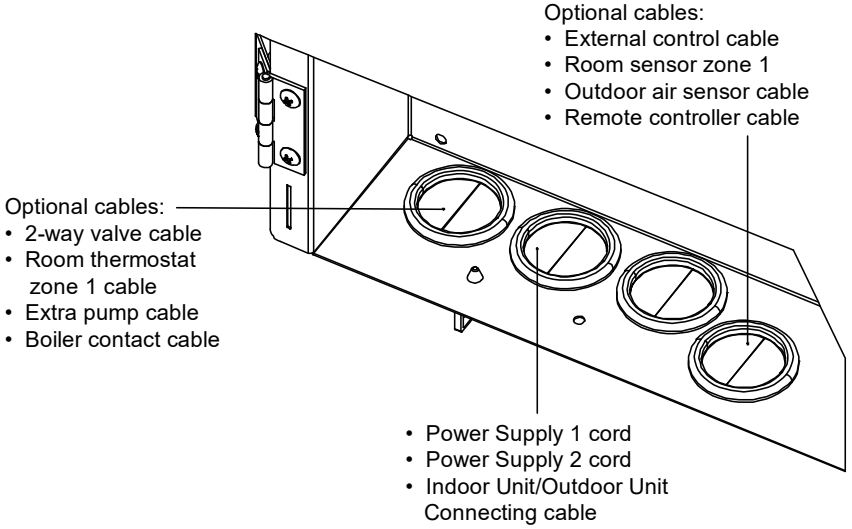
12.5.2.1 Connecting with External Device (Optional)

- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB ⑦

1. Two-way valve shall be spring and electronic type, refer to “Field Supply Accessories” table for details. Valve cable shall be (3 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
   *note: - Two-way Valve shall be CE marking compliance component.
   - Maximum load for the valve is 12VA.
2. Room thermostat cable must be (4 or 3 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
3. Extra pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
4. Boiler contact cable shall be (2 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
5. External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 × min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
   * note: - Switch used shall be CE compliance component.
   - Maximum operating current shall be less than $3A_{rms}$.
6. Room sensor zone 1 cable shall be (2 × min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.
7. Outdoor air sensor cable shall be (2 × min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.

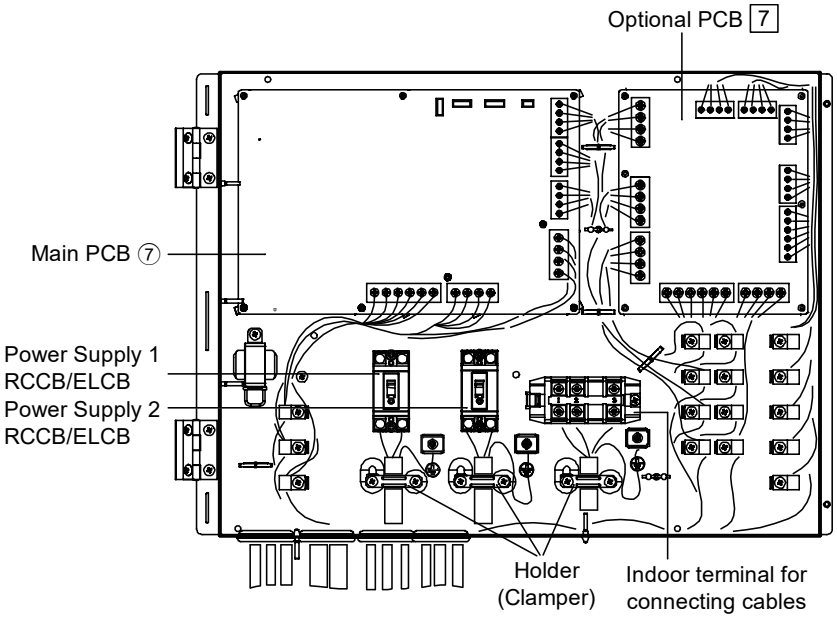


How to guide the optional cables and power supply cord
(view without internal wiring)

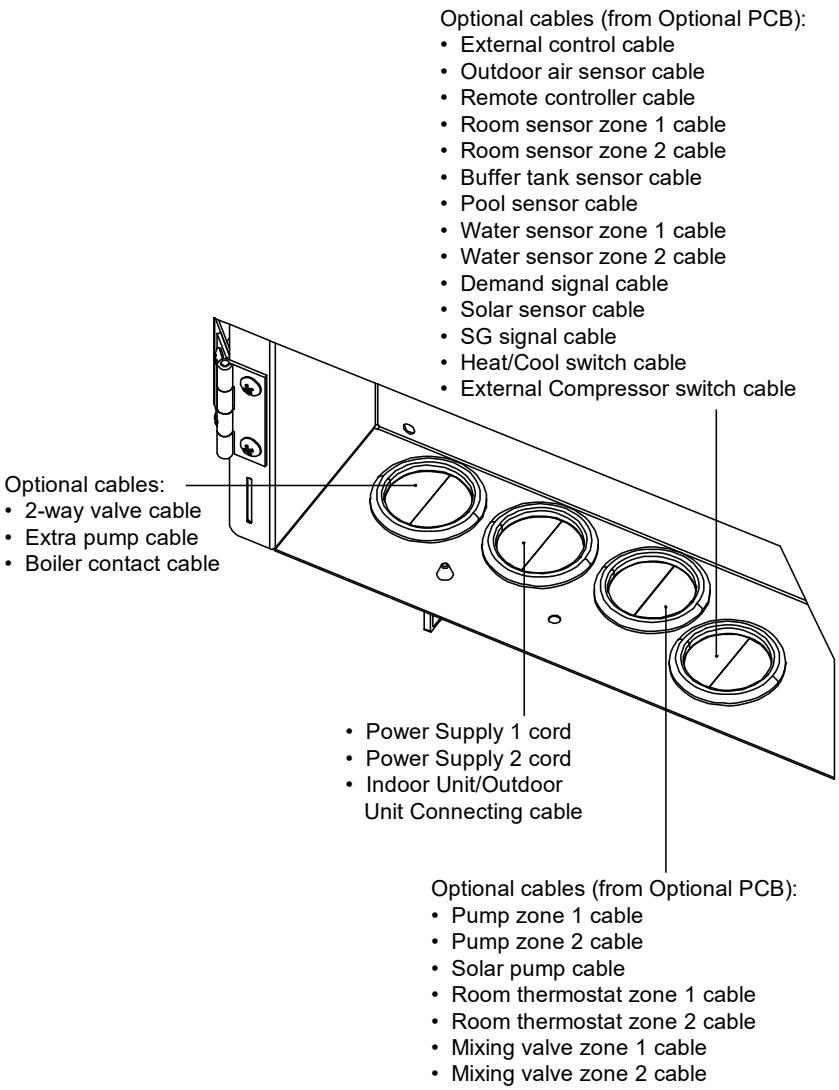


- For connection to Optional PCB 7

1. By connecting Optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in Optional PCB. Temperature of each zone can be controlled independently by remote controller.
2. Pump zone 1 and zone 2 cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
3. Solar pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
4. Pool pump cable shall be (2 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
5. Room thermostat zone 1 and zone 2 cable shall be (4 × min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
6. Mixing valve zone 1 and zone 2 cable shall be (3 × min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
7. Room sensor zone 1 and zone 2 cable shall be (2 × min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
8. Buffer tank sensor, pool water sensor and solar sensor cable shall be (2 × min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
9. Water sensor zone 1 and zone 2 cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
10. Demand signal cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
11. SG signal cable shall be (3 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
12. Heat/Cool switch cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
13. External compressor switch cable shall be (2 × min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide the optional cables and power supply cord
(view without internal wiring)

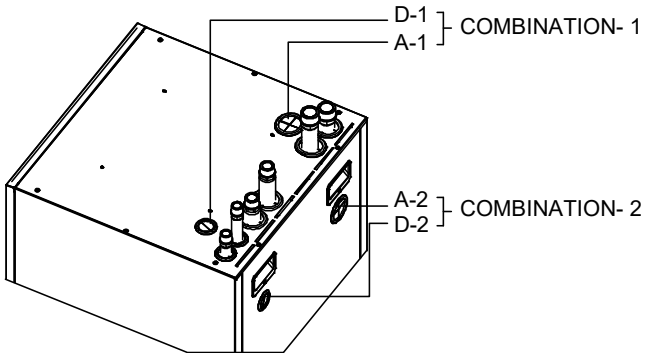


Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

12.5.2.2 Guide Optional Cables and Power Supply Cords to Bushings

⚠ CAUTION
Wire guiding shall be free from hot surfaces. Else, cable insulator damage and electrical shock may happen.
Wire ways shall be smooth and free from sharp edges. Else, cable insulator damage and electrical shock may happen.

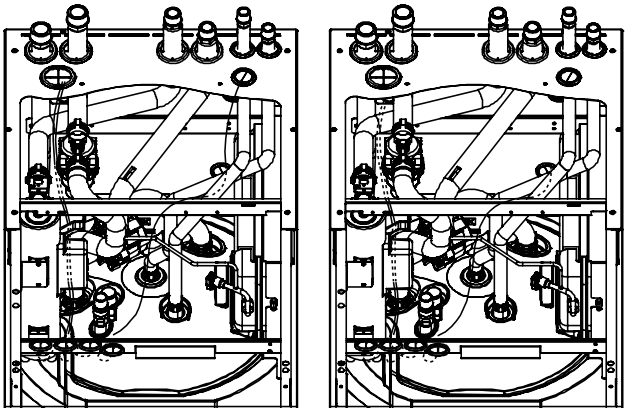
- Use either “COMBINATION-1” or “COMBINATION-2” for guiding Optional Cables and Power Supply Cords to Bushings.



- A-1 and A-2 Bushings are for:
  - Power Supply 1 cord
  - Power Supply 2 cord
  - Indoor Unit/Outdoor Unit Connecting cable
  - Pump zone 1 cable
  - Pump zone 2 cable
  - Solar pump cable
  - Room thermostat zone 1 cable
  - Room thermostat zone 2 cable
  - Mixing valve zone 1 cable
  - Mixing valve zone 2 cable
  - 2-way valve cable
  - Extra pump cable
  - Boiler contact cable

- D-1 and D-2 Bushings are for:
  - External control cable
  - Outdoor air sensor cable
  - Remote controller cable
  - Room sensor zone 1 cable
  - Room sensor zone 2 cable
  - Buffer tank sensor cable
  - Pool sensor cable
  - Water sensor zone 1 cable
  - Water sensor zone 2 cable
  - Demand signal cable
  - Solar sensor cable
  - SG signal cable
  - Heat/Cool switch cable
  - External Compressor switch cable

- Ensure all sensor cables are not touching with Front Panel
- Guide the wiring inside the unit like below figure. Once all wiring work done, tie the cable / cord with the banding strap (field supply), to prevent them touching with hot surfaces such as Heater Assembly, bare copper pipes and etc.



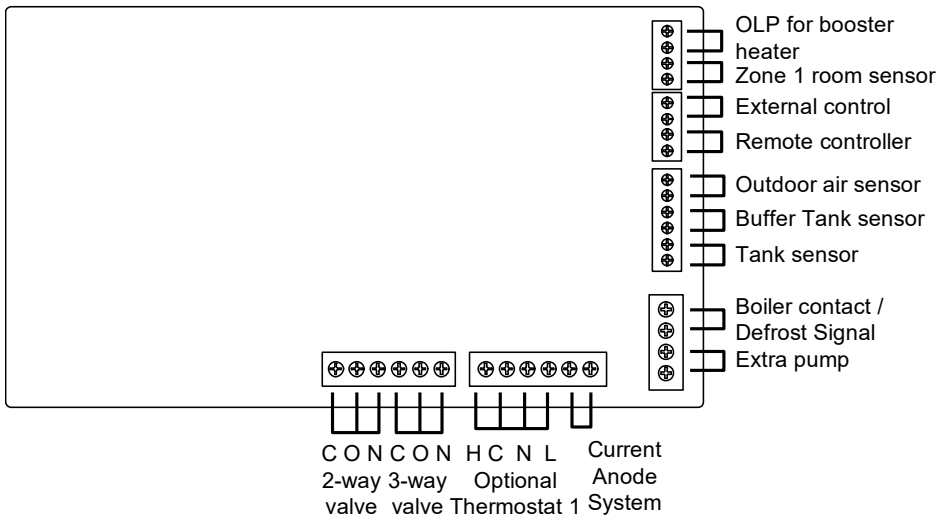
Wiring for “COMBINATION-1” Wiring for “COMBINATION-2”

12.5.2.3 Connecting Cables Length

When connecting cables between Tank Unit and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Mixing valve	50
Room thermostat	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Room sensor	30
Outdoor air sensor	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

12.5.2.4 Connection of the Main PCB



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

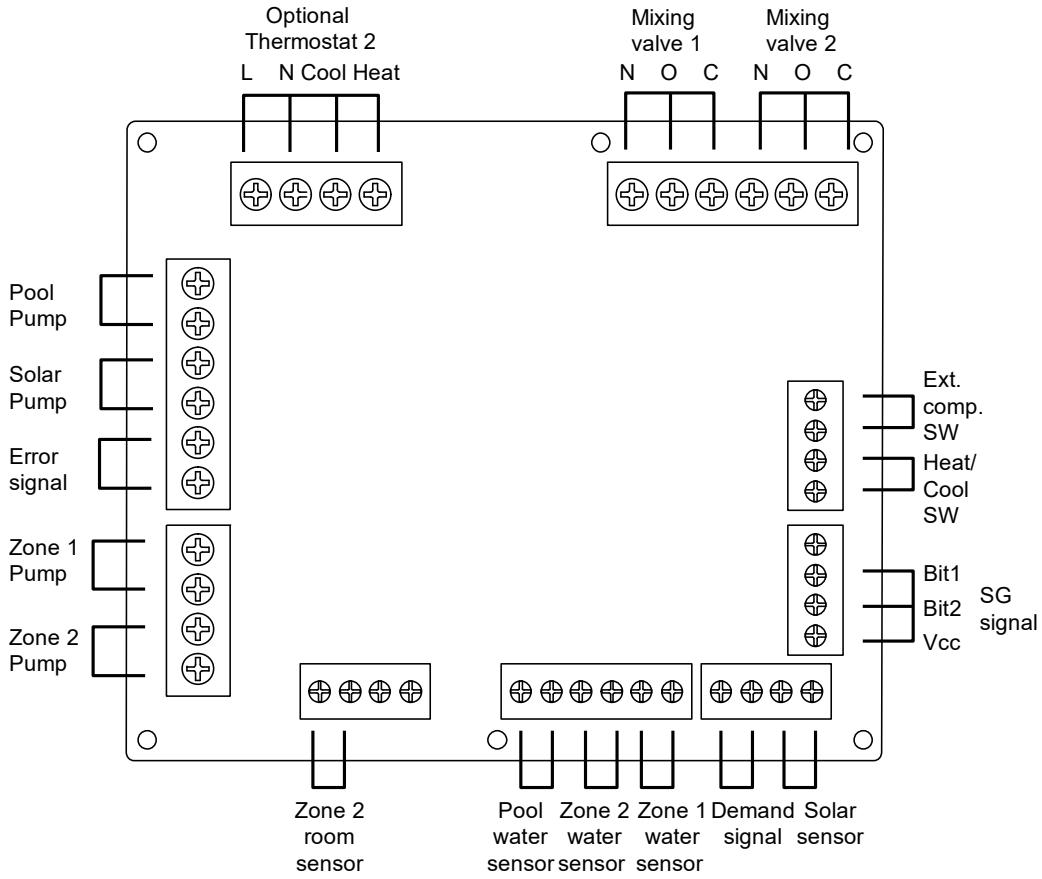
■ Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)	AC230V, 12 VA
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)	AC230V, 12 VA
Extra pump	AC230V (Used when Tank Unit pump capacity is insufficient)	AC 230V, 0.6 A max
Boiler contact / Defrost signal	Dry contact (System setup necessary)	

■ Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT
Outdoor air sensor	PAW-A2W-TSOD (Total cable length shall be 30m or less)

12.5.2.5 Connection of Optional PCB (CZ-NS5P)



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.OFF, Short=Comp.ON (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

■ Outputs

Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s	AC230V, 6VA
Pool pump	AC230V	AC 230V, 0.6 A max
Solar pump	AC230V	AC 230V, 0.6 A max
Zone pump	AC230V	AC 230V, 0.6 A max

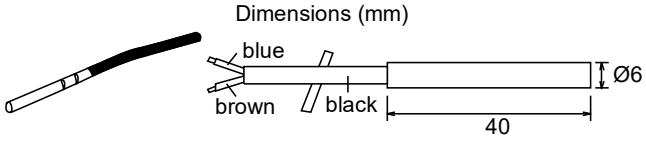
■ Thermistor inputs

Zone room sensor	PAW-A2W-TSRT
Buffer tank sensor	PAW-A2W-TSBU
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

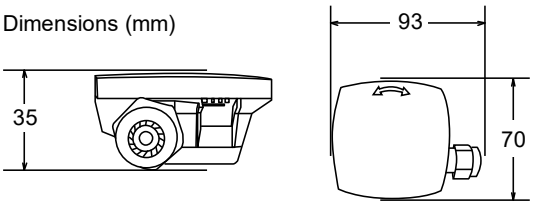
12.5.2.6 Recommended External Device Specification

- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.

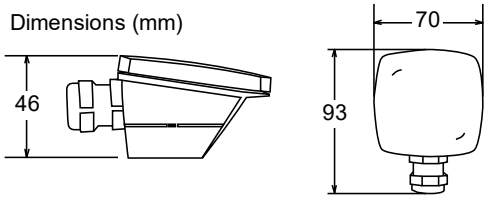
1. Buffer tank sensor: PAW-A2W-TSBU
   Use for measurement of the buffer tank temperature.
   Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



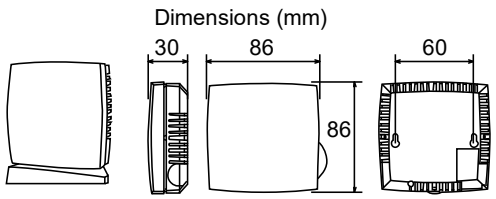
2. Zone water sensor: PAW-A2W-TSHC
   Use to detect the water temperature of the control zone.
   Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



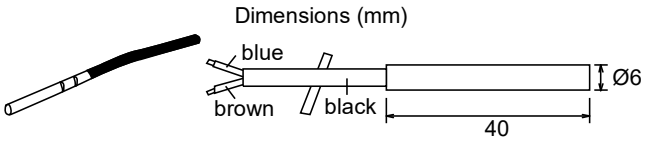
3. Outdoor sensor: PAW-A2W-TSOD
   If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
   In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



4. Room sensor: PAW-A2W-TSRT
   Install the room temperature sensor to the room which requires room temperature control.



5. Solar sensor: PAW-A2W-TSSO
   Use for measurement of the solar panel temperature.
   Insert the sensor into the sensor pocket and paste it on the solar panel surface.

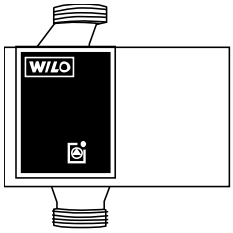


6. Please refer to the table below for sensor characteristic of the sensors mentioned above.

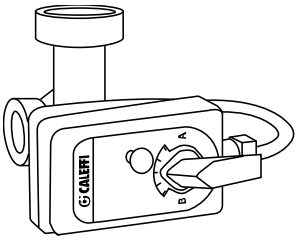
Temperature (°C)	Resistance (kΩ)
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo



- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

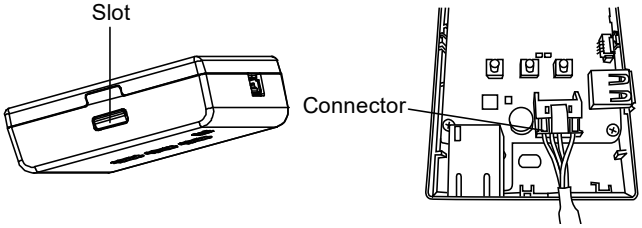


⚠ WARNING
This section is for authorized and licensed electrician/water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

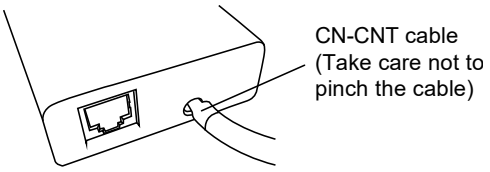
12.5.2.7 Network Adaptor [4] Installation

1. **Remove the Control Board Cover ⑤, then connect the cable included with this adaptor to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Tank Unit so that there is no pinching.
   - If an Optional PCB has been installed in the Tank Unit, connect to the CN-CNT connector of the Optional PCB.

2. **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**

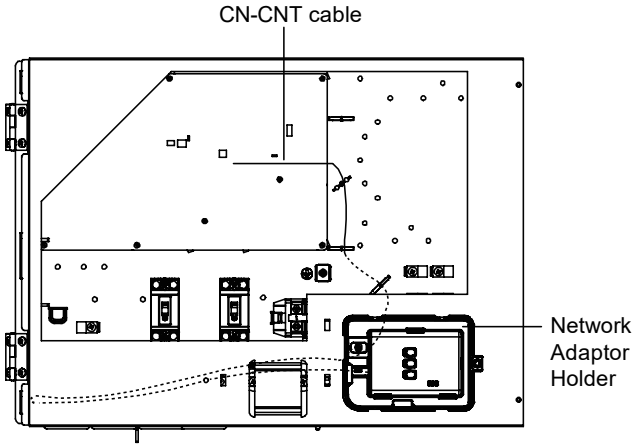


3. **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**

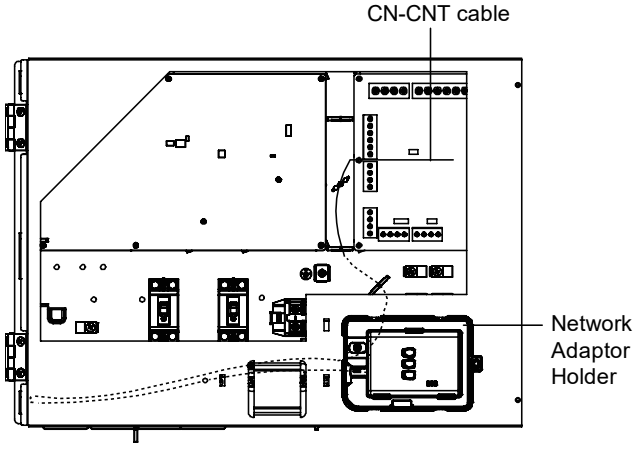


4. **Fix the Network Adaptor [4] to Network Adaptor Holder.**
   Guide the cable as shown in the diagram so that external forces cannot act on the connector in the adaptor.

Connection examples:



Without Optional PCB

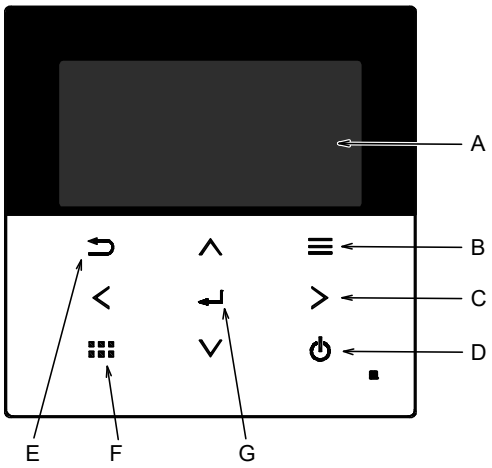


With Optional PCB

12.5.3 System Installation

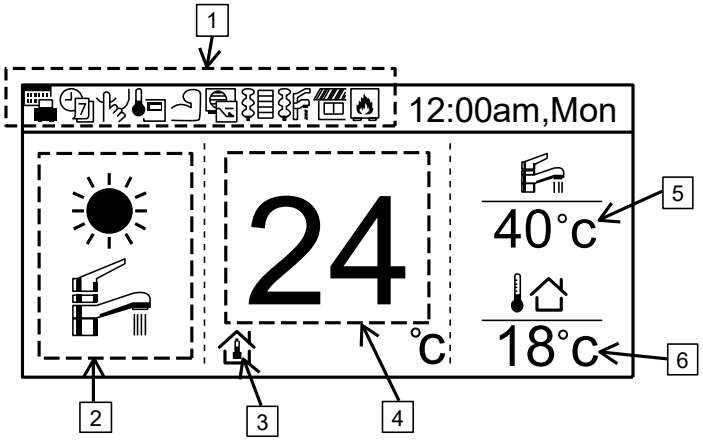
12.5.3.1 Remote Controller Outline

The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.



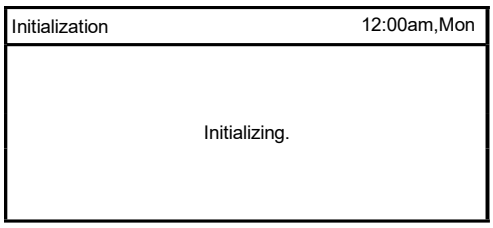
LCD Display
(Actual - Dark background with white icons)

Name	Function
A: Main screen	Display information
B: Menu	Open/Close main menu
C: Triangle (Move)	Select or change item
D: Operate	Start/Stop operation
E: Back	Back to previous item
F: Quick Menu	Open/Close Quick menu
G: OK	Confirm

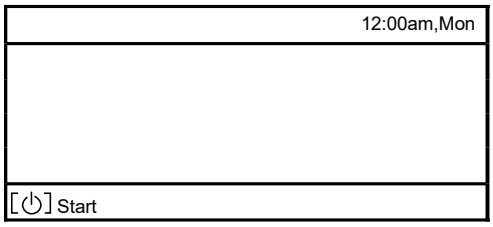


	Name	Function
1:	Function icon	Display set function/status
	Holiday mode	
	Weekly timer	
	Quiet mode	
	Remote controller room thermostat	
	Powerful mode	
	Demand control	
	Room heater	
	Tank heater	
	Solar	
	Boiler	
2:	Mode	Display set mode/current status of mode
	Heating	
	Cooling	
	Auto	
	Hot water supply	
	Auto heating	
	Auto cooling	
	Heat pump operating	
3:	Temp setting	
	Set room temp	
	Compensation curve	
	Set direct water temp	
	Set pool temp	
4:	Display Heat temp	Display current heating temperature (it is set temperature when enclosed by line)
5:	Display tank temp	Display current tank temperature (it is set temperature when enclosed by line)
6:	Outdoor temp	Display outdoor temp

First time of power ON (Start of installation)



When power is ON, firstly initialization screen appears (10 sec)



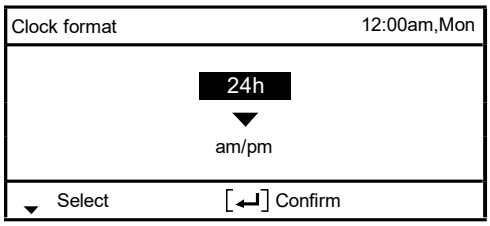
When initialization screen ends, it turns to normal screen.



When any button is pressed, language setting screen appears.
(NOTE) If initial setting is not performed, it does not go into menu.

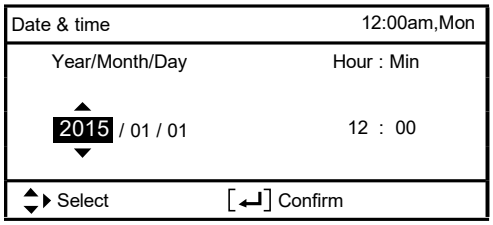
When there are two remote controllers installed from the beginning, the first remote controller to set and confirm language will be recognised as main remote controller.

Set language & confirm



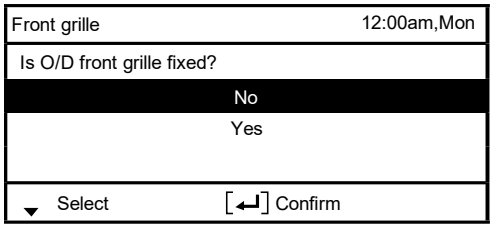
When language is set, setting screen of time display appears (24h/am/pm)

Set time display & confirm

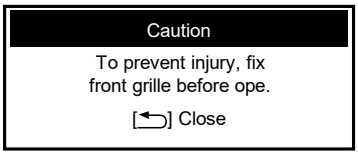


YY/MM/DD/Time setup screen appears

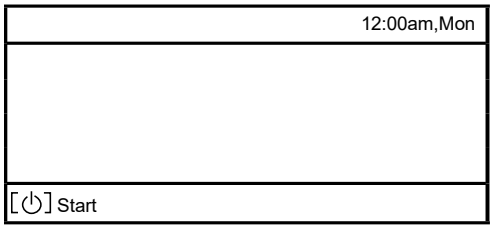
Set YY/MM/DD/Time & confirm



If set No & confirm, a caution message will be displayed to ensure outdoor front grille is installed before proceed to operate the unit.

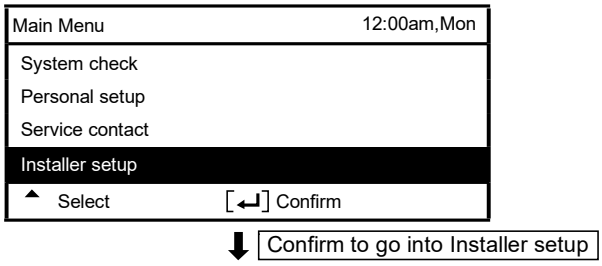


Set Yes & confirm if outdoor front grille has been installed

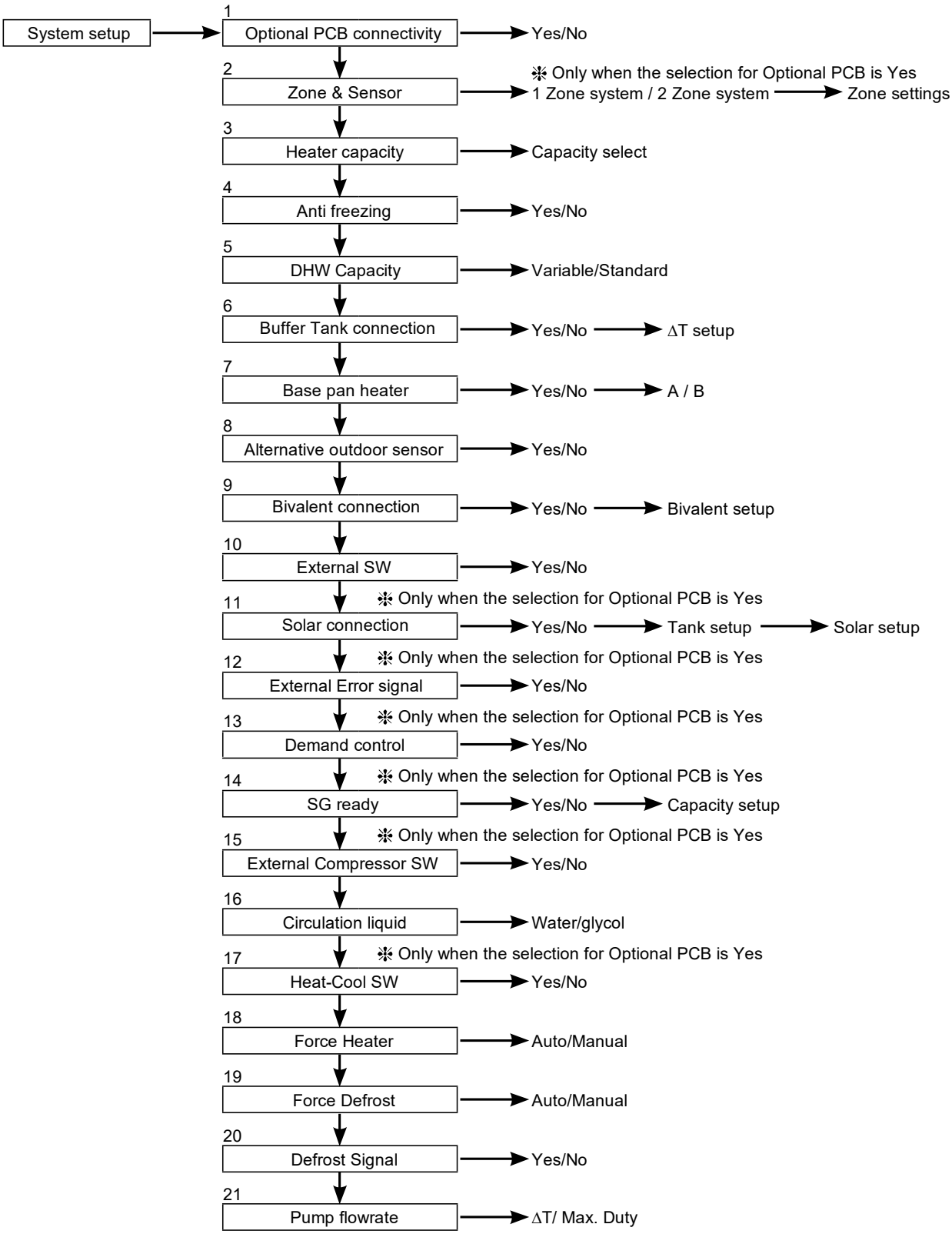


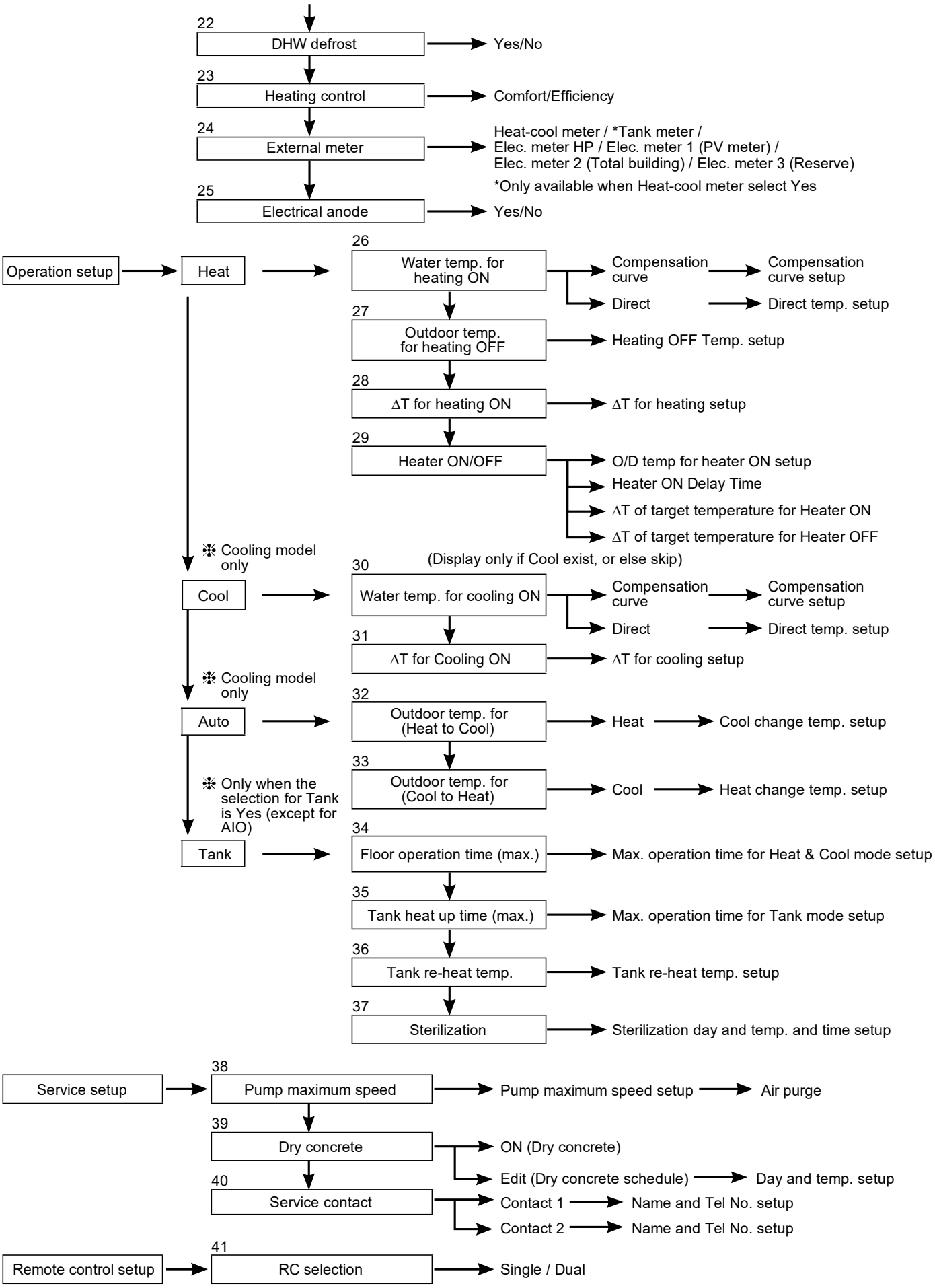
Back to initial screen

Press menu, select Installer setup



12.5.3.2 Installer Setup



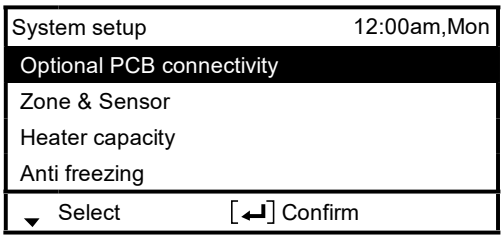

22
DHW defrost
Yes/No
23
Heating control
Comfort/Efficiency
24
External meter
Heat-cool meter / *Tank meter /
Elec. meter HP / Elec. meter 1 (PV meter) /
Elec. meter 2 (Total building) / Elec. meter 3 (Reserve)
*Only available when Heat-cool meter select Yes
25
Electrical anode
Yes/No
Operation setup
Heat
26
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
27
Outdoor temp. for heating OFF
Heating OFF Temp. setup
28
ΔT for heating ON
ΔT for heating setup
29
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
❇ Cooling model only
Cool
(Display only if Cool exist, or else skip)
30
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
31
ΔT for Cooling ON
ΔT for cooling setup
❇ Cooling model only
Auto
32
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
33
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
❇ Only when the selection for Tank is Yes (except for AIO)
Tank
34
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
35
Tank heat up time (max.)
Max. operation time for Tank mode setup
36
Tank re-heat temp.
Tank re-heat temp. setup
37
Sterilization
Sterilization day and temp. and time setup
Service setup
38
Pump maximum speed
Pump maximum speed setup
Air purge
39
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
40
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup
Remote control setup
41
RC selection
Single / Dual

12.5.3.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install Optional PCB.
Please select Yes after installing Optional PCB.

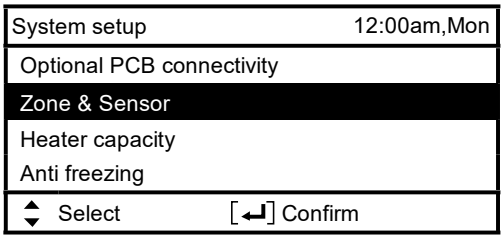
- 2-zone control
- Pool
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

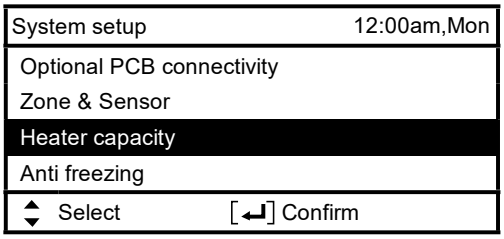
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

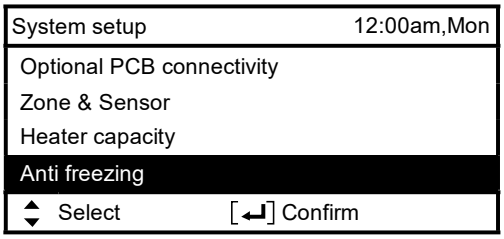
(NOTE) There are models which cannot select Heater capacity.



4. Anti freezing | Initial setting: Yes

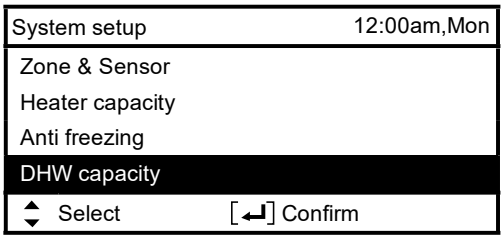
Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



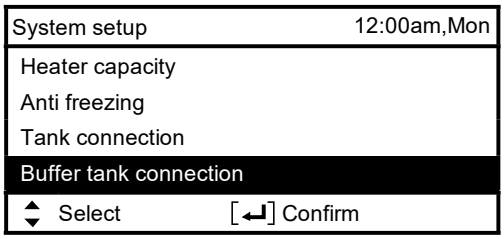
5. DHW Capacity | Initial setting: Variable

Variable DHW capacity setting normally run with efficient boiling which is energy saving heating. But while hot water usage high and tank water temperature low, variable DHW mode will run with fast heat up which heat up the tank with high heating capacity.
If standard DHW capacity setting is selected, heat pump run with heating rated capacity at tank heat up operation.



6. Buffer Tank connection | Initial setting: No

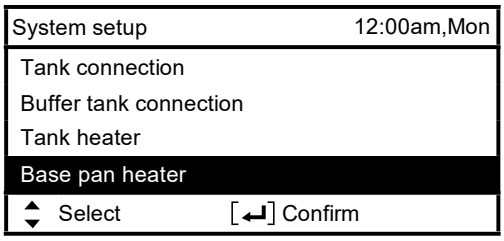
Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
If the buffer tank capacity is not so large, please set larger value for ΔT.



7. Base pan heater | Initial setting: No

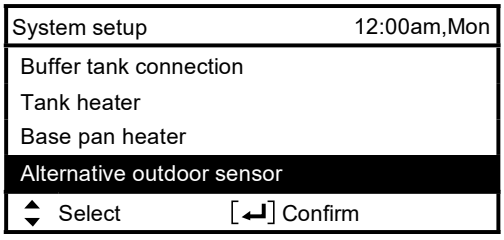
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



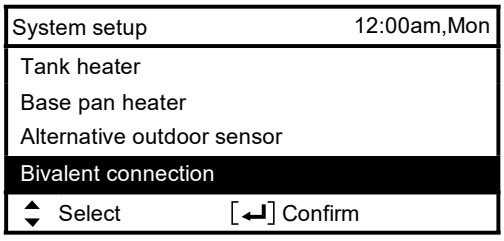
8. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



9. Bivalent connection | Initial setting: No

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.



After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)

1) SG ready (Only available to set when optional PCB set to YES)
   - SG Ready input from optional PCB terminal control ON/OFF of boiler and heat pump as below condition

SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

* This bivalent SG ready input is sharing same terminal as [14. SG ready] connection. Only one of these two setting can be set at the same time.
When one is set, another setting will reset to not set.

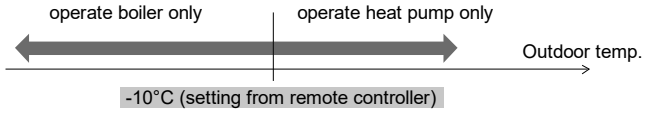
2) Auto

There are 3 different modes in the boiler auto pattern operation. Movement of each modes are shown below.

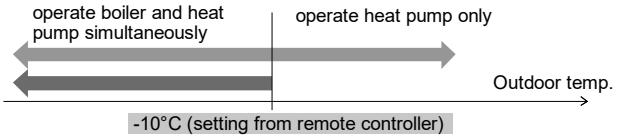
① Alternative (switch to boiler operation when drops below setting temperature)
② Parallel (allow boiler operation when drops below setting temperature)
③ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is “ON”, “boiler contact” is “ON”, “_”(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

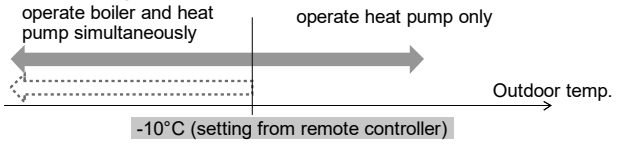


Parallel mode



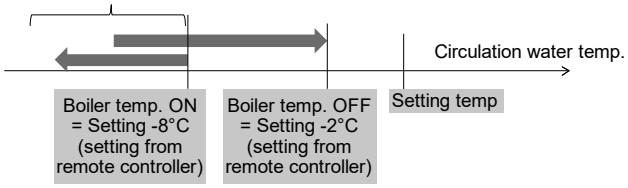
Advanced Parallel mode

For heating

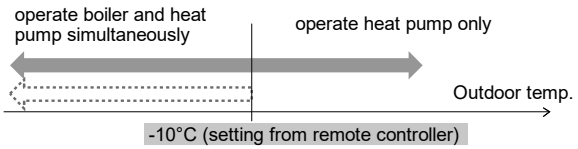


Although heat pump operates but water temperature does not reach this temperature for more than 30 mins (setting from remote controller)

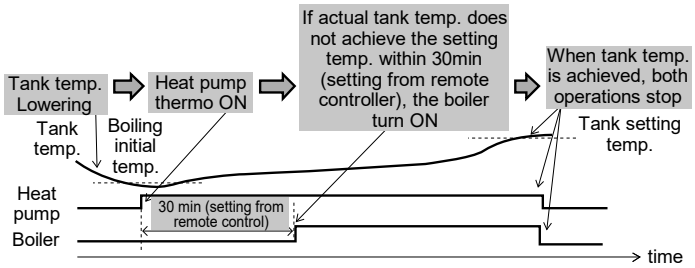
AND



For DHW tank



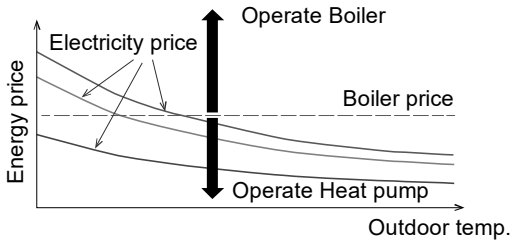
AND



In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of “Heating/Tank” mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

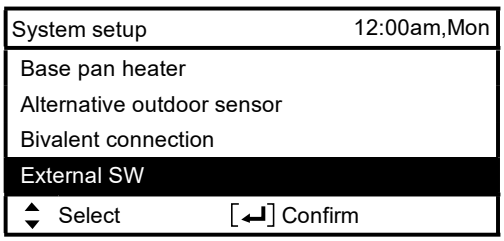
3) Smart

There are Energy price (both electricity and boiler) and Schedule to be set on remote controller.
Operation setting of Energy price and Schedule shall be responsible by installer.
Based on these settings, system will calculate the final price for both electricity and boiler.
When final price of Electricity is lower than Boiler’s, heat pump will operate.
When final price of Electricity is higher than Boiler’s, boiler will operate.



10. External SW | Initial setting: No

Able to turn ON/OFF the operation by external switch.

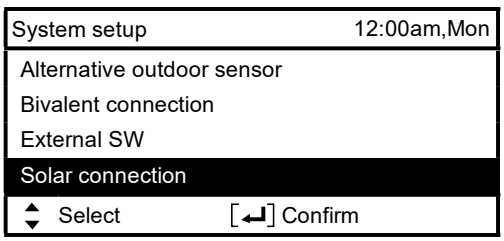


11. Solar connection | Initial setting: No

Set when solar water heater is installed.

Setting include items below.

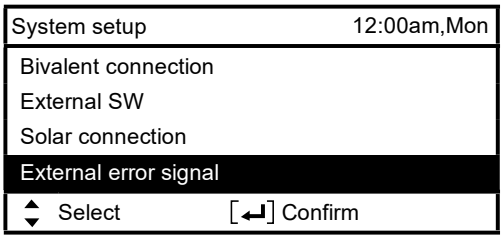
① Set either buffer tank or DHW tank for connection with solar water heater.
② Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to operate the solar pump.
③ Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to stop the solar pump.
④ Anti-freezing operation start temperature (please change setting based on usage of glycol.)
⑤ Solar pump stop operation when it exceeds high limit temperature (when tank temperature exceed designated temperature (70~90°C))



12. External Error Signal | Initial setting: No

Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

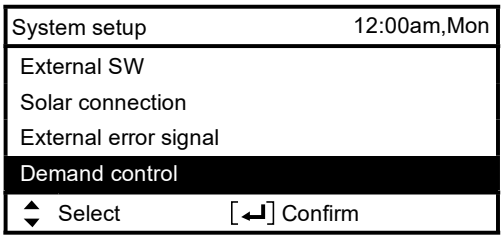
(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off “close” from the display, error signal will still remain ON.



13. Demand control | Initial setting: No

Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.



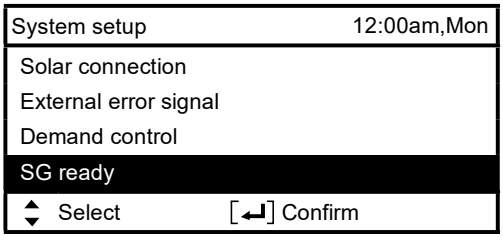
Analog input [v]	Rate [%]
0.0	not activate
0.1 ~ 0.6	
0.7	10
0.8	not activate
0.9 ~ 1.1	10
1.2	15
1.3	10
1.4 ~ 1.6	15
1.7	20
1.8	15
1.9 ~ 2.1	20
2.2	25
2.3	20
2.4 ~ 2.6	25
2.7	30
2.8	25
2.9 ~ 3.1	30
3.2	35
3.3	30
3.4 ~ 3.6	35
3.7	40
3.8	35

Analog input [v]	Rate [%]
3.9 ~ 4.1	40
4.2	45
4.3	40
4.4 ~ 4.6	45
4.7	50
4.8	45
4.9 ~ 5.1	50
5.2	55
5.3	50
5.4 ~ 5.6	55
5.7	60
5.8	55
5.9 ~ 6.1	60
6.2	65
6.3	60
6.4 ~ 6.6	65
6.7	70
6.8	65
6.9 ~ 7.1	70
7.2	75
7.3	70

Analog input [v]	Rate [%]
7.4 ~ 7.6	75
7.7	80
7.8	75
7.9 ~ 8.1	80
8.2	85
8.3	80
8.4 ~ 8.6	85
8.7	90
8.8	85
8.9 ~ 9.1	90
9.2	95
9.3	90
9.4 ~ 9.6	95
9.7	100
9.8	95
9.9 ~	100

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

14. SG ready | Initial setting: No



Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2

Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

> Set by SG ready setting of remote controller

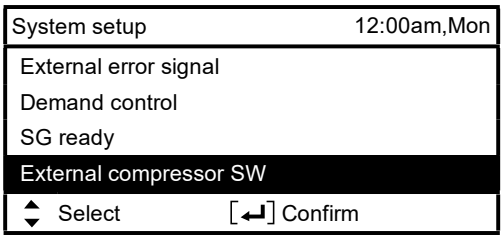
(When SG ready set to YES, Bivalent control pattern will set to Auto.)
(NOTE) Does not display if there is no Optional PCB.

15. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

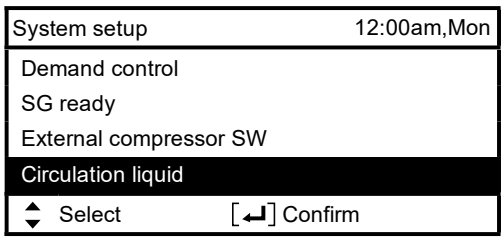


16. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

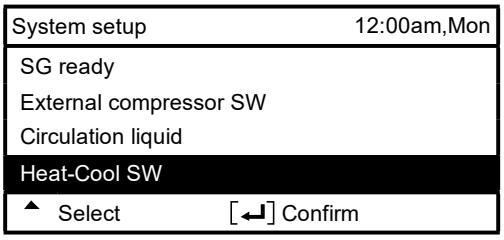


17. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

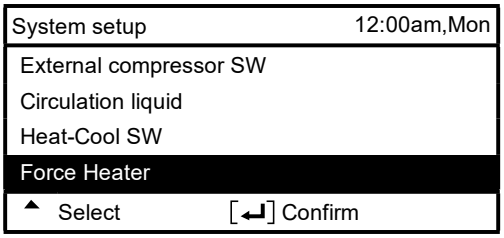


18. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

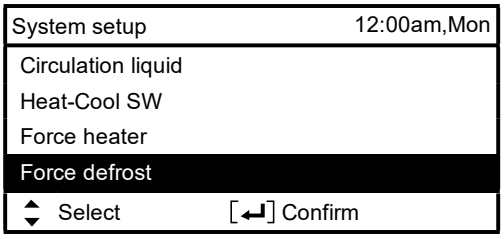
Heater source will ON during force heater mode.



19. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

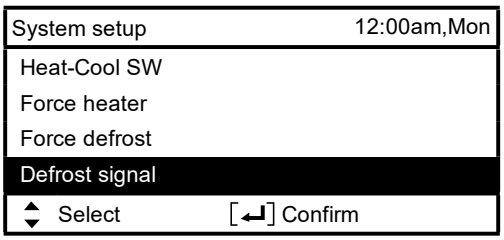
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition. (Even auto is selected, user still can turn on force defrost through quick menu)



20. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).

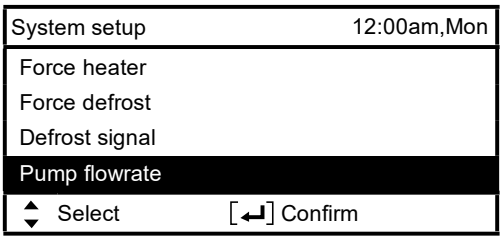


21. Pump flowrate

Initial setting: ΔT

If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

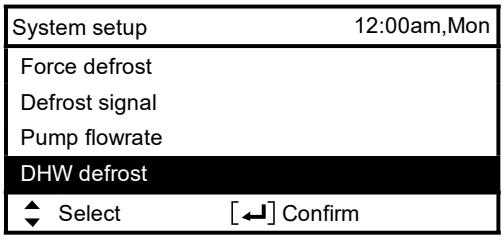
If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



22. DHW defrost

Initial setting: Yes

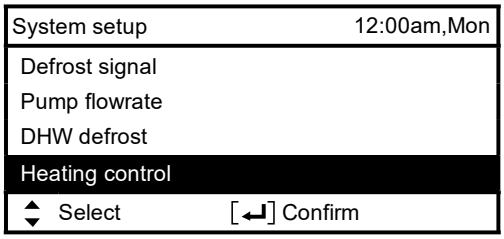
When DHW defrost set to YES, hot water of domestic hot water tank will be used during defrost cycle.
When DHW defrost set to NO, hot water of floor heating circuit will be used during defrost cycle.



23. Heating control

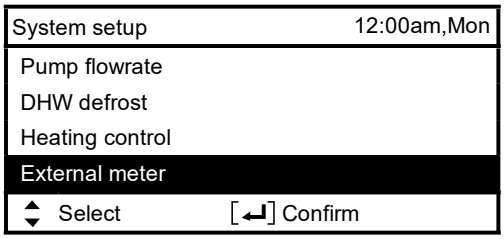
Initial setting: Comfort

There are two modes to select for compressor frequency control : Comfort or Efficiency.
When set to Comfort mode, compressor will run at zone limit maximum frequency to reach set temperature faster.
When set to Efficiency mode, compressor will run at part load frequency at initial stage for energy saving.



24. External meter

Initial setting : [Heat-cool meter : No]
[Tank meter : No] *only available when Heat-cool meter select Yes
[Elec. meter HP : No]
[Elec. meter 1 (PV meter) : No]
[Elec. meter 2 (Total building) : No]
[Elec. meter 3 (Reserve) : No]



There are two systems for generation meter connection : one generation meter system (Heat-cool meter) or two generation meter system (Heat-cool meter and Tank meter)
Both systems can provide all generation data of heating, cooling and DHW directly from external meter.
If Heat-cool meter is set to Yes, it will read from external meter for heat pump's energy generation data during heating, cooling and DHW operation[*1].
If Heat-cool meter is set to No, it will base on unit's calculation for heat pump's energy generation data during heating, cooling and DHW operation.
If Tank meter is set to Yes, it will read from external meter for heat pump's energy generation data during DHW operation[*1].
If Elec. meter HP is set to Yes, it will read from external meter for heat pump's energy consumption data.
If Elec. meter HP is set to No, it will base on unit's calculation for heat pump's energy consumption data.
If Elec. meter 1 (PV meter) is set to Yes, it will read from external meter for energy generation data of solar system and display it on Cloud system.
If Elec. meter 2 (Building) is set to Yes, it will read from external meter for energy consumption data of the building and display it on Cloud system.
If Elec. meter 3 (Reserve) is set to Yes, it will read from external meter for energy consumption data obtained from reserved electricity meter and display it on Cloud system.

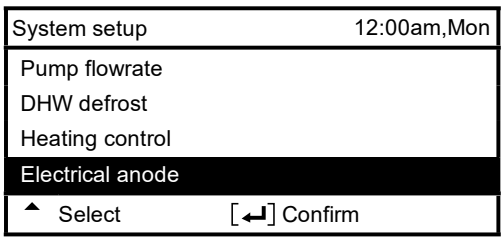
[*1] Set Heat-cool meter to Yes and set Tank meter to No when 1 generation meter system is installed.
Set Heat-cool meter to Yes and set Tank meter to Yes when 2 generation meter system is installed.

Remark : Elec. meter HP refers to Electricity meter that measures Heat Pump unit's consumption.
Elec. meter 1 / 2 / 3 refers to Electricity meter no. 1 / no. 2 / no. 3

25. Electrical anode

Initial setting : No

When Electrical anode set to YES, anode will be turned on.
When Electrical anode set to NO, anode will not be turned on.



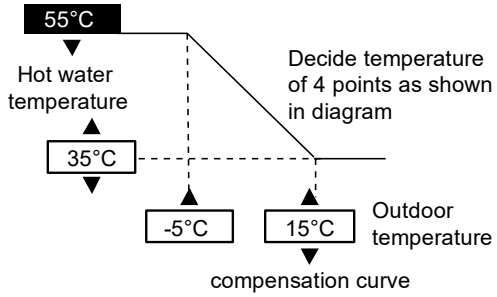
12.5.3.4 Operation Setup

Heat

26. Water temp. for heating ON | Initial setting: compensation curve

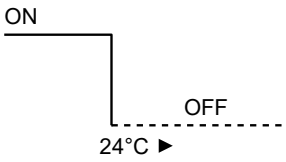
Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



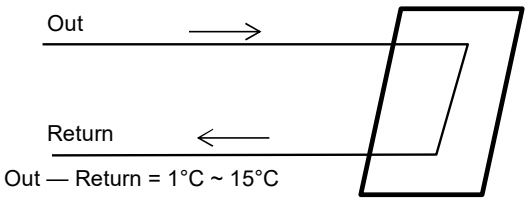
27. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C



28. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

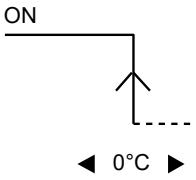


29. Heater ON/OFF

a. Outdoor temp. for heater ON | Initial setting: 0°C

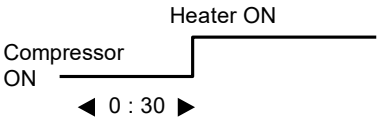
Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.



b. Heater ON delay time | Initial setting: 30 minutes

Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

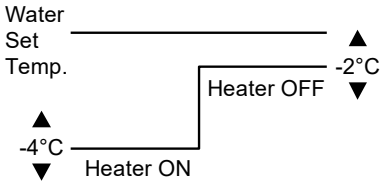


c. Heater ON: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp | Initial setting: -2°C

Set water temperature for heater to turn off at heat mode.
Setting range is -3°C ~ 0°C

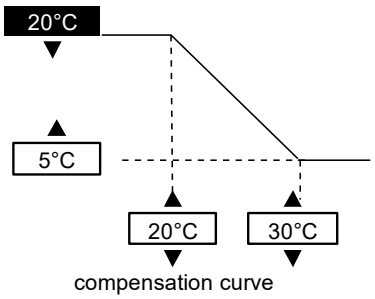


Cool

30. Water temp. for cooling ON | Initial setting: compensation curve

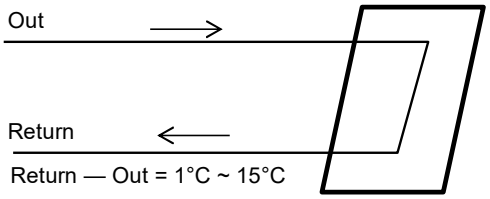
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



31. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

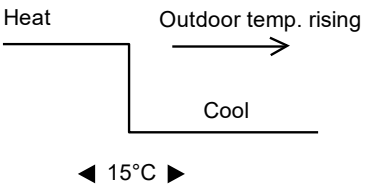


Auto

32. Outdoor temp. for (Heat to Cool) | Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 11°C ~ 25°C

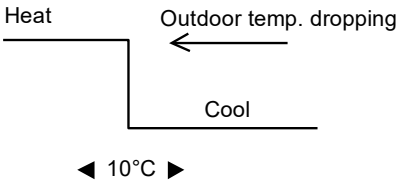
Timing of judgement is every 1 hour



33. Outdoor temp. for (Cool to Heat) | Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 14°C

Timing of judgement is every 1 hour

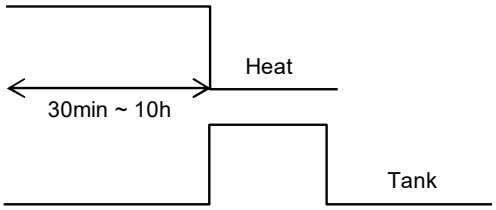


Tank

34. Floor operation time (max) Initial setting: 8h

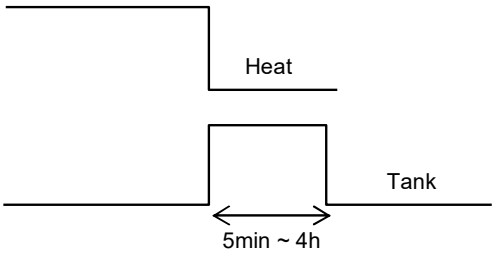
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



35. Tank heat up time (max) Initial setting: 60min

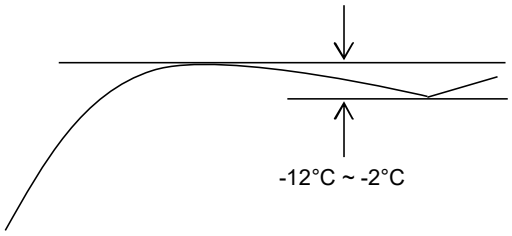
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



36. Tank re-heat temp. Initial setting: -8°C

Set temp to perform reboil of tank water.

Setting range is -12°C ~ -2°C

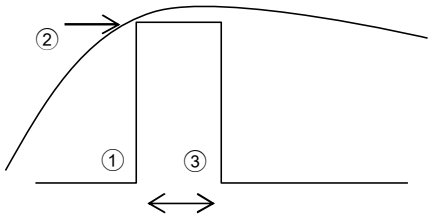


37. Sterilization Initial setting: 65°C 10min

Set timer to perform sterilization.

① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.



12.5.3.5 Service Setup

38. Pump maximum speed | Initial setting: Depend on model

Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.

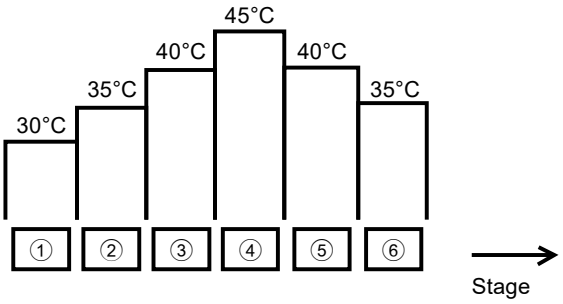
Service setup		12:00am,Mon
Flow rate	Max. Duty	Operation
88:8 L/min	0xCE	▲ Air Purge
◀▲ Select		

39. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



40. Service contact

Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)

Service setup	12:00am,Mon
Service contact:	
Contact 1	
Contact 2	
▲ Select	[↵] Confirm

Contact-1: Bryan Adams	
ABC/ abc	0-9/ Other
A B C D E F G H I J K L M N O P Q R	
S T U V W X Y Z a b c d e f g h i	
j k l m n o p q r s t u v w x y z	
▼▶ Select	[↵] Enter

12.5.3.6 Remote Control Setup

41. RC selection | Initial setting: Single

Set to “Single” when only one remote controller is installed.
Set to “Dual” when two remote controllers are installed.

RC selection	12:00am,Mon
Single	
▼	
Dual	
▼ Select	[↵] Confirm

12.6 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

12.6.1 Maintenance Menu

Setting method of Maintenance menu

Maintenance menu	12:00am,Mon
Actuator check	
Test mode	
Sensor setup	
Reset password	
▼ Select	[↵] Confirm

Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

12.6.2 Custom Menu

Setting method of Custom menu

Custom menu	12:00am,Mon
Cool mode	
Back-up heater	
Reset energy monitor	
Reset operation history	
Smart DHW	
▼ Select	[↵] Confirm

Please press ☰ + ▼ + ◀ for 5 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it.
In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer.
When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

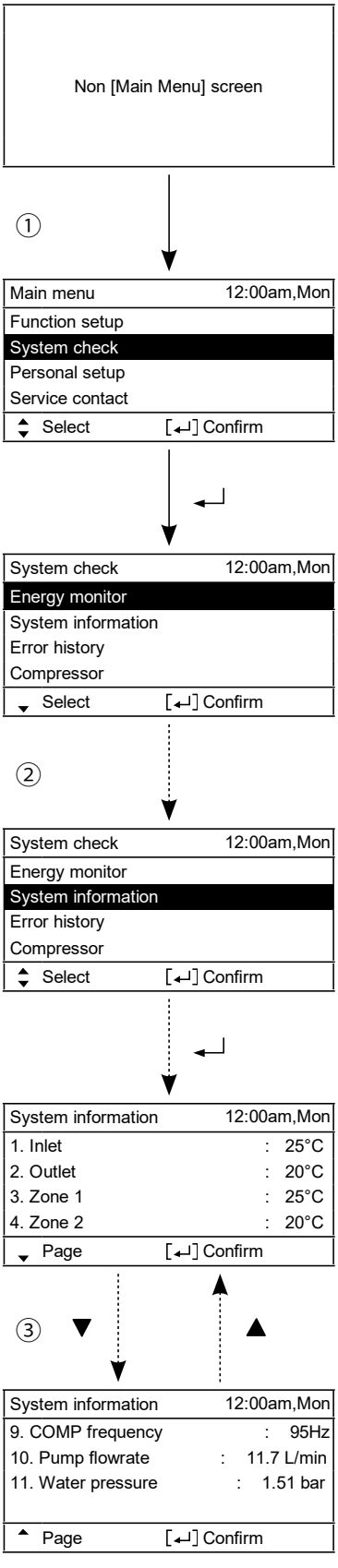
③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

12.6.3 Check Water Pressure from Remote Controller

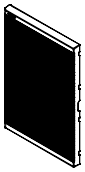
1. Press ☰ SW and scroll to “System check”.
2. Press ↵ and scroll to “System information”.
3. Press ↵ and search for “Water pressure”.



Screens shown are for illustration purposes only.

12.7 Outdoor Unit

Attached accessories

No.	Accessories part	Qty.
1	Drain elbow 	1
2	Rubber cap 	3
3	Discharge grille (left side) 	1
4	Discharge grille (right side) 	1
5	Screw 	4

Optional accessories

No.	Accessories part	Qty.
6	Base Pan Heater CZ-NE4P	1

- It is strongly recommended to install a Base Pan Heater (optional) if the outdoor unit is install in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.

12.7.1 Select the Best Location

- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- For WH-WDG05LE5, WDG07LE5 and WDG09LE5, avoid installations in areas where the ambient temperature may drop below -25°C.
- A protective zone is defined for the area close around the product. See section “12.7.2 Protective zone”.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- If outdoor unit installed near sea, region with high content of sulphur or oily location (e.g. machinary oil, etc), it lifespan maybe shorten.
- Please follow the table below for water piping specifications.

Model	Water piping between outdoor unit and indoor unit			
	Inner diameter	Maximum length	Insulator thickness	Maximum Elevation
WH-WDG05LE5	ø20	30 m	30 mm or more	10 m
WH-WDG07LE5	ø25			
WH-WDG09LE5				

12.7.2 Protective Zone

This outdoor unit is filled with R290(Extremely flammable gas, safety A3 group per ISO 817). Note that this refrigerant has a higher density than air. In case of a refrigerant leak, the leaked refrigerant may accumulate near the ground.

Prevent accumulation of refrigerant in any way that is potentially dangerous, explosive or risk suffocation. Prevent refrigerant from entering the building through building openings. Prevent accumulation of refrigerant in the drain grooves.

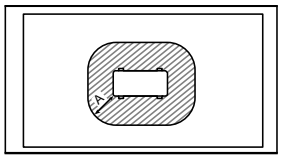
A protective zone is defined around this outdoor unit. There must be no building openings, windows, doors, light shafts, cellar entrances, escape hatches, flat-roof windows or ventilation openings in the protective zone.

There must be no ignition sources, such as heat above 360°C, sparks, open flame, plug sockets, light switches, lamps, electrical switches or other permanent ignitions sources, in the protective zone.

The protective zone must not extend to adjacent buildings or public traffic areas (boundaries of neighbors, the public road, neighbor's private roads, subsidence area, depressions, pump shafts, sewers intakes, waste water shafts and so on.).

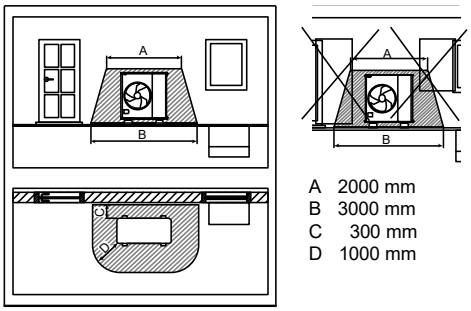
In the protective zone, you are not permitted to make any subsequent structural alterations which infringe the stated rules for the protective zone.

1) Protective zone for ground installation (or flat-roof installation) at the open areas



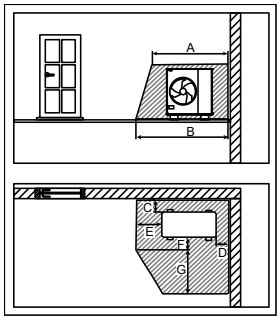
A 1000 mm

2) Protective zone for ground installation in front of a building wall



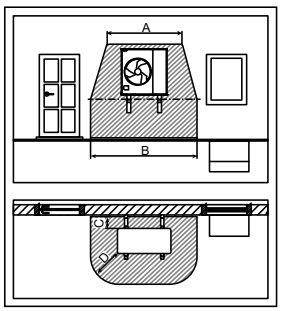
A 2000 mm
B 3000 mm
C 300 mm
D 1000 mm

3) Protective zone for ground installation in a building corner



A 2000 mm
B 2500 mm
C 300 mm
D 500 mm
E 1000 mm
F 500 mm
G 1800 mm

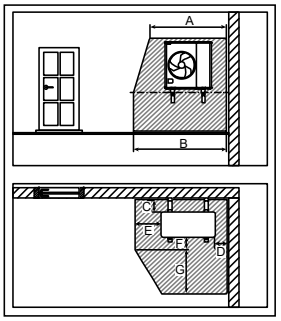
4) Protective zone for wall installation in front of a building wall



A 2000 mm
B 3000 mm
C 300 mm
D 1000 mm

The protective zone under the product extends to the floor.

5) Protective zone for wall installation in a building corner

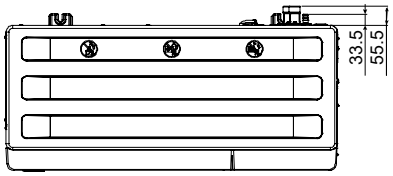


A 2000 mm
B 2500 mm
C 300 mm
D 500 mm
E 1000 mm
F 500 mm
G 1800 mm

The protective zone under the product extends to the floor.

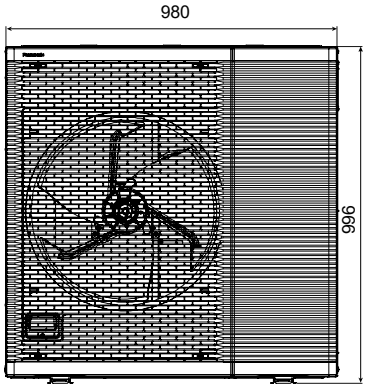
12.7.3 Install the Outdoor Unit

12.7.3.1 Dimension Diagram

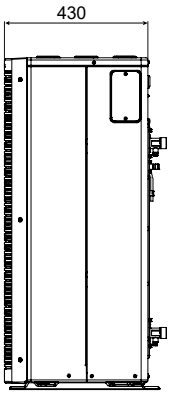


Top view

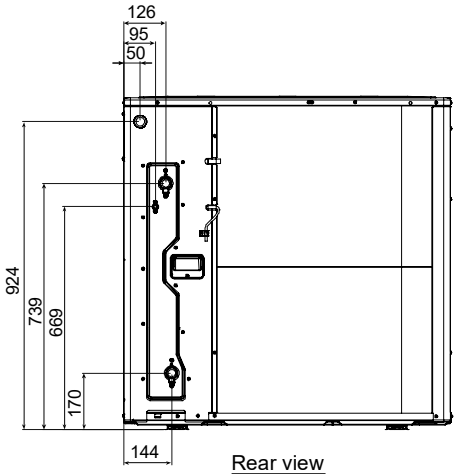
It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.



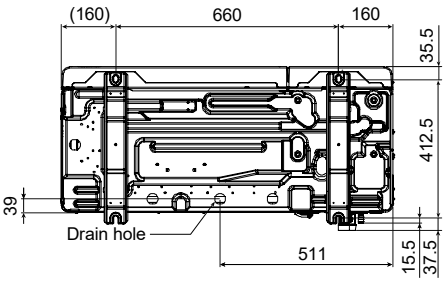
Front view



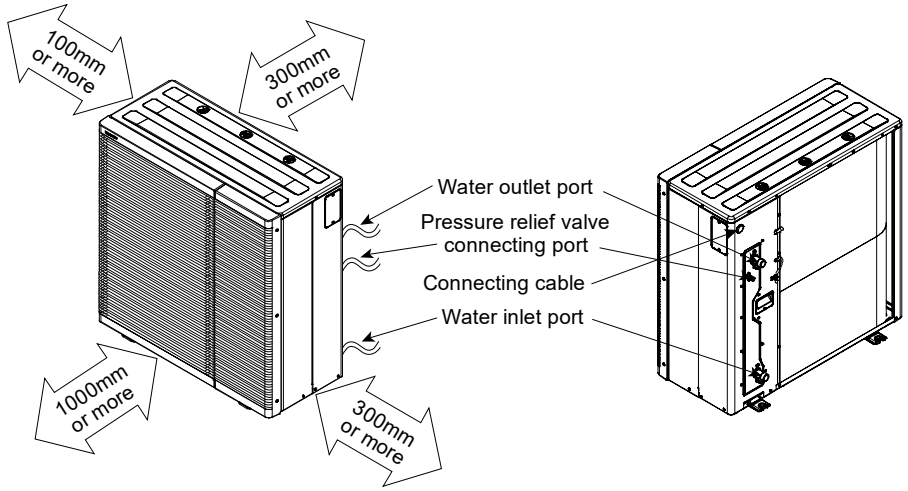
Side view



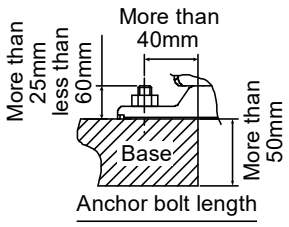
Rear view



Bottom view



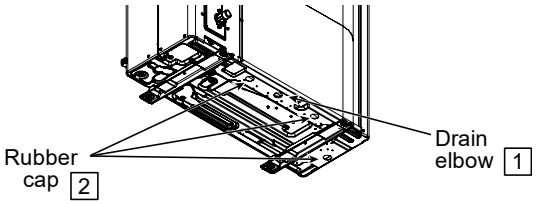
- After selecting the best location, start installation according to Installation Diagram.
  1. When installing at roof, please consider strong wind and earthquake. Please fasten the installation stand firmly with bolt or nails.
  2. When installing to a concrete or solid surface, use M10 or a W 3/8 bolts and nuts to secure the unit. Ensure that it installed upright on a horizontal plane. (Use an anchor bolt for the installation as shown in the diagram below.)



Anchor bolt length

12.7.3.2 Disposal of Outdoor Unit Drain Water

- When a Drain elbow [1] is used, please ensure to follow below:
  - the unit should be placed on a stand which is taller than 50 mm.
  - cover the ø32mm holes with Rubber cap [2] (refer to illustration below).
  - use a tray (field supply) when necessary to dispose the outdoor unit drain water.
- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow [1] and Rubber cap [2] , for the drain water freezes and the fan will not rotate.



Pressure relief valve drainage pipe-work

- Use a drain hose with inner diameter of 15mm.
- The hose must be installed in a continuously downward direction and left open to the frost-free atmosphere.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain hose.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulfuric gas, etc.
- If necessary, use a hose clamp to tighten the hose at drain hose connector to prevent it from leaking.

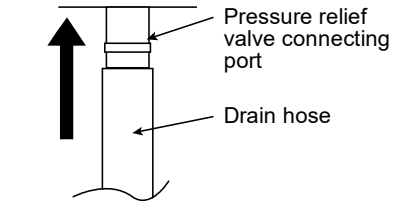
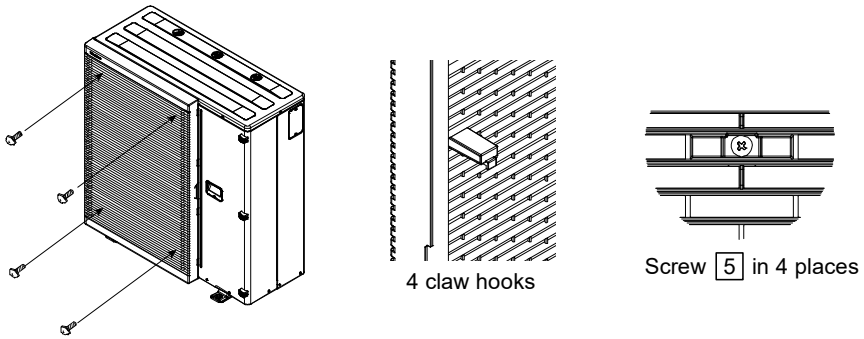


Illustration of how to fix drain hose to unit

12.7.4 Connect the Cable to the Outdoor Unit

(FOR DETAIL REFER TO WIRING DIAGRAM AT UNIT)

1. Attach the discharge grille (left side) [3] for safety before connecting the cable.



2. Connecting cable between indoor unit and outdoor unit shall be approved polychloroprene sheathed flexible cable (see below table), type designation 60245 IEC 57 or heavier cable. Sheath diameter of connecting cable shall be within specification (see below table) to fit cable gland.

Flexible cable specification	Cable diameter
4 × min 2.5 mm^2	ø12.0-14.0

3. Route the cable as follows
   *1 Locally procured cabtyre cables with the specified diameter.
   *2 Must be fastened with the specified torque Viewpoint of gas intrusion prevention.

① Remove the front plate (Fig. 4-1)
② Remove the terminal cover and cable gland cap (Fig. 4-2)
③ Insert the cabtyre cable *1 (Fig. 4-3, 4-4)
(Pulled in from the rear panel and inserted from the bottom of the electric controller BOX)
④ Connect to terminal block (Fig. 4-5)
⑤ Fasten the cable gland *2 (Fig. 4-4)
⑥ Set the terminal cover *2 (Fig. 4-2)
⑦ Set the front plate (Fig. 4-1)

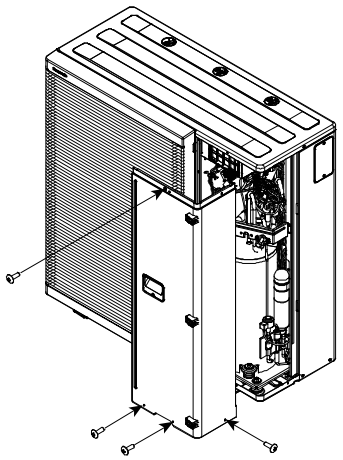


Fig. 4 - 1

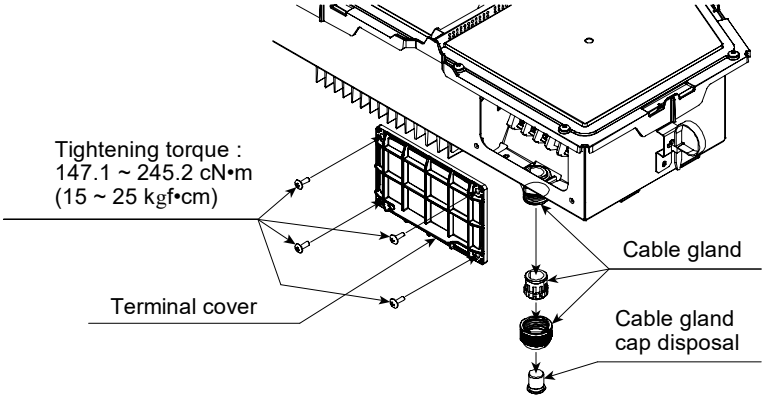
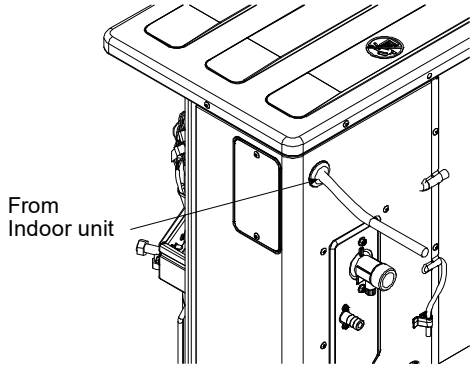


Fig. 4 - 2

The cabinet top plate does not need to be removed.



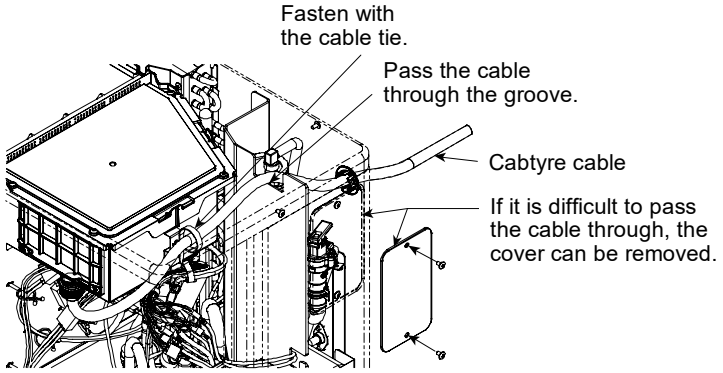
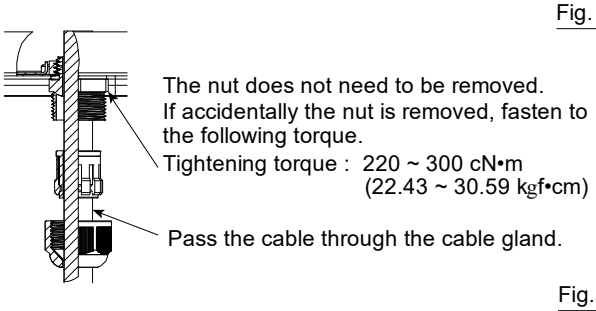


Fig. 4 - 3



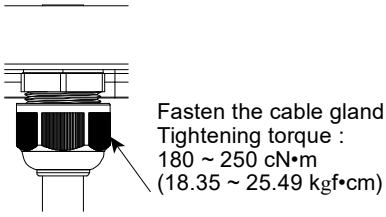
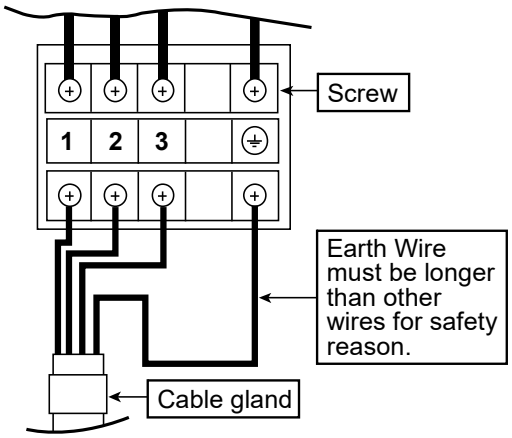


Fig. 4 - 4



⚠ WARNING

This equipment must be properly earthed.

Terminals on the indoor unit	1	2	3	⏚
Colour of wires				
Terminals on the outdoor unit	1	2	3	⏚

Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157 ~ 196 {16 ~ 20}
M5	196 ~ 245 {20 ~ 25}

12.7.4.1 Wire Stripping and Connecting Requirement

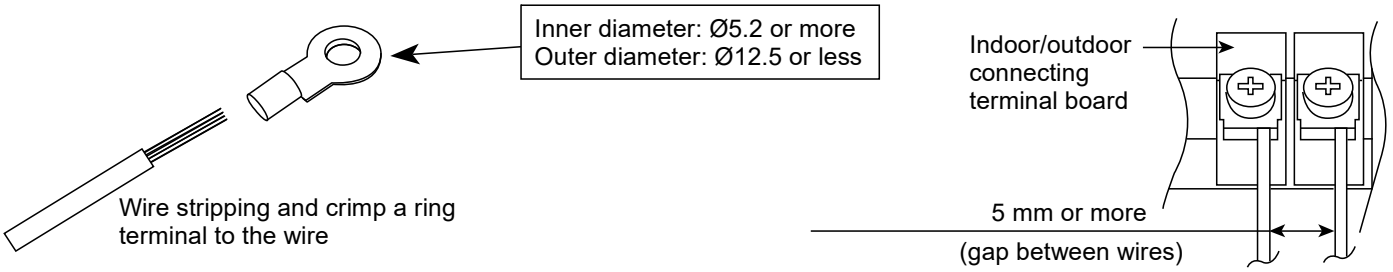


Fig. 4 - 5

12.7.5 Pipe Insulation

1. Please carry out insulation at pipe connection portion as mentioned in Indoor/Outdoor Unit Installation Diagram. Please wrap the insulated piping end to prevent water from going inside the piping.

12.7.6 Install the Discharge Grille

1. Remove the 3 screws fastened to the cabinet front plate. (Fig. 6-1)
2. Insert 4 claws of the discharge grille (right side) 4 and fasten the 3 screws. (Fig. 6-2)

⚠ CAUTION

Make sure to install the discharge grille to the outdoor unit before powering ON to protect againts a rotating fan.

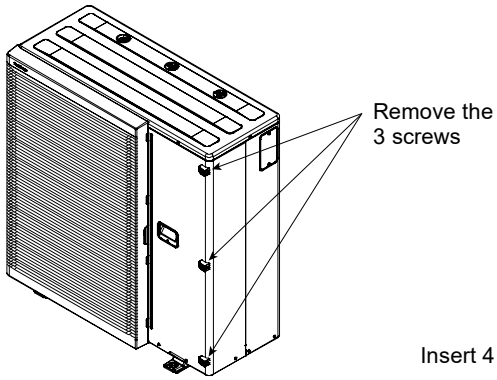


Fig. 6-1

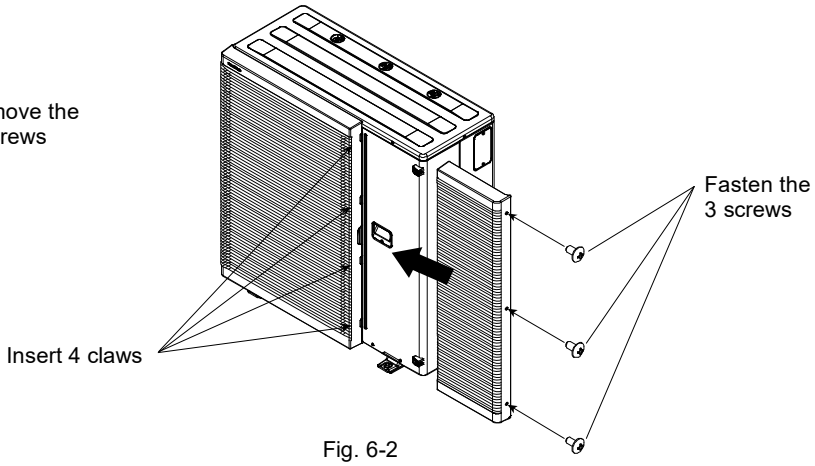


Fig. 6-2

⚠ CAUTION

If cleaning of the outdoor unit is necessary during installation or servicing, do not clean the outdoor unit with any hydrocarbon based solvent.

13. Installation and Servicing Air-to-Water using R290

WARNING	This symbol shows that this equipment uses a flammable refrigerant with safety A3 group per ISO 817. If the refrigerant is leaked, together with an external ignition source, there is a possibility of fire / explosion.

13.1 About R290 Refrigerant

Hydrocarbon is a class of organic chemical made up only with the element Carbon (C) and Hydrogen (H). R290 is the type of Hydrocarbon refrigerant which is environmentally good-natured and odorless refrigerant. Under Kigali Amendment to the Montreal Protocol, 80% reduction of greenhouse gas emission by next 30 years is required, and due to this requirement, further reduction in the emission of high greenhouse effect gas is required. Therefore, the conversion of air-conditioning refrigerant into one which has no greenhouse effect, even if it is dissipated into the atmosphere became our responsibility.

Nevertheless, in case of air-conditioning refrigerant, it would be the best if there is a refrigerant which has no impact on global warming but ensures good energy efficiency and performance, and is safe; however, there is no such refrigerant which satisfies all these conditions. As a result, we have been considering the practical usage, within the safety frame-work, of R290 refrigerant which has no effect of global warming but highly flammable.

13.2 Characteristics of R290 Refrigerant

1. Chemical Characteristics

R290 (Propane) is refrigerant grade propane, which is natural, non toxic, and chemically stable compound formed by hydrogen.

R290 is one of natural refrigerant, therefore it has almost zero greenhouse gas effect. R-290 is a single-component hydrocarbon substance and the most hydrocarbon properties as it is highly flammable.

Chemical Characteristic Table of R290 and R32

	R290	R32
Chemical Formula	C3H8	CH2F2
Composition	Single Composition	Single Composition
Boiling point (°C)	-42.1	-51.7
50°C vapor pressure (MPa)	1.71	3.14
Ozone Depletion Potential	0	0
Global Warming Potential (GWP)	3	675
Inflammability	Highly Inflammable (A3)	Slightly Inflammable (A2L)
Toxicity	None	None

2. Characteristic of Pressure

As shown in Table 2, R290 has half the vapor pressure of R32 at the same refrigerant temperature.
As such, it can be installed and maintained with the same high-pressure tools and components as the R32.

Table 2. Saturated vapor pressure comparison table

(Unit: MPaG)

Temperature (°C)	Refrigerant	
	R290	R32
-20	0.14	0.30
0	0.37	0.71
20	0.74	1.37
40	1.27	2.38
60	2.03	3.84
65	2.23	4.29

Reference : Thermal properties table of Japan Society of Refrigerating and Air Conditioning Engineers (60, 65°C)
NIST REFPROP V8.0 (-20 ~ 40°C)

3. Flammable characteristic

As shown in below table, R290 is highly flammable and explosive when heated. The installation must be equipped with ATEX (Atmospheres Explosible) certified equipment and must always turn on the combustible gas detector during servicing and when entering the service area. Service must also be performed in a well-ventilated area, especially if the refrigerant system is being accessed.

3.1 Safety class

Increasing Flammability	SAFETY GROUP	
Higher Flammability	A3	B3
Lower Flammibility	A2	B2
	A2L	B2L
No Flame Propagation	A1	B1
	Lower Toxicity	Higher Toxicity

Increasing Toxicity

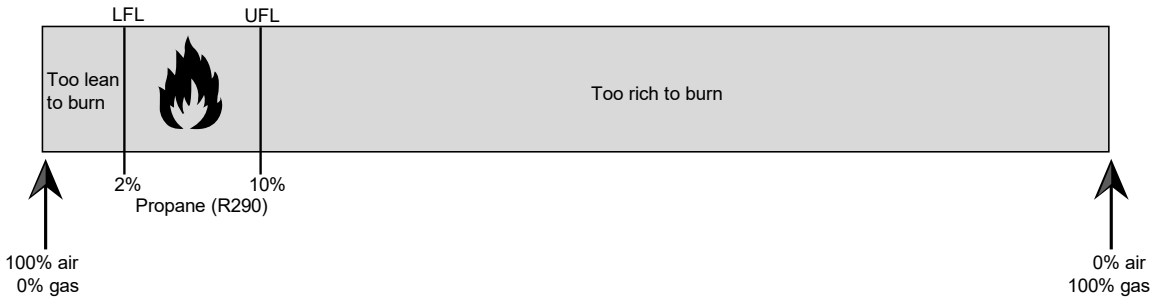
Refrigerant	Burning Speed cm/s
R32	6.7
R290	**38.7**

3.2 Concentration control for R290

Because R290 is highly flammable, it can burn or explode if there is enough product concentrated in one space and the refrigerant comes in contact with an ignition source.

Control measurement;

Descriptions	Specifications
Lower flammability limit (LFL) [kg/m^3]	0.038
Lower flammability limit (LFL) [%]	2.1
Practical limit (PL) [kg/m^3]	0.008
Density of vapour [kg/m^3]	1.83



The concentration of R290 between the lower flammable limit (2%) and upper flammable limits (10%) is enough to ignite fire.

Note:
LFL – Lower flammable limit whereby the concentration of flammable gas, vapour or mist in the air below which an explosive gas atmosphere will not be formed.
UFL – Upper flammable limit whereby the concentration of flammable gas, vapour or mist in the air below which an explosive gas atmosphere will not be formed.
PL – Defined as concentration used for simplified calculation to determine the maximum acceptable amount of refrigerant in an occupied space (20% of LFL)

3.3 Material classification and Hazard statement

H280	**Contain gas under pressure; may explode if heated**
CGA-HG01	**May cause frostbite**

13.3 Refrigerant piping installation • Tools used in services

13.3.1 Required Tools

R290 refrigerant air conditioners must use ATEX (Atmosphere Explosible) certified equipment. The common parts as R32 air conditioners for two-way valves and three-way valves (diameters of service ports); thus, they maintain commonality in the maintenance of the compressive strength, the size of pipe flaring, and the size of flare nuts as R32. However refrigerant pipe installation and services must use tools certified for highly flammable gas.

However, mixing of refrigerants is not allowed, so that you have to separate the cylinders for the recovery of refrigerants.

Tools used for installation • relocation • replacement of air conditioning units

Works	R290	R32
Connecting of refrigerant pipes	Not applicable for Monobloc	
	Not applicable for Monobloc	
Manifold gauge charging hose	HC Manifold gauge to avoid refrigerant contamination	R32 & R410A Common
Air purging	Vacuum pump complied with ATEX (Atmosphere Explosible)	Vacuum pump + Reducer / expander
Gas leakage test	Combustible gas detector	Detection liquid or soup water, HFC detector

For other installation, you can use general tools such as screw drivers (+, -), metal saws, long-nose pliers, hole core drills, linen tape, levels, temperature gauges, clamp meters, electric knives, nippers, pipe cutters, reamers or scrapers, spring benders, monkey wrenches, fixing wrenches, feeler gauges, hexagon wrenches (4 mm), testers, megohm testers, etc.

Tools used for services.

Works	R290	R32
Insertion of refrigerant	Digital scale for refrigerant charging, refrigerant cylinders, cylinder adopters and packing *a	
Recovery of refrigerant	Refrigerant recovery devices, refrigerant cylinders, manifold gauges, charging hoses *b	

*a. Use cylinder for each refrigerant, cylinder adopter and packing.
*b. Use refrigerant recovery cylinder separately for each refrigerant **(no mixture of refrigerant allowed)**.

13.3.2 Tools for R290

1. Manifold gauges

R32 gauge can be used for R290 pressure.

Each port of manifold has different shapes in order to prevent inserting wrong refrigerant.
*However, the port shape for R290 and R32 is the same; therefore, attention need to be paid not to insert wrong refrigerant.

Differences in high/low pressure gauges

	R290 (common R32)
High pressure gauges (red)	-0.1 ~ 5.3 MPa -76 cmHg ~ 53 kgf / cm^2
Low pressure gauges (blue)	-0.1 ~ 3.8 MPa -76 cmHg ~ 38 kgf / cm^2

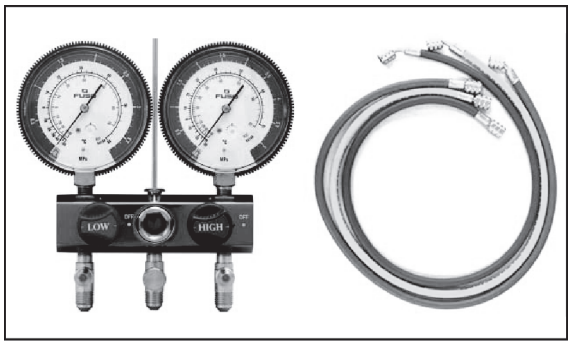
Difference in manifold port sizes

	R290 (common R32)
Port sizes	1/2 UNF20

2. Charging hoses

The material is changed to HC resistant, and the size of each manifold adopter is common, as the R32 port size of manifold gauge.
Further, some hoses are with anti-gas pressure backflow valves placed near the adopters. (hoses with the valves recommended)

Manifold gauges / Charging hoses



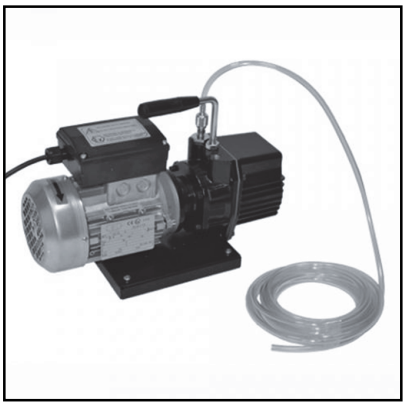
Differences in charging hoses

		R290 (common R32)
Pressure Resistance	Normal operation pressure	5.1 MPa (52 kgf / cm^2)
	Burst pressure	27.4 MPa (280 kgf / cm^2)
Material		HNBR rubber Internal nylon coating

3. Vacuum pump and isolator.

When using a vacuum pump, it is compulsory to use an ATEX certified vacuum pump.
When connecting service equipment (such as vacuum pumps) to a power source, the connection should be made in outside the Temporary Danger Zone.
It is recommended to use an ATEX Isolator switch to facilitate a safe shutdown in the danger zone.
It is necessary to set a solenoid valve in order to prevent backflow of vacuum pump oil into the charge hoses and use a vacuum pump with oil backflow prevention function.
If vacuum pump oil (mineral oil-based) mixes with R290, it may cause damage to the machine.

Vacuum pump



Isolator



4. Leak Detector

HC refrigerant Electric gas leakage tester is used for R290. The usage of existing HFC detectors cannot be use as they can produce spark. We recommend to use detectors specifically designed for combustible gas.

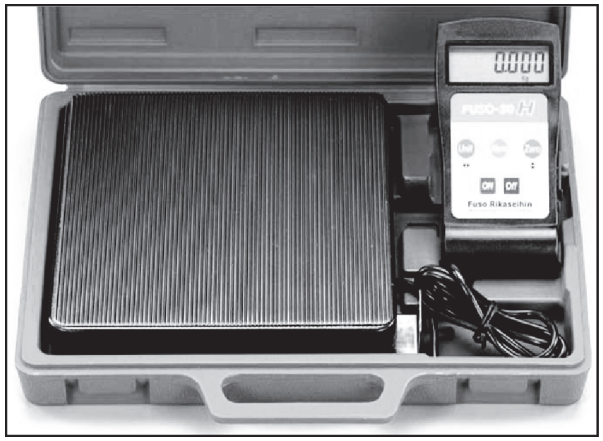
Combustible gas leak detector



5. Digital scale for refrigerant charging

R290 has lower pressure level and the evaporates speed is slow. Thus, the digital scale for refrigerant charging can be used in common with R32. The charging port for R32 is (1/2 UNF20), common with R290

Digital scale for refrigerant charging



6. Refrigerant cylinders

Refrigerant cylinders for R290 are painted in other colors that might subject to change according to the international standards. R290 is a single refrigerant, so that both liquid and gas insertion are possible. Additional charging is also possible.

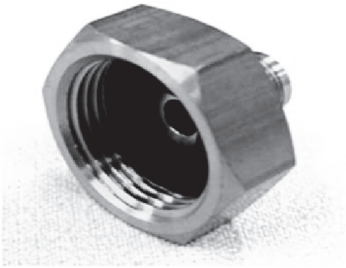
Refrigerant cylinders



7. Connection ports of refrigerant cylinders and packing

Charging ports which fit to the charging hose connection port size (1/2 UNF20) is needed. At the same time, the packing has to be of HC resistant materials.

Connection ports and packing



8. Tools used for refrigerant piping installations and services

	Common tools	R290	R32
1.	Pipe cutters, reamers or scrapers	Not applicable for Monobloc	Not applicable for Monobloc
2.	Flare tools (clutch type)	Not applicable for Monobloc	Not applicable for Monobloc
3.	Torque wrench (1/4, 3/8)	Not applicable for Monobloc	Not applicable for Monobloc
4.	Torque wrench (1/2, 5/8)	Not applicable for Monobloc	Not applicable for Monobloc
5.	Manifold gauges, charging hose	○	○
6.	Vacuum pump, vacuum pump isolator *2	Connection 5/16 [ATEX certified] *1	
7.	Electric gas leakage detectors	Combustible gas detector	HFC detector
8.	Digital scale for refrigerant charging	○	○
9.	Recovery devices (connection port 5/16) *2	ATEX certified	HFC recovery devices
10.	Refrigerant cylinder color	Other (colors that might subject to change according to the international standards)	Other (colors that might subject to change according to the international standards)
11.	Refrigerant cylinder connection port and packing	x	○
12.	Allen wrench (4mm) Electric knives	x	○
*1	Those testers only for HC only cannot be for common use with HFC		
*2	Recovery devices which are certified by Atmosphere Explosible (ATEX)		

[Knowledge for the common usage of tools for R290 & R32]
R290 and R32 machines use different compressor oils.

- If unregulated compressor oil gets mixed into, it may cause damage to the machine function.

[Inserting wrong refrigerant]

- It may cause "not cooling" and "not heating" customer claims because each component (expansion valve, compressor, PCB) of the refrigerant cycle is specially adjusted for R290.
- At the same time, it is not subject to product warranty, if wrong refrigerant was inserted into system.

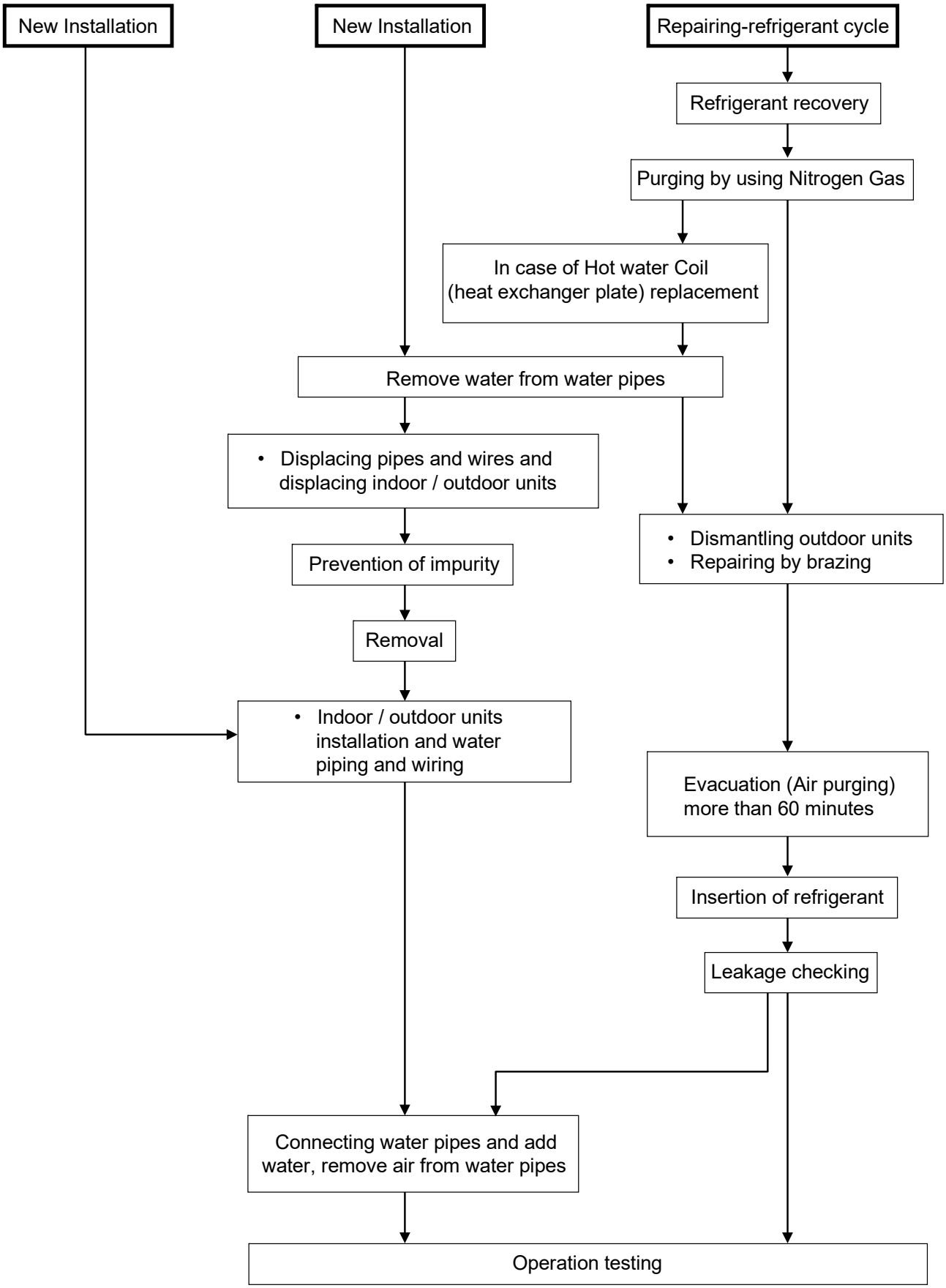
Reference:-

- ASHRAE Standard 34-2016
- ISO Standard ISO 5149
- ISO 817:2014

13.4 New installation, and Repairing of Refrigerant Cycle System Procedures

Personnels working on A3 systems may be subject to applicable occupational hazard or regulations required by local or national law.

For safe servicing and disposal, technicians must have detailed knowledge and skills in handling of flammable refrigerants, prevention of refrigerant leaks, leak detection, personal protective equipment, cylinder handling and loading. A dry powder or $C0_2$ fire extinguisher must be available at the place of service.

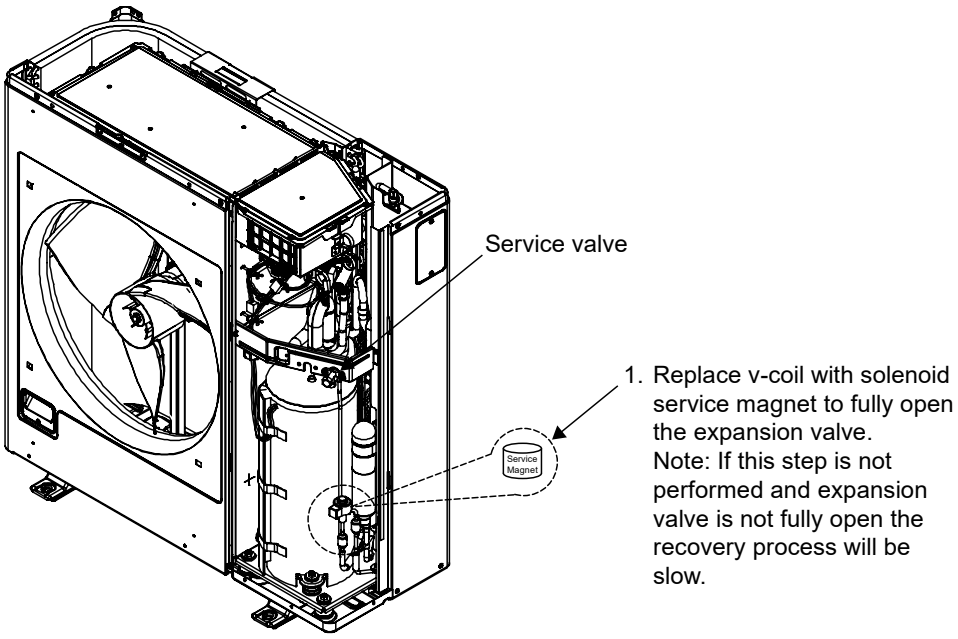


13.5 Servicing

13.5.1 Recover R290 refrigerant with refrigerant recovery machine

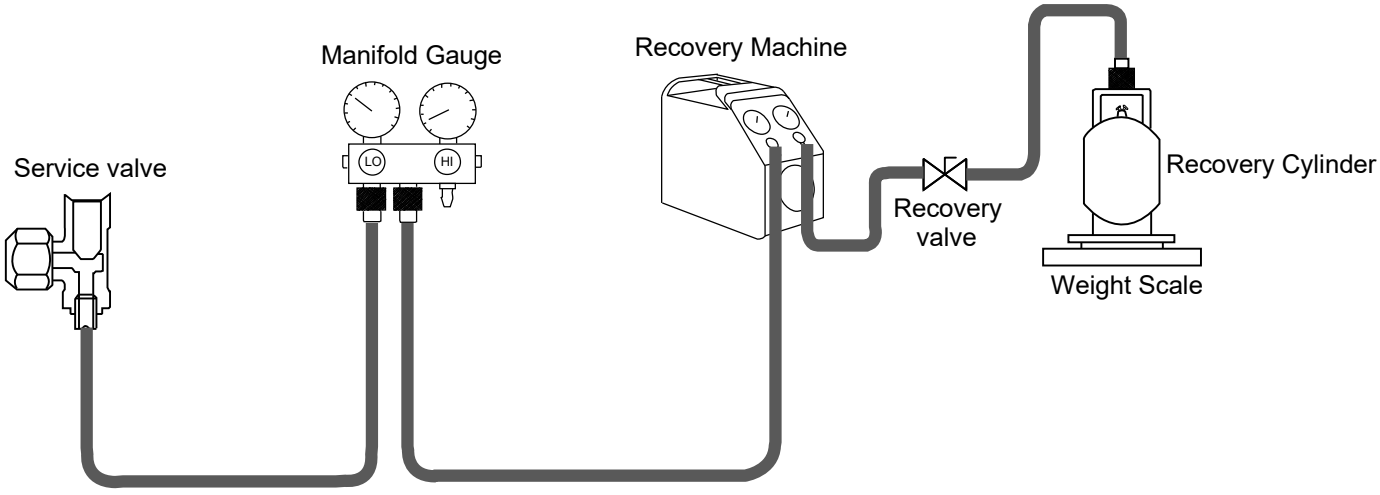
CAUTION!

- Always turn ON the combustible leak detector.
- Keep all ignition sources, hot surface, and open flames 3 meter away from the product.
- Ensure the servicing area is well ventilated.
- Ensure the product is service by certified serviceman.
- Ensure to always have the approved fire extinguisher.



CAUTION!

All equipment and material must be ATEX certified to be allowed to operate within Atmosphere Explosible zone.



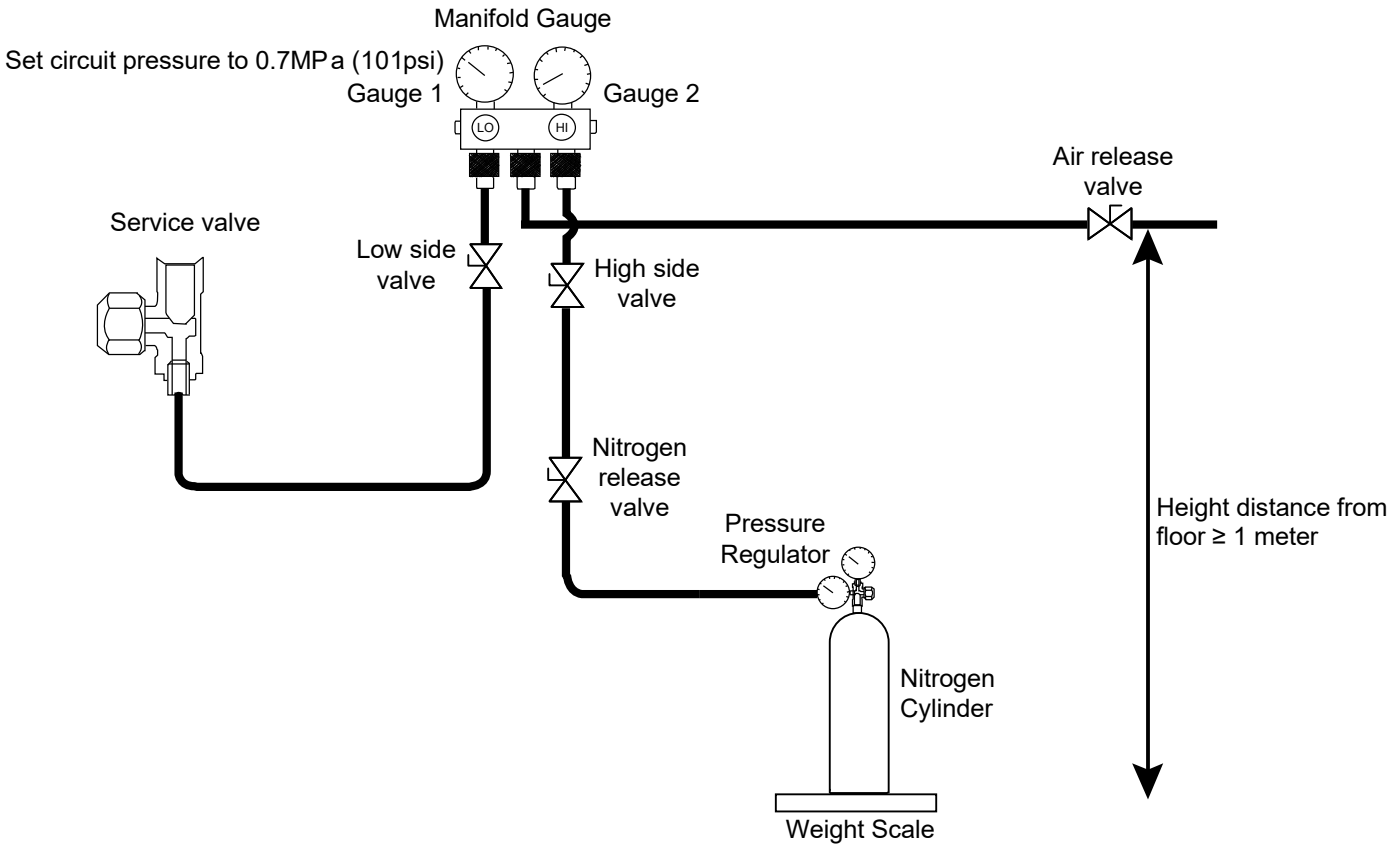
CAUTION!

Do not refill R290 refrigerant more that 50% of recovery cylinder capacity to avoid overpressure. (Safety risk)

Refrigerant Type	Recovery Cylinder Capacity
R32	80% of the weight
R290	50% of the weight

2. Connect the Low side charging hose of the manifold gauge to the service valve (2-way valve) in the outdoor unit.
3. Connect the center hose of the manifold gauge to the recovery machine.
4. Attached the manifold gauge correctly and tightly. Ensure both valve (Low side and High side) is in close position.
5. Connect the hose from recovery machine to the recovery cylinder.
6. Turn ON the recovery machine, turn the low side valve, service valve and recovery valve to open position.

13.5.2 Purging by using Nitrogen Gas Before Servicing and Disposal



1. Connect nitrogen cylinder to the high side at the manifold gauge. Ensure nitrogen cylinder is connected to pressure regulator.
2. Close air release valve and open service valve.
3. Open High side valve and nitrogen release valve. (Set the pressure regulator to 200psi).
4. Charge nitrogen gas up to 0.7MPa (101psi) or charge 400 grams of nitrogen into the unit.
5. Close high side valve and open air release valve (half open) to release remaining propane to the atmosphere. Air release valve must be directed 1 meter from the floor to enable the propane gas to spread properly in the atmosphere. [Beware not to fully open the air release valve to avoid high pressure and compressor oil discharge].
6. When gauge 1 pressure reach near atmospheric (15psi) point the combustible leak detector to air release valve to detect the presence of propane. If presence of propane can still be detected, repeat procedure 2 to 5 until presence of propane cannot be detected.
   [Then refrigerant circuit repairment or unit decommissioning should be done].
7. Before charging new refrigerant, vacuum the system until it reaches to 500micron (67Pa) to remove foreign gas in the system and hold for 15 minutes.

14. Operation and Control

14.1 Basic Function

Inverter control, which equipped with a microcomputer in determining the most suitable operating mode as time passes, automatically adjusts output power for maximum comfort always. In order to achieve the suitable operating mode, the microcomputer maintains the set temperature by measuring the temperature of the environment and performing temperature shifting. The compressor at outdoor unit is operating following the frequency instructed by the microcomputer at indoor unit that judging the condition according to internal water setting temperature and water outlet temperature.

14.1.1 Internal Water Setting Temperature

Once the operation starts, control panel setting temperature will be taken as base value for temperature shifting processes. These shifting processes are depending on the Air-to-Water Heat pump settings and the operation environment. The final shifted value will be used as internal water setting temperature and it is updated continuously whenever the electrical power is supplied to the unit.

14.1.2 Heating Operation

14.1.2.1 Thermostat Control

- Compressor is OFF when Water Outlet Temperature – Internal Water Setting Temperature > 3°C for continuously 3 minutes.
- Compressor is ON after waiting for 3 minutes, if the Water Outlet Temperature – Water Inlet Temperature (temperature at thermostat OFF is triggered) <-3°C.

14.1.2.2 Thermostat Control (Outdoor Ambient Temperature)

Stops provide heating to room side during high outdoor ambient condition.
Control content:

- Heating operation and water pump will turn OFF when outdoor ambient temperature > outdoor thermo off temperature + 3°C.
  (Outdoor thermo off set temperature is set by control panel. Thermo off set temperature is between 5°C ~ 35°C)
- Heating operation will resume when Outdoor ambient temperature < Outdoor thermo OFF set temperature + 1°C.

14.1.2.3 Heat Mode Operation

Operation of heat pump provide heating capacity to room side by hot water through heating panel, floor heating or fan coil unit.

1 3 ways valve control:
   - 3 ways valve switch and fix to heating side.
2 Heat pump operates follow normal heating operation.
3 Back up heater operate follow normal operation.
4 2 ways valve control:
   - 2 ways valve opens.

14.1.3 Cooling Operation

14.1.3.1 Thermostat Control

- Compressor is OFF when Water Outlet Temperature – Internal Water Setting Temperature > -1.5°C for continuously 3 minutes.
- Compressor is ON after waiting for 3 minutes, if the Water Outlet Temperature – Water Inlet Temperature (temperature at thermostat OFF is triggered) >3°C.

14.1.3.2 Cool Mode Operation

1 3 ways valve control:
   - 3 ways valve switch and fix to cooling side.
2 Heat pump operates follow normal cooling operation.
3 Room heater DOES NOT operate during cool mode.
4 2 ways valve control:
   - 2 ways valve is closed.

14.1.4 Target Water Temperature Setting

14.1.4.1 Target Water Temperature Control of Standard System (Optional PCB not Connected)

There are 2 types of temperature control selection which are Compensation and Direct.

- Temperature control type selection by installer:
  1 Compensation : Wlo, WHi, ODLo, ODHi can be set at installer menu.
  2 Direct : Direct Water Temperature Set
- Remote control setting by user:
  1 Compensation : Shift value ±5°C from the compensation curve
  2 Direct : Direct water temperature set change

*This setting only able to set when room sensor select as Water Temperature.
*Instead of water temperature, user will set target room temperature when room sensor select as Room Thermistor OR Internal Room Thermostat.

- Target water temperature is calculated as below condition.
  - Target water temperature = A (Base temperature) + B (shift temperature)

A (Base Temperature)	Compensation	Direct
	Value from the curve + User shift value set	Direct value from user setting

  - B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

B (Shift Temp.)	B shift value depend on the room sensor selection at remocon as table below
Sensor selection	
Water temperature	B = 0
External Room thermostat	B = 0
Internal Room thermostat & Room Thermistor	**Cool Mode:** B = 0 ; when Zone OFF or Zone Room Thermo OFF B = B = 1* (room set temp (R/C) – actual room temp) Max/Min Regulation of B: (Max = 5 ; Min = -5)
	Heat Mode: B = 0 ; when Zone OFF or Zone Room Thermo OFF B = follow Heating Room Temperature PI control logic

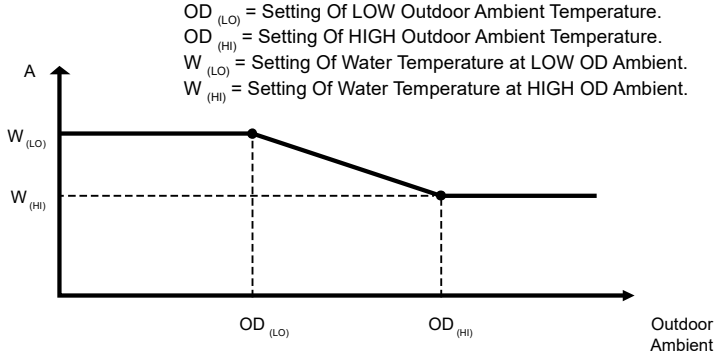
- Maximum/minimum regulation of Target Water Temperature

	Heating	Cooling
Max	55°C (Below Ambient -15°C) * 75°C (Below Ambient -10°C) *	20°C
Min	20°C	5°C

* Between outdoor ambient -10°C and -20°C, max target decreases gradually from 75°C to 55°C.

Compensation Type: (Operation under Heat Mode and Cool Mode)

- The set temperature defines the parameters for the ambient (Outdoor temperature) dependent operation of the unit. The water temperature is determined automatically depending on the outdoor temperature. Default setting is the colder outdoor temperature will result in warmer water and vice versa. The user has the possibility to shift up and shift down the target water by remote control setting.



- Outdoor ambient is updated every 30 minutes when operation ON.
- Setting water outlet temperature always follow $W_{(LO)}$ or $W_{(HI)}$ whenever is higher if outdoor ambient sensor or indoor communication error happen.

However, when powerful mode is requested by remote control during heating mode, the higher value of HLo or Whi will be used for A calculation.

* There are 2 compensation curves (for heating and cooling). During heating mode, the heating curve is used and during cooling mode, the cooling curve is use.

- Compensation curve set shift value:



14.1.5 Target Water Temperature at Extension System (Optional PCB is Connected)

Target water temperature is calculated as below.

- Heat Mode:

- When buffer tank selection is “YES:”
  Target water temperature = Target buffer tank temperature + [2°C]

- When buffer tank selection is “NO”
  - If both zone 1 and zone 2 is active
    Target Water Temperature = Higher zone target water temperature of Zone 1 and Zone 2.
  - If only one zone is active
    Target Water Temperature = Zone target water temperature of active zone.

- Cool mode:

- When buffer tank selection is “YES”
  - If both zone 1 and zone 2 active
    Target Water Temperature = Lower Zone Target Water Temperature of Zone 1 and Zone 2
  - If only one zone is active
    Target Water Temperature = Zone Target Water Temperature of active zone

- When buffer tank selection is “NO”
  - If both zone 1 and zone 2 active
    Target Water Temperature = Lower Zone Target Water Temperature of Zone 1 and Zone 2
  - If only one zone is active
    Target Water Temperature = Zone Target Water Temperature of active zone

*Cool Mode does not have SG ready control

14.1.6 Target Zone Water Temperature Control

Purpose:- To control zone mixing and zone pump according to the zone sensor temperature

14.1.6.1 Target Zone 1 Water Temperature Setting Control

- Start condition
  - Heating zone 1 is ON by remote control or Timer or Auto Mode **OR**
  - Cooling zone 1 is ON by remote control or Timer or Auto Mode.

- Cancel condition
  - Heating zone 1 is OFF by remote control or Timer or Auto mode **AND**
  - Cooling zone 1 is OFF by remote control or Timer or Auto mode.

- Target Zone 1 water temperature is calculated as below condition.
  - Target Zone 1 water temperature = A (Base temperature) + B (shift temperature)

A (Base Temperature)	**Compensation**	**Direct**
	Value from the curve + User shift value set	Direct value from user setting

* During heat mode and compensation select, if powerful mode is activated, higher value of WLo or WHi will be use as curve value.

  - B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

B (Shift Temp.)	B shift value depend on the room sensor selection at remocon as table below
Sensor selection	
Water temperature	B = 0
External Room thermostat	B = 0
Internal Room thermostat & Room Thermistor	**Cool Mode:** B = 0 ; when Zone OFF or Zone Room Thermo OFF B = B = 1* (room set temp (R/C) – actual room temp) Max/Min Regulation of B: (Max = 5 ; Min = -5)
	Heat Mode: B = 0 ; when Zone OFF or Zone Room Thermo OFF B = follow Heating Room Temperature PI control logic
Pool Function Selected	B = Delta value setting from remocon

* B = 0 regardless of which sensor selection, if SHP control bit is enable except Pool function select (maintain Pool “B” value)

** Pool function also can be select at Zone 1 when optional PCB is connected and Zone 1 system is select.

- Maximum/minimum regulation of Target Water Temperature.

	Heating	**Cooling**
Max	55°C (Below Ambient -15°C) * 75°C (Below Ambient -10°C) *	20°C
Min	20°C	5°C

* Between outdoor ambient -10°C and -20°C, max target decreases gradually from 75°C to 55°C.

- Target Zone 1 Water Temperature (Heat mode only) during SG ready control
  - If buffer tank selection is “NO” then following shift is carried out.
    - While digital input is “10” or “11” then,
      Final Target Zone 1 water temperature
      = Target Zone 1 water temperature* (SG ready % setting (remote control menu))%
  - If buffer tank selection is “YES”
    - No shift of Target Zone 1 Water Temperature. Target Buffer Tank Temperature will change accordingly.
      * Refer to “Buffer tank temperature control”

14.1.6.2 Target Zone 2 Water Temperature Setting Control

- Start condition
  - Heating zone 2 is ON by remote control or Timer or Auto Mode **OR**
  - Cooling zone 2 is ON by remote control or Timer or Auto Mode.

- Cancel condition
  - Heating zone 2 is OFF by remote control or Timer or Auto mode **AND**
  - Cooling zone 2 is OFF by remote control or Timer or Auto mode.
- Target Zone 2 water temperature is calculated as below condition.
  - Target Zone 2 water temperature = A (Base temperature) + B (shift temperature)

A (Base Temperature)	Compensation	Direct
	Value from the curve + User shift value set	Direct value from user setting

* During heat mode and compensation select, if powerful mode is activated, higher value of WLo or WHi will be use as curve value.

  - B (shift temperature) value is depend on the room sensor selection at remote controller as below table:

B (Shift Temp.)	B shift value depend on the room sensor selection at remocon as table below
Sensor selection	
Water temperature	B = 0
External Room thermostat	B = 0
Internal Room thermostat & Room Thermistor	**Cool Mode:** B = 0 ; when Zone OFF or Zone Room Thermo OFF B = B = 1* (room set temp (R/C) – actual room temp) Max/Min Regulation of B: (Max = 5 ; Min = -5)
	Heat Mode: B = 0 ; when Zone OFF or Zone Room Thermo OFF B = follow Heating Room Temperature PI control logic
Pool Function Selected	B = Delta value setting from remocon

* B = 0 regardless of which sensor selection, if SHP control bit is enable except Pool function select (maintain Pool "B" value)

** Pool function also can be select at Zone 2 when optional PCB is connected and Zone 2 system is select.

- Maximum/minimum regulation of Target Water Temperature.

	Heating	Cooling
Max	55°C (Below Ambient -15°C) * 75°C (Below Ambient -10°C) *	20°C
Min	20°C	5°C

* Between outdoor ambient -10°C and -20°C, max target decreases gradually from 75°C to 55°C.

- Target Zone 2 Water Temperature (Heat mode only) during SG ready control
  - If buffer tank selection is "NO" then following shift is carried out.
    - While digital input is "10" or "11" then,
      Final Target Zone 2 water temperature
      = Target Zone 1 water temperature* (SG ready % setting (remote control menu))%
  - If buffer tank selection is "YES"
    - No shift of Target Zone 2 Water Temperature. Target Buffer Tank Temperature will change accordingly.
      * Refer to "Buffer tank temperature control"

14.1.6.3 Zone Temperature Control Contents

- During Standard System (Optional PCB not connected)
  - Only 1 zone temperature control is available
  - This zone room temperature is control by either one of the 4 room sensor (Room Th, Int/Ext Room Thermostat, Water temperature)

- Target Zone Water Temperature is calculated based on selected temperature control type (Compensation or Direct) and selected room sensor. Target Water Temperature will set same as Target Zone Water Temperature
- Target Water Temperature is the temperature for heat pump to operate refer to indoor water outlet sensor.
- Heat pump and water pump OFF when ROOM Thermo OFF (Zone thermo OFF by Room Th or Room Thermostat).

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

- During Extension System (Optional PCB connected)

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

- During Extension System (Optional PCB connected)

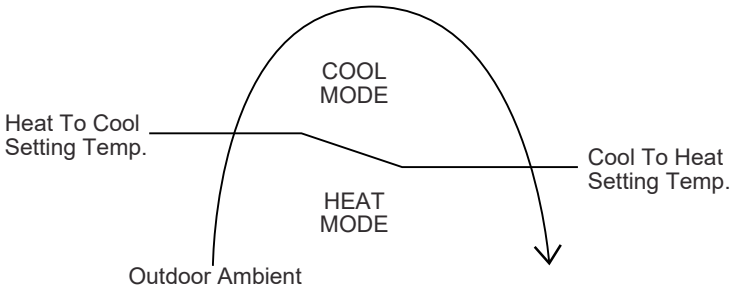
- Buffer Tank connection select "NO" &
  - One zone system is select
    - This zone room temperature control by either one of the 4 room sensor (Room Th, Int/Ext Room Thermostat, Water temp.)
    - Target Zone Water Temperature calculate base on selected temperature control type (Compensation or Direct) and selected room sensor.
    - Target Water Temperature will set same as Target Zone Water Temperature
    - Target Water Temperature is a temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM Thermo OFF (Zone thermo OFF by Room Th or Room Thermostat).

* There will be no zone sensor connected to zone 1 (No zone sensor error), mixing valve and zone pump will not operate.

  - 2 zone system select
    - Each zone room temperature is control by each sensor which select from either one of the 4 room sensor
    - Target Zone 1 & 2 Water Temperature is calculated based on selected temperature control type (Compensation or Direct) and selected room sensor.
      * Zone Mixing Valve & Zone pump will operate to achieve Target Zone Water Temperature which refer to zone sensor.
      * Zone Sensor will detect if zone sensor is open or short.
    - Target Water Temperature will set same as the active & higher zone water temperature setting. (When cooling mode, lower zone water temp setting)
    - Target Water Temperature is the temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM Thermo OFF (Both Zone thermo OFF by Room Th or Room Thermostat).

- Buffer Tank Connection select "YES" &
  - 1 zone system or 2 zone system select
    - Each zone control by each sensor which select from either one of the 4 room sensor
    - Target Zone 1 & 2 Water Temperature calculate base on selected temperature control type (Compensation or Direct) and selected room sensor. Each zone have their own Target Zone Water Temperature.
      * Zone Mixing Valve & Zone pump will operate to achieve each Target Zone Water Temperature which refer to zone sensor
      * Zone Sensor will detect if zone sensor is open or short.
    - Target Buffer Tank Temperature will be set as active & higher zone water temperature setting + Buffer Delta T. (Heating) Target Buffer Tank Temperature will be set as active & lower zone water temperature setting (Cooling).
    - Target Water Temperature will set as Target Buffer Tank Temperature + [2°C] (Heating) Target Water Temperature will set as Lower or Active Target Zone Water Temp. + [-3°C] (Cooling)
    - Target Water Temperature is a temperature for heat pump to operate refer to indoor water outlet sensor.
    - Heat pump and water pump OFF when ROOM thermo OFF
      - Heat mode: ROOM thermo OFF (Buffer Tank Temperature > Target Buffer Tank + [0°C]
      - Cool mode: ROOM Thermo OFF (Both Zone thermo OFF by Room Th or Room Thermostat)

14.1.7 Auto Mode Operation

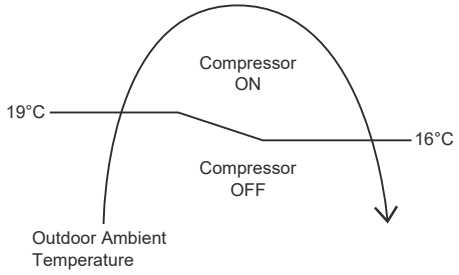


- Control details:
  - To enable the unit to operate either heat or cool mode automatically, heat to cool set temperature and cool to heat set temperature can be set by control panel.
  - Automatic operation is judged based on control panel setting temperature and outdoor ambient temperature.

  * Minimum setting of heat to cool set temperature is 1°C higher than cool to heat set temperature.

- Judgement control:
  - If outdoor ambient temperature < Heat to Cool Set Temperature, unit will operate in Heat Mode or else the unit will operate in Cool Mode.
  - If current operation is Cool mode, outdoor ambient temperature > Cool to Heat Temperature, unit will maintain Cool mode operation or else the unit will operate Heat mode.
  - If current operation is Heat mode, outdoor ambient temperature >Heat to Cool Temperature, unit will maintain Heat mode operation or else the unit will operate Cool mode.
  - Every 60 minutes the outdoor ambient temperature is judged.
  - When Auto + Tank mode is selected, operation mode switching is judged by both outdoor ambient temperature and indoor air temperature.

14.1.8 Auto Cooling Mode Operation Limit



- Auto Mode Cooling Only operation will start once the outdoor ambient temperature reaches 19°C and compressor will continue to run until the outdoor ambient temperature drops to 16°C.
- Due to this limitation,
  If Heat to Cool temperature is set lower than 19°C, the compressor will not operates until the outdoor ambient temperature reaches 19°C or higher.

14.1.9 Tank Mode Operation

- 3 ways valve direction
  - 3 ways valve switch to tank side during Tank Thermo ON condition. Switch 3 ways valve to room side when tank achieve Tank Thermo OFF temperature.

- Tank Thermo ON/OFF Characteristic
  - Tank Thermo OFF

  Case 1: <u>Internal Tank Heater is select and Tank Heater ON</u>
  - Tank temperature > Tank Set Temperature continuously for 15 seconds.
  - Water outlet >75°C

- Case 2: Tank Heater OFF OR External Heater is select
  - When heat pump OFF due to water thermos & Tank temperature > Tank water set temperature for continuously 20 seconds. **OR**
  - Tank temperature > Tank set temperature + 1°C for continuously 20 seconds.

  - Tank Thermo ON

  Case 1: Tank Heater ON (Internal Tank Heater)
  - Tank temperature < Tank set temperature + R/C (Tank re-heat temperature)

  Case 2: Tank Heater OFF (Internal Tank Heater)
  - Tank temperature < Tank water set temperature + R/C (Tank re-heat temperature)

  * When tank thermo ON, water pump will ON for 3 minutes then only heat pump turn ON.
  * Tank water set temperature = tank set temperature or 65°C whichever lower.

- 2 ways valve close
- Heat pump Thermostat Characteristic
  - Heat pump Water Outlet set temperature is set to below table:

Outdoor ambient temperature	Heat pump water outlet temperature
< -20°C	57°C
> -20°C	65°C
> -10°C	75°C

Characteristic of heat pump thermos ON/OFF under tank mode condition:

Water Outlet Thermo Condition
- Heat pump thermos OFF temperature:
  1 Heat pump thermo OFF temperature = Target Water outlet temperature + (2°C)
  2 Water outlet temperature > heat pump thermo OFF temperature for continuously 3 minutes, heat pump OFF but water pump continue ON.

- Heat pump thermo ON temperature
  1 Heat pump thermo ON temperature = water inlet during thermo OFF time + [-3°C]
  2 When water outlet temperature < heat pump thermo ON temperature, heat pump ON.

Water inlet thermo protection condition
- Heat pump thermo OFF temperature:
  1 Water inlet temperature > [75°C/55°C] for continuously 30 seconds, heat pump OFF, water pump continue ON.

- Heat pump thermo ON temperature:
  1 Heat pump thermos ON temp = water inlet temperature < [75°C/55°C].

Outdoor ambient temperature	Water inlet temperature
< -10°C	55°C
> -10°C	75°C

Thermo ON/OFF for Heat Pump in Tank Operation:

When tank temperature achieve heat pump OFF condition, refer below condition:
Conditon 1 : Tank Heater ON (Internal Tank Heater)
- Heat pump will turn OFF, water pump continue ON and room heater will continue ON if tank temperature below tank heater thermo ON condition. 3 ways valve will only switch to room side after tank temperature reach tank heater thermo OFF condition.

Conditon 2 : Tank Heater OFF (Internal Tank Heater)

- If tank temperature achieve tank thermo OFF, heat pump turn OFF, water pump turn OFF, room heater OFF and 3 ways valve switch to room side.

When tank temperature achieve heat pump ON condition, water pump ON, heat pump ON and room heater turn OFF.

Heat pump OFF condition at Tank Mode

- Tank temperature > tank water set temperature continuously for 20 seconds after heat pump thermos OFF due to water thermo. (Heat pump turn OFF but water pump continue ON and room heater turn ON to achieve tank set temperature) **OR**
- Tank temperature > tank set temperature + [1°C] for continuously 20 seconds. (Heat pump OFF, water pump OFF, room heater OFF and 3 ways valve switch to room side)

Heat pump ON condition at Tank Mode

- Tank temperature < tank water set temperature + R/C setting (Tank re-heat temp) (Water pump turn ON OR continue ON, heat pump ON and 3 ways valve switch to tank side or maintain at tank side)

Tank heater control

- Internal heater only operates to tank side if Tank heater ON and backup heater is enable.

Internal heater turn ON condition:

- Tank temperature < tank set temperature **AND**
- Heat pump thermos OFF **AND**
- 20 minutes from previous heater off **AND**
- Internal tank heater selects USE from control panel.

Internal heater turn OFF condition:

- Tank temperature > tank set temperature for continuously 15 seconds **OR**
- Heat pump thermo ON **OR**
- Mode change or operation is off by control panel.

14.1.10 Heat + Tank Mode Operation

1. 3 ways valve control:
   - 3 ways valve switch to room side during room heat-up interval and switch to tank side during tank heat-up interval. Both modes will switch alternately. Tank mode is the initial running mode of Heat + Tank mode.

2. Heat pump operation control:
   - During room heat-up interval
     - Follow normal heating operation.
       Switching to tank side depends to below cases:
       Case 1:
       [Previous switch from tank interval to room interval due to thermo OFF]
     - Switch to tank heat-up interval when Tank temp < Tank thermos ON temp (Room heat-up interval ends)
       Case 2:
       [If heating operation at room side is less than 30 minutes and switch to tank side 3 times consecutively]
     - Maintain at room heat-up interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** Tank temperature < Tank thermo ON temperature.
       Case 3:
       [Previous switch from tank interval to room interval due to tank interval timer is complete]
     - Maintain at room heat-up interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** tank temperature < Tank thermo ON temperature.
   - During Tank heat-up interval
     - Tank interval is the first mode running when heat + tank mode is select.
     - Switch to room interval only when tank achieve tank thermo OFF **OR** tank heat-up interval timer is complete.
     - Heat pump operates according to normal tank mode operation.

3 Room heater control:
   - During heating heat-up interval
     - Follow normal room heater control operation.

4 Tank heater control:
   - During heating heat-up interval
     - Internal tank heater will not function under heating heat-up interval.
   - During tank heat-up interval
     - Internal tank heater will turn ON after heat pump thermo off to boil tank temperature to tank set temperature.
   - 2 ways valve control is open
   - Indoor water pump control:
     - Indoor water pump always turn ON if room heat pump thermo ON **OR** Tank thermo ON.

14.1.11 Cool + Tank Mode Operation

1 3 ways valve control:
   - 3 ways valve switch to room side during room cooling interval and switch to tank side during tank heat-up interval. Both mode will switch alternately. Tank mode is the initial mode of cool + tank mode.

2 Heat pump operation control:
   - During room heat-up interval
     - Follow normal cooling operation.
       Switching to tank side depends to below cases:
       Case 1:
       [Previous switch from tank interval to room interval due to thermo OFF]
     - Switch to tank heat-up interval when Tank temperature < Tank Thermo ON temperature (Room interval will ends)
       Case 2:
       [If cooling operation at room side is less than 30 minutes and switch to tank side for 3 times consecutively]
     - Maintain at room cooling interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** Tank temperature < Tank thermo ON temperature.
       Case 3:
       [Previous switch from tank interval to room interval is due to tank interval timer is complete]
     - Maintain at room cooling interval regardless of the tank temperature. Switch to tank heat-up interval only when (Room Interval Timer is complete **OR** Room heat pump thermo OFF) **AND** tank temperature < Tank thermo ON temperature.

       *Tank Thermo ON temperature:

Internal Tank Heater select USE	<Tank set temperature + R/C setting (Tank re-heat temperature)
Others	<Tank water set temperature + R/C setting (Tank re-heat temperature)

   - During Tank heat-up interval
     - Tank interval is the first mode running when the cool + tank mode is select.
     - Switch to room interval only when tank achieve tank thermo OFF **OR** tank heat-up interval timer is complete.
     - Heat pump operates according to normal tank mode operation.

3 Room heater control:
   - During room cooling interval
     - Room heater is OFF and not operates.

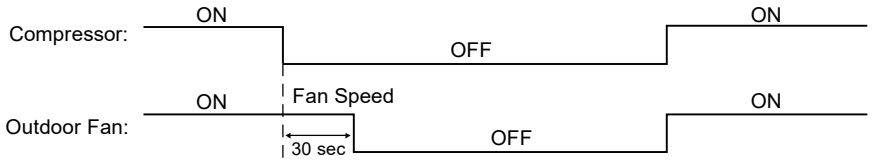
4 Tank heater control:
   - During room cooling interval
     - Internal tank heater will not function under room cooling interval.
   - During tank heat-up interval
     - Internal tank heater will turn ON after heat pump thermos off to boil tank temperature to tank set temperature.

5 2 ways valve is close.

6 Indoor water pump control:
- Indoor water pump always turn ON if room heat pump thermo ON **OR** Tank thermo ON.

14.1.12 Outdoor Fan Motor Operation

Outdoor fan motor is adjusted according to operation condition. It starts when compressor starts operation and it stops 30 seconds after compressor stops operation.

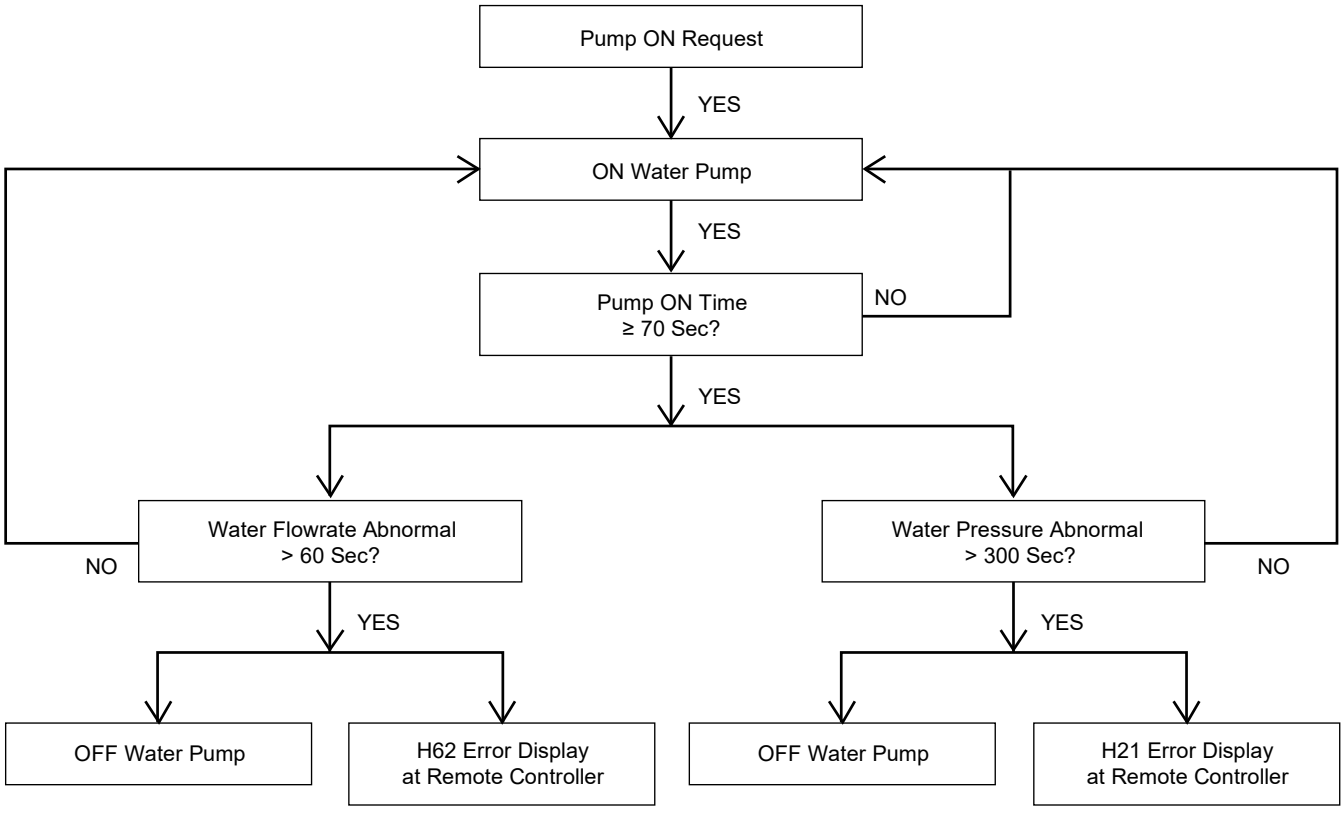


14.2 Water Pump

The system will start checking on the water flow level after operation start for 70 seconds. If water pressure level is detected low or high continuously 300 seconds, the water pump and the compressor will be OFF permanently and OFF/ON control panel LED will blink (H21 error occurs).

14.2.1 Water Pump Control (For WH-ADC0509L3E5, WH-ADC0509L6E5, WH-ADC0509L3E5AN, WH-ADC0509L6E5AN, WH-ADC0509L3E5UK only)

- Once the indoor unit is ON, the water pump will be ON immediately and no error judgment for 70 seconds. However, during this 70 seconds operation, if there is any abnormality cause at outdoor or malfunction, the compressor should be OFF immediately and restart delay after 3 minutes.
- The system will start checking on the water flow level after operation start for 70 seconds. If water flow level is detected low continuously 60 seconds, the water pump and the compressor will be OFF permanently and OFF/ON control panel LED will blink (H62 error occurs).
- When error happens, the power has to be reset to clear the error.
- If there is no error indication, the water pump shall be continuously running.
- The water pump will remain ON when compressor OFF due to thermostat OFF setting is reached.
- Water pump will OFF when room thermo **OR** tank thermo **OR** buffer tank thermo OFF.
- Water pump will delay 15 seconds to turn OFF when request to OFF except during anti-freeze deice activate or air purge mode.



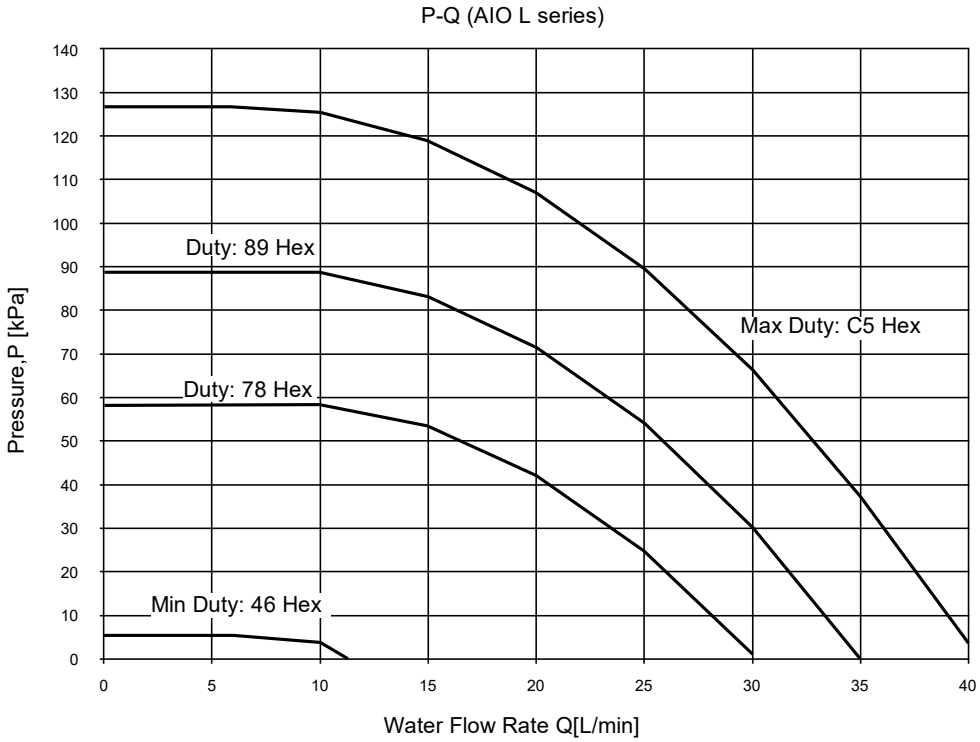
Maximum pump speed setting on remote control

- Standard pump speed is automatically controlled to get the designed water temperature different between water inlet and outlet (ΔT). Instead of setting the standard pump speed, maximum pump speed is manually adjusted by the installer according to water circuit pressure drop.

However, the following sequences do not follow maximum pump duty setting by remote control.

- Pump down mode
- Air purge mode
- Normal deice

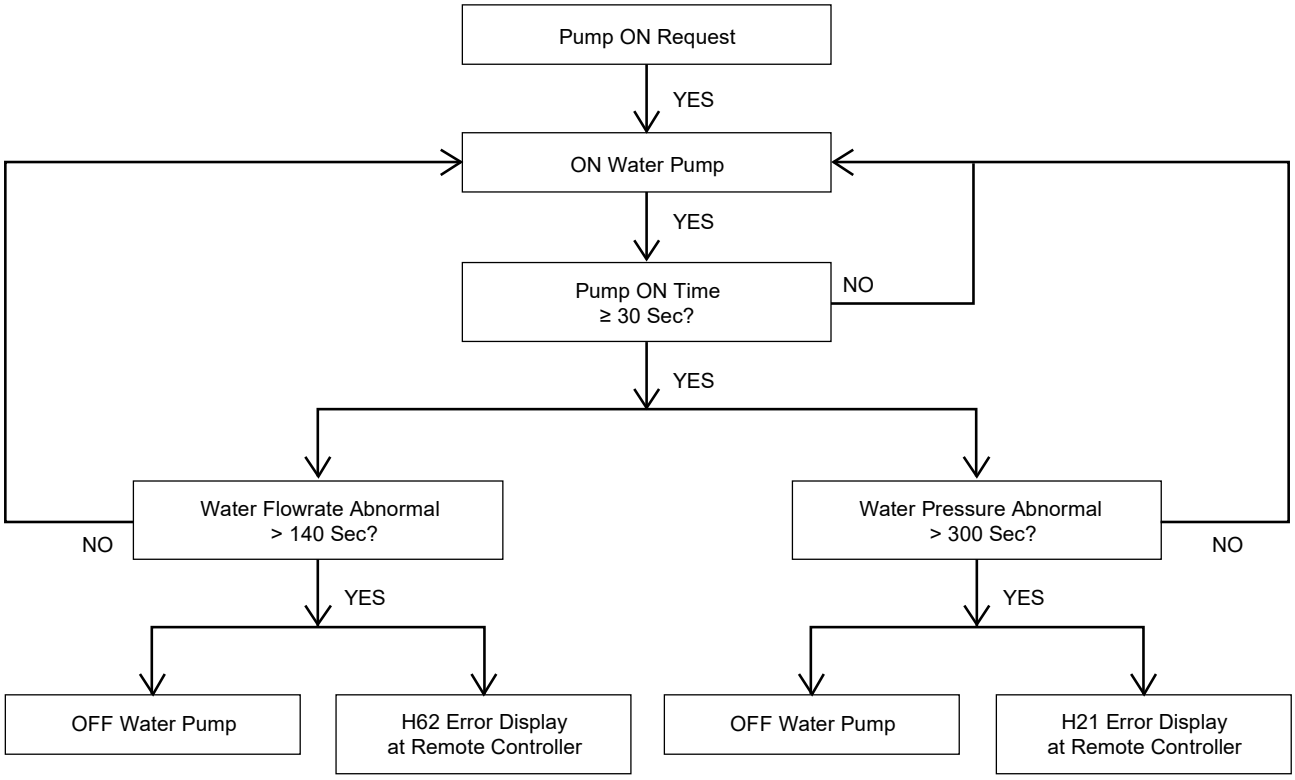
1) P-Q graph for different pump HEX duty



※ This condition is under connecting ODU and connecting 5m piping (inner diameter Φ25) between ODU and IDU (AIO).

14.2.2 Water Pump Control (For WH-ADC0509L3E5B only (No change for other models))

- Once the indoor unit is ON, the water pump will be ON immediately and no error judgment for 30 seconds. However, during this 30 seconds operation, if there is any abnormality cause at outdoor or malfunction, the compressor should be OFF immediately and restart delay after 3 minutes.
- The system will start checking on the water flow level after operation start for 30 seconds. If water flow level is detected low continuously 140 seconds, the water pump and the compressor will be OFF permanently and OFF/ON control panel LED will blink (H62 error occurs).
- When error happens, the power has to be reset to clear the error.
- If there is no error indication, the water pump shall be continuously running.
- The water pump will remain ON when compressor OFF due to thermostat OFF setting is reached.
- Water pump will OFF when room thermo **OR** tank thermo **OR** buffer tank thermo OFF.
- Water pump will delay 30 seconds to turn OFF when request to OFF except during anti-freeze deice activate or air purge mode.



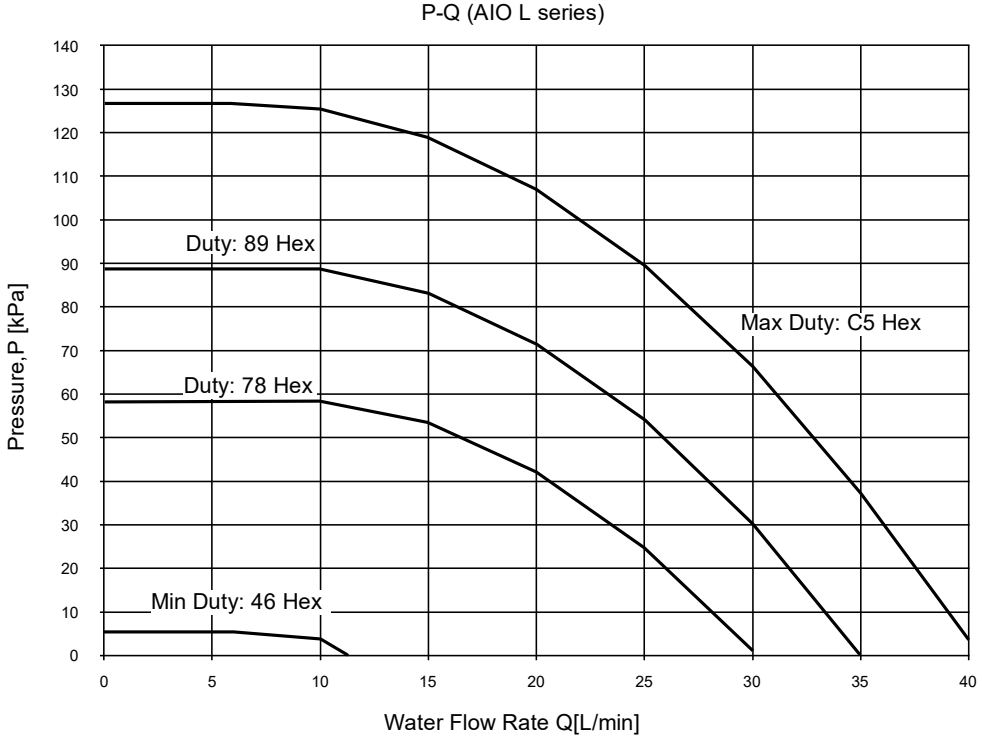
Maximum pump speed setting on remote control

- Standard pump speed is automatically controlled to get the designed water temperature different between water inlet and outlet (ΔT). Instead of setting the standard pump speed, maximum pump speed is manually adjusted by the installer according to water circuit pressure drop.

However, the following sequences do not follow maximum pump duty setting by remote control.

- Pump down mode
- Air purge mode
- Normal deice

1) P-Q graph for different pump HEX duty

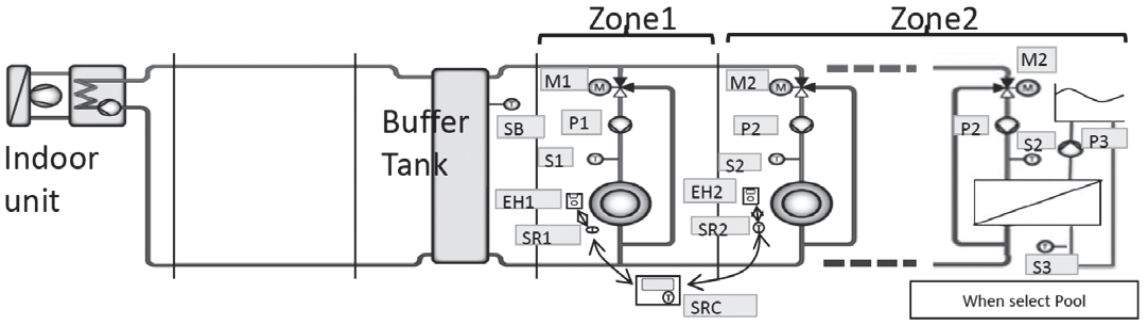


※ This condition is under connecting ODU and connecting 5m piping (inner diameter Φ25) between ODU and IDU (AIO).

14.2.3 Zone Water Pump Control

Purpose:

- Water pump install at each zone to circulate the water inside each zone during buffer tank connection selected “YES” or 2 zone systems.



Content:

- AC type water pump install for this zone water pump control. When optional PCB connected, 230V output will drive this zone pump.
- There are three pump can be connected through Optional PCB. (Zone 1 Pump, Zone 2 Pump, & Pool Pump)

  * Zone 1 pump [P1] use to circulate zone 1 water circuit & Zone 1 mixing valve [M1] adjust to control the Zone 1 target water temperature.

  * Zone 2 pump [P2] use to circulate zone 2 water circuit & Zone 2 mixing valve [M2] adjust to control the Zone 2 target water temperature.

  * When Pool Function select as Zone 2 circuit, [P2] use to circulate water to heat exchanger which use to transfer heat to pool water.

  * Pool pump [P3] circulates the pool water through the heat exchanger to get warm water.

- Zone 1 and Zone 2 water pump start condition:
  - Zone room request ON
    (eg. Zone 1 thermo ON, only zone 1 pump will turn ON)

- Zone 1 and Zone 2 water pump stop condition:
  - Zone room request OFF

- Pool water pump start condition:
  - Pool Zone request ON **AND**
  - Pool function is selected

- Pool water pump stop condition
  - Pool zone: Zone room request OFF **OR**
  - Pool function is cancel

    * Zone 1 & Zone 2 water pump need to turn OFF when antifreeze deice pump stop control activate and turn ON back after the antifreeze deice pump stop control end under setting of "NO" buffer tank connection.

Zone Pump Prohibit ON control:

- Start condition:
  Zone 1 water temperature ≥ 80°C continuously for 5 minutes
  *stop zone water pump operates if the zone water fulfilled.
- Cancel condition:
  After 30 minutes from start condition fulfilled.
  *zone water pump operates according to normal condition.

Zone Pump Control during Anti-Freeze

Zone pump control during Zone Anti-Freeze Control:

- When Zone Anti-Freeze Flag=1, Zone Pump Turn ON.
- When Zone Anti-Freeze Flag=0, Zone Pump Turn OFF.

Zone pump control during Indoor Anti-Freeze Control:

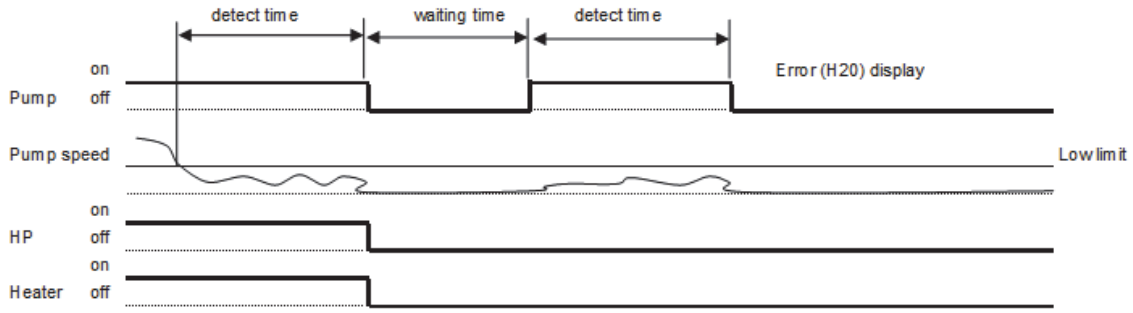
- Zone pump only ON/OFF if the Extension PCB connected and Buffer Tank select "NO" condition

  When Indoor Anti-Freeze flag=1, Zone Pump Turn ON
  When Indoor Anti-Freeze flag=0, Zone Pump Turn OFF

* Pool Water Pump will not affected by both Indoor anti-freeze control or zone anti-freeze control.

14.2.4 Water Pump Speed Feedback Error

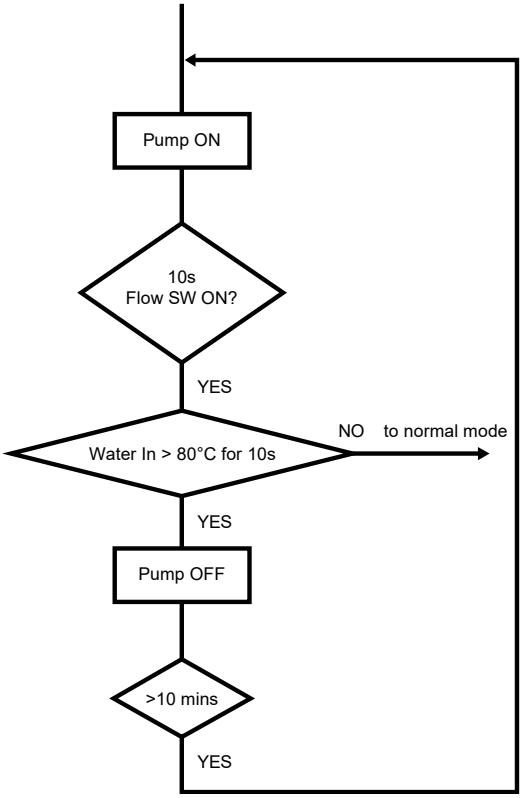
- Basically pump speed feedback is control by micon.
- When pump speed is below low limit or over high limit for a few seconds, micon detect pump error and system is stopped.
- Error detection conditions:
  - Detect abnormal water pump speed for continuous 10 secs.
  - Current pump speed < 700 rpm or
  - Current pump speed > 6000 rpm for 10 seconds.
- Control contents:
  - When error occurs, water pump, heating and heater is stopped for 30 seconds then restart again (Retry control).
  - When micon detect error again, system is stopped and error code [H20] is displayed at control panel.



14.3 Indoor Unit Safety

14.3.1 Indoor Unit Safety Control

1 When water pump is ON, the system will start checking flow switch status (ON/OFF).
2 If the flow switch ON for 10 seconds, the system will check on the water inlet temperature for 10 seconds.
If the water inlet temperature not exceeds 80°C, the water pump shall be continuously running with normal mode.
If the water inlet temperature exceeds 80°C for continuously 10 seconds, the water pump will be OFF immediately.
3 After water pump OFF for more than 10 minutes, it will be ON back and the indoor unit safety control checking is restarted.



14.4 Auto Restart Control

1 When the power supply is cut off during the operation of Air-to-Water Heatpump, the compressor will reoperate after power supply resumes.

14.5 Indication Panel

LED	Operation
Color	Green
Light ON	Operation ON
Light OFF	Operation OFF

Note:
- If Operation LED is blinking, there is an abnormality operation occurs.

14.6 Indoor Back-Up Heater Control

14.6.1 Indoor Electric Heater Control

1 Normal Heating Mode
- Heater On condition:
  a. Heater switch is ON
  b. After Heatpump thermo ON for **[30]** mins
  c. After water pump operate **[9]** mins
  d. Outdoor air temperature < Outdoor set temperature for heater
  e. When water outlet temperature < Water set temperature + **[-4°C]**
  f. **[20]** minutes since previous Backup heater Off
     * When heatpump cannot operate due to error happens during normal operation, heater will go into force mode automatic
     * Heater need to operate during deice operation

- Heater Stop Condition:
  a. When outdoor set temperature > outdoor set temperature + **[+2°C]** for continuous 15 secs OR
  b. When water out temp> water set temperature + **[-2°C]** for continuous 15 secs OR
  c. Heater switch is Off OR
  d. Heatpump thermo-off or OFF condition

2 Force Heater Mode
- Heater On condition:
  a. After water pump operate **[9]** mins
  b. When water outlet temperature < water set temperature + **[-4°C]**
  c. **[20]** minutes since previous Backup heater Off

- Heater Stop condition
  a. Force mode off **OR**
  b. When water outlet temperature > water set temperature + **[-2°C]** for continuous 15 secs

* Do not operate heater at the following situation
1 Water outlet temperature sensor, and water inlet sensor abnormal
2 Flow switch abnormal
3 Circulation pump stop condition

14.6.2 Room Heater Operation during Deice

Purpose:
- To protect the indoor Heat Exchanger from ice forming and prevent heat exchanger plate breakage.

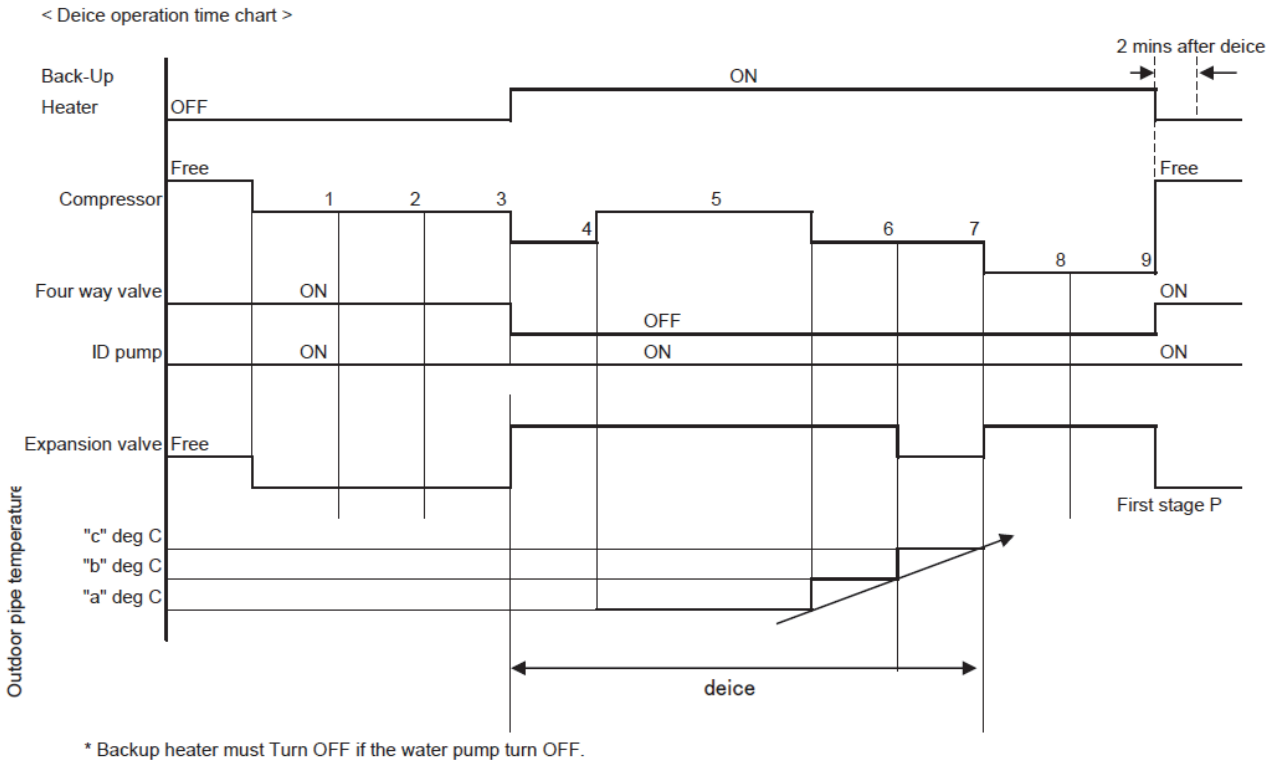
Control content:
- This Heater protection control will activate only if the backup heater is enable at custom setup by remote controller. Once fulfil the start condition, room heater will turn ON together (base on max heater capacity selection) and stop together if stop condition is fulfilled.
  * If the heater is request to turn ON OLP feedback will be detected.

Starting conditions:
- During normal deice operation 4~9
- Water outlet temperature < 7°C or
  Water outlet 2 temperature < 7°C

Heater operates when 1 ~2 fulfilled.

Stop condition:
- When normal deice end or
- Water outlet temperature > 45°C

However, room heater keeps ON if indoor electric heater control activate.



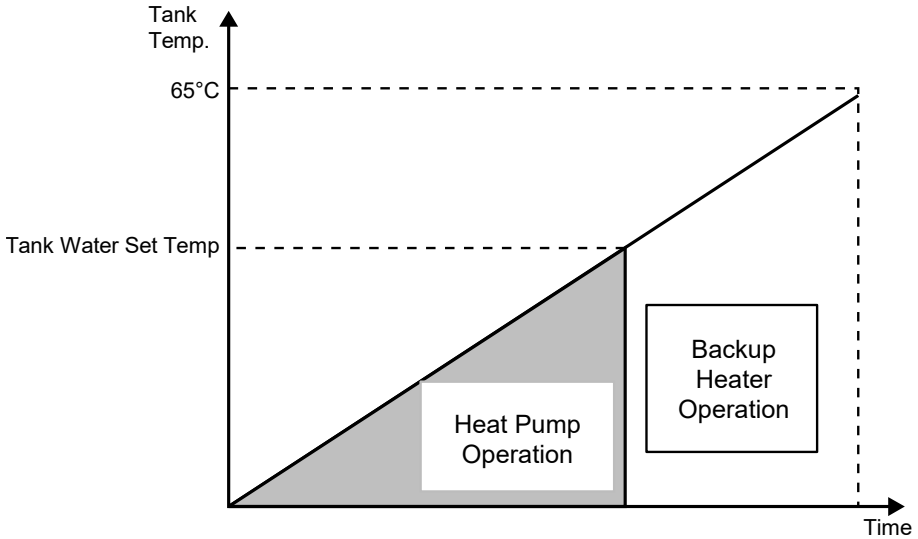
14.7 Tank Heater Control

14.7.1 Internal Heater Control at Tank Mode

- Internal heater turn ON condition:
  1. Internal Heater select for Tank heater by remote controller
  2. Tank Heater select ON by user.
  3. Backup Heater Enable
  4. Tank Temperature < Tank Set Temperature
  5. Heat Pump Thermo OFF
  6. 20 minutes from previous heater off.

- Room heater turn OFF condition:
  1. Tank Temperature > Tank Set Temperature + [0°C] for continuous 15 seconds. **OR**
  2. Heat Pump Thermo ON **OR**
  3. Mode Change or Operation OFF by remote controller **OR**

  * Backup Heater Turn ON/OFF all together according to the selected heater capacity.



14.8 Base Pan Heater Control (Optional)

- To enable the base pan heater function, control panel initial setting has to be manually adjusted by activating Base Pan Heater menu.
- There are 2 optional start condition can be selected, Type A or Type B.
- Control details:

1 Type A: (Default Auto Mode)

Start conditions:

- When outdoor air temperature ≤ 3°C during heating and deice operation is ON.

Control contents:

- Base pan heater is ON during deice operation and continues ON for 10 minutes after deice operation ends.

Cancel condition:

- When outdoor temperature > 6°C after deice end or
- When operation is not at heating mode or
- Base pan heater ON timer count is completed.

2 Type B: (ON Mode)

Start conditions:

- When outdoor air temperature is ≤ 5°C and operates in heating mode, base pan heater is ON.

Cancel conditions:

- When outdoor air temperature is > 7°C or
- When operation is not at heating mode.

14.9 Force Heater Mode

Purpose of Force Heater Mode:

- As a backup heat source when heat pump error. Force heater Mode only control backup heater to heat up the room circuit, and turn ON back up heater or booster heater to boil up tank water base on the tank heater selection (internal or external).

Force Heater Control start condition:

- Force heater request ON by user during error OR auto turn ON by remote controller during error **AND** (Force Heater mode can be operate regardless of mode selection, remocon will send the latest mode selection force bit by bit to indoor. Indoor will judge to turn ON heater to room side if it is heat mode selected, and turn ON heater to heat tank water base on tank heater selection)
- During Error Happen (exclude the error list below)

Error List which not allow Force Heater operation

H12	Capacity Mismatch	H76	Indoor-Remote Controller Communication Error
H20	Abnormal Water Pump	H95	Abnormal Voltage Connection
H62	Abnormal Water Flow	F37	Abnormal Water Inlet sensor
H70	Abnormal Back-up Heater OLP	H45	Abnormal Water Outlet sensor
H74	PCB Communication Error		
[When tank mode operate with external heater selected & tank heater select ON]			
H73	Abnormal tank sensor	H91	Abnormal tank heater OLP

Force Heater Control Stop Condition:

- Force Heater request OFF **OR**
- Operation OFF request **OR**
- Power reset **OR**
- Error of above list happens during force heater operation.

Control contents:

- After fulfill start condition, indoor will operate the force heater operation according to below mode condition
  Heat mode Only: Turn ON backup heater to achieve room heat pump target water temperature.
  Heat + Tank mode: Turn ON backup heater to heat up room **OR** Turn ON Heater to Boil up tank water.
  Cool mode Only: Water pump and backup heater will OFF in force heater mode.
  Cool + Tank mode: Operate pump and internal Heater OR External heater to Boil up tank water.
  Tank mode Only: Operate pump and internal Heater OR External heater to Boil up tank water.

* For heat mode condition, backup heater will only turn ON if the backup heater is enable regardless of Room Heater Selection.
* For tank mode condition, If internal heater selected backup heater will turn ON to boil up tank water.
If external heater selected , booster heater will turn ON to boil up tank water regardless of tank heater selection.

Room Side: (Heat Mode):
- When force heater mode start condition fulfilled, turn ON water pump and turn ON backup heater follow below control.
- Operate the 3 ways valve at room side only and turn ON 2 ways valve as heat mode operation.
- Turn ON the zone pump and mixing valve if system select 2 zone system or Buffer tank connect YES, control according to normal zone pump and mixing valve control.
- When Force heater mode stop condition fulfilled, turn OFF heater as below condition and turn OFF water pump after pump delay time.

Backup Heater On Condition:
- When Force Heater Control start condition fulfill **AND**
- After water pump operate 2 minutes **AND**
- When water outlet temperature < water set temperature + [-4°C] **AND**
- 20 minutes since previous Backup heater Off **AND**
- Backup Heater Enable

Backup Heater Stop condition:
- Force mode off **OR**
- Operation off **OR**
- When water outlet temperature > water set temperature + [-2°C] for continuous 15 secs **OR**
  * ON/OFF follow normal heater sequence.

Tank side (Tank mode):
- When tank mode select and force heater bit received, turn ON backup heater (INTERNAL) or Booster Heater (External) depend on the tank heater selection.
- If tank heater selection is INTERNAL, follow normal thermo judgement to switch 3 ways valve to tank side and room side.
- If tank heater selection is EXTERNAL, only turn ON booster heater according to tank thermo.

<u>Tank Heater selection is INTERNAL:</u>
Backup Heater ON Condition:
- After water pump operate 2 mins **AND**
- When tank temperature < Tank set temperature - [Remocon Set Tank Re-heat Temp] **AND**
- 20 minutes since previous Backup heater OFF **AND**
- Backup Heater Enable

Backup Heater OFF condition:
- Force mode off **OR**
- When tank temperature > Tank set temperature for continuous 15 secs **OR**
- Tank Operation OFF

<u>Tank Heater selection is EXTERNAL:</u>
Booster Heater ON condition:
- Force Heater mode ON **AND**
- Tank temperature < tank set temperature + [Remocon Set Tank Re-heat Temp] - 1°C, **AND**
- 20 minutes since previous heater off.

Booster Heater OFF condition:
- Tank temperature > tank set temperature for continuous 15 secs.
- Force mode OFF
- Tank Mode Operation OFF
  (During tank interval or tank mode condition, water pump and 3 ways valve will OFF)

14.10 Powerful Operation

Powerful mode is use to increase the capacity of heat pump to achieve higher target temperature. Powerful mode is applicable when heat mode is operating.

Remote control setting:
On quick menu of remote control, there is 4 options of powerful mode can be select.
- OFF : Cancel powerful mode
- 30 minutes : Set powerful for 30 minutes
- 60 minutes : Set powerful for 60 minutes
- 90 minutes : Set powerful for 90 minutes

Control contents:
During the time set by remote control, powerful will activate according to 2 shift up controls. However, this function is applicable only for heating. Remote control will transmit the signal to indoor unit once this function is select then transmit OFF signal to indoor when the timer is complete. Indoor will transmit signal to outdoor for frequency control.

Indoor setting temperature shift

- If system is standard system (Optional PCB is not connected)
  - Target water temperature will shift up to Wlo or Whi whichever higher.
- If system is extension system (Optional PCB is connected)
  - Target water Zone 1 and Zone 2 temperature will shift up to Wlo or Whi whichever higher.

  * If “Direct Type” temperature control is select, this powerful shift up setting is not effective.

- **Start condition**
  - Powerful function is select by remote control.

- **End Condition**
  - OFF/ON button is pressed.
  - Powerful function is OFF by remote control.

14.11 Quiet Operation

Quiet mode is use to reduce the noise of outdoor unit by reducing the frequency or fan speed.

Quiet level
There are 3 level (Level 1, Level 2, Level 3) to set by quick menu function on remote control.

Control content
Once the quiet function is select, the remote control will transmit the signal to indoor and outdoor unit.

Start condition
- Quiet mode is set on remote control.
- Quiet mode is request ON by weekly timer.

Stop condition
If any of below condition is achieve.
- OFF/ON button is pressed.
- Quiet mode is OFF by remote control.
- Quiet mode is request OFF by weekly timer.

14.12 Sterilization Mode

- Purpose:
  - To sterilize water tank by setting the required boiling temperature.

- Remote control setting
  - Days for sterilization function to start can be select.
  - Time of selected day to start sterilization function.
  - Boiling temperature (Internal heater is 55°C ~ 65°C)
  - Maximum operation time is 5 minutes to 1 hour.

- Start condition
  - Tank connection set to “YES” by remote control
  - Sterilization function selects “YES”.
  - Sterilization signal received from remote controller by timer.
  - Tank mode request ON.

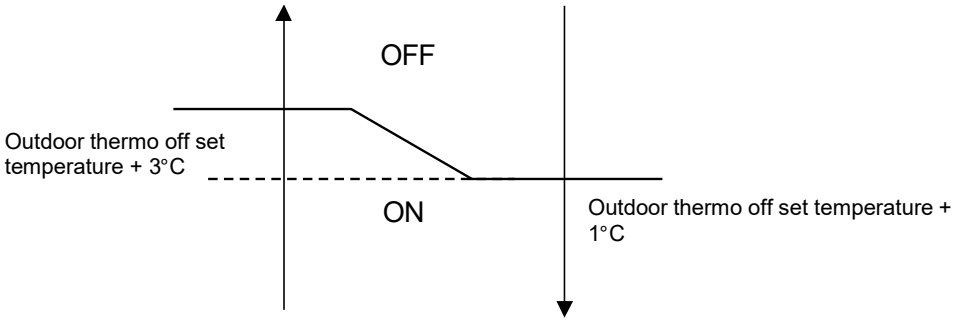
- Stop condition
  - When boiling timer is completed. Boiling timer (Remote control set maximum operation time) start counting once tank achieve boiling set temperature **OR**
  - After 8 hours of operation since sterilization start.
  - Tank mode request OFF.

- Control content:
  - During sterilization function activation time, target tank set temperature will internally change to boiling set temperature.
  - During sterilization activates, heat pump and heater (external or internal) will operate as normal tank mode to achieve the boiling set temperature.
  - Sterilization operation will end when stop condition is fulfill.
  - After sterilization is complete, tank set temperature will resume to normal operation.

* Tank temperature may not achieve boiling set temperature if tank heater is select OFF **OR** external compressor switch.

14.13 Outdoor Ambient Thermo OFF Control

- Purpose:
  - Stop provides heating to room side during high outdoor ambient condition.



- Control content:
  - Heating outdoor ambient thermos OFF control only applicable when heat pump operate in heat mode. (This control will not activate when running in tank side)
  - Heat pump and water pump will turn OFF when outdoor ambient is higher than outdoor thermo OFF set temperature.
  - Heat pump thermos ON when outdoor ambient < outdoor thermos OFF set temperature + 1°C.

14.14 Alternative Outdoor Ambient Sensor Control

Purpose of the Alternative Outdoor Ambient Sensor:

- It is some possibility that the air to water heat pump unit will install at a location where the original ambient sensor is expose to direct sunlight. Therefore, another optional ambient sensor can be connect to indoor PCB and locate at new and better reading location to improve the heat pump performance.

Control Detail:

- Remocon can select either the extra outdoor ambient sensor is connected or not. (YES/NO)
- The alternative outdoor ambient sensor will connect to indoor unit main PCB terminal.

    - when alternative sensor select NO
        - Original Outdoor temperature sensor will use for Indoor & Outdoor heat pump operation reference sensor.
        - Data communication direction : OUTDOOR send outdoor temperature reading to INDOOR.
        - Error judge : OUTDOOR will judge the original outdoor sensor error (F36 display if error detect). No judge error on alternative outdoor sensor

    - when alternative sensor select YES
        - Alternative Outdoor temperature sensor will use for Indoor & Outdoor heat pump operation reference sensor.
        - Data communication direction : INDOOR send outdoor temperature reading to OUTDOOR.
        - Error judge : INDOOR will judge the Extra outdoor sensor error only after operation ON request received from remocon.
          (F36 display if error detect). No judge error on original outdoor sensor.

14.15 Force DHW mode

Purpose:
When user want to use hot water now, user can press this force DWH mode under the quick menu to operate tank only mode to boil up the tank temperature.

Remocon setting:
Force DHW function can be activate under quick menu.

Control Content:

- when press the Force DHW function during operation OFF condition:
    - When receive this Force DHW bit from remocon, indoor will run tank only mode regardless of the mode selection.
    - After tank temperature achieve tank thermo off temperature, turn OFF force DHW bit and return to operation OFF with previous mode selection.
- When press the Force DHW function during operation ON condition:
    - When receive this Force DHW bit from remocon, indoor will memories the running mode and run tank only mode regardless of the mode selection.
    - After tank temperature achieve tank thermo off temperature, turn OFF force DHW bit and return to previous memories running mode.

      * when operation OFF or mode change request from remocon during force DHW mode operation, End force DHW mode and follow the new request operation.

      * Once receive force DHW mode from remocon, indoor direct start tank mode and consider tank thermo ON. Thermo OFF only when achieve tank thermo OFF depend on the Tank System Setting.

14.16 SMART DHW mode

Panasonic All In One model provide the option to choose STANDARD DHW Mode or SMART DHW Mode for Tank Heat Up according to requirement. SMART DHW mode comparatively consume lower tank heat up power but longer re-heat time than STANDARD DHW Mode.

SMART DHW control

- During SMART DHW start time 20:00 (Default Setting) to SMART DHW stop time 05:00 (Default setting)
  Heat pump re-heat the tank water only when tank temperature drop below 20°C (Default setting)

- Time between 05:00 to 20:00
  Heat pump reheat the tank water when tank temperature as below condition

Condition 1: Tank Heater ON
Reheat when tank temperature below tank set temperature + R/C (Tank re-heat Temperature) - 3°C

Condition 2: Tank Heater OFF
Reheat when tank temperature below Tank set temperature or 51°C (Whichever lower) + R/C (Tank re-heat Temperature) -3°C

* SMART DHW start time, stop time and SMART ON Temperature can change in CUSTOM menu.

14.17 Anti Freeze Control

- Anti freeze protection control menu can be set YES or NO by control panel.
- In heatpump system, there are 3 types of anti freeze control:
  1. Expansion tank anti-freeze control
     - Expansion tank anti freeze heater ON condition:
       - Outdoor ambient temp. < 3°C
     - Expansion tank anti freeze heater OFF condition:
       - Outdoor ambient temp. > 4°C

  2. Water pump circulation anti freeze control
     - Water pump turns ON when **ALL** below conditions are fulfilled:
       - Heatpump OFF (stand by) OR error occurs.
       - Water flowing flag is ON.
       - Water flow switch is not abnormal.
       - Outdoor ambient temp. < 3°C OR outdoor ambient temp. sensor is abnormal.
       - Water inlet/outlet temp. < 6°C.
       - After 5 minutes from previous water pump OFF.
     - Water pump turn OFF when **ANY** of below conditions is fulfilled:
       - Outdoor ambient temperature ≥ 4°C.
       - During -5°C < outdoor ambient temp. < 4°C
         - After water pump ON for 4 minutes, and water inlet temp. ≥ 8°C.
         - Else, shift to back up heater anti freeze control.
       - During outdoor ambient temp. < -5°C
         - After water pump ON for 4 minutes, and water inlet/outlet ≥ 20°C.
         - Else, shift to back up heater anti freeze control.
     - However, if flow switch is abnormal (H62), then water pump circulation anti freeze control will not activate.

  3. Back up heater anti freeze control:
     - Back up heater turn ON when **ALL** below conditions is fulfilled:
       - Water inlet/outlet temp. < 6°C.
       - Water pump circulation anti freeze control activated and water pump ON for 4 minutes.
     - Back up heater turns OFF when ANY of below conditions is fulfilled:
       - When OD temp < -5°C, Water inlet/outlet temp. > 28°C.
       - When OD temp >= -5°C, Water inlet/outlet temp. > 13°C.
     - However, if back up heater is abnormal (H70) then back up heater anti freeze control will not activate.

14.17.1 Zone Anti-Freeze Control

- If buffer tank selection is “NO” and Anti- Freeze function select “NO” from remote control, this control cannot activate.

Start condition:

- After [5] min from previous Zone pump off. **AND**
- Outdoor air temp < [3] °C **OR** Outdoor sensor is abnormal. **AND**
- Zone water temperature < [6]˚C **OR** Zone Sensor Short or Open

Cancel condition:

- After water Zone pump ON [4] min **AND**
- Outdoor air temp ≥ [4]°C **OR**
- During -5 °C ≤ Outdoor air temp < [4] °C **OR**
  Zone water temperature sensor > [8] °C
- During Outdoor air temp < [-5] °C
  Zone water temperature sensor > [20] °C
  *However, Zone water temperature sensor is Open or Short, Condition C and D is ignored.

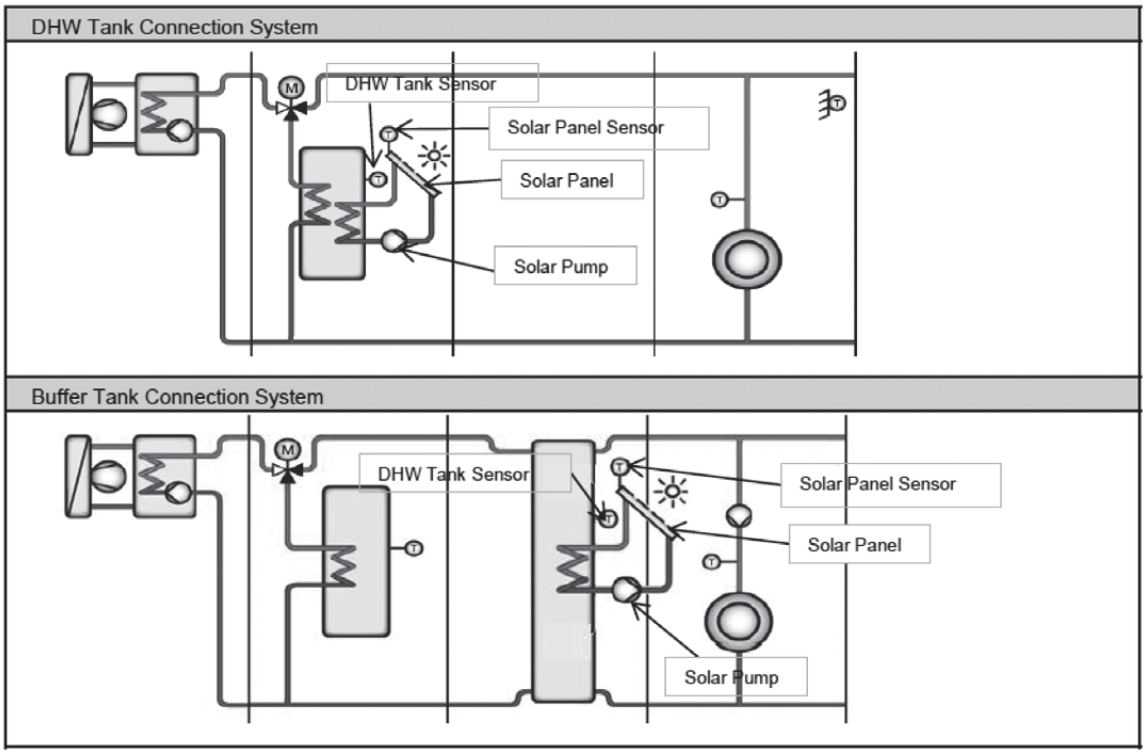
14.18 Solar Operation (Optional)

14.18.1 Solar Operation:

Solar function:

- This function allow user to control the solar pump to operate depend on the solar sensor reading compare to the tank installed. Solar pump will circulate the hot water energy store at solar panel to heat up the DHW Tank or Buffer Tank.

Solar Connection Diagram:



- Solar function can only enable when the Optional PCB is connected.
- Few part as below need to install to control the solar operation:
  - Solar Panel
  - Solar Pump
  - Solar Panel Sensor
  - Tank Sensor (Buffer tank sensor OR DHW Tank sensor depend on the connection direction)
    * During Solar Connection to the system, installer need to alert on the high water temperature may flow to the zone circuit or DHW piping circuit. Therefore pipe which withstand higher water temperature need to be installed.

- Solar remote control setting
  1 Solar Setting can only be set when the optional PCB connection is select "YES"
  2 By remote controller, Setting as below list can be set for solar function operation (Installer Menu)
    - Solar Function ("YES" or "NO)
    - Tank Connection Direction ("DHW TANK" or "BUFFER TANK")
    - Delta T turns ON: Difference temperature setting between solar panel sensor and Tank to turn on solar pump. (Range :5 ~ 15°C)
    - Delta T turns OFF: Difference temperature setting between solar panel thermistor and Tank to turn off solar pump. (Range :2 ~ 10°C)
    - Outdoor temp for Anti-Freeze : Outdoor temp to start Anti-Freeze control for solar circuit. (Range : -20 ~ 10°C)
    - Tank Temperature HI Limit Set (Range : 70 ~ 90°C)

14.18.2 Solar Operation Control

- Solar function can only be activate if the solar function selection “YES” from remote control. To achieve hot water from solar panel, indoor need to control the solar pump and circulate hot water from solar panel.

  Under normal case:
  - Solar pump start condition:
    - Solar panel temperature > Delta T turn on setting temp (R/C) + Tank temperature (depend on selection, DHW or Buffer) **AND**
    - Tank temperature (DHW or Buffer) < Solar HI Limit Temp (R/C) **AND**
    - Operation ON with heat mode (apply to solar connect to "Buffer Tank" case)

    * Condition c) ignore if the solar system is connect to DHW tank (control active under operation OFF time for Tank connection case)

  - Solar pump stop condition:
    - Solar panel temperature < Delta T turn OFF setting temp (R/C) + Tank temperature (depend on selection, DHW or Buffer) **OR**
    - Tank hot water temp >= Solar HI Limit Temp (R/C) + [2]°C

  Under solar Anti-freeze protection control:
  - Solar pump start condition:
    - Outdoor temp < Outdoor temp setting for Anti-Freeze (R/C)
  - Solar pump stop condition:
    - Outdoor temp > Outdoor temp setting for Anti-Freeze + [2]°C

**However, During Cool mode this function cannot activate if Tank selection is “Buffer Tank".
**Solar pump can operate even if Heat pump is under error stop.

- Solar operation during error:
  - During Tank sensor (DHW or Buffer depend on selection) abnormal, Solar operation will not able to function.
  - During Solar Panel sensor detect OPEN (not include SHORT), Solar operation will not able to function too.

14.19 Boiler Bivalent Control

- Boiler is an additional or alternative heat source to heat up the room when necessary.
- Purpose of this control is to turn ON and turn OFF the Boiler output signal when boiler heating capacity needed in the system.
- Boiler is possible to connect to DHW Tank and Buffer Tank depends on the installer.
- Boiler operation parameter need to be set on Boiler itself, indoor do not control the boiler operation direction and operation.
- There are Alternative mode, Parallel mode, & Advance Parallel mode available to select by installer to fit to the total system.

Bivalent control selection by remote controller

Remote control setting value:
1 Outdoor Ambient Set = (Range: -15°C ~ 15°C)

- **Alternative Mode**
  - Only one heat source operates at one time, either heat pump or boiler depends on condition.

Control detail:
During Operation ON at Heat mode or Tank mode or Heat + Tank Mode
  - Boiler signal turn ON and heat pump and water pump turn OFF when:
    - Outdoor ambient < Outdoor Ambient Set **AND**
    - Boiler prohibit flag = 0

** However indoor water pump can operate when Anti-freeze control condition fulfilled.

  - Boiler signal turn OFF and heat pump and water pump turn ON when:
    - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
    - Boiler prohibit flag = 1

- **Parallel Mode**
  - Parallel mode allows heat pump and boiler ON at the same time. Boiler operates as an additional heating capacity when low heat pump capacity at low ambient condition.

Control detail:
During operation ON at Heat mode or Tank mode or Heat + Tank mode
  - Boiler signal turns ON when:
    - Outdoor ambient < Outdoor Ambient Set **AND**
    - Boiler prohibit flag = 0
  - Boiler signal turns OFF when:
    - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
    - Boiler prohibit flag = 1

- **Advance Parallel Mode**
  - Advance parallel mode allow heat pump to operate and turn ON boiler only when ambient and temperature condition is fulfilled.

Remote control setting value:
1 Outdoor Ambient Set = (Range : -15°C ~ 15°C)
2 Selection of boiler connection direction. (Heat only, DHW only, Heat & DHW)
3 Setting data under Heat Direction
  - Start Temperature | START_TEMP |
  - Start Delay Timer | START_TIMER |
  - Stop Temperature | STOP_TEMP |
  - Stop Delay Timer | STOP_TIMER |
4 Setting data under DHW Direction
  - Delay Timer | DELAY_TIMER |

Control detail:
<u>During operation ON at Heat Mode</u>
- Boiler signal turns ON when
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Buffer tank temperature < Target Buffer Tank Temperature + [START_TEMP] for [START_TIMER] **AND**
  - Heat pump operate at room side AND
  - Connection of Boiler to Heating Select "YES" From installer menu **AND**
  - Buffer Tank connection select "YES" **AND**
  - Boiler prohibit flag = 0
- Boiler signal turns OFF when
  - Outdoor ambient > Outdoor Ambient Set + [-2°C] **OR**
  - Buffer Tank temperature > Target Buffer Tank temperature + [STOP_TEMP] for [STOP_TIMER] **OR**
  - Heat pump not at room side. **OR]**
  - Boiler prohibit flag = 1

<u>During operation ON at Tank Mode</u>
- Boiler signal turns ON when
  - Outdoor ambient < Outdoor Ambient Set **AND**
  - Heat pump operate at tank side for continuous | DELAY_TIMER | **AND**
  - Connection of Boiler to DWH Tank select "YES" from installer menu. **AND**
  - Boiler prohibit flag = 0
- Boiler signal turns OFF when
  - Outdoor ambient > Outdoor Ambient Set + [2°C] **OR**
  - Heat pump not operates at tank side. **OR**
  - Boiler prohibit flag = 1

<u>Boiler prohibit flag control</u>
Purpose:
- For product safety. Boiler signal is OFF when water temperature is too high.

Start condition:
- Water outlet ≥ 85°C continues for 5 minutes.
- Water inlet ≥ 85°C continues for 5 minutes.
- Zone1 water temp ≥ 75°C continues for 5 minutes.
- Zone2 water temp ≥ 75°C continues for 5 minutes.

<u>Contents:</u>
- After start condition fulfilled, set boiler prohibit flag = 1

Cancel condition:
- After 30 minutes from start condition fulfilled.

<u>Contents:</u>
- Set boiler prohibit flag = 0

14.20 External Room Thermostat Control (Optional)

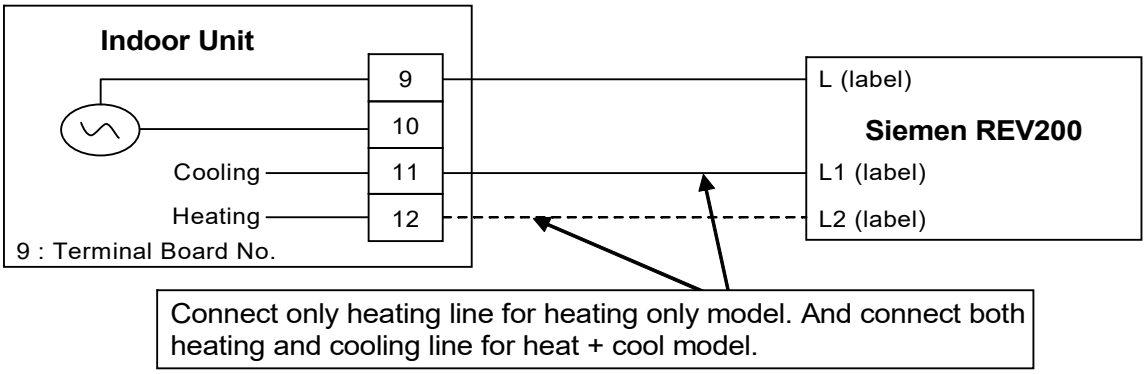
Purpose:

1 Better room temperature control to fulfill different temperature request by external room thermostat. Recommended external room thermostat:

Maker	Characteristic
Siemen (REV200)	Touch panel
Siemen (RAA20)	Analog

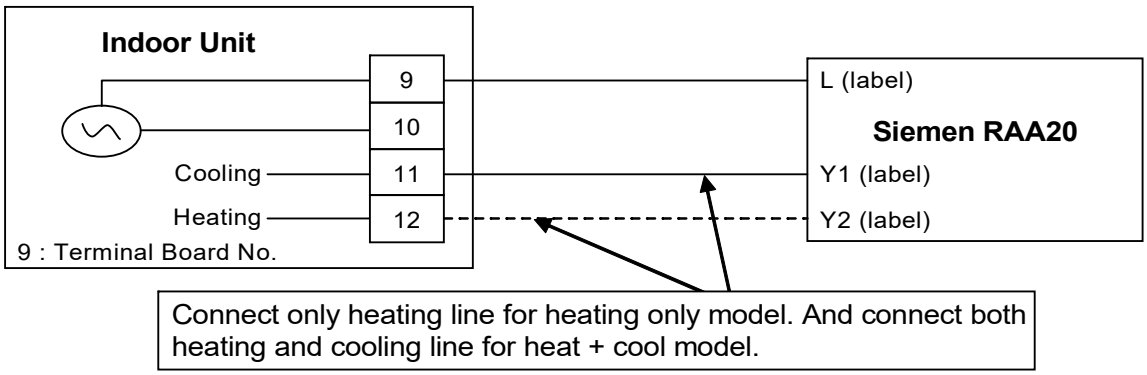
Connection of external room thermostat:
Wire Connection and thermo characteristic of Siemen REV200:

Setting	L/L1 (H)	Heat Thermo	L/L2 (C)	Cool Thermo
Set Temp < Actual Temp	Open Circuit	OFF	Short Circuit	ON
Set Temp > Actual Temp	Short Circuit	ON	Open Circuit	OFF



Wire Connection and thermo characteristic of Siemen RAA20:

Setting	L/Y1 (H)	Heat Thermo	L/Y2 (C)	Cool Thermo
Set Temp < Actual Temp	Open Circuit	OFF	Short Circuit	ON
Set Temp > Actual Temp	Short Circuit	ON	Open Circuit	OFF



Control Content:

- External room thermostat control activate only when remote thermostat connection select YES by Indoor control panel.
- When indoor running heat mode, refer thermo On/Off from heating line feedback. And when indoor running cool mode, refer thermo On/Off from cooling line feedback.
- Heat pump Off immediately when receive thermo off feedback.

14.21 Three Ways Valve Control

Purpose:

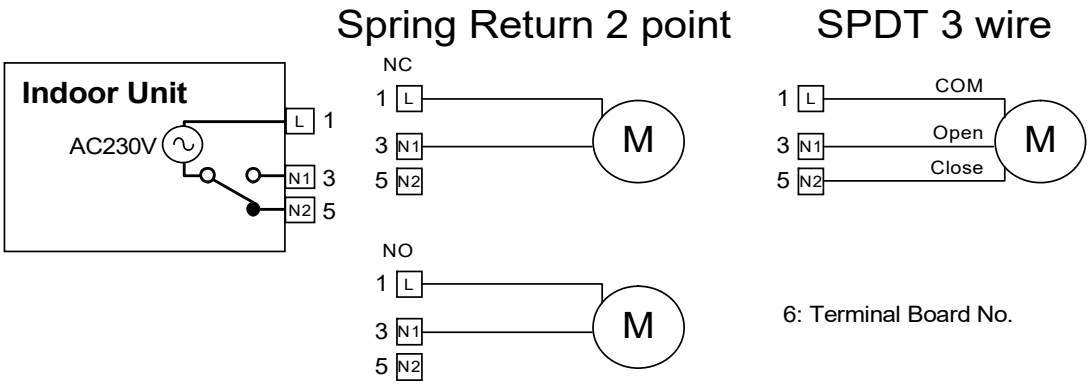
- 3 ways valve is used to change flow direction of hot water from heat pump between heating side and tank side.

Control contents:

1 3 ways valve switch Off:
   - During 3 ways valve switch Off time, the hot water will provide heat capacity to heating side.

2 3 ways valve switch On:
   - During 3 ways valve switch On time, the hot water will provide heat capacity to tank side.

3 Stop condition:
   - During stop mode, 3 ways valve will be in switch off position.



* During pump down and force mode, fix 3 ways valve in close condition.

* Recommended Parts : SFA 21/18 (Siemens)

14.22 Two Ways Valve Control

Functionality of 2 ways valve:

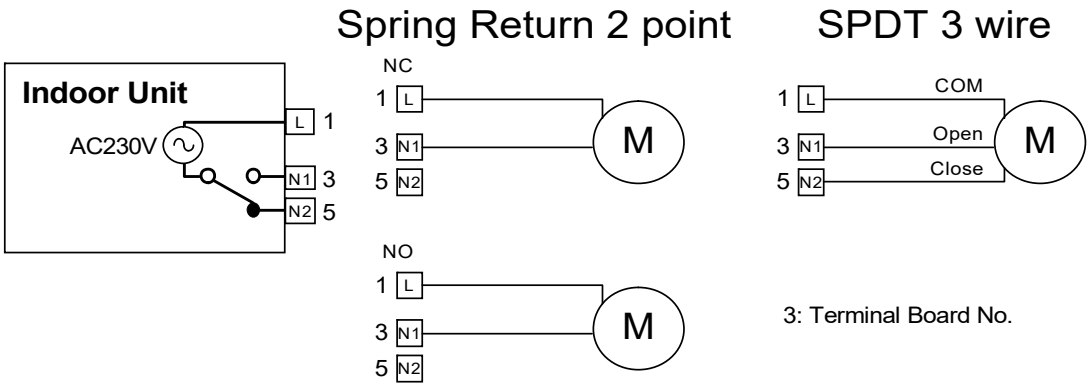
- Use to allow hot water to floor heating panel or block cold water to floor heating panel.

Control contents:

1 When indoor running in heat mode, OPEN the 2 ways valve.

2 When indoor running in cool mode, CLOSE the 2 ways valve.

3 Stop condition:
   a. During stop mode, fix 2 ways valve in close condition.

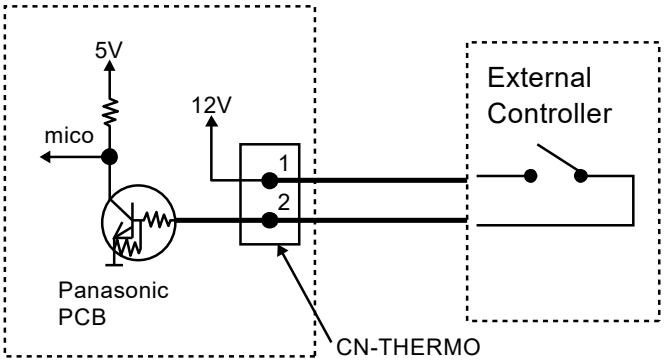


* During pump down mode, fix 2 ways valve in close condition.
* During force mode, open 2 ways valve.

* Recommended Parts : SFA 21/18 (Siemens)

14.23 External OFF/ON Control

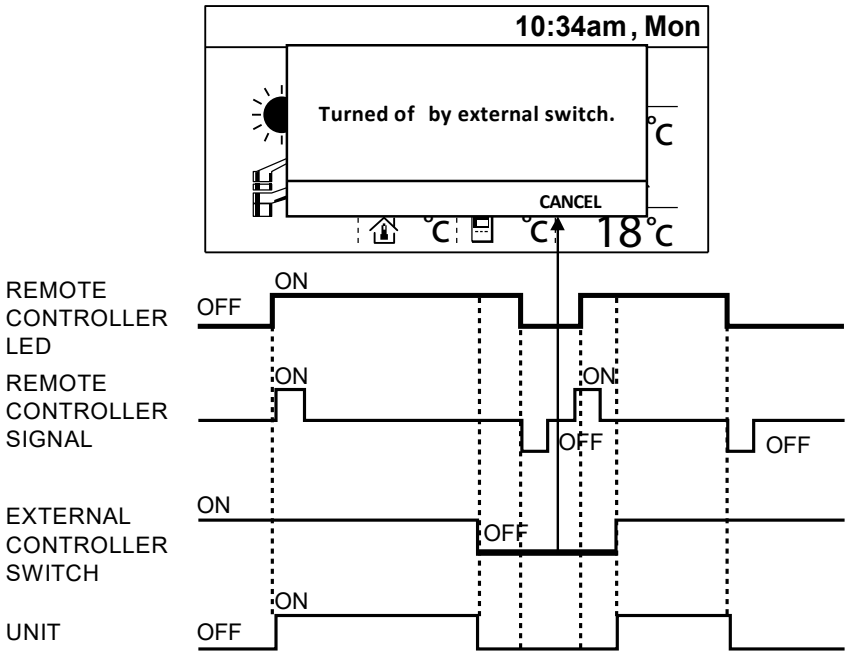
- Communication circuit between indoor unit and external controller is as per below.



- Maximum length of communication cable is 50 meter.
- Control content:

External Control Switch	Control Panel OFF/ON	Control Panel Power LED	System Status
ON	ON	ON	ON
ON	OFF	OFF	OFF
OFF	ON	ON	OFF
OFF	OFF	OFF	OFF

Remocon Screen Display and Control Detail:



When External SW connection select "YES" from remocon installer menu:

- Heating or Cooling system will operate normally if the External Switch signal is ON.
- Once the External Switch turn OFF, System Turn OFF (Heat pump, water pump, heater etc…)
- Remocon LED remain ON or OFF according to the current operation request.
- Pop up menu at remocon main screen as above screen to inform customer system stop by External Switch.
- It is possible to press cancel and return to main screen to do change of operation setting while waiting the External Switch turn ON back.
- Remocon LED will always follow the latest changes from remocon.
- If no action on remocon for continuous 5 minutes, the pop up screen will show again on the screen.
- But once the External Switch Turn ON back, pop up screen will disappear and system can operate normally according to the latest operation setting and request.

14.24 External Compressor Switch (Optional PCB)

External compressor switch port can have two purpose of control as below:

- Heat source ON/OFF function (Dip switch Pin 3 on PCB "OFF")
- Heater ON/OFF function (Dip switch Pin 3 on PCB "ON")

- Heat source ON/OFF function

Purpose:

- Heat pump ON/OFF function is use to turn OFF the high power consumption device (Heat pump, & Heater) when there is energy or electric current limitation. Other optional function still can be operate under heat pump and heater OFF condition.

Control Detail:

- This External Compressor Switch is possible to connect to Optional PCB only.
- Once the remocon select External Compressor Switch connection "YES", & Dip Switch on PCB "OFF" This heat pump ON/OFF function will activate
- The ON/OFF signal of this External Compressor Switch is same as External Switch.
- When the External Compressor Switch is ON:
  - Heat pump system operate normally
- When the External Compressor Switch is OFF:
  - Heat pump, Indoor water pump & Heater (Booster heater & Backup Heater) need to turn OFF
  - Solar, Boiler and zone control can be operate follow normal control condition.
    * pump delay OFF also included in this control

  (There is NO pop up screen like External Switch when this External Compressor Switch is OFF.)

- Heater ON/OFF function

Purpose:

- Heater ON/OFF function is use to turn OFF the heater (backup heater & booster heater) when there is energy or electric current limitation. Heat pump and other optional function still can operate.

Control detail:

- This External Compressor Switch is possible to connect to Optional PCB only.
- Once the remocon select External Compressor Switch connection "YES", & Dip Switch on PCB "ON" This heater ON/OFF function will activate
- When the External Compressor Switch is ON:
  - Heat pump and heater operate normally
- When the External Compressor Switch is OFF:
  - Backup heater and booster heater cannot operate even heater request is ON.
  - Heat pump and option function (Solar, Boiler and zone control) can be operate follow normal control condition.

  (There is NO pop up screen like External Switch when this External Compressor Switch is OFF.)

14.25 Heat/Cool Switch (Optional PCB)

Purpose:

- User can switch the running mode from heat to cool or cool to heat through external installed Heat/Cool switch. This kind of heat / cool switch may built in inside the field supply room remocon as well.

Control contents:

- Heat/Cool Switch can only be set when Cool Function is "enable" at custom menu setting, & Extension PCB select "YES" & Zone 1 not set "Pool" condition.
- This heat/cool switch control will be activate only when installer set the Heat/Cool Switch "USE" through remocon.
- Once the Heat/Cool Switch Set "USE", remocon will check indoor send Signal to judge the option of mode select.
  - When Heat/Cool Switch Contact Open : Remocon only can select Heat Mode, or Heat + Tank Mode, or Tank Mode
  - When Heat/Cool Switch Contact Close : Remocon only can select Cool Mode, or Cool + Tank Mode, or Tank Mode

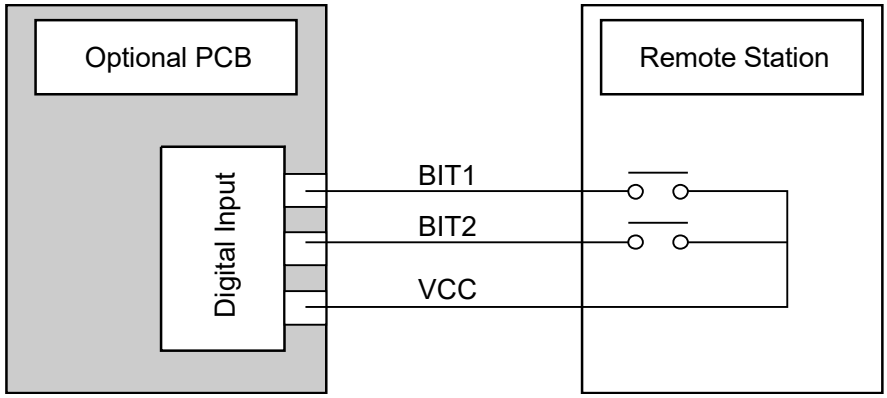
- Operation ON/OFF will depend on remocon request.
- When Heat Mode is running with Contact Open, user change this setting to contact close, indoor will this signal to remocon judge and change mode to cool and send back to indoor. And it is same as from cool mode change to heat mode.
  * This switch have higher priority, remocon follow indoor send signal when control activated.
  * There is no effect to the operation when the mode running is only Tank Mode.

  (Weekly Timer are ignored and cannot be set during Heat / Cool Switch is "Enable" Condition.)

14.26 SG Ready Control (Optional PCB)

Purpose:

- To set ON/OFF of heat pump and target temperature by digital input of third party device if necessary in field.



Remote control setting

For this function, following items need to be set on R/C (installer menu) –

- SG control = YES or NO
- Capacity up setting 1
  - Heating capacity [50 ~ 150 %]
  - DHW capacity [50 ~ 150 %]
  - Cooling capacity [-15 ~ 0 %]
- Capacity up setting 2
  - Heating capacity [50 ~ 150 %]
  - DHW capacity [50 ~ 150 %]
  - Cooling capacity [-15 ~ 0 %]

Control contents:

If SG control on remote control = "Yes", then following control only activate by digital input.

- **While Digital input is " 00 " (Normal operation)**
  - Normal operation. Once detect '00' system will operate back to normal condition.
    (All the target set temperature for heating side and DHW side will return back to previous set temperature when digital signal change from "10' or"11" back to "00".)
- **While digital input is detected " 01 " (HP stop)**
  - Heat pump & room heater & tank heater cannot operate.
    (Solar control and Boiler back up and 2 Zone control can activate.)
- **While digital input is detected " 10 " (Capacity 1)**
  - Target temperature for heating and DHW Tank is changed according to the percentage set by Remote control setting. However, which setting temperature is change depend on system setting.
  - Target temperature of cooling is change according to the adjustment value set by remocon setting.
- **While digital input is detected " 11 " (Capacity 2)**
  - Target temperature for heating and DHW Tank is changed according to the percentage set by Remote control setting. However, which setting temperature is change depend on system setting.
  - Target temperature of cooling is change according to the adjustment value set by remocon setting.
- **While digital input is detected " 10 " (Capacity 1)**
  - Setting temperature for heating and Tank is changed.
    However, which setting temperature is change depend on system setting.

If Buffer selection is "YES"
Room side
New Target Buffer tank temperature = Current Target Buffer Tank Temperature * Remote Control setting (" capacity 1) %
* Max Min regulation is follow Target Buffer tank temperature control specification
** No change of Target zone water temperature, only set higher buffer tank temperature.

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 1) % * (Max regulation depend on the tank max setting limit)

If Buffer selection is "NO"
Room side
New Target Zone Water Temperature = Current Target Zone Water Temperature * Remote Control Setting (*Capacity 1) %
(Zone 1 and Zone 2 will change according to its own target zone water temperature.)
(Max regulation depend on the temperature control type select)

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 1) % * (Max regulation depends on the tank max setting limit)

- **While digital input is detected " 11 " (Capacity 2)**
  - Setting temperature for heating and Tank is changed.
    However, which setting temperature is change depend on system setting.

If Buffer selection is "YES"
Room side
New Target Buffer tank temperature = Current Target Buffer Tank Temperature * Remote Control setting (" capacity 2) %
* Max Min regulation is follow Target Buffer tank temperature control specification
** No change of Target zone water temperature, only set higher buffer tank temperature.

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 2) % * (Max regulation depends on the tank max setting limit)

If Buffer selection is "NO"
Room side
New Target Zone Water Temperature = Current Target Zone Water Temperature * Remote Control Setting (*Capacity 2) %
(Zone 1 and Zone 2 will change according to it's own target zone water temperature.)
(Max regulation depend on the temperature control type select)

DHW Tank side
New Tank Set Temperature = Tank Set Temperature * Remote Control setting for DHW ("Capacity 2) % * (Max regulation depends on the tank max setting limit)
** This function is not applicable for Cooling mode.

14.27 Demand Control (Optional PCB)

Remote control setting:
- When Optional PCB connection select 'YES", Demand Control function can select "YES" or "NO".

Purpose:
- After the demand control select YES, below control will activated.
  - 0-10V Demand control

0-10V Demand control
- Demand control is use to reduce the current usage of heat pump unit by third party device.

Control start condition:
- Select "YES" at Demand control at installer menu.
- 0-10V input for this electrical current control is detected.

Control content:

- If start condition is fulfilled, indoor will receive the voltage signal from optional PCB. Indoor will send the rate value to outdoor unit.
- Outdoor will change the current limit according to the percentage receive from indoor unit.

14.28 Holiday Mode

- Purpose:
  Promotes energy saving by allowing the user to stop the system during holiday and enables the system to resume at the preset temperature after holiday.

- Control details:
  - Indoor operate the unit according running mode request. Target temperature will follow holiday setting temperature.
    - If heat mode request is receive, Target Water Out Temperature will change according to holiday shift temperature set.
      [If heat is set OFF at holiday, unit, water pump and zone control will OFF]
    - If tank mode request is receive, Target Tank Set Temperature will change according to the holiday tank shift temperature set.
      [If tank is set OFF at holiday, heat pump and tank heater will OFF]
  - After days of holiday have been set, heat pump will stop and only resume operation at the end of holiday countdown.

- Start condition:
  - Holiday timer set and the holiday timer start
    * The day holiday mode was set is counted as day 1.

- Stop condition:
  - OFF/ON button is pressed.
  - Holiday timer is reached.

14.29 Dry Concrete

- Purpose
  Provide heat to floor heating panel and dry the wet concrete during installation.

- Setting condition:
  - Dry concrete parameter can be set through remote control under system setup.
  - Parameters are possible to set up to 99 days with different target set temperature

- Control details:
  - Dry concrete mode will be activates when select ON from service setup.
  - Once start dry concrete function, remote control will send step 1 setting temperature to indoor unit.
    * This temperature is set at zone temperature. If system is 2 zones, both zone target temperature is set as same temperature.
  - Heat pump will start heat mode operation to room side with received target water outlet temperature.
    * Heat pump will operate according to Heat pump Target Water Temperature.
  - After complete day 1 setup operation, day 2 data will be send to indoor at 12.00am on the second day.
  - Each preset data will be send every day until dry concrete mode is complete, unit will turns OFF and exit dry concrete function.
  - 3 ways valve and booster heater will turn OFF and 2 ways valve will turns ON.

- Cancel condition:
  - Dry concrete mode is complete and OFF signal is received.
  - OFF signal is received by pressing OFF/ON button.

14.30 Flow Sensor

- The water flow sensor serves as an overload protector that shuts down the unit when the water level is detected to be low.
- Abnormal flow detection:

Sequence	Abnormal flow	Normal flow
Normal case	Flow rate < 7 l/min or ≥ 69 l/min	≥ 7 l/min
During status 2~6 on Anti-freeze deice	Flow rate ≥ 7 l/min	< 7 l/min

15. Protection Control

15.1 Protection Control for All Operations

15.1.1 Time Delay Safety Control

1 The compressor will not start for three minutes after stop of operation.

15.1.2 30 Seconds Forced Operation

1 Once the compressor starts operation, it will not stop its operation for 30 seconds.
2 However, it can be stopped using control panel at indoor unit.

15.1.3 Total Running Current Control

1 When the outdoor running current exceeds X value, the compressor frequency will decrease.
2 If the outdoor running current does not exceed X value, the compressor frequency will return to normal operating frequency.
3 If the outdoor running current continue to increase till exceed Y value, compressor will stop, and if this occurs 3 times within 20 minutes, system will stop operation and OFF/ON control panel LED will blink (F16 error occurs).

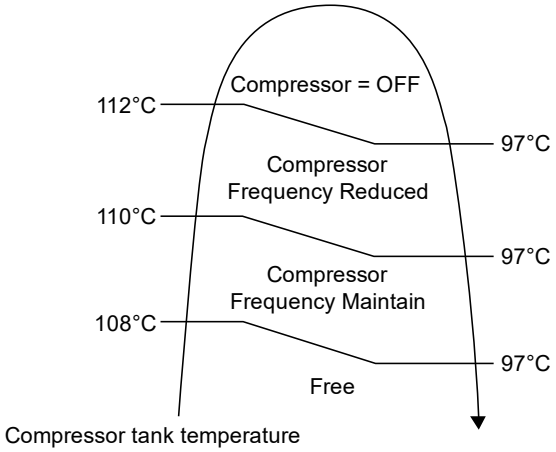
	WH-WDG05LE5		WH-WDG07LE5 / WH-WDG09LE5	
Operation Mode	X (A)	Y (A)	X (A)	Y (A)
Heating	12.0	15.0	14.8	18.0
Cooling	12.0	15.0	14.0	18.0

A. DC Peak Current Control

1 When the current to IPM exceeds set value of 26 A, compressor will stop. Compressor will restart after three minutes.
2 If the set value exceeds again for more than 30 seconds after the compressor restarts, operation will restart after two minutes.
3 If the set value exceeds again for within 30 seconds after the compressor restarts, operation will restart after one minute. If this condition repeats continuously for seven times, system will stop operation and OFF/ON control panel LED will blink (F23 error occurs).

15.1.4 Compressor Overheating Prevention Control

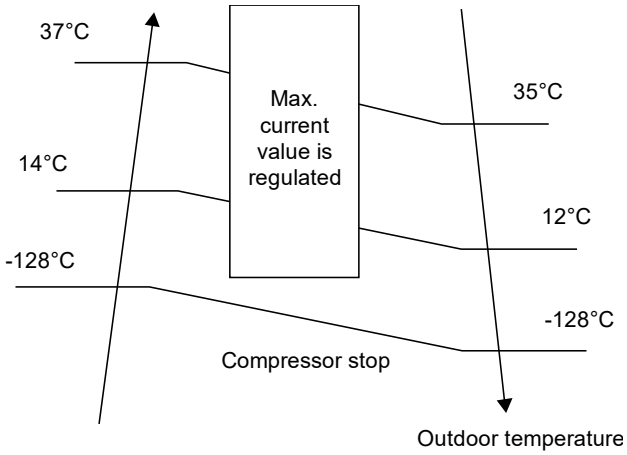
- The compressor operating frequency is regulated in accordance to compressor tank temperature as shown in below figures. When the compressor tank temperature exceeds 107°C, compressor will stop, and if this occurs 4 times within 30 minutes, system will stop operation and OFF/ON control panel LED will blink (F20 error occurs).



15.1.5 High Pressure Sensor Control

- Purpose:
  - To protect the system operation.
- Detection period:
  - After compressor on for 1 minute.
- Detection conditions:
  - When abnormal high voltage detection, 5 V or when open circuit detection 0V for 5 seconds continuously.
- After detection:
  - When abnormality is detected 4 times within 120 minutes, unit stop operation.
  - OFF/ON control panel LED will blink (H64 error occurs).

15.1.6 Outside Temperature Current Control



15.1.7 Pre-Heat Control

- Purpose:
  - For compressor protection during low outdoor ambient operation (during heating low temperature operation).
- Control content:
  a. Trigger Pre-Heat Control condition
     - When the outdoor air temperature is below than 4°C, and discharge temperature is 6°C or below.
  b. Resetting Pre-Heat Control condition
     1. When the outdoor air temperature exceeds entry condition (2°C)
     2. When the discharge temperature exceeds entry condition (2°C)

15.2 Protection Control for Heating Operation

15.2.1 Outdoor Air Temperature Control

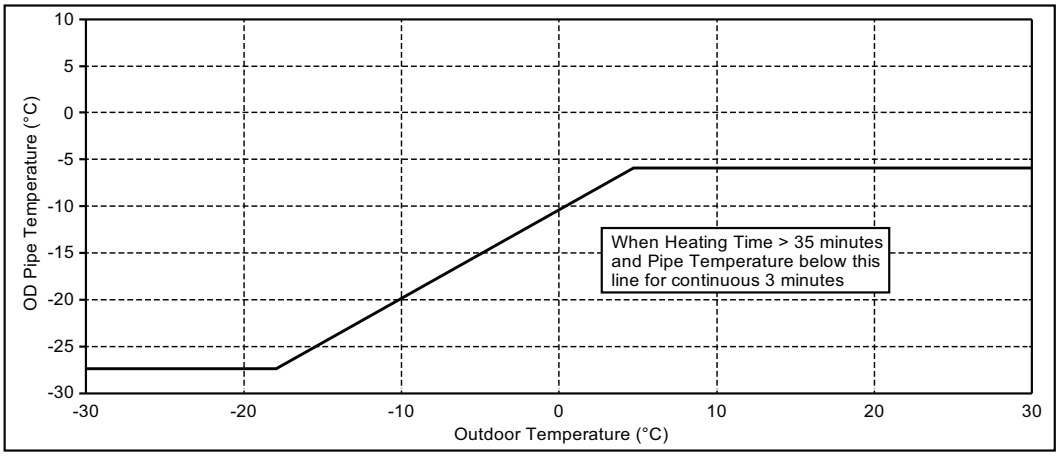
The maximum current value is regulated when the outdoor air temperature rises above 14°C in order to avoid compressor overloading.

15.2.2 Deice Operation

When outdoor pipe temperature and outdoor air temperature is low, deice operation start where outdoor fan motor stop.

- Deice judging condition

Outdoor Unit Deice Control

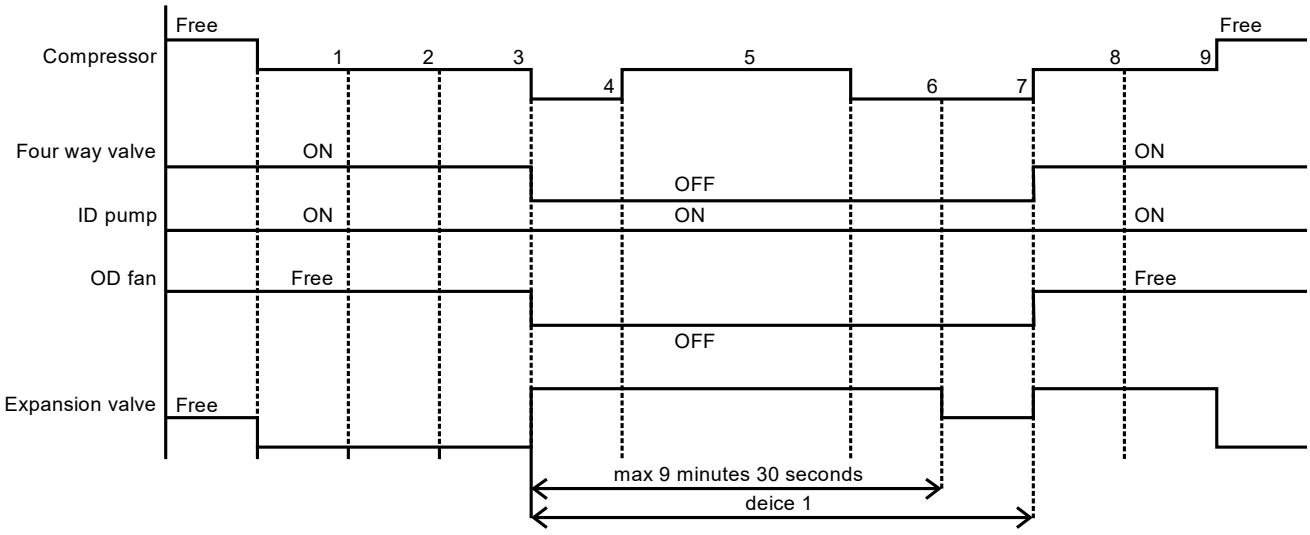


Deice start depends:

1. Outdoor air sensor temp.
2. Outdoor pipe sensor temp.
3. Heating accumulation time

- Deice operation time diagram

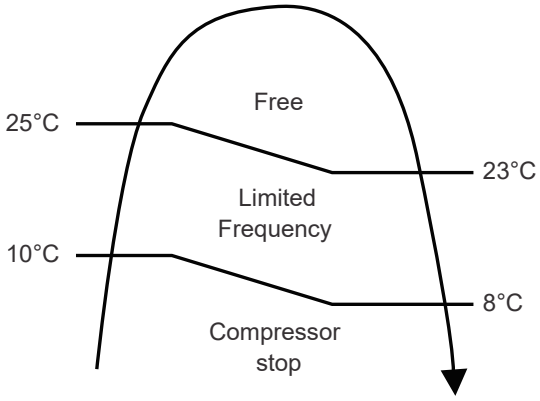
a. Deice mode 1 control:



15.3 Protection Control for Cooling Operation

15.3.1 Outdoor Air Temperature Control

- The Compressor operating frequency is regulated in accordance to the outdoor air temperature as shown in the diagram below.
- This control will begin 1 minute after the compressor starts.
- Compressor frequency will adjust base on outdoor air temperature.



15.3.2 Freeze Prevention Control 1

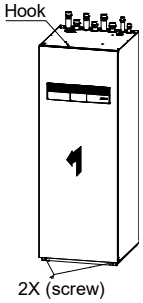
1 When indoor heat exchanger temperature is lower than 0°C continuously for 10 seconds, compressor will stop operating.
2 Compressor will resume its operation three minutes after the indoor heat exchanger is higher than 1°C.
3 Indoor heat exchanger freeze prevention (H99) will memory in error history.

16. Servicing Guide

16.1 How to Take Out Front Plate

Open and Close Front Plate ⑱

1 Remove the 2 mounting screws of Front Plate ⑱.
2 Slide it upwards to unhook the Front Plate ⑱ hook.
3 Reverse above steps 1~2 for close it.



⚠ CAUTION
Open or close the Front Plate carefully. The heavy Front Plate may injures the fingers.

* The remote control cable is connected to the front panel, so be careful when removing the panel.

16.2 Test Run

1 Before test run, make sure below items have been checked:-
   a) Pipework are properly done.
   b) Electric cable connecting work are properly done.
   c) Tank Unit is filled up with water and trapped air is released.
   d) Please turn on the power supply after filling the tank until full.
2 Switch ON the power supply of the Tank Unit. Set the Tank Unit RCCB/ELCB to “ON” condition. Then, please refer to the Operation Instruction for operation of Remote Controller ①.

Note:
- During winter, turn on the power supply and standby the unit for at least 15 minutes before test run. Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.

3 For normal operation, Water Pressure reading should be in between 0.5 bar and 3 bar (0.05 MPa and 0.3 MPa) If necessary, adjust the Water Pump ④ SPEED accordingly to obtain normal water pressure operating range. If adjust Water Pump ④ SPEED cannot solve the problem, contact your local authorized dealer.
4 After test run, please clean the Magnetic Water Filter Set ⑨ and Water Filter Set ㉚. Reinstall it after finish cleaning.

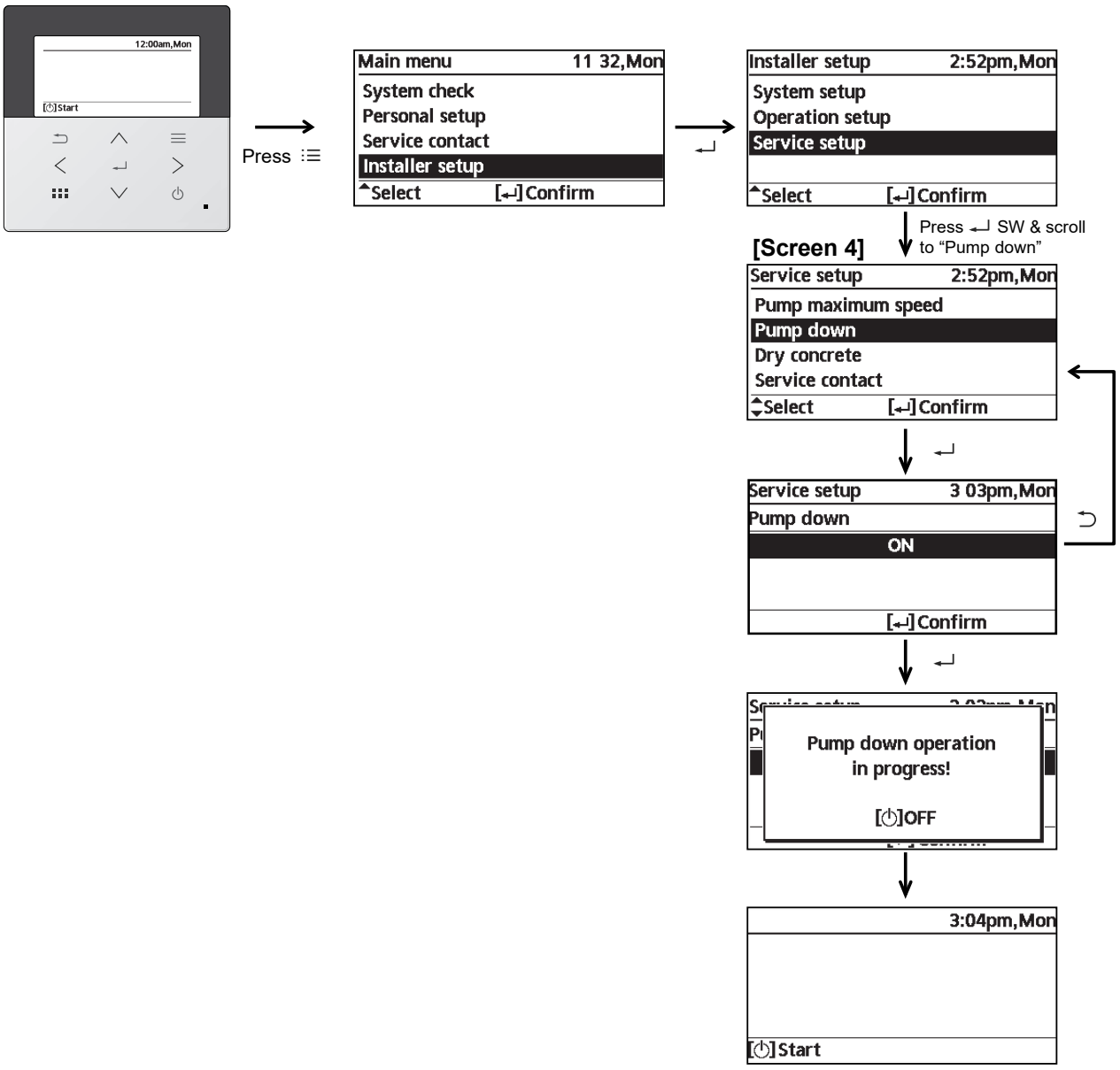
16.3 Expansion Vessel ⑬ Pre Pressure Checking

For Space Heating / Cooling

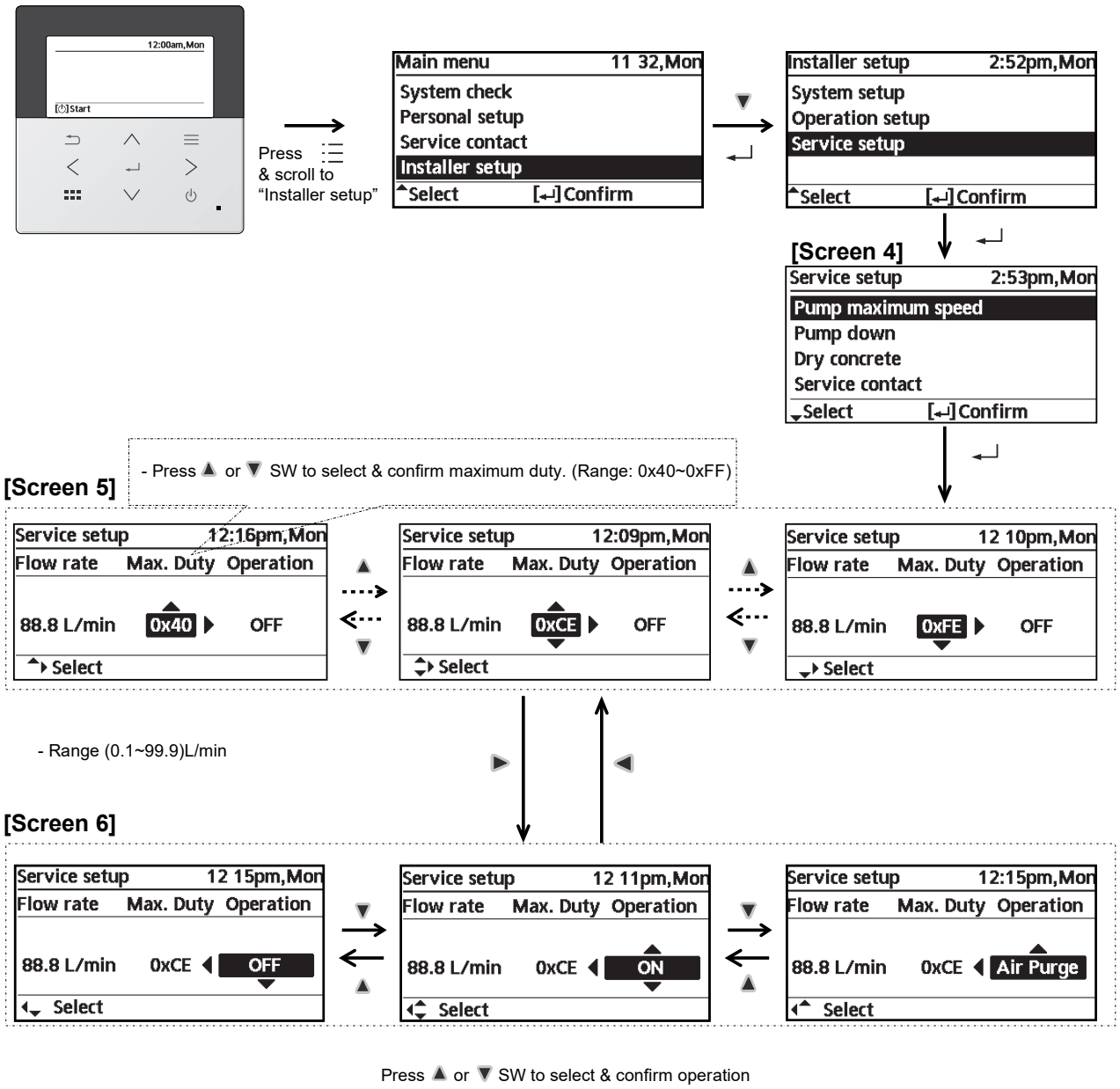
- Expansion Vessel ⑬ with 10 L air capacity and initial pressure of 1 bar is installed in this Tank Unit.
- Total amount of water in system should be below 200 L.
  (Inner volume of Tank Unit’s piping is about 5 L)
- If total amount of water is over 200 L, please add another expansion vessel. (field supply)
- Please keep the installation height difference of system water circuit within 10 m. (Extra pump may be required)

16.4 Pump Down Procedures

Refer below steps for proper pump down procedure.



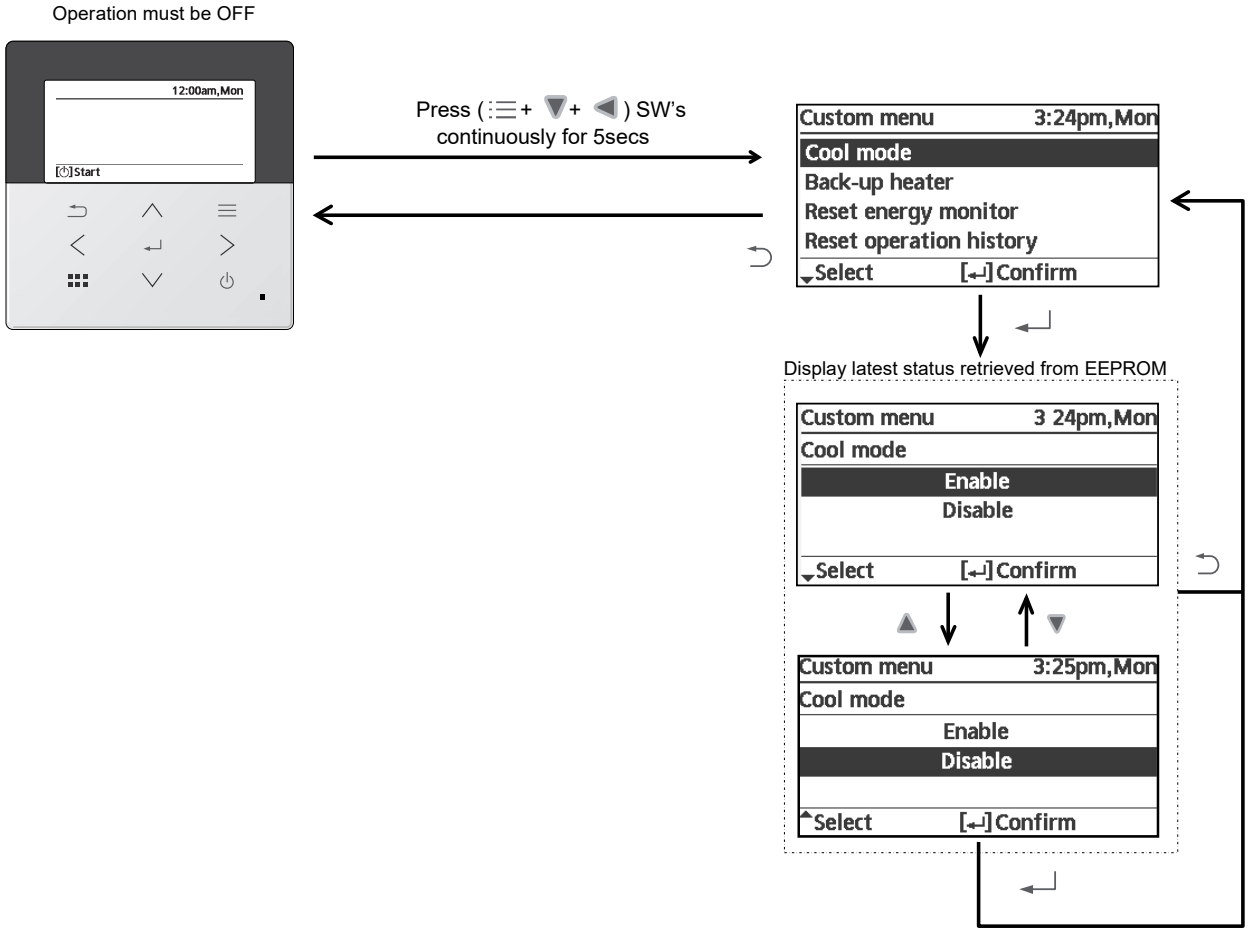
16.5 How to Adjust Pump Speed



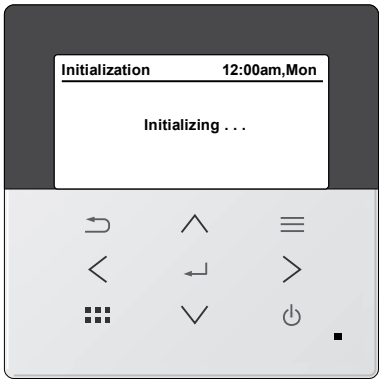
NOTE:

1. Whenever at [Screen 5], if press ⏻ SW to OFF, pump operation should be turned OFF.
2. Whenever at [Screen 6], if press ⏻ SW to OFF, pump operation should be turned OFF.

16.6 How to Unlock Cool Mode

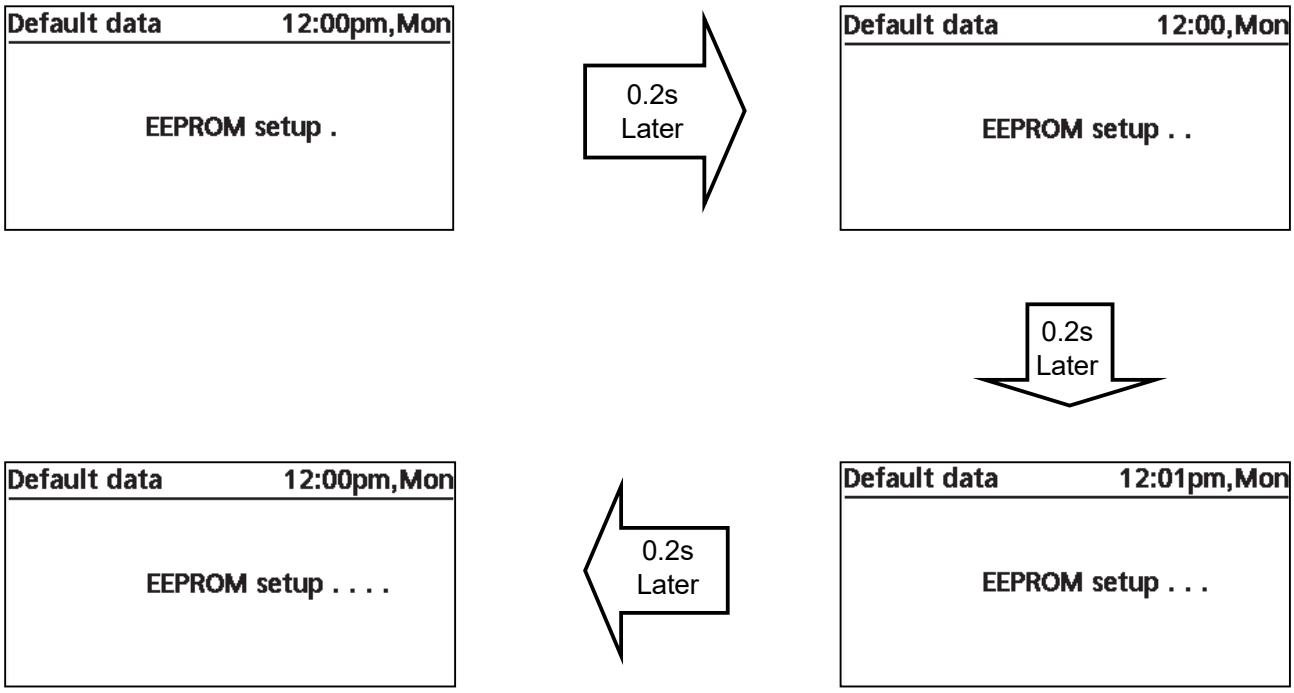


16.7 EEPROM Factory Default Data Setup Procedure

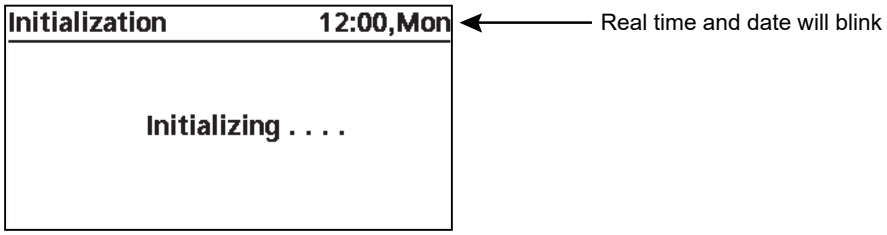


- EEPROM default data setup is only possible during initialization process.
- Press (▲ , ▼ , ◀ , ▶) simultaneously for 5secs continuously, initialization process will stop & EEPROM default data setup process will start.

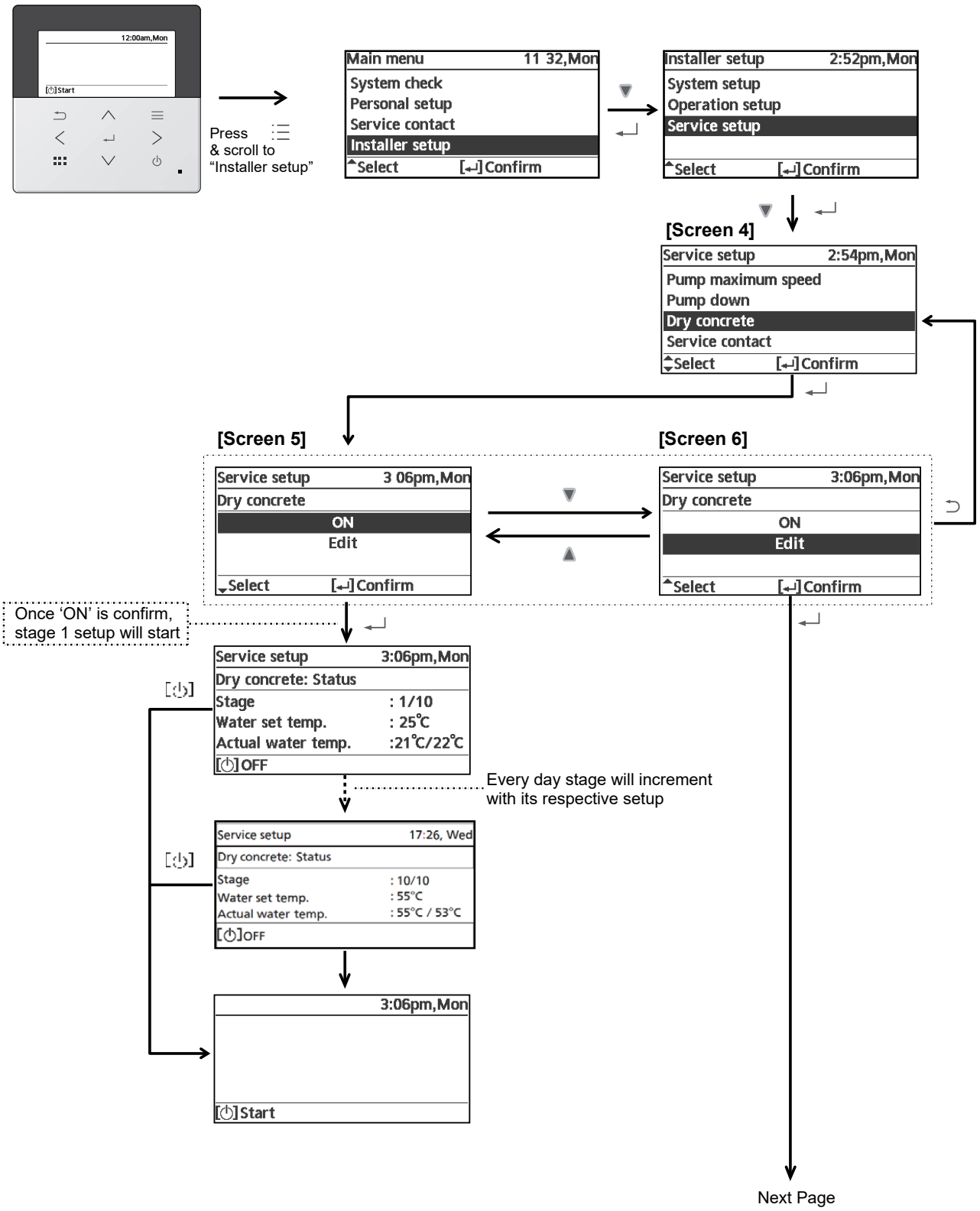
During EEPROM default data setup process, display should be as shown below.

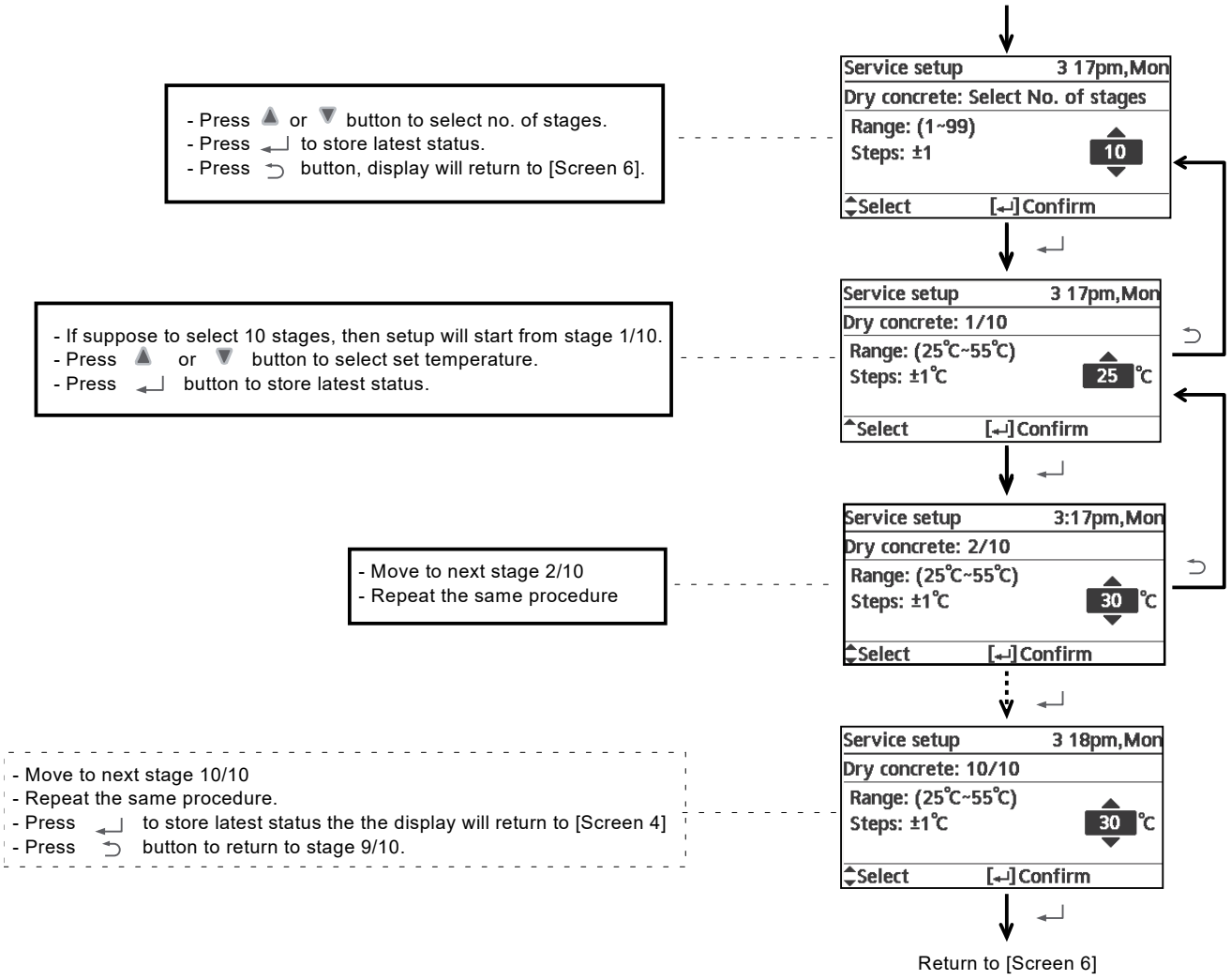


- Once EEPROM default data setup process is complete, initialization process will re-start from beginning.



16.8 Dry Concrete Setup



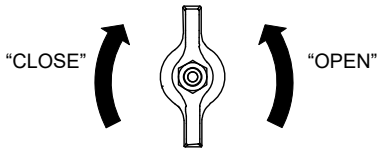

- Press ▲ or ▼ button to select no. of stages.
- Press ↵ to store latest status.
- Press ↩ button, display will return to [Screen 6].
Service setup 3 17pm,Mon
Dry concrete: Select No. of stages
Range: (1~99)
Steps: ±1
10
Select [↵]Confirm
- If suppose to select 10 stages, then setup will start from stage 1/10.
- Press ▲ or ▼ button to select set temperature.
- Press ↵ button to store latest status.
Service setup 3 17pm,Mon
Dry concrete: 1/10
Range: (25℃~55℃)
Steps: ±1℃
25 ℃
Select [↵]Confirm
- Move to next stage 2/10
- Repeat the same procedure
Service setup 3:17pm,Mon
Dry concrete: 2/10
Range: (25℃~55℃)
Steps: ±1℃
30 ℃
Select [↵]Confirm
- Move to next stage 10/10
- Repeat the same procedure.
- Press ↵ to store latest status the the display will return to [Screen 4]
- Press ↩ button to return to stage 9/10.
Service setup 3 18pm,Mon
Dry concrete: 10/10
Range: (25℃~55℃)
Steps: ±1℃
30 ℃
Select [↵]Confirm
Return to [Screen 6]

17. Maintenance Guide

In order to ensure safety and optimal performance of the Tank Unit, seasonal inspections on the Tank Unit, functional check of RCCB/ELCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer. Contact dealer for scheduled inspection.

1 Charging and Discharging the Water
Make sure all the piping installations are properly done before carry out below steps.
Charge the Water

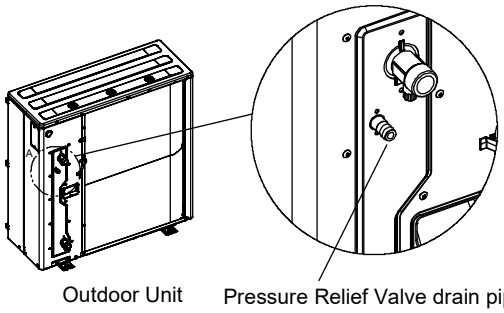
- For Domestic Hot Water Tank
  a. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

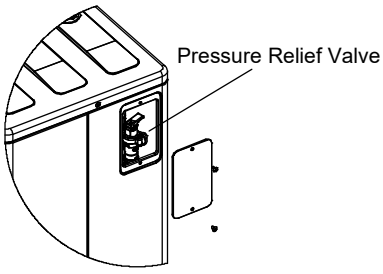


Domestic Hot Water Tank Discharge (Drain Tap) ⓖ

  b. Set all Tap / Shower “OPEN”.
  c. Start filling water to the Domestic Hot Water Tank via Tube Connector ⓒ.
  After 20~40min, water should flow out from Tap / Shower.
  Else, please contact your local authorized dealer.
  d. Check and make sure no water leaking at the tube connecting points.
  e. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “OPEN” for 10 seconds to release air from this pipeline. Then set it “CLOSE”.
  f. Turn the Safety Relief Valve ㉕ knob counterclockwise slightly and hold for 10 seconds to release air from this pipeline. Then recover the knob to original position.
  g. Ensure Step 5 & 6 is carried out each time after charging water to Domestic Hot Water Tank.
  h. To prevent back pressure from happening to the Safety Relief Valve ㉕, do turn the Safety Relief Valve ㉕ knob counterclockwise.

- For Space Heating / Cooling
  a. Start filling water (with pressure more than 1 bar (0.1MPa)) to the Space Heating/Cooling circuit via Tube Connector ⓐ.
  b. Stop filling water if the free water flow through Pressure Relief Valve drain pipe. (Check the Outdoor Unit)
  c. Turn ON the Tank Unit.
  d. Remote control menu → Installer setup → Service setup → pump maximum speed → Turn on the pump.
  e. Make sure Water Pump ④ is running.
  f. Check and make sure no water leaking at the tube connecting points.
  g. Reinstall the Cabinet front plate f by tightening the 2 mounting screws.



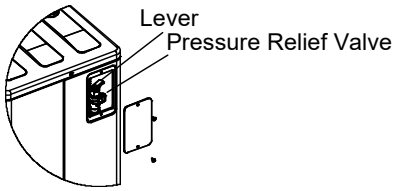


Discharge the Water

- For Domestic Hot Water Tank
  a. Turn OFF power supply.
  b. Set the Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “OPEN”.
  c. Open Tap / Shower to allow air inlet.
  d. Turn the Safety Relief Valve ㉕ knob counterclockwise slightly and hold it until all air is released from this pipeline. Then recover the knob to original position after ensured the pipeline is emptied.
  e. After discharge, set Domestic Hot Water Tank Discharge (Drain Tap) ⓖ to “CLOSE”.

2 Check Water Pressure *(1 bar = 0.1 MPa)
Water pressure should not lower than 0.5 bar (with inspects the Water Pressure from Remote Controller). If necessary add water into Tank Unit (via Tube Connector ⓐ).

3 Check Pressure Relief Valve
*Pressure Relief Valve is mounted in the Outdoor Unit.
- Confirm that the pressure relief valve is working properly, Pull the lever horizontal direction.
- Release the lever when water comes out of the drain pipe of the pressure relief valve. (While the air continues to come out of the drain pipe, keep raising the lever to completely discharge the air.)
- Confirm that the water from the drain pipe stops.
- If water is leaking, pull the lever several times and return it to make sure the water stops.
- If water keeps coming out of the drain, drain water.
- Turn off the system and contact your local authorized dealer.



4 Check Air Accumulation
- Open the air vent plugs on the heating panel, fan convector, etc., and remove the air accumulated in the equipment and piping.
- If the outdoor unit and the indoor unit are installed on different floors, open the air vent plug on the water plug of the outdoor unit and the air vent plug on the heater bottle inside the indoor unit to remove the air. (be careful, water will come out)

5 Indoor Unit Control Board Area
Thorough visual inspection of the control board and look for defects, i.e. loose connection, melting of wire insulator and etc.

6 RCCB/ELCB
Ensure the RCCB/ELCB set to “ON” condition before check RCCB/ELCB.
Turn on the power supply to the Tank Unit.
This testing could only be done when power is supplied to the Tank Unit.

⚠ WARNING
Be careful not to touch parts other than RCCB/ELCB test button when the power is supplied to Tank Unit. Else, electrical shock may happen. Before obtaining access to terminals, all supply circuits must be disconnected.

- Push the “TEST” button on the RCCB/ELCB. The lever would turn down and indicate “0”, if it functions normal.
- Contact authorized dealer if the RCCB/ELCB malfunction.
- Turn off the power supply to the Tank Unit.
- If RCCB/ELCB functions normal, set the lever to “ON” again after testing finish.

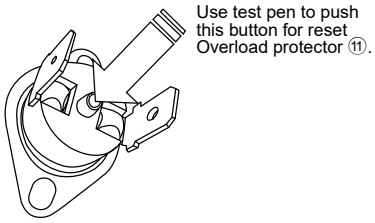
7 Reset Overload Protector

Overload Protector serves the safety purpose to prevent the water over heating. When the Overload Protector trip at high water temperature, take below steps to reset it.

a. Take out the cover.

b. Use a test pen to push the centre button gently in order to reset the Overload Protector.

c. Fix the cover to the original fixing condition.



8 Maintenance for Magnetic Water Filter Set ⑦

a. Turn OFF power supply.

b. Place a container below Magnetic Water Filter Set ⑦.

c. Turn to remove the Magnet Bar at bottom of Magnetic Water Filter Set ⑦.

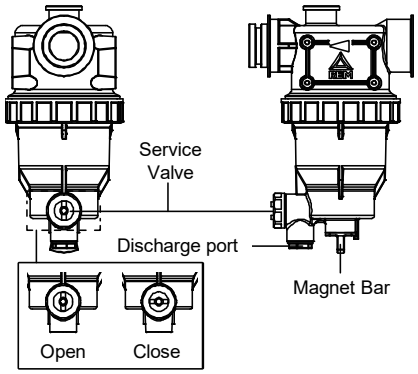
d. By using Allen key (8mm), remove the Cap of Discharge Port.

e. By using Allen Key (4mm), open the Service Valve to release the dirty water from the Discharge Port into a container. Close the service valve when the container is full to avoid spillage in the tank unit. Dispose the dirty water.

f. Reinstall the Cap of Discharge Port and Magnet Bar.

g. Re-charging the water to Space Heating / Cooling circuit if necessary (refer Section 5 for details.)

h. Turn ON power supply.



9 Maintenance for Safety Relief Valve ㉕

It is strongly recommended to operate the valve by turn the knob counter clockwise to ensure free water flow through discharge pipe at regular intervals to ensure it is not blocked and to remove lime deposit.

Stagnant water in Tank Unit should be drained if it is not going to be operated for more than 60 days.

17.1 Maintenance for Magnetic Water Filter Set

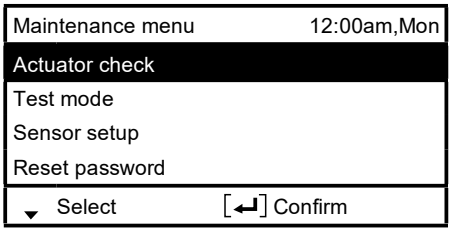
17.1.1 Service and Maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(CAUTION) Only display when it is locked by password.

17.1.2 Maintenance Menu

Setting method of Maintenance menu



Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

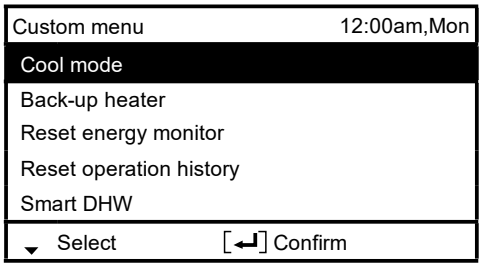
② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated. It affects temperature control.

④ Reset password (Reset password)

17.1.3 Custom Menu

Setting method of Custom menu



Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it.
In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer.
When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history)
Please use when moving house and handover the unit.

⑤ Smart DHW (Set Smart DHW mode Parameter)
a) Start time: Tank reboil at lower ON Temp. onward.
b) Stop time: Tank reboil at normal ON Temp. onward.
c) ON Temp.: Tank Reboil Temp when Smart DHW start.

17.1.4 Specifications

17.1.4.1 Specifications of Fresh Water was Heat Transfer Medium in Brazed Heat Exchanger

Parameter	Quality Limits for Tap Water on the Secondary Side
Temperature	Below 60°C
pH	7 to 9
Alkalinity	60mg/l <HCO_3 <300mg/l
Conductivity	< 500µS/cm
Hardness	[Ca^+, Mg^+] / [HCO_3^-] > 0.5
Chloride	< 200mg/l at 60°C
Sulphate	[SO_4^{2-}] > 100mg/l and [HCO_3^-] / [SO_4^{2-}] > 1
Nitrate	NO_3 < 100mg/l
Chlorine	< 0.5mg/l

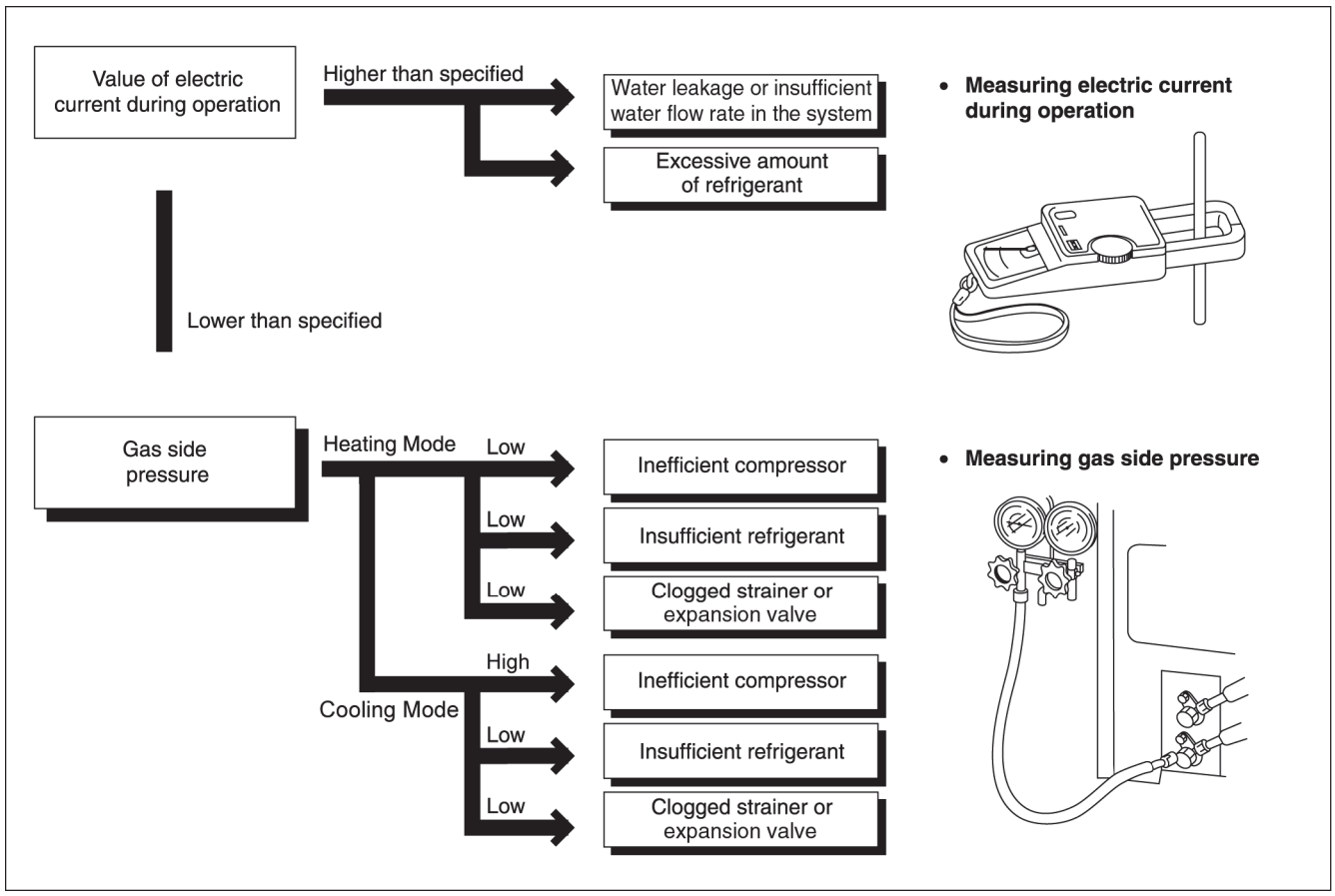
17.1.4.2 External Filter

Solids in the water must be filtered.
Minimum filter mesh size required for the field supply external filter in the water inlet is 20 mesh.

18. Troubleshooting Guide

18.1 Refrigeration Cycle System

In order to diagnose malfunctions, make sure that there are no electrical problems before inspecting the refrigeration cycle.
Such problems include insufficient insulation, problem with the power source, malfunction of a compressor and a fan.



18.2 Relationship between the Condition of the Air-to-Water Heatpump Indoor and Outdoor Units and Pressure and Electric Current

Condition of the Air-to-Water Heatpump indoor and outdoor units	Heating Mode			Cooling Mode		
	Low Pressure	High Pressure	Electric current during operation	Low Pressure	High Pressure	Electric current during operation
Water leakage or insufficient water flow rate in the system						
Excessive amount of refrigerant						
Inefficient compression						
Insufficient refrigerant (gas leakage)						
Outdoor heat exchange deficiency						
Clogged expansion valve or Strainer						

- Carry out the measurements of pressure, electric current, and temperature fifteen minutes after an operation is started.

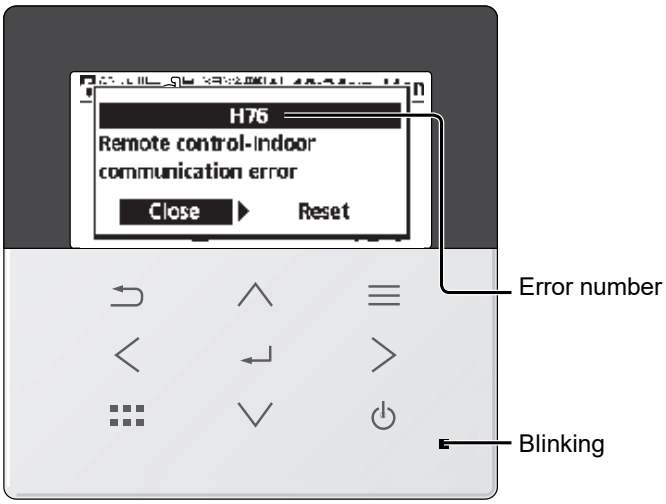
18.3 Breakdown Self Diagnosis Function

18.3.1 Self Diagnosis Function (Three Digits Alphanumeric Code)

- When abnormality occur during operation, the system will stop operation, and OFF/ON control panel LED will blink and error code will display on the control panel.
- Even error code is reset by turning OFF power supply or by selecting ERROR RESET, if the system abnormality is still unrepaired, system will again stop operation, and OFF/ON control panel LED will again blinks and error code will be display.
- The error code will store in IC memory.

- **To check the error code**
  1 When an abnormality occurs, system will stop operation and OFF/ON control panel LED will blink.
  2 Error code of the abnormality will be display on the control panel.
  3 To determine the abnormality description, the error code table needs to be referred.

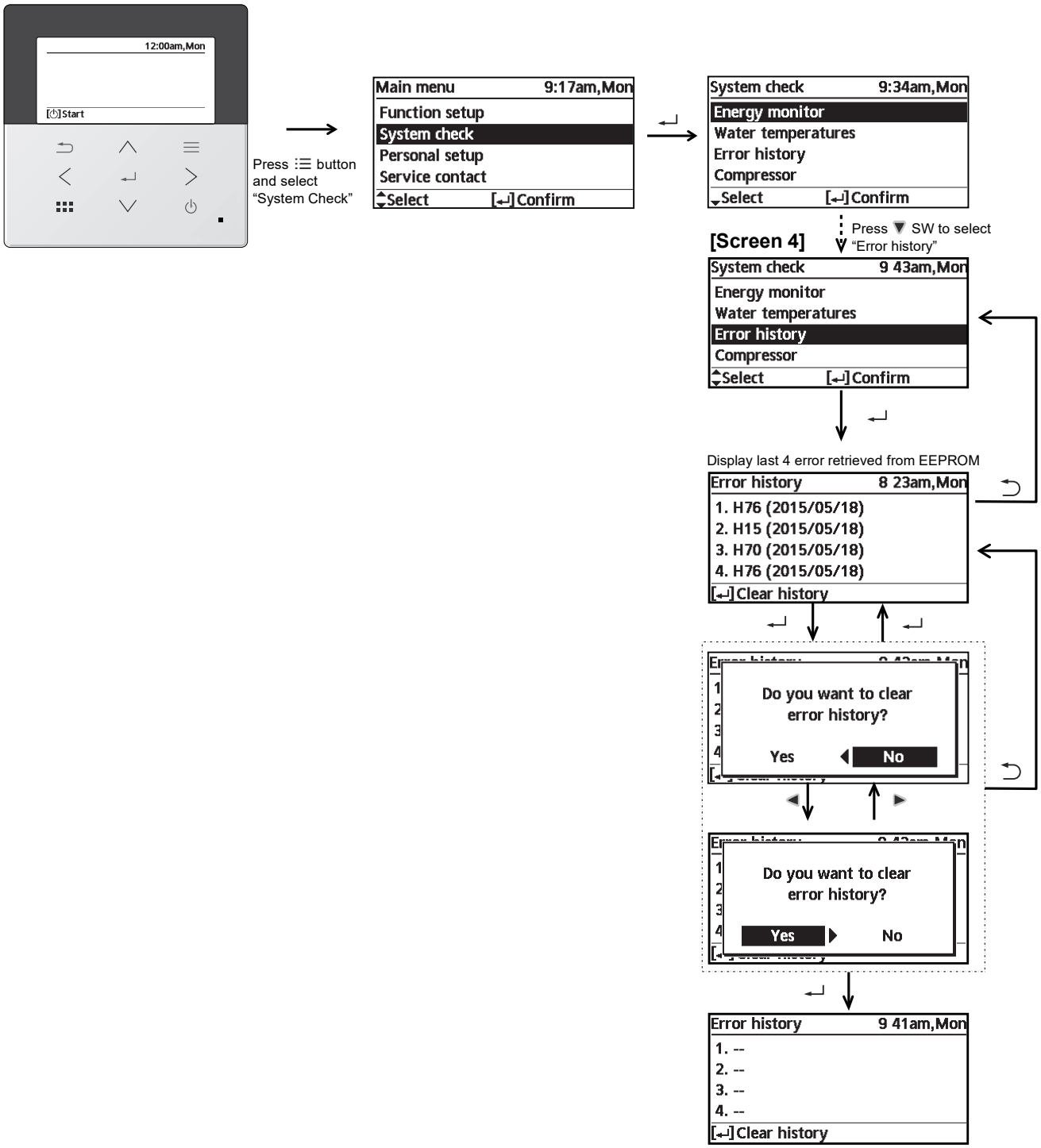
eg:



Press < > to select Close / Reset then press ↵

- **To display past/last error code**
  1 Turn ON power supply.
  2 Refer below procedure to retrieve the error code history.

- **To permanently delete error code from IC memory**
  1 Turn ON power supply.
  2 Refer below procedure to clear error history.


12:00am,Mon
Start
Press ☰ button and select "System Check"
Main menu 9:17am,Mon
Function setup
System check
Personal setup
Service contact
Select [↵]Confirm
System check 9:34am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Press ▼ SW to select "Error history"
[Screen 4]
System check 9 43am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Display last 4 error retrieved from EEPROM
Error history 8 23am,Mon
1. H76 (2015/05/18)
2. H15 (2015/05/18)
3. H70 (2015/05/18)
4. H76 (2015/05/18)
[↵]Clear history
Do you want to clear error history?
Yes No
Do you want to clear error history?
Yes No
Error history 9 41am,Mon
1. --
2. --
3. --
4. --
[↵]Clear history